

Futuretesting Quantitative Strategies

Daniel Bloch

28th of November 2023

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Abstract

Building quantitative strategies with a Backtesting engine leads to overfitting, which produces results that cannot be replicated in the future. To assess the robustness of a strategy and avoid overfitting, it is necessary to perform multiple backtests on different datasets. However, there is no market dataset spanning the whole time-space domain as we only observe a single realisation per security. We must therefore rely on generating synthetic data that is statistically close to the market data. We present the Futuretesting engine as a framework for generating realistic synthetic market data used for testing and optimising trading strategies, performing risk management, and analysing risk models. We utilise it to compute probabilistic Risk/Return profiles for a set of trading strategies everywhere in the time-space domain. We perform multiple futuretests on different datasets and under different scenarios to obtain a distribution of performance metrics. Given some accuracy measures, such as the mean-variance criterion, we can then find the optimum strategy maximising returns and minimising risk. Finally, we test this optimum strategy on real market data on different securities and over different periods of time.

First, we introduce quantitative principles and give an overview of quantitative trading strategies, focusing on trading with moving averages as well as trading with options. Next, we formulate the optimum trading problems, present backtesting strategies, discuss the calibration of trading rules, and introduce the modelling of trailing stops. Then, we present our probabilistic framework, detail the creation of market scenarios, present the tools used for measuring the performance of the strategies, and provide quantitative trading algorithms. To conclude, we consider a stochastic model to generate market returns, perform a Futuretesting on a few quantitative strategies, and display results.

Systematic Trading, Quantitative Strategies, Optimisation, Risk Analysis, Generating Data, Futuretesting

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1 Introduction

1.1 Overview

Trading is the process of repetitively taking positions in the markets by buying and selling securities, such as stocks, bonds, currencies or commodities, over short periods of time, to earn profits (see Harris [2002]). Quantitative trading consists in trading strategies based on quantitative analysis which rely on computer algorithms and programs based on mathematical models to (1) identify trading opportunities, and (2) capitalise on available trading opportunities (see Murphy [1999], Chan [2021]). The objective of quantitative trading is to calculate the optimal probability of executing a profitable trade. These models are first expressed as functions of information known about the future distribution of asset returns. Then, they are used for taking rational trading decisions. In general, we draw a distinction between:

- **Alpha:** Given the time t , it is the expected value of future asset returns conditioned on information up to time t . Practically, it is some function of prior information, designed by the trader or his team, that describes the expected future asset returns. Given time t , let $r(t, T)$ be the return of the traded asset over the time interval $[t, T]$, then $\alpha(t, T) = E[r(t, T) | \mathcal{F}_t]$ is the conditional expectation of those returns given the information set available to the trader. While the trader can take a guess at $\alpha(t, T)$, future returns are uncertain and we need a mathematical model to estimate their values.
- **Trading strategy:** It is the process used to map an alpha to a desired position. A trading strategy is a recipe (set of rules) in an algorithm, also designed by the trader or his team, that takes as an input an alpha, together with information on the current position, risk limits, etc., and outputs a recommended new position.

There is a large number of trading strategies taking advantage of market inefficiencies. Some of them use econometric arguments to forecast financial variables, some use fundamental and accounting information to price securities, while others use mathematical models. Even though strategies can be very heterogeneous in nature, tactics are relatively homogeneous. The trading rules for a strategy provide the algorithm with a set of decisions that must be followed to enter and exit a position. Conditions for entering and exiting a position are often defined through reaching values and thresholds for profit-taking and stop-losses. These entry and exit rules rely on parameters that are usually calibrated via historical simulations (backtesting) to maximise performances. However, it leads to the problem of backtest overfitting, where the parameters target specific observations and become unfit for the future.

Practically, the quantitative trading cycle of an agent is as follows:

- Perform market analysis to identify trading opportunities
 - Determine the probability of an outcome happening
 - Estimate the direction of returns
 - Estimate the magnitude of returns
 - Infer the level of stock uncertainty
 - Assess the duration of trends or mean reversion
- Define trading rules
 - Identify entry/exit points
 - Determine stop-loss and profit taking strategies
- Identify a trading signal
- Optimise portfolio allocation

- Decision process: decide to trade or not
- Execute the trade and risk manage it

Identifying trading opportunities (alphas) and mapping these opportunities to a desired position (trading rules) require some type of indicators or mathematical statistics with price forecasting capability. Anything that indicates what the future may hold is an indicator. We use these indicators to infer the trading rules that maximise the probability of executing a profitable trade. Hence, we need some consistent method or trigger to determine the entry and exit thresholds that will maximise future performances for some underlying instruments. Some of the indicators and mathematical models used to estimate the alpha include

1. Technical analysis: It is mostly involved with pattern recognition, regardless of performance. These patterns are then used to uncover pricing trends. It is about market traders studying market price history with the view of predicting future price changes in order to enhance trading profitability (see Lo et al. [2010]). Technical trading rules involve the use of technical analysis to design indicators that help a trader determine whether current behaviour is indicative of a particular trend, together with the timing of a potential future trade. Hence, it assumes that the historical data in the markets forms appropriate indications about the market future performance. See Section (3).
2. Stochastic methods: Use mathematical models of returns to estimate the first-passage time. In some models, a formula is derived to compute the parameters. See Section (7).
3. Option Theory: Infer the distribution of future returns via linear combination of vanilla options. We can also use options to estimate the first-passage time. See Section (6).

Financial models to detect patterns or estimate future return distribution are parametric and add up to the parameters from the trading rules that need to be estimated. For instance, in a trend-following moving average (MA) strategy, the researcher might try alternative sample lengths on which the moving averages are computed, entry thresholds, exit thresholds, stop losses, holding periods, and sampling frequencies. As another example of a strategy (see experience at Lawrence Berkeley National Laboratory described by Bailey et al. [2013]), one can identify the optimal monthly trading rule, given four customary parameters: entry day, holding period, stop-loss, and side (hold long or short positions). For illustration, we could explore all nodes that span the interval $[1, \dots, 22]$ for the entry day, the interval $[1, \dots, 20]$ for the holding period, the interval $[0, \dots, 10]$ for the stop-loss, and $\{1, -1\}$ for long or short positions. The parameters combinations involved form a four-dimensional mesh of 8,800 elements. Hence, a backtest will run several alternative model configurations (or trials) over a given period of time.

Having identified the trading opportunities and devised the trading rules, we need to estimate the model and trading rules parameters that maximise the probability of executing a profitable trade. These parameters must be calibrated in view of maximising future profit. This is a hard problem because we don't know a priori the future realisation of market prices and therefore cannot compute the future profit and loss (Pnl) of a given strategy. Several approaches have been proposed to estimate the best model and trading rules parameters, including

1. Brute force optimisation using real market data in a backtesting engine. See Section (7.2).
2. Abstract probabilist approach on real market data. See Section (7.3).
3. Optimisation with the first-passage time. Assuming a specific dynamic for the traded security, we can use stochastic calculus for deriving optimal statistical arbitrage strategies. See Section (7.4).
4. Futuretesting: It is a framework that generates realistic synthetic market data used for testing and optimising trading strategies, performing risk management, and analysing risk models. See Section (9).

In the method [1], we optimise both the model and the trading rules parameters on a backtesting engine (BE). A BE seeks to estimate the performance of a strategy or model if it had been employed during a past period. That is, on a single trajectory of prices realised in the past. To be realistic, this approach requires detailed historical datasets with all the past events and conditions. For each of those backtests, we can compute a performance metric (such as the Sharpe ratio) to identify the best combination of parameters. In general, practitioners estimate the parameters of an investment strategy by maximising its performance. This is a Combinatorial Optimisation problem: There is an objective function to be minimised, with a very large discrete configuration space. Practically, we define a suitable metric (or distance between the solutions) as well as a loss function to maximise profit. An optimisation engine is selected, and the outcome is expressed as a combinatorial structure of parameters. However, this approach leads to the problem of backtest overfitting, where the parameters target specific observations and become unfit for the future. Since we are applying the test on the same data multiple times, this approach is prone to false positives, leading to biased conclusion. As long as the researcher tries more than one strategy configuration, overfitting is always present.

In general, practitioners prevent overfitting by withholding a portion of the available data sample for separate testing and validation as out-of-sample (OOS) performance. However, the IS dataset is usually too short to fit, and the OOS dataset too short to conclude anything with sufficient confidence. In the method [2], Bailey et al. [2013] defined overfitting as occurring when the strategy configuration that delivers maximum performance in-sample (IS) systematically underperform the remaining configurations out-of-sample (OOS). There are several approaches to measuring a backtest overfit: the authors modelled it by using an abstract probability space in which the sample space consists of pairs of IS and OOS test results. Another approach consists in modelling the underlying variable by generating pseudorandom scenarios and measuring the performance of the resulting investment strategy for those scenarios. It relies on generating a distribution of outcomes, rather than relying on a single OOS performance estimate.

One of the main goal of statistical arbitrage trading in method [3] is to develop highly automated trading strategies that take a probabilistic approach to trading. It uses algorithms based on stochastic methods to identify price inefficiencies in the market. A common approach when performing this type of trading is to construct a stationary, mean reverting synthetic asset as a linear combination of traded securities. One example is the method of pairs trading (see Vidyamurthy [2004]). It allows for the construction of a trading strategy where trades are entered when the process reaches an extreme value and exited when the process reverts to some equilibrium value. Bertram [2009] provided a mathematical framework for the construction and analysis of statistical arbitrage methods in the case of continuous time trading strategies. He showed that when considering statistical based strategies, the time between trades (trading frequency) becomes an important quantity to consider. In such an environment there is a need for analytic and approximate solutions. One solution is to assume that the traded security is described by an Ornstein-Uhlenbeck process. The resulting analysis provides a mathematical model for calculating optimal statistical arbitrage strategies. We can construct a continuous trading strategy and express the trade length and return of the strategy in terms of the first-passage time of the process. Optimal trading strategies can be found by constructing objective functions that are expressed in terms of the expected value and variance of the return. One can then derive pseudo-analytic solutions for the strategy maximising expected return. In general, the solution satisfies a real valued integral equation.

For trading and risk management purposes, we need to build quantitative strategies, and perform market risk analysis on trading portfolios. Financial markets are some of the most dynamic entities that exist. Therefore, quantitative trading models must be as dynamic to be consistently successful. However, many quantitative traders develop models (and strategies) that are temporarily profitable for the market condition for which they were developed, but they ultimately fail when market conditions change. In the method [4], our main objective is not to estimate the parameters of a quantitative strategy from a single historical time-series in view of optimising its performance and spend our time recalibrating our model. Instead, we seek to estimate robust model parameters in all market configurations. That is, we are interested in generating synthetic data, somehow equivalent in the description of the actual stock price evolution, and use this data either to build quantitative strategies or perform risk analysis, and test them under a wide variety of market conditions.

1.2 From backtesting to futuretesting

We present the Futuretesting engine as a framework that generates realistic synthetic market data used for testing and optimising trading strategies, performing risk management, and analysing risk models.

We define a backtesting engine as a framework for testing trading hypothesis/strategies on historical data. The engine is fed with real market data and the trading strategy is evaluated (see Chan [2013]). It is designed to build quantitative strategies, or perform market risk on trading portfolios on historical data. This way, we can estimate the performance of a strategy if it had been employed during a specific period of time. The strategy achieving the maximum profit on past data is used in future trades. The simulation is run using historical data from stocks, bonds, and other financial instruments. In a backtest, one can perform a walk-forward (historical simulation) to simulate the historical performance of an investment strategy, as if it had been run in the past. However, backtesting relies on the assumption that stocks move in similar patterns as they did historically. Another limitation is the inability to model strategies that would affect historic prices. Further, we cannot use historical data alone as we only observe a single realisation of the underlying process. It may lead to overfitting of the trading strategy. Models only fitting a single realisation produce results that cannot be replicated in the future (see Lopez [2020]). Consequently, we cannot infer any meaningful insight from a single time series.

Forward performance testing (or paper trading) provides traders with another set of out-of-sample data on which to evaluate a system. It is a simulation of actual trading and involves following exactly the system's logic in a live market. In general, trade entries and exits are documented along with any profit or loss for the system, but no real trades are executed. One can also use a simulated trading account where trades can be placed and the corresponding profit and loss calculated.

Both backtesting and forwardtesting use a single trajectory to assess the performance of trading strategies. However, to assess the robustness of a strategy and avoid overfitting, it is necessary to perform multiple backtests on different datasets. Each dataset should contain a different time period, with different market conditions (regimes), and different assets' universe. For instance, we can create ordered sequences of prices at different entry points by stitching together subsamples of the real data series in a varying time-order. However, it can introduce spurious autocorrelations, or overlook important dependencies in the data series. In addition, it is challenging to produce a synthetic series that looks substantially different from the original one. Alternatively, one can consider resampling trades techniques by breaking down the temporal sequence and generating a re-ordered sequence. Resampling can be deterministic (jackknife (leave-one-out), cross-validation (one-fold-out), and combinatorial cross-validation (permutation tests)) or random (subsampling (random sampling without replacement), bootstrap (random sampling with replacement)). Another approach is to create a synthetic dataset that characterise statistically the observed market historical data, and then use those statistics to generate synthetic data. Define a statistical model (such as GARCH models and other (E, CO, F, I, M, Q, T, ZD)-GARCH models) of the underlying asset processes (see Appendix (14)), estimate it statistically on historical data, and use it to simulate a large number of trajectories (see Bollerslev [1986], Hentsche [1995], Li et al. [2018], Otto et al. [2018]). One can also rely on portfolio-level Monte Carlo (MC) analysis if there is strong correlation between underlying assets. It randomly samples new data sets from an estimated population or data-generating process. Assuming a parametric MC, one has to assume a dynamical model and statistically estimate its parameter on historical time series. Stress testing generates synthetic data corresponding to different market scenarios such as bull markets, bear markets, side markets, crises, bubbles, etc. It tests the resilience of investment portfolios against possible future financial situations.

Rather than defining a particular stock price process from a model, or using a single historical time-series, and then assess trading performances or perform risk analysis in a backtesting engine, we can use a range of forecasting models to predict the future underlying asset prices and test the performance of trading strategies or risk models on these predictions. For example, Zabcic-Matic [2019] used NNs to predict whether or not the stock price move over a day will be larger than straddle option with one day maturity. This information was used for devising trading strategies.

Similarly, Letteri et al. [2022] presented the DNN-forwardtesting, which uses predicted prices with DNN to determine the most effective technical indicators as the basis of trading strategies. More generally, we can either consider a family of stochastic/statistical processes, or data-driven methods, somehow equivalent in the description of the actual stock price evolution, and carry out performance optimisation and risk analysis. For instance, Buehler et al. [2020a] presented a statistically driven market simulator based on generative modelling. In such a framework, we can build a financial theory and focus on out-of-sample predictability to optimise trading strategies and improve risk models. That is, the strategies and risk models are inferred from the possible future predicted by (1) process-driven methods, (2) data-driven methods, or both. Even though the different approaches for generating realistic synthetic data have pros and cons, this data covers a wide range of stock trajectories representing different market scenarios. Further, it allows to simulate market moves to unexplored territories. Thus, it can be better used for building quantitative strategies and testing them under a wide variety of simulated market conditions, including tail events. This data can also be used for performing risk management. We briefly discuss the generation of realistic synthetic market data in Section (8).

In Section (9), we propose the **Futuretesting engine** (as opposed to backward and forward testing which only use a single trajectory) as a framework for discovering economic and financial theories. We generate realistic synthetic market data and use it to build quantitative strategies and test them under a wide variety of market conditions. For instance, we define some performance criterion in incomplete markets, such as the mean-variance criterion, and devise strategies that optimise the performance of the portfolio over a predefined range of market conditions. We can also use this synthetic data to perform market risk analysis on trading portfolios. The trading strategies and risk models will be evaluated on all of the data series simultaneously. We can then use all of the real market data for out-of-sample backtesting. As a result, quantitative strategies and risk management developed with the Futuretesting engine are less prone to overfitting and more likely to prove robust across a wider range of plausible market scenarios in the real world.

2 Quantitative trading principles

2.1 Overview

Trading refers to purchasing and selling securities or other assets in view of making a profit. On one hand, traders make profit from buying low and selling high (going long) or selling high and buying low (going short), usually over the short or medium term. They use mathematical models to infer the direction of future market prices and decide to go long or short. On the other hand, investors aim to buy shares at a favourable price and make profit from holding the stock and selling it at a higher premium. They hope that the share price will increase over the long-term and that they will make a profit from this movement. This is illustrated in Figure (1) with the history of the SP500 Index, which consists in a long-term trend together with major stock-market booms and busts ¹.

¹A bust is characterised by decreasing economic growth, decreasing inflation, and increasing deflation.

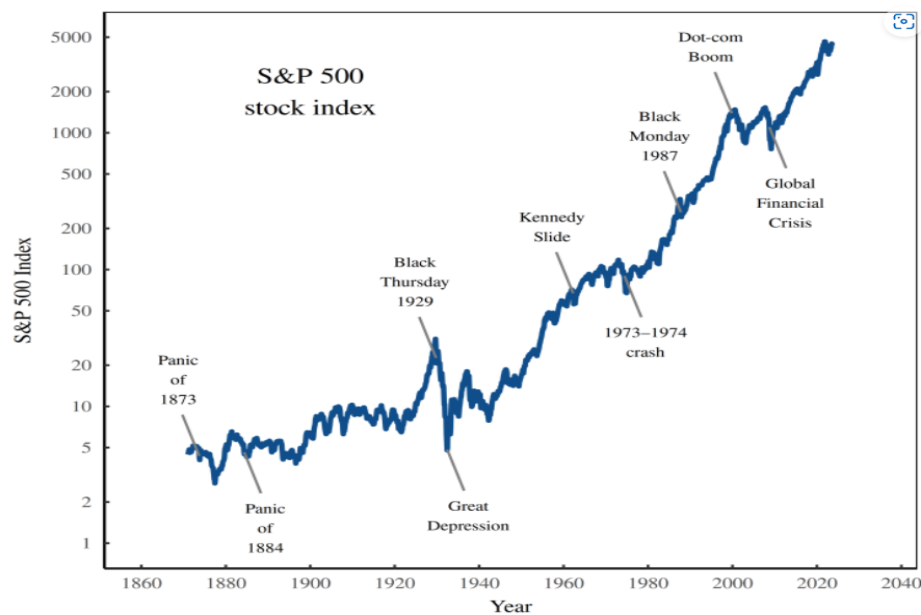


Figure 1: Long-term history of the SP500 Index.

The price of a stock indicates a company's current value (or its market value) to buyers and sellers. However, the stock's intrinsic value may be higher or lower. When trading, we profit when the market price of our position moves in the right direction, and conversely, we lose money when it moves in the wrong direction. Hence, it is crucial to understand the formation of market prices. A financial market is influenced by multiple factors, such as investors' psychological expectations, dynamics of economic development, changes in the industries, national policies, exchange rate, etc. These factors are expressed in different information vehicles, such as news, multimedia, social networks, financial reports, etc. While stock prices are driven by a variety of factors, ultimately the price at any given moment is due to the supply and demand: When demand is greater than supply, prices rise, while when supply is greater than demand, prices fall. Sometimes, prices will move sideways as both supply and demand are in equilibrium. Thus, markets can do one of three things: (1) go up, (2) go down, or (3) move sideways. In general, stock prices behave in a mix of random and cyclical manners. Therefore, every trader and investor must ask himself where is the overall market (or a specific security price) headed over a short or medium term.

Quantitative trading is about using mathematical models to predict the next direction of the overall market or of a specific security. The main goal of quantitative trading is to identify market moves over a given period of time. This include:

1. Direction of market move
2. Order of magnitude of market move
3. Financial patterns

In all cases, we can use two approaches to identify these moves: (1) data-driven methods or (2) process-driven methods.

In approach (1), we rely on points, lines, and charts to identify the direction of market prices. For example, a **trend** is the general direction of the price of a market, asset, or metric. The market trend has three primary directions: bull trend, bear trend, and sideways market.

1. Uptrends are characterised by higher peaks and troughs² over time and imply bullish sentiment among investors. It is a series of higher swing highs³ and higher lows.
2. Downtrends are characterised by lower peaks and troughs and mimic changes in the perception of investors. It is a series of lower highs and lower lows.
3. A sideways market occurs when the price of a security trades within a fairly stable range without forming any distinct trends over some period of time.

These three categories are recurring and determine market behaviour for a given asset or pair over a certain length of time. A **reversal** is a change in the price direction of an asset and can occur to the upside or downside. An uptrend reverses into a downtrend by changing to a series of lower highs and lower lows. A downtrend reverses into an uptrend by changing to a series of higher highs and higher lows.

In financial markets, we observe high frequency price action of specific underlying assets. Over a given fixed period of time (such as a day or less), they are sampled and expressed as open, high, low, and close (OHLC) prices. This leads to multivariate stock time series represented by candlestick charts (see Section (2.7)). The open-high-low-close prices, candlestick charting can reflect not only the changing balance between supply and demand, but also the sentiment of the investors in the market (see Marshall et al. [2006]). The opening position of the day reflects the accumulation of sentiments during non-trading period.

In general, market prices consist of several patterns which appear repeatedly at time intervals. Patterns can be based on seconds, minutes, hours, days, months or even ticks and can be applied to bar, candlestick and line charts. Chart patterns are repetitive lines and shapes formed on price charts and are used to help traders predict future price movements, such as breakouts and reversals. For instance, trendlines are straight lines drawn on a chart by connecting a series of descending peaks (highs) or ascending troughs (lows). They are usually used in candlestick trading, which makes it slightly easier to see the previous opens and closes of the market. They are capable of highlighting areas of support and resistance (see Figure (7)), which can help traders decide whether they should open a long or short position, or whether they should close out their open positions in the event of a possible trend reversal.

Stock trading is the process of buying and selling company shares listed on a stock exchange in view of making a profit. There are many different stock trading strategies we can use depending on our preferred approach to the market and how long we want to keep positions open. Some of these strategies include:

- News trading strategy: It is based on responding to news announcements and events. We must understand market expectations, assess whether a news event has already been priced into a stock, and trade it accordingly.
- End-of-day trading strategy: Focus on trading at the end of a trading session when liquidity is often higher. Closing prices represent the willingness to hold a position overnight (or over a weekend or market holiday).
- Swing trading strategy: It takes advantage of share price swings by buying stocks before they swing higher and selling them before they swing lower. It requires indicators to detect when a price trend is likely to change direction.
- Day trading strategy: It focusses on opening and closing positions on the same day to take advantage of stock price fluctuations during a trading session. We can open more than one position during a session, but we need to close them before the end of the day.

²In economic terms, a trough can refer to a stage in the business cycle where activity is bottoming, or where prices are bottoming, before a rise.

³A swing high refers to a peak reached by an indicator or a security's price before a decline. A swing high forms when the high reached is greater than a given number of highs positioned around it. A series of consecutively higher swing highs indicates that the given security is in an uptrend. The opposite of a swing high is a swing low.

- **Trend trading strategy:** It involves using technical analysis tools to determine price trends and trading stocks in line with the direction of a trend.
- **Scalping trading strategy:** It involves opening and closing trades within very short time periods of hours or even minutes to accumulate small profits. It requires indicators to exit trades quickly before the price suddenly move against our position and trigger losses.
- **Position trading strategy:** It consists in holding positions open for months or years to speculate on long-term price trends. It places more emphasis on analysis of a stock's fundamentals rather than technical analysis.

No matter which strategy we consider, we must risk managed it. The recent market turbulence has led to high volatility and large drawdowns in equity prices. This makes it crucial to incorporate risk management tools into trading systems to avert significant losses. In risk management, the cardinal rule of investing is to protect and preserve our principal. To this end, the stop loss and profit order (profit taking) are the two end of the spectrum:

1. A stop loss strategy is necessary to cap the maximum loss we are willing to make on a position on a stock.
2. A take profit order determines a specific value at which we are willing to close an open position on a stock for a profit.

By combining stop losses and take profit orders, it is possible to put together a risk ratio formula that works for long-term investments. Ideally, we want to keep the potential downside much lower than the potential upside.

A large number of strategies exist in between these two extreme cases. The simplest strategy to protect upside gains is to take profits off the table. However, it is often the case that winning stocks are simply taking a rest before continuing higher. In this case, rather than selling the stock, we should lock in some of the realised gains. This is called preservation-of-capital. Preservation-of-capital techniques include diversifying holdings over different asset classes and choosing assets that are non-correlating (that is, they move in inverse relation to each other). Put options and stop-loss orders can stem the bleeding when the prices of our investments start to drop. Dividends buttress portfolios by increasing your overall return. Principal-protected notes safeguard an investment in fixed-income vehicles.

In this section, we are first going to define some terminology on stock trading, we will then briefly introduce profit protection strategies and stop loss strategies. At last, we will give an overview of chart analysis before introducing support and resistance.

2.2 Some terminology

2.2.1 Stock trading

2.2.1.1 Market order The simplest and most common type of stock trade is carried out with a market order. The order is executed as quickly as possible at the current market price. All orders are processed within present priority guidelines. Transactions are subject to the availability of given stocks and can vary significantly based on the timing, the size of the order, and the liquidity of the stock. Hence, there is always the threat of market fluctuations occurring between the time the broker receives the order and the time the trade is executed. Market orders indicate that we are willing to take whatever price is presented to us when our order is executed. It offers a greater likelihood of a trade being executed, but there is no guarantee that it will actually go through.

2.2.1.2 Limit order A limit order allows us to limit either the maximum price we will pay or the minimum price we are willing to accept when buying or selling a stock, respectively. The primary difference between a market order and a limit order is that the latter order may not be executed.

There are three considerations we should take into account before placing a limit order:

1. A limit order is only triggered when the limit price meets the buyer specifications. Hence, the stock price may never fall (or rise) to the limit we have established. As a result, our order may never be executed.

2. Limit orders are executed in the order in which they are received. It is possible that the stock we are interested in buying (or selling) will reach our limit price, yet our trade will not be filled because the price fluctuated above (or below) our limit before the trade could be carried out. This problem is far less common now with online trading than it was when people used to call their broker to place trading orders.
3. If there is a sudden drop in the stock price, our order will be executed at our limit price. For example, imagine that Bancorp's stock is trading at \$51 and the bank's CEO resigns unexpectedly or some other type of bad news is reported. As a result, the U.S. Bancorp's stock drops to \$45. As the stock was falling in price, our order was executed. We are now sitting on a loss of \$6 a share.

The difference between the highest price a buyer is willing to pay for an asset and the lowest price a seller is willing to accept is called the bid-ask spread. Limit orders are particularly valuable when the stock price exhibits high volatility or when it has a wide bid-ask spread.

2.2.1.3 All-or-None (AON) When we purchase a substantial amount of a company's stock, it may take a while for the order to be completed, and so we might end up paying different prices for different parts of the order. One way of avoiding that situation is to place an all-or-none (AON) order, which requires the stock to be purchased in a single transaction or not at all. AON orders that cannot be executed immediately remain active until they are executed or cancelled. Thus, they can take longer to execute than normal orders, especially for larger order sizes. Further, an order may not be executed at all if there are not enough shares available to fill it. Unlike the next two similar types of trading orders, an AON order is in effect until we cancel it or it is executed.

2.2.1.4 Fill-or-Kill (FOK) A fill-or-kill (FOK) order combines AON and immediate or cancel (IOC): it must be filled immediately in its entirety at the market or a specified price, or it is killed (cancelled) if not filled. That means FOK orders may never be partially executed. These orders ensure that an entire position is executed at prevailing prices in a timely manner. They are most often used by active traders and are usually for a large quantity of stock. They avoid prolonging the period of time to complete the order, which can cause significant changes to stock's prices resulting in market disruption.

2.2.1.5 Immediate-or-Cancel (IOC) An immediate-or-Cancel (IOC) order is an order to buy or sell a security that attempts to execute all or part immediately and then cancels any unfilled portion of the order. The key difference between this kind of trade order and the FOK is that this order allows partial amounts of the order to be completed. When shares are no longer available at the limit or a better price, buying or selling ends immediately and the order is cancelled. These orders are used when submitting a large order to avoid having it filled at an array of prices. They are also used when markets are volatile and investors want to fill as much as possible at current market prices.

2.2.1.6 Day order A day order is an order to a broker to execute a trade at a specific price that is only good until the end of the regular trading day, 4:00 p.m. Eastern Time, at which point it is cancelled. It can be a limit order to buy or sell a security, but its duration is limited to the remainder of that trading day. If the trade isn't triggered, the order goes unfilled and is cancelled at the end of the session. In general, these orders serve as the default order duration on trading platforms. It helps intraday traders monitor and trade multiple securities at one time.

2.2.1.7 Good-Till-Cancelled (GTC) A good-till-cancelled (GTC) order is an order to buy or sell a stock that remains active until either the order is filled or the investor cancels it. The order remains open until one of three things occurs:

1. It is completely filled.
2. It is cancelled.

3. A certain time period determined by our online broker has passed.

Investors can place buy or sell orders at specific price points and keep them for several weeks. If the market price hits the price of the GTC order before it expires, the trade will execute. Investors may also place GTC orders as stop orders, which set sell orders at prices below the market price and buy orders above the market price to limit losses. Some of the risks in using these orders are:

- We may forget we placed the order.
- Execution of orders at inopportune moments, such as the brief rally in prices or temporary volatility. The consequent raise/fall in prices could leave traders with losses.
- If we place a large trade with GTC status, we may pay a commission each day the order is partially filled. If, on the other hand, our order is filled by multiple transactions in a single day, our broker should charge us only a single commission.

2.2.1.8 Stop There are three main types of stop orders: stop-loss, stop-entry, and trailing stop-loss. In common parlance, stop and stop limit orders are known as 'stop loss' orders because day traders and other investors use them to lock in profits from profitable trades. In the case of a stop order, it automatically converts into a market order when a predetermined price (the stop price) is reached. At that point, the ordinary rules of market orders apply: the order is guaranteed to be executed, but we will not know the price. A stop order is always executed in the direction that the price is moving: If the market is moving lower, the stop order is set to sell at a pre-set price below the current market price. Conversely, if the price is moving higher, the stop order will be to buy once the security reaches a pre-set price above the current market price.

2.2.1.9 Stop limit Stop-limit orders are a conditional trade that combine the features of a stop loss with those of a limit order to mitigate risk. The investor must select two price points: the stop price and the limit price. The stop price dictates the price whether the order is triggered, then the limit price dictates the price at which the order is filled. The stop limit order automatically converts into a limit order (to buy or sell at the limit price) when the stop price is reached. These orders offer risk management, automation, and flexibility in trading. However, as with other limit orders, the stop limit order may or may not be executed depending upon the price movement of the security. They are mainly used to lock in profits or limit downside losses.

2.2.1.10 Short Sell Order Selling short (or shorting a stock) is a practice that can enable traders to profit if they correctly predict that the price of a stock they do not own will fall. They borrow (from the broker-dealer) a security on margin for a specified time, whose price they think is going to fall, and sell it on the open market when either the price is reached or the time period expires. For example, if we think that General Electric (GE) stock is overvalued at a price of \$12.50, we can try to take advantage of this situation by selling borrowed shares of the stock at the price we believe to be inflated. Short sellers aim to sell shares while the price is high, and then buy them later after the price has dropped. However, short sales is a risky trading strategy because losses are unlimited (stock price rises rather than declines). Thus, short sellers must use a stop-loss order, so that if the stock price begins to rise, the short sale will be automatically covered with only a small loss.

2.2.1.11 Buy to cover Buy to cover refers to a buy order made on a stock or other listed security to close out an existing short position. It involves purchasing an equal number of shares to those borrowed, to cover the short sale and allow the shares to be returned to the original lender.

Back on our previous example, let's say GE stock did as we predicted and fell to \$10.50 per share. We would place a buy to cover order to complete the short sale. Our buy to cover order would repurchase the 1,000 shares for \$10,500 and return the borrowed shares to our online broker. Since we bought the shares for \$2,000 less than we sold them for, we have made a \$2,000 gain.

Buying and selling stock can be made with cash or on margin with funds and securities borrowed from brokers. Thus, a short sale is inherently a margin trade, as investors are selling something they do not already own. Here are a few important rules regarding short selling:

- In order to sell short, we must have margin privileges in our brokerage account. That means we can trade with more money than we have in our account if we wish.
- We must maintain enough purchasing power in our account to carry out a buy to cover order on our short sale. If the price of the shorted stock increases and we don't have enough money in our account to buy the shares back at the higher price, we will face a margin call: a demand by our broker to put more cash or securities in our account to be able to cover the trade.

2.2.1.12 Trailing stop order One way of protecting gains and limiting losses automatically is by placing a trailing stop order. With this kind of order, we set a stop price as either a spread in points or a percentage of current market value. Imagine we purchased 500 shares of Coca-Cola at \$50 per share. The current price is \$58. We want to lock in at least \$5 of the per share profit we have made, but wish to continue holding the stock, hoping to benefit from any further increases. To meet this objective, we could place a trailing stop order with a stop value of \$3 per share. If the stock price increases, our order will sit on our broker's books and automatically adjust upwards as the price of Coca-Cola's common stock increases. At the time our trailing stop order is placed, the broker knows to sell our shares if the stock price falls below \$55 (\$58 current market price - \$3 trailing stop loss = \$55 sale price). If Coca-Cola increases steadily to \$62 per share, the trailing stop order automatically keep pace and will convert to a market order at \$59 (\$62 current market price - \$3 trailing stop loss = \$59 sale price). That would provide a gain of \$9 per share.

2.2.1.13 Bracketed Orders Bracketed orders go one step further than trailing stop orders. Just like the latter type of order, with a bracketed order, we set a trailing stop as either a percentage or fixed amount below the stock price. However, we can also establish an upper limit that, when reached, will result in the stock being sold. Going back to our Coca-Cola example, let's now assume we placed a bracketed order with a trailing stop level of \$3 per share and an upper limit of \$65 per share. The bracketed order will behave the same as the trailing stop order, with the \$3 trailing stop automatically ratcheting up as the price increases. The only difference is that if and when Coca-Cola hits \$65, the bracketed order will automatically convert into a market order and will be immediately executed.

2.2.2 Profit protection strategies

Preservation-of-capital techniques can protect our portfolio from the inevitable volatility that exists in the investment world. We list below some of the most popular strategies for profit protection. Putting some of them in place will help preserve principal.

2.2.2.1 Diversification One of the cornerstones of modern portfolio theory (MPT) is diversification. It consists in spreading investments across different asset classes, industries, and geographic regions to reduce the overall risk of an investment portfolio. In a market downturn, MPT disciples believe that a well-diversified portfolio will outperform a concentrated one. Investors create deeper and more broadly diversified portfolios by owning a large number of investments in more than one asset class, thus reducing **unsystematic risk**. This is the risk that comes with investing in a particular company (as opposed to systematic or market risk which is generally unavoidable). Stock portfolios that include 12, 18, or even 30 stocks can eliminate most, if not all, unsystematic risk, according to some financial experts. However, diversification can result in missing out on company- or industry-specific above-average returns that may be captured by a more active monitoring but less diversified approach. Further, diversification can be expensive, so that traders wanting to avoid the transaction fees of buying and selling that many stocks might prefer getting index funds.

2.2.2.2 Non-correlating assets The opposite of unsystematic risk is **systematic risk**, which is the risk associated with investing in the markets generally. Common causes include inflation rates, exchange rates, political instability, war, and interest rates. Unfortunately, systematic risk is always present and cannot be eliminated or reduced through diversification. It can be mitigated through hedging or by using the correct asset allocation strategy. In the latter, it can be reduced by adding non-correlating asset classes such as bonds, commodities, currencies, and real estate to the equities in our portfolio. Non-correlating assets often react differently to changes in the markets compared to stocks, and they generally move in inverse ways. When one asset is down, another is up, so that they smooth out the volatility of our portfolio's worth overall. Ultimately, the use of non-correlating assets eliminates the highs and lows in performance, providing more balanced returns. We can identify the systematic risk of a particular security, fund, or portfolio by looking at its beta. It measures the volatility of that investment compared with the overall market. A beta greater than one means that the investment has more systematic risk than the market, while a beta less than one means less systematic risk than the market.

2.2.2.3 Put options One strategy to protect upside gains is to buy put options, which is a bet that the underlying stock will go down in price. Different from shorting the stock, the put gives us the option to sell at a certain price at a specific point in the future. For example, let's assume we own 100 shares of Company A, which has risen by 80% in a single year and now trades at \$100. We are convinced that its future is excellent but that the stock has risen too quickly and is likely will decline in value in the near term. To protect our profits, we buy one put option of Company A with an expiration date six months in the future at a strike price of \$105, or slightly in the money. The cost of buying this option is \$600 or \$6 per share, which gives us the right to sell 100 shares of Company A at \$105 sometime before its expiry in six months. If the stock drops to \$90, the cost to buy the put option will have risen significantly. At this point, we sell the option for a profit to offset the decline in the stock price. Investors looking for longer-term protection can buy long-term equity anticipation securities (LEAPS) with terms as long as three years. Note, we are not trying to make money off the options, but are instead trying to ensure our unrealised profits don't become losses. Investors interested in protecting their entire portfolios instead of a particular stock can buy index LEAPS that work in the same manner.

2.2.2.4 Stop Losses In a long position, stop-loss orders protect against falling share prices. Conversely, in a short position, stop-loss orders protect against raising share prices. These orders will remove you from your position at a pre-set level if the market is moving against you. There are several types of stops we may use such as

- Hard stops involve triggering the sale of a stock at a fixed price that doesn't change. For example, when we buy Company A's stock for \$10 per share with a hard stop of \$9, the stock is automatically sold if the price drops to \$9.
- A trailing stop is different in that it moves with the stock price and can be set in terms of dollars or percentages. Using the previous example, suppose we set a trailing stop of 10%. If the stock appreciates by \$2, the trailing stop will move from the original \$9 to \$10.80. If the stock then drops to \$10.50, using a hard stop of \$9, we will still own the stock. However, in the case of the trailing stop, our shares will be sold at \$10.80. What happens next determines which is more advantageous. If the stock price then drops to \$9 from \$10.50, the trailing stop is the winner. However, if it moves up to \$15, the hard stop is the better call.

Proponents of stop-losses believe they protect us from rapidly changing markets. Opponents suggest that both hard and trailing stops make temporary losses permanent. It is for this reason that stops of any kind need to be well-planned.

2.2.2.5 Dividends Investing in dividend-paying stocks is probably the least known way of protecting a portfolio. Historically, dividends account for a significant portion of a stock's total return. In some cases, it can represent the entire amount. Owning stable companies that pay dividends is a proven method for delivering above-average returns. In addition to the investment income, studies have shown that companies that pay generous dividends tend to grow

earnings faster than those that don't. Faster growth often leads to higher share prices which, in turn, generates higher capital gains. It protects our portfolio by increasing our overall return. When stock prices are falling, the cushion dividends provide is important to risk-averse investors and usually results in lower volatility. In addition to providing a cushion in a down market, dividends are a good hedge against inflation. By investing in blue-chip companies that both pay dividends and possess pricing power, we provide our portfolio with protection that fixed-income investments-with the exception of Treasury inflation-protected securities (TIPS) can't match. Furthermore, if we invest in 'dividend aristocrats', those companies that have been increasing dividends for 25 consecutive years, we can be virtually certain that these companies will up the yearly payout while bond payouts remain the same.

2.2.2.6 Principal-protected notes Investors who are worried about maintaining their principal might want to consider principal-protected notes with equity participation rights. They are similar to bonds in that they are fixed-income securities that return principal investment if held until maturity. However, they differ in the equity participation that exists alongside the guarantee of principal.

For example, let's say we wanted to buy \$1,000 in principal-protected notes tied to the S&P 500. These notes will mature in five years. The issuer would buy zero-coupon bonds that are maturing around the same time as the notes at a discount to face value. The bonds would pay no interest until maturity when they are redeemed at face value. In this example, the \$1,000 in zero-coupon bonds is purchased for \$800, and the remaining \$200 is invested in S&P 500 call options. The bonds would mature and, depending on the participation rate, profits would be distributed at maturity. If the index gained 20% over this period and the participation rate was 90%, we would receive our original investment of \$1,000 plus \$180 in profits. We are forfeiting \$20 in profits in return for the guarantee of our principal. But let's say the index loses 20% over the five years. we would still receive our original investment of \$1,000. In contrast, a direct investment in the index would be down \$200.

2.2.2.7 Chart patterns Chart patterns occur when the price of an asset moves in a way that resembles a common shape, like a triangle, rectangle, head and shoulders, or a cup and handle (see Section (2.6)). These patterns are a visual way to trade providing a logical entry point, a stop-loss location for managing risk, and a price target for exiting a profitable trade. For instance, we can improve our risk-reward ratio (see Section (2.4)) by setting liquidation points based on the pattern's size and shape. However, using chart patterns can lead to false breakouts or breakdowns, where the price breaks the pattern's boundary but then reverses and moves in the opposite direction (see Section (2.6.3)). This can trigger liquidation points prematurely or incorrectly, and cause traders to lose money or miss opportunities. Furthermore, we may have to adjust our liquidation points dynamically due to changing market conditions. Note, additional indicators such as volume, momentum, or moving averages should support the pattern and its direction.

2.2.3 Trading periods

In trading, a period refers to the time duration of a given trading session. A lookback (LB) period is the number of periods of historical data used for observation and calculation. It is how far into the past a system looks when trying to calculate the variable under review. Look back period is simply the number of candles we wish to view before taking a trading decision. When measuring days, long-term lookback windows typically range from 30 to 90+ days, while short-term windows range from 1 to 15 days.

Whatever the form of momentum, there are different ways traders measure the momentum of a stock. Here are some of them:

- **3-month momentum:** Check the momentum of the stock over a 3-month lookback period by comparing the price of the stock now to what it was 3 months ago or checking the position of the stock relative to its 63-day moving average.
- **6-month momentum:** Here, the lookback period is 6 months or 126 trading days. The momentum can be done by comparing the price of the stock to what it was 6 months ago, or by checking the position of the stock relative to the 126-day moving average.

- 12-month momentum: In this case, the lookback period is 12 months.
- 12-1 momentum calculation: This is the ratio of the return of a share price over the past year but excluding the last month. Given $R_{t-n} = \ln \frac{S_t}{S_{t-n}}$, then $MPI(12m - 1m) = R_{t-12} - R_{t-1} = \ln \frac{S_{t-1}}{S_{t-12}}$, where MPI is the momentum price index.
- 12-2 momentum strategy: In this ratio, the most recent 2 months are excluded from the index.

2.3 Entry and exit points

In the fast-paced world of electronic markets, buying and selling stocks is easy. However, knowing when to buy and when to sell those stocks is an art. This is called market timing. An entry point is the price at which we buy or sell a security, while the exit point is the price at which we unwind our position. The entry and exit levels play a crucial role in the success of a trade. As such, no trading strategy can be built without a proper entry and exit model.

When searching for an entry point, we are looking for a spot that indicates momentum is going up or down, but hasn't hit its peak or trough yet. In general, traders use technical indicators to make their decisions. Technical analysis (TAs) (see Appendix (13)) is a method of analysing stocks to predict future price movements that assumes the vast majority of available information about that stock is almost instantaneously incorporated in the price by market forces. Technical traders (TTs) try to divine how stocks will move on a short-term basis by looking at the behaviour of markets in similar, past situations. For instance, TTs identify both the resistance and support level (see Section (2.6.2)) so that they can time their buying and selling of a stock to capitalise on any breakouts or trend reversals (see Section (2.6.3)).

2.3.1 Calculating entry points

2.3.1.1 Pullbacks, support, and resistance Examples of trendlines are support for rising markets, or resistance for declining markets (see Section (2.6.2)). One can look at the stock's moving average (MA), which is a calculation of a stock's average price over a given number of time units (see Appendix (11.1)). As long as the price doesn't move through that line, the trend remains intact.

- Pullback: If you have found an up-trending stock for which prices are outperforming the MA, look for an occasion when they go down in price closer to the line of the MA, or even go below it. It is a simple downward correction before prices go back up again. In other words, it is an entry point.
- Support: Upward trending price lines are indications of support in the marketplace. Even if the prices dip a little bit from day-to-day, if they don't fall too far too long, there will still be support.
- Resistance: A resistance level is the price at which the price of an asset meets pressure on its way up by the emergence of a growing number of sellers who wish to sell at that price. The simple resistance level can be charted by **drawing a line along the highest highs for the time period being considered**. One can also incorporate bands, trendlines, and moving averages to identify that level. It is the opposite of support indicating a possible exit indicator.

Support and resistance are complementary concepts. Resistance establishes the current price ceiling for the stock, commodity, or currency, and support forms the floor. When the price action breaches either support or resistance, it is considered to be a trading opportunity.

2.3.1.2 Trading volume Another way of calculating an entry point is to look at how many people are trading in the stock at the moment. Trading volume is the total number of shares of a security traded during a given period of time. It is generally used to confirm the existence or continuation of a trend, or a trend reversal. For instance, when there is a lot of volume and the stock price is going up, that is as close as we can get to a confirmed bull market for the stock. It may be time to enter into a position before the profit margin disappears. Conversely, high volume accompanied

by a declining stock price means people are selling their shares in the droves. Volume is used by many traders for confirmation because rising prices accompanied by rising volume are typically viewed as a bullish confirmation signal. It can also signal investors that they should take profits and sell a security due to low activity. In that case, it indicates weakness in the current trend and a possible reversal. As an example, chart patterns such as the head and shoulders and the flag and pennant (see Section (2.8)) use volume to confirm trends, reversals, and breakouts.

2.3.1.3 Crossover with two Moving Averages A moving average crossover (MAC) (see Section (3.1)) is a popular trading strategy that uses two or more moving averages of different lengths to identify potential buy and sell signals when they crossover. It is used by traders to gain insight into the direction of the market and the strength of the trend. It helps them making informed trading decisions, such as buying or selling assets at the right time. For example, one can monitor two different sets of moving averages: say a 20-day MA and a 100-MA, and execute on positions when the shorter-term MA crosses the longer one. The longer MA indicates a more stable pattern, its rises and falls are very mild, while the shorter MA is slightly more subject to volatility. Conversely, when the shorter MA crosses downward and goes south of the longer MA, it is time to exit. MAC can be used on multiple time frames, allowing traders to identify both short-term and long-term trends. However, they are lagging indicators, providing signals after the trend has already begun. Further, in whipsaw markets they can generate a large number of false signals, leading to losses.

2.3.1.4 Financial patterns While the price chart of a stock appear to be only driven by random movements, within those price movements are patterns. Financial patterns are successions of trend periods, continuation, and consolidation periods (see Section (2.6)). They are geometric shapes found in the price data which help traders understand the price action and make predictions about the direction of market prices.

- **Consolidation:** It describes a stock's price movement within a given support and resistance range for a period of time. The asset neither continues nor reverses, but oscillates between a well-defined pattern of trading levels. The consolidation zone is illustrated in Figure (2). For instance, a stock emerges from a long flat or downward trend with a sudden jump in buying volume and price at the end of a stock market day. Then, the stock recedes a little bit and bumps up mildly, then drops a bit, then goes back up a bit, and maybe goes through the cycle another time or two. These oscillations are generally caused by traders' indecisiveness. Traders prefer to wait out the consolation period until they see signs of continued, sustained momentum. This occurs when the asset's price moves above or below the trading pattern. A breakout above resistance means the price will keep raising (buy signal), while a breakout below the support level indicates the price will keep falling (sell signal). In general, once the price breaks through the identified areas of support or resistance, volatility quickly increases.
- **Continuation pattern:** It is a signal that a price trend is likely to remain in play. This pattern occurs mid-trend and represents a pause in the price action of varying durations, indicating that the trend is likely to resume after the pattern completes. Usually, the pattern has formed and then breakout or reverse (in opposite direction). This is illustrated in Figure (2). For example, it can be a series of up-and-down price movements immediately following a big vertical movement. It is a resting period where traders take a few profits. Continuation patterns are used to time entry, and occasionally exits, from a stock position in anticipation of an eventual return to the trajectory.



Figure 2: Continuation pattern with a consolidation zone.

Triangles, flags, pennants, and rectangles are examples of continuation patterns (see Section (2.8)). They are defined as follows:

- **Flag:** It is a price pattern that, in a shorter time frame, moves counter to the prevailing price trend observed in a longer time frame on a price chart. The initial stock price jump resembles a flagpole (upward or downward) (see Figure (2)). Flag patterns are areas of tight consolidation in price action showing a counter-trend move that follows directly after a sharp directional movement in price. Hence, they signify trend reversals or breakouts after a period of consolidation.
- **Pennant:** It is a continuation pattern formed when there is a large movement in a security, known as the flagpole, followed by a consolidation period with converging trend lines (the pennant), followed by a breakout movement in the same direction as the initial large movement, which represents the second half of the flagpole. A bullish pennant pattern can help identify an upside breakout, a brief pause or congestion period (which forms the pennant), and then a thrust higher as the stock move continues. A typical pennant pattern forms with an initial quick, strong rally (the flagpole). Then the action turns sideways in a short-term consolidation, which forms the pennant. The entry trigger is a breakout higher from the pennant.
- **Triangle:** It is a converging of the price range, with higher lows and lower highs, such that the converging price action creates a triangle formation. That is, the upper and lower trendlines ultimately meet at the apex on the right side, forming a corner. There are three variations of triangles: ascending, descending, and symmetrical. If validated, it is a continuation pattern on a chart that forms a triangle-like shape, otherwise it is a reversal pattern.

As seen in Figure (2), after consolidation, if the stock price eventually crosses the top line of the flag, or the nose of the pennant, it may indicate a great time to enter into the stock (breakout). Flag and pennant patterns also work in reverse: If a stock endures a steep vertical drop, there may be a period of consolidation where trading goes up and down. In this scenario, if the stock price drops below the bottom line of the Flag, it is be time to exit (breakdown). In general, traders expect to use the opposite side the flag pattern as a stop-loss point. In addition, traders use the difference, measured in price, between the flag pattern's parallel trend lines to set a profit target.

2.3.2 Calculating exit points

While calculating a stock exit point is an entry calculation in reverse, figuring out an exit point is more complex than buying into a stock. Most investment experts strongly advise setting an exit plan as far enough in advance as one can. We list below some of the most popular exit strategies. For some of these strategies, the readers should refer to Section (2.2).

2.3.2.1 Limit orders The main idea of limit orders is to set a limit order for certain underlying, so that we can take profits and keep our investments more liquid. In a long position, a limit order is simply a direction to the brokerage to sell the security after it reaches a certain higher price point. Conversely, in a short position, the limit order is a direction to the brokerage to buy the security after it reaches a certain lower price point.

2.3.2.2 Take-profit order In a long position, a take-profit order (profit taking) is known as a limit order, which guarantees that a position is closed at or greater than a predefined price point. If a position on a stock moves in the right direction to the take-profit level, the position is closed for a profit. The position is closed for a profit once it reaches a predetermined percentage of profit or monetary value. However, in volatile markets, it can be tricky to know when to take our profits. Generally more popular with short-term traders that monitor daily or even hourly price moves, take-profit orders are the most efficient way of executing a trade and taking the human element out of managing an open position.

2.3.2.3 Setting take-profit order There are various strategies we can adopt with take-profit orders. We list a few of the most popular methods used by experienced traders below:

- Place orders at major support or resistance levels: Many traders will use technical analysis (TA) to pinpoint support or resistance points and **use previous highs and lows to set take-profit orders at similar levels.**
- Understand the peaks and troughs of Fibonacci trends: Fibonacci extensions can give us an idea behind a stock's profit potential based on how far its trends extend.
- Place orders based on the moving average: During trending market periods, some traders will also place a take-profit order in and around the price of the moving average, particularly when a stock has dipped well below its medium-term moving average.
- Where price action suggests a switch in market sentiment: It is also possible to use signals based on price action alone to determine changing market sentiments such as engulfing bars and rejections at highs or lows.

2.3.2.4 Stop-loss orders A stop-loss order works just like a limit order, but for stocks that are headed for a loss. We direct our brokerage to sell our shares after they have declined to a certain point, so that we can cut our losses. Every position we own should have a stop-loss order attached to it, even the value stocks. It is a form of protection against drastic losses. In general, practitioners use violations of technical features-like trendlines, round numbers, and moving averages to establish the natural stop-loss price. A typical strategy is to **keep raising our stop-loss limits** on stocks that are consistently rising in value (it is called trailing stop loss). Once the stop-loss limit passes our entry price, we have guaranteed that no matter when the sale executes, we will break even. Continuing to raise our stop-loss limit as the price goes up guarantees we will turn a profit when it sells.

Practically, a stop-loss order is triggered in the market once the price of an asset drops beneath the stop price specified by the trader. Then, the market order is executed, selling at the next available price below the stop loss level. As an example, we bought 10 shares in Company ABC at \$50 apiece, an investment totalling \$500. We set a stop loss order at \$40. The share price of Company ABC goes on to tumble in the days after we open our position and it breaches the \$40 stop loss. The stop-loss order is triggered, and our position is closed at \$39.95 for a loss of \$10.05 per share.

2.3.2.5 Protective stop A protective stop (similar to profit taking) is a stop-loss order deployed to guard against losses, usually on profitable positions, beyond a specific price threshold. It is a strategy designed to protect existing gains or thwart further losses by means of a stop-loss order or limit order. It is set to activate at a certain price level and normally guarantees that an investor will make a predetermined profit or limit their losses by a predetermined amount. A protective stop is a popular strategy for risk-averse investors. Popular tools for measuring risk include downside deviation and semivariance.

2.3.2.6 Trailing stop loss A trailing stop loss is a kind of order that is intended to help you lock in profits while protecting you from day trading losses. It caps the amount that will be lost if the trade doesn't work out but doesn't cap the potential gain if the trade works in your favour. This type of order converts into a market order when the security price reaches the stop price. The main difference between a regular stop loss and a trailing stop loss is that the trailing one moves whenever the price moves in your favour. It moves higher as the price of the stock moves higher. The scenario for a short trade is similar except that you are expecting the price to drop, so the trailing stop loss is placed above the entry price.

Note, they can get you out of a trade too soon, such as when the price is only pulling back a bit, not actually reversing. To try to prevent that scenario, trailing stops should be placed at a distance from the current price (usually as a percentage) that we do not expect to be reached unless the market changes its direction. If we intend to buy (go long) on a stock we will place a trailing stop loss beneath the market value. On the other hand, if we intend to sell (short) a stock we will place a trailing stop loss above the market value. It allows for natural price volatility while still protecting the majority of the profit. One can also use the Average True Range (ATR) of the stock to set trailing stops. The main alternative to a trailing stop-loss order is the trailing stop limit order. It differs only in that once the stop price is reached, the trade is executed at the limit price we have set or a better price rather than at the then available market price. This type of order is meant to lock in profits while protecting you from significant losses. It converts into a market order when the security price reaches the stop price and is executed at the then-available price.

2.3.2.7 Scaling exit strategies For a scaling exit approach, raise your stop to break even as soon as a new trade moves into a profit. This can build confidence because you now have a free trade. Then sit back and let it run until the price reaches 75% of the distance between risk and reward targets. You then have the option to exit all at once or in pieces. Larger positions benefit from a tiered exit strategy, exiting one-third at 75% of the distance between risk and reward targets and the second third at the target. Place a trailing stop behind the third piece after it exceeds the target, using that level as a rock-bottom exit if the position turns south.

2.4 The risk/reward ratio

Returns are characterised by non-Gaussian random variables and are event driven – spikes for rare events. Unexpected news produces abnormally large returns. This is illustrated in Figure (3).

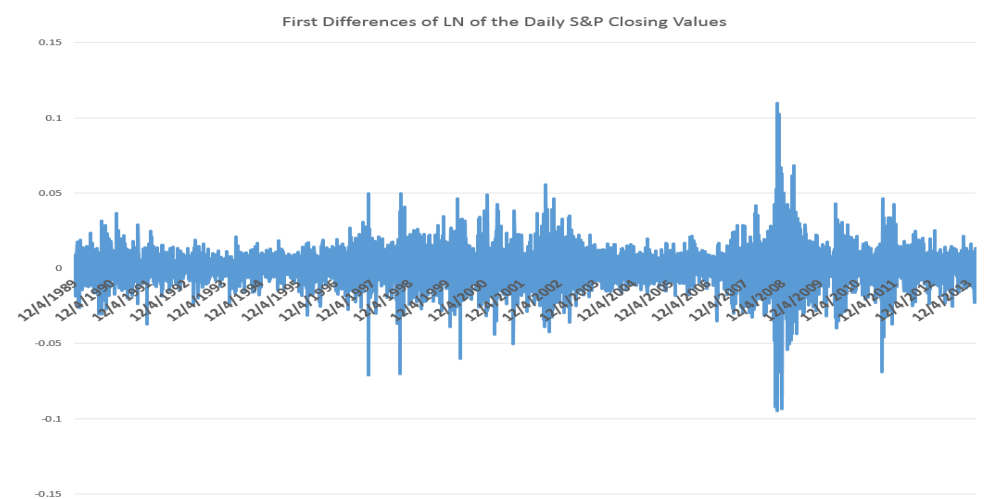


Figure 3: Daily market returns.

They can be simply decomposed as

$$\text{Return} = \text{Trend} + \text{Volatility} + \text{Jumps}$$

Future returns are unknown and their distribution is stochastic and non-stationary. This is illustrated in Figure (4). As a result, traders need to optimise their Risk/Return profiles dynamically.

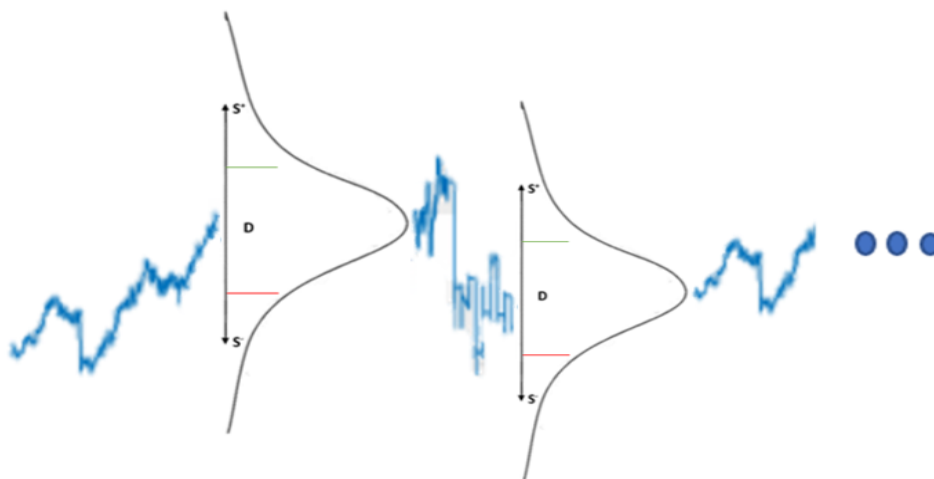


Figure 4: Returns distribution.

The **risk/reward ratio** is used by traders and investors to manage their capital and risk of loss during the life of a trade. It assesses the relationship between the expected return and expected risk of that trade. It marks the prospective reward an investor can earn for every dollar risked on an investment. It can be used to compare the expected returns of an investment (see Equation (7.41)) with the amount of risk they must undertake to earn these returns. For example, an investment with a risk-reward ratio of 1 : 5 suggests that an investor is willing to risk \$1, for the prospect of earning

§5. Investors can manage risk/reward more directly through the use of stop-loss orders and derivatives such as put options.

Remark 2.1 *The main problem with this approach is the difficulty in computing the PnL distribution and assessing what should be the values of the expected return and expected risk of a given trade. These are cutoff values corresponding to the tails of the distribution (see Figure (4)).*

Given an opportunity $i \in \mathcal{I}$ and a strategy S , the risk/reward ratio is calculated by dividing the amount (in cash) a trader stands to lose if the price of an asset moves in an unexpected direction (the risk) by the amount (in cash) of profit the trader expects to have made when the position is closed (the reward). To compute this ratio (over the life of a trade), we must estimate the distribution of the PnL defined in Equation (7.40), and then identify a risk target and a reward target. In a long position, traders let the risk target corresponds to the stop loss and the reward target corresponds to profit taking. In this sense, the risk/reward targets can be seen as barriers to hit. Hence, the first hitting time of a barrier becomes crucial. Again, we can use (1) data-driven methods or (2) process-driven methods to estimate the expected PnL and the risk/reward targets. In the approach (1), the chartists use support and resistance lines to estimate the risk/reward targets (see Section (2.4.4)). In the approach (2), we are interested in computing the distribution of the stock price at maturity T . The mean value of that distribution is the forward price $F(t, T) = E[S_T | \mathcal{F}_t]$ and the variance is $\text{var}(S_T | \mathcal{F}_t)$. These moments can be used to estimate the PnL distribution and identify the two cutoff values. We can also use linear combination of European options to estimate the risk/reward targets and barrier options to compute the first hitting time (see Section (6)).

2.4.1 Description

Given the initial time t and the trading horizon T , we let the entry price for the opportunity $i \in \mathcal{I}$ be S_t . For simplicity we only consider a long position. We let $S_{t,T}^+$ be the reward target and $S_{t,T}^-$ be the risk target seen at time t for the maturity T . The expected reward (gain) and expected risk (loss), seen at time t for the maturity T , are defined as follows:

$$\Delta_{t,T}^+ = S_{t,T}^+ - S_t \text{ and } \Delta_{t,T}^- = S_t - S_{t,T}^- \quad (2.1)$$

Note, $S_{t,T}^+$ is also called the take profit price and $\Delta_{t,T}^+$ is the take profit distance. Similarly, $S_{t,T}^-$ is also called the stop-loss price and $\Delta_{t,T}^-$ is the stop-loss distance. The distance between the risk and reward targets is

$$D_{t,T}^\pm = \Delta_{t,T}^+ + \Delta_{t,T}^- = S_{t,T}^+ - S_{t,T}^- \quad (2.2)$$

Note, these metrics can be normalised by the entry price S_t . We let the normalised reward target $S_{t,T}^{N,+}$ and the normalised risk target $S_{t,T}^{N,-}$ be

$$S_{t,T}^{N,+} = \frac{S_{t,T}^+}{S_t} \text{ and } S_{t,T}^{N,-} = \frac{S_{t,T}^-}{S_t} \quad (2.3)$$

Thus, the expected normalised return (gain) and expected normalised risk (loss) are

$$R_{t,T}^+ = \Delta_{t,T}^{N,+} = \frac{1}{S_t} \Delta_{t,T}^+ = \frac{S_{t,T}^+ - S_t}{S_t} \text{ and } R_{t,T}^- = \Delta_{t,T}^{N,-} = \frac{1}{S_t} \Delta_{t,T}^- = \frac{S_t - S_{t,T}^-}{S_t} \quad (2.4)$$

Then, the normalised distance is

$$D_{t,T}^{N,\pm} = R_{t,T}^+ + R_{t,T}^- = S_{t,T}^{N,+} - S_{t,T}^{N,-} \quad (2.5)$$

The risk/reward ratio is given by

$$RR^{\mp}(t, T) = \frac{R_{t,T}^{-}}{R_{t,T}^{+}} = \frac{\Delta_{t,T}^{-}}{\Delta_{t,T}^{+}}$$

Note, the reverse is the reward/risk ratio given by

$$RR^{\pm}(t, T) = \frac{R_{t,T}^{+}}{R_{t,T}^{-}} = \frac{\Delta_{t,T}^{+}}{\Delta_{t,T}^{-}}$$

2.4.2 Market practice

In practice, traders **arbitrarily** decide on the values of the expected reward and expected risk. Assuming a given reward target $S_{t,T}^{+}$, they select the risk target $S_{t,T}^{-}$ that gives them the desired risk/reward ratio. This is equivalent to assuming that the normalised distance $D_{t,T}^{N,\pm}$ is known, and choosing a level of risk, $\lambda \in [0, 1]$, such that the expected normalised reward and risk become

$$\Delta_{t,T}^{N,+} = (1 - \lambda)D_{t,T}^{N,\pm} \text{ and } \Delta_{t,T}^{N,-} = \lambda D_{t,T}^{N,\pm} \quad (2.6)$$

As a result, the risk/reward ratio $RR^{\mp}$ and the reward/risk ratio $RR^{\pm}$ become

$$RR^{\mp}(t, T) = \frac{\lambda}{1 - \lambda} \text{ and } RR^{\pm}(t, T) = \frac{1 - \lambda}{\lambda}$$

Equivalently, we get

$$\lambda = \frac{RR^{\mp}(t, T)}{1 + RR^{\mp}(t, T)} \text{ and } \lambda = \frac{1}{1 + RR^{\pm}(t, T)}$$

Hence, given the risk/reward ratio $RR^{\mp}$, or the level of risk λ , we can either assume that the normalised distance $D_{t,T}^{N,\pm}$ is known or that the normalised reward target $S_{t,T}^{N,+}$ is known.

At last, the normalised reward target $S_{t,T}^{N,+}$ and the normalised risk target $S_{t,T}^{N,-}$ are given by

$$S_{t,T}^{N,+} = \Delta_{t,T}^{N,+} + 1 \text{ and } S_{t,T}^{N,-} = 1 - \Delta_{t,T}^{N,-} \quad (2.7)$$

In a long position, we can then set the global stop loss equal to the normalised risk target $S_{t,T}^{N,-}$, while in a short position we set it to the normalised risk reward $S_{t,T}^{N,+}$. This way, we can capture the upward/downward trends in the market.

2.4.3 Examples

At time t , a trader purchases 100 shares of XYZ Company at price $S_t = \$20$, and places a stop-loss order at $SL = \$15$ to ensure that losses will not exceed \$500. That is, $S_{t,T}^{-} = 15$ and the stop-loss distance is $\Delta_{t,T}^{-} = 5$. Also, assume that this trader **believes** that the price of XYZ will reach the value $S_{t,T}^{+} = \$30$ in the next few months. In this case, the trader is willing to risk $\Delta_{t,T}^{-} = \$5$ per share to make an expected gain of $\Delta_{t,T}^{+} = \$10$ per share after closing the position. The distance is $D_{t,T}^{\pm} = \$15$ and the risk/reward ratio is $RR^{\mp} = \frac{1}{2}$. Since the trader stands to make double the amount that he has risked, he should be said to have a 1 : 2 risk/reward ratio on that particular trade. Note, a put option can be used to obtain a similar effect.

We can also compute the normalised reward target $S_{t,T}^{N,+} = \frac{3}{2}$ and the normalised risk target $S_{t,T}^{N,-} = \frac{3}{4}$, so that the expected normalised return is $\Delta_{t,T}^{N,+} = \frac{1}{2}$ and the expected normalised risk is $\Delta_{t,T}^{N,-} = \frac{1}{4}$. Hence, the normalised distance is $D_{t,T}^{N,\pm} = \frac{3}{2} - \frac{3}{4} = \frac{3}{4}$, the level of risk is $\lambda = \frac{1}{3}$, and the risk/reward ratio is $RR^{\mp} = \frac{1/3}{2/3} = \frac{1}{2}$. Alternatively,

given the level of risk λ and the normalised distance $D_{t,T}^{N,\pm}$, we recover the normalised reward target $S_{t,T}^{N,+}$ and the normalised risk target $S_{t,T}^{N,-}$.

Note, if an investor prefers to seek a 1 : 5 risk/reward ratio for a specified investment (five units of expected return for each additional unit of risk), then he can modify the stop-loss order to $SL = \$18$, and thus adjust the risk/reward ratio to $RR^{\mp}(t, T) = \frac{2}{10}$. However, by doing so the investor has changed the probability of success in his trade because the expected gain is still $\Delta_{t,T}^+ = \$10$ but the expected loss is now $\Delta_{t,T}^- = \$2$, leading to a distance of $D_{t,T}^{\pm} = \$12$. By doing so he would certainly reduce the size of the potential loss (assuming no change to the number of shares), but he will have increased the likelihood that the price action will trigger his stop loss order. This is because the stop order is proportionally much closer to the entry S_t than the target price S_T^+ is. Hence, although the investor may stand to make a proportionally larger gain (compared to the potential loss), he has a lower probability of receiving this outcome. The reverse is true if the investor seeks less return per unit of risk. For example, suppose an investor set a stop-loss order at $SL = \$\frac{40}{3}$, instead of $\$15$, leading to an expected loss of $\Delta_{t,T}^- = \$\frac{20}{3}$, and continued to target a $\Delta_{t,T}^+ = \$10$ profit-taking exit. In that case, the risk/reward ratio, $RR^{\mp}(t, T) = \frac{20/3}{10}$, becomes 2 : 3.

2.4.4 Estimating the risk/reward targets

Given the initial time t and trading horizon T , we want to estimate the reward target $S_{t,T}^+$ and risk target $S_{t,T}^-$ before entering each trade. One solution is to consider chart analysis:

- Mark the reward target $S_{t,T}^+$: We look at the chart and find the **next resistance level** likely to come into play within the time constraints of our holding period $\tau = T - t$.
- Mark the risk target $S_{t,T}^-$: We find the price where we will be proven wrong if the security turns and hits it. It can be the **next support level**.

Note, it is not clear how to estimate the next resistance level and the next support level. We can then calculate the **risk/reward ratio**, looking for a ratio of at least 1 : 2 in our favour. Anything less favourable, and we should skip the trade, moving on to a better opportunity.

Alternatively, we can use technical indicators (TIs) such as the Bollinger bands (BB) (see Appendix (13.1.2)), which identifies asset prices that have deviated from the mean. It is assumed that when a price goes beyond two standard deviations (above or below the mean) it should revert to the MA. Hence, assuming stationarity, we can use the upper band (TopBand) and the lower band (BotBand) as the reward target and the risk target, respectively. We can also use the Bollinger double bands (BDB) and play with the two upper bands (TopBand1) and (TopBand2) to control the reward target as well as with the two lower bands (BotBand1) and (BotBand2) to control the risk target.

The next step is to quantify the **speed** at which the stock price S_t will reach the reward target $S_{t,T}^+$. The faster the stock price gets to the reward target, the more flexibility we will have in choosing a favourable exit point. In essence, we are trying to estimate the probability for the stock of hitting an up or down barrier at $S_{t,T}^+$ or $S_{t,T}^-$, respectively. One solution from the chartist is to use the price rate of change (ROC) (see Appendix (13.1.2)). Then, when the price is strongly trending in our favour, we can let it exceed the reward target, placing a protective stop at that level $SL = S_{t,T}^+$, while we attempt to add to gains. Next, look for the next obvious barrier, staying positioned as long as it doesn't violate our holding period.

Practically, we need a **profit protection strategy** that kicks into gear once the price has traversed, say 75%, of the distance between our risk and reward targets $D_{t,T}^{\pm}$ in Equation (2.2). For instance, place a **trailing stop** that protects partial gains (see Section (2.5)). The trick is to stay positioned until a **price action** gives us a reason to get out. As discussed above, it can be done by using the Bollinger bands (BB) or the Bollinger double bands (BDB).

2.5 A few facts about trailing stop loss

2.5.1 Description

A trailing stop is a modification of a typical stop order that can be set at a defined percentage or dollar amount away from a security's current market price. For a long position, an investor places a trailing stop loss below the current market price, while for a short position it is placed above the current market price. It is designed to protect gains by enabling a trade to remain open and continue to profit as long as the price is moving in the investor's favour. The order closes the trade if the price changes direction by a specified percentage or dollar amount. A trailing stop is typically placed at the same time that the initial trade is placed, although it may also be placed after the trade.

A trailing stop is more flexible than a fixed stop-loss order, as it automatically tracks the stock's price direction and does not have to be manually reset like the fixed stop-loss. In a long position, there are two ways to enter a stop loss order. We can enter a dollar amount, for example, if our stock is selling at \$40 per share, we might enter a stop loss order for \$37.50 per share. When the stock price drops to \$37.50, it trips the stop loss order, and the broker sells it. Alternatively, we might tell our broker we want a trailing stop 10% below the market price. In that case, if a 10% trailing stop loss is added to a long position, a sell trade will be issued if the price drops 10% from its peak price after purchase. The trailing stop only moves up once a new peak has been established. Once the trailing stop has moved up, it cannot move back down. The trick is setting the percentage at a level that will pick up a true price drop as opposed to normal daily price fluctuations.

However, establishing the ideal trailing stop distance is difficult. A short-term fluctuation in a stock's price can activate the stop price. In general, it is very difficult to pick the proper stop-loss percentage that allows a stock to fluctuate, while also preventing as much downside risk as possible. There is no ideal distance since markets and the way that stocks move are always changing. A simple rule consists in using a wider trailing stop in more volatile periods, and consider a tighter trailing stop loss during quieter times, or with a very stable stock. In fact, there is no ideal distance because markets and the way that stocks move are always changing. The ideal trailing stop loss will change over time. During more volatile periods, a wider trailing stop is a better bet. Conversely, during quieter times, or in a very stable stock, a tighter trailing stop loss may be effective. More formally, it is an optimal stopping problem which can be modelled in some cases (see Section (7.6)).

2.5.2 Managing risk

Trailing stop-loss rules provide dynamic protection in that the sell trigger price is adjusted upwards if the price moves higher following a purchase. As a result, a position is closed if the price subsequently declines a given percentage below the new high price (see trailing stop in Equation (7.49)). With a stop-loss we no-longer need to micro-manage our portfolio. We can specify the size of our stop-loss, enter the order into the market and it will sit and wait until it needs to be triggered. However, the main drawbacks of a stop-loss order is that in volatile markets the stop loss can be triggered following a particularly volatile short-term spike in the markets, only to watch the value recover or move into profitable territory over the medium-to-long term. Hence, this means there are only certain trading scenarios where stop-loss orders are best employed, including:

- Stocks experiencing a bullish uptrend with no apparent risk on the horizon.
- Trading opportunities based on fundamental analysis such as press releases or news stories using high leverage. These stories will often be published in the news feeds of reputable traders.
- Uncertain markets whereby companies have solid fundamentals.
- Stocks experiencing bearish downtrends or shares heading for a market correction based on technical analysis.

By using stop loss and take profit orders in tandem, we can control the risk versus reward ratio of any trade we enter. Understanding stop-loss orders and take-profit orders, we can determine our reward/risk ratio on any trade we enter in the market. It is computed by dividing our target net profit by the price of our maximum risk, that is, where our stop loss order is placed (see Section (2.4)). Determining where to place a stop-loss order is based on our risk threshold. A stop-loss should be viewed as a worst-case scenario, a signal in the market that our judgement on a share price is wrong, whilst limiting our losses. Once we have defined the minimum reward/risk ratio that we are comfortable with using, it is simple to position our stop-loss order. For example, if we have a 4 : 1 reward/risk ratio, we may be prepared to risk 25% of our initial investment to hit our take-profit.

2.5.3 Setting trailing stop loss

We saw that in a long position, setting fixed bounds for profit taking and stop loss was a challenge as they could be disproportionately large or narrow. For instance, placing a conservative stop-loss can either close positions prematurely or lead to missing out on potential gains. Conversely, placing an aggressive exit condition will result in substantial losses, especially when the closed position reverses back into gains which will no longer be realised. Implicitly, this approach consists in estimating the probability of reaching an up and down barrier to infer the quantiles of the return distribution. However, returns are not Gaussian and the market conditions keep changing. That is, the market distribution for returns is stochastic and non-stationary so that we cannot decide in a static way what should be the first and third quantile where to cut the distribution. As a result, risk levels should be assessed dynamically both at the individual asset and overall market levels. That is, we need to make stop loss and target gains dynamic.

First, we briefly review some existing approaches for making stop loss and target gains dynamic. Then, we present a few techniques used to set the trailing stop loss.

2.5.3.1 Existing approaches

Some well known approaches are:

- The naive fixed threshold: It assigns upper and lower thresholds at the time of entering a position. These bounds remain static until one of them is crossed, thereby exiting the position.
- The Trailing Threshold: Once calculated, the dollar value of the upper and lower thresholds are reapplied on a rolling basis according to price. Even though the distance between the upper and lower bounds is static, the value of these bounds is adjusted day to day.
- The ATR threshold: It assigns upper and lower thresholds at the time of entering a position according to the stock's idiosyncratic volatility expressed by its ATR (average true range) calculated over a recent lookback period (see Appendix (13.1.1)). The bounds are therefore decided from historical price action information and not determined arbitrarily. Once computed, the upper and lower thresholds remain static until one of them is crossed. However, arbitrarily chosen ATR cut offs are suboptimal and can cause investors to lose money when tighter bounds are chosen, or endure unnecessary risk when looser ones are selected.
- The ATR trailing threshold: The bounds are calculated in a data-driven manner according to historical price information. Once calculated, the upper and lower thresholds are reapplied on a roll forward basis each time a new high is reached. While the distance between the upper and lower bounds is static, the value of these bounds is adjusted to protect the gains already incurred.
- The dynamic ATR threshold: The bounds are computed according to ATR over a historical lookback period. These bounds are static and set based on price for the day on which they are calculated. After N days, the thresholds are recomputed according to ATR over a historical lookback period, and the process repeats.
- The dynamic ATR trailing threshold: The bounds are determined according to ATR calculated over a historical lookback period. Once computed, the upper and lower thresholds are reapplied on a roll forward basis according to price for N days. On day $N + 1$, the thresholds are recalculated according to ATR over a historical lookback period, and the process repeats. This is illustrated in Figure (5).

- The Moving Average stop loss measure: Surges in volatility frequently create large fleeting price changes that result in positions exiting prematurely. This phenomenon can deplete an investor of profitable trades. The effects of sudden price deviations can be tempered by exiting trades according to a Moving Average of historical prices, rather than the price itself. The smoothing effect caused by averaging makes instances of the depleting phenomenon unlikely.
- Decelerating stop loss: It reduces the stop loss levels gradually as returns are further cemented until we ultimately exit by hitting a trailing stop loss but at a much lower threshold compared to when the trade was started. It takes the trailing stop loss to a new level as the trailer still adjusts when a new high is reached (or a new low if we are targeting a short trade) but will also contract the stop limit accordingly.
- Deep Reinforcement Learning for exit conditions: Here, the actions are represented by the bound values set as a function of ATR. Instead of setting bounds that are disproportionately large or narrow, it produces thresholds that are just right.

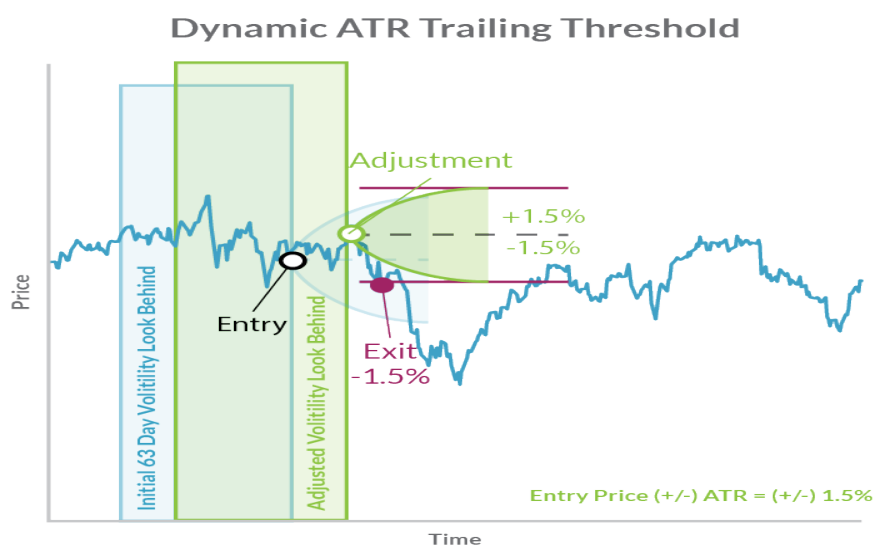


Figure 5: Dynamic ATR trailing threshold.

2.5.3.2 Using moving average The Moving Average (MA) (see Appendix (11.1)) is an indicator that averages out the past prices. We can use it to identify trends, filter out noise. We can also use it to trail our stop loss (see Section (3)) as follows:

1. Decide on the type of trend we want to ride
2. Use the appropriate Moving Average
3. Exit when the price closes beyond it (crosses the average from above)

For example, if we want to ride a short-term trend, we can trail our stop loss with a 20-period MA and exit our trade when the price closes beyond it. Similarly, we can use the 50-period MA to ride the medium-term trend and the 200-period MA to ride the long-term trend.

2.5.3.3 The zero indicator method to trail stop loss Since an uptrend consists of higher highs and higher lows, we can use the swing low to trail our stop loss because if the trend holds, it shouldn't close below it. The steps are as follows:

1. Identify the previous swing low
2. Set our trailing stop loss below the swing low
3. If the price closes below it, exit the trade

The market tends to hunt stop losses below support or swing low. It can be avoided by setting our stop loss 1 ATR below the market structure.

2.5.4 Problems

The approaches for setting the trailing stop loss presented in the previous section are based on the principle that market prices follow a diffusion process. They assume a continuous-time Markov process such as a Gaussian process. Given an initial starting price, the process diffuses in time, creating a cone of diffusion. This idea has been widely used in Technical Analysis where a few technical indicators use volatility or ATR to calculate the upper and lower bands around an averaged price process (see Appendix (13.1.1)). For instance, the Bollinger Bands use standard deviation to calculate the upper and lower bands, while the Keltner Channels use ATR. In general, the averaged process is the daily average price (or typical price) or the exponential moving average (EMA) price. They assume that prices are Gaussian $N(\mu, \psi)$, where $\psi = \sigma$, ATR is a measure of deviation from the mean. Given $P(t)$ the price seen at time t and the maturity $T = t + \Delta t$, we get

$$P(T) = P(t) + \mu\Delta t \pm \alpha \times \psi\sqrt{\Delta t}$$

where α is the number of standard deviation. It gives the dispersion for the price distribution. However, in financial models, prices are log-normally distributed and returns are normally distributed. Further, the drift is set to zero, $\mu = 0$, so that the cone of diffusion is symmetric around each starting price $P(t)$. That is, at any point in time, prices are as likely to go up or down. This means that, as we diffuse, the price distribution is always centred on the current price, which is not the case. In reality, we only observe one trajectory in the market making financial time series sensitive to both their values and their sequential order. One solution is to cluster temporal features along the time series by grouping observations such that they share the same distribution within a cluster (see Bemporad et al. [2018]). That is, we can build a price distribution across time in a sliding window. If the current price happen to be in one the tail of the distribution, then the likelihood to go up or down is clearly not symmetric, which leads to a misspecified risk/reward ratio and potential losses.

2.6 Chart analysis

2.6.1 Overview

Patterns are the distinctive formations created by the movements of security prices on a chart. A pattern is identified by a line that connects common price points, such as closing prices or highs or lows, during a specific period of time. They provide some logic on what the market may potentially do. As a result, chartists seek to identify patterns as a way to anticipate the future direction of a security's price. They can then create a trading plan to take advantage of these patterns.

Chart patterns fall broadly into three categories:

1. Continuation patterns: A continuation signals that an ongoing trend will continue. They occur mid-trend and represent a pause in the price action of varying durations. Triangles, flags, pennants, and rectangles are examples of continuation patterns.

2. Reversal patterns: They indicate that a trend may be about to change direction (from going up to going down, or vice-versa). Reversal refer to large price changes, where the trend changes direction, while pullbacks (or consolidations) refer to small counter-moves against the trend. A reversal keeps going and forms a new trend, while a pullback ends and then the price starts moving back in the trending direction. Examples of common reversal patterns include head and shoulders, double tops, double bottoms.
3. Bilateral patterns: They let traders know that the price could move either way, meaning the market is highly volatile

Thomas [2000] identified 47 different chart patterns revealing different market trends. Examples of such patterns are consolidation, cup with handle, head-and-shoulders, double bottom, saucer. Figure (6) display typical chart patterns. However, any stock chart may exhibit countless different pattern combinations, making it very difficult identify and predict.

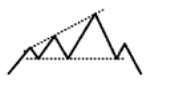
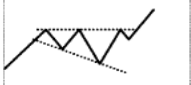
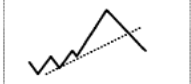
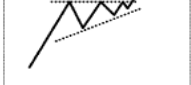
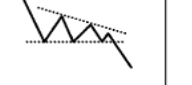
 Broadening Bottoms	 Broadening Formations, Right-Angled and Ascending	 Broadening Formations, Right-Angled and Descending	 Broadening Tops
 Broadening Wedges, Ascending	 Bump-and-Run Reversal Bottoms	 Bump-and-Run Reversal Tops	 Cup with Handle
 Diamond Bottoms	 Diamond Tops	 Double Bottoms	 Double Tops
 Head-and-Shoulders, Top	 Head-and-Shoulders, Bottoms	 Triangles, Ascending	 Triangles, Descending

Figure 6: Chart patterns

Since chart pattern identification is highly subjective, pattern analysis led to the development of technical indicators (TIs) defined as mathematical transformations of market returns (see Murphy [1999], Lo et al. [2000], Wang et al. [2007]). See Appendix (13) for more details on TIs. For example, technical analysts use support and resistance levels to identify price points on a chart where the probabilities favour a pause or reversal of a prevailing trend.

- Support (floor) occurs where a downtrend is expected to pause due to a concentration of demand.
- Resistance (ceiling) occurs where an uptrend is expected to pause temporarily, due to a concentration of supply.
- Support and resistance areas can be identified on charts using trendlines and moving averages (see Appendix (11.1)).

Hence, by analysing financial data and studying charts, we can anticipate which way the market is most likely to go. That is, even though we do not know when we pick a specific stock if its price is going to rise or fall, we can use technical indicators to give us a future perspective on its behaviour in order to determine the best choice when building a portfolio. Technical indicators try to capture the behaviour and investment psychology in order to determine if a stock is under or overvalued.

2.6.2 Support and resistance

Assuming a long position, the basics of Support and Resistance consist of a support level, which can be thought of as the floor under the realised prices, and a resistance level, which can be thought of as the ceiling above the realised prices. This is illustrated in Figure (7). In a downtrend, market prices fall and test the support level, which will either hold, and the price will reverse to the upside, or be violated, and the price will drop through the support and likely continue lower to the next support level. The logic reverses in an uptrend. Hence, Support and Resistance are bilateral patterns. They are defined as follows:

- **Support:** In a downtrend, prices fall because there is an excess of supply over demand. The lower prices go, the more attractive prices become to those waiting on the sidelines to buy the shares. At some level, demand that would have been slowly increasing will rise to the level where it matches supply. At this point, prices will stop falling. This level is called a support.
- **Resistance:** It is the opposite of support. Prices move up because there is more demand than supply. As prices move higher, there will come a point when selling will overwhelm the desire to buy. This happens for a variety of reasons. It could be that traders have determined that prices are too high or have met their target. It could be the reluctance of buyers to initiate new positions at such rich valuations. It could also be for a number of other reasons. But a technician will clearly see on a price chart a level at which supply begins to overwhelm demand. This level is called resistance. Like support, it can be a level or a zone.

Once an area (or zone) of support or resistance has been identified, those price levels can serve as potential entry or exit points because, as price reaches a point of previous support or resistance, it will do one of two things: (1) bounce back away from the support or resistance level (breakdown), or (2) violate the price level (break line) and continue in its prior direction until it hits the next support or resistance level (breakout). Figure (7) illustrates first a breakout and then a breakout. See Section (2.6.3) for more details on breakout and breakdown.



Figure 7: Support and resistance curves.

The timing of some trades is based on the belief that support and resistance zones will not be broken. Whether the price is halted by or breaks through the support or resistance level, traders can bet on the direction of price and can quickly determine if they are correct. If the price moves in the wrong direction (breaks through prior support or resistance levels), the position can be closed at a small loss (Risk). However, if the price moves in the right direction (with respect to prior support or resistance levels), the move may be substantial (Gain).

2.6.2.1 Time scales Support and resistance can be found in all charting time periods; daily, weekly, monthly. Traders also find support and resistance in smaller time frames like one-minute and five-minute charts. But the longer the time period, the more significant the support or resistance. To identify support or resistance, we need to look back at the chart to find a significant pause in a price decline or rise. Then look forward to see whether a price halts and/or reverses as it approaches that level. Many experienced traders will pay attention to past support or resistance levels and place trades in anticipation of a future similar reaction at these levels.

2.6.2.2 Risk Technical analysis (TA) relies on historical price data and patterns to predict future market movements. However, past performance does not always indicate future results. Hence, technical analysis indicators can give false signals, which can lead to poor trading decisions. As an example, price will sometimes dip below support levels or reverse before it gets to the prior support level. The same is true for resistance: Price may reverse before it gets to the prior resistance level or break above it. In each case, flexibility is required in interpreting these chart patterns. This is why support and resistance levels are sometimes referred to as zones. Further, TA can be less effective during times of high market volatility where sudden price movements can trigger stop losses and result in unexpected losses.

2.6.2.3 Trendlines Since the price series of financial assets generally trends upward or downward, the support and resistance (floor and ceiling) lines change over time. This is why the concepts of trending and trendlines are important when learning about support and resistance.

Historically, when the price of a security falls toward the broader support of the (down) trendline, it has been an area that has prevented the price of the asset from moving substantially lower. A trendline can provide support for an asset for a period of time. For example, an upward sloping floor and ceiling. On the other hand, when the market is trending

to the downside, traders will watch for a series of declining peaks and will attempt to connect these peaks together with a trendline (downward sloping floor and ceiling). When the price approaches the trendline, most traders will watch for the asset to encounter selling pressure and may consider entering a short position because this is an area that has pushed the price downward in the past. To be a valid trendline, price needs to touch the trendlines at least three times. Sometimes with stronger trendlines, price will touch the trendline several times over longer time periods. Also, in an uptrend, the trendline is drawn below price, while in a downtrend, the trendline is drawn above price.

The support/resistance of an identified level, whether discovered with a trendline or through any other method, is deemed to be stronger the more times that the price has historically been unable to move beyond it. Many technical traders will use their identified support and resistance levels to choose strategic entry/exit points because these areas often represent the prices that are the most influential to an asset's direction. Most traders are confident at these levels in the underlying value of the asset, so the volume generally increases more than usual, making it much more difficult for traders to continue driving the price higher or lower.

2.6.2.4 Trading range These are areas where support and resistance levels are relatively close and price bounces between two levels for a period of time. Experienced traders will sometimes trade within these trading ranges, which are also known as **sideways trends**. One strategy used is to place short trades as the price touches the upper trendline and long trades as price reverses to touch the lower trendline. This strategy is extremely dangerous, and it is much better to wait to see in which direction price will break out of the range and then place your trades in that direction.

2.6.2.5 Identify support and resistance Some of the rules for identifying support and resistance are as follows:

- **Number of Touches:** The more times that the price tests a support or resistance area, the more significant the level becomes. When prices keep bouncing off a support or resistance level, more buyers and sellers notice and will base trading decisions on these levels.
- **Preceding Price Move:** Support and resistance zones are likely to be more significant when they are preceded by steep advances or declines. For example, a fast, steep advance or uptrend will be met with more competition and enthusiasm and may be halted by a more significant resistance level than a slow, steady advance. A slow advance may not attract as much attention. This is a good example of how market psychology drives TIs.
- **Volume at Certain Price Levels:** The more buying and selling that has occurred at a particular price level, the stronger the support or resistance level is likely to be. This is because traders and investors remember these price levels and are apt to use them again. When strong activity occurs on high volume and the price drops, a lot of selling will likely occur when price returns to that level, since people are far more comfortable closing out a trade at the breakeven point than at a loss.
- **Time:** Support and resistance zones seen in longer time frame charts such as weekly or monthly charts are often more significant than those seen in shorter time frame charts such as the one-minute or five-minute chart.

2.6.3 Breakout and breakdown

A **breakout** occurs when the price moves above a resistance level or moves below a support level (see Section (2.6.2)). It indicates the potential for the price to start trending in the breakout direction. A breakout to the upside indicates traders to possibly get long or cover short positions. Conversely, a breakout to the downside indicates traders to possibly get short or to sell long positions. A **breakdown** is a downward move in a security's price, usually through an identified level of support, that portends further declines. Traders can draw trendlines on a chart that connect several swing lows ⁴ to find areas where prices may be susceptible to breaking down (see Figure (7)). It usually occurs on

⁴Swing low refers to the troughs reached by a security's price or an indicator during a given period of time, usually fewer than 20 trading periods. It is created when a low is lower than any other surrounding prices in a given period of time.

heavy volume and the subsequent move lower tends to be quick in duration and severe in magnitude. When a break down occurs, traders seek confirmation from several indicators and other chart time-frames to ensure the move is not a head-fake. A breakdown is the bearish counterpart of a breakout.

A typical breakout or breakdown unfolds in three distinct phases: (1) action, (2) reaction, and (3) resolution. Each phase generates specific trading strategies. Breakouts and breakdowns start with an action impulse that carries price above or below a support or resistance level (a horizontal high or a trendline). The market trend evolves, stretching away from the prior barrier, and then stalls, printing a notable high (low in a breakdown) and entering a reaction impulse. A reaction is a sudden but usually short-lived upwards or downwards movement in a stock's price. For instance, when the market is trending to the upside, resistance levels are formed as the price action slows and starts to move back toward the trendline. An action-reaction breakout is illustrated in Figure (8) and an action-reaction breakdown is illustrated in Figure (9).

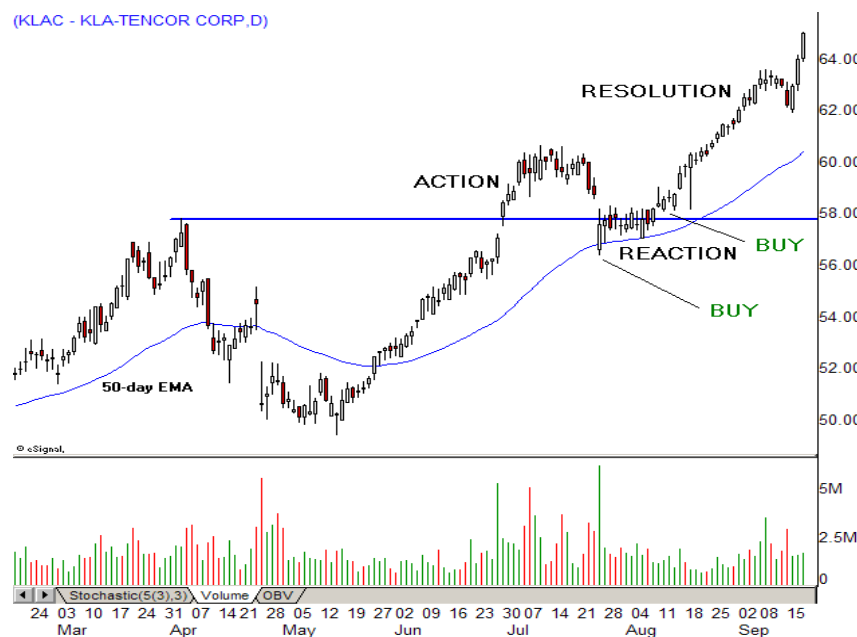


Figure 8: Action and reaction breakout signal.



Figure 9: Action and reaction breakdown signal.

Technical analysts call the moment when a price is moving against the prevailing trend a reaction. Reactions can occur for a large variety of reasons, including profit taking or near-term uncertainty for a particular issue or sector. The resulting price action undergoes a plateau effect, or a slight drop-off in stock price, creating a short-term top. It may last for only a few sessions before reverting back to the prevailing trend. A reaction is similar to a correction or a reversal but lacks the same intensity or longevity. A downward reaction is often a response to negative news. An overreaction is an extreme response to new information.

The reaction usually marks the testing period, often characterised by pullbacks to new support or resistance. Price action eventually completes a consolidation pattern, yielding one of two outcomes: (1) it pushes beyond the action impulse high or low swing, confirming the new trend (see Figure (8)), or (2) the trend fails, yielding a reversal that traps trend followers (see Figure (9)). In both cases, it marks the resolution impulse. As a result, traders can (1) play the breakout when price pushes above resistance (or the breakdown when it fails support), or (2) take a narrow range entry, entering in low volatility before the move occurs and, placing a tight stop, getting out for a small loss if proved wrong. In both cases, we can use a trailing stop, locking in profits by raising or lowering it as the security swings in our direction. During the reaction, we can buy the dip or sell the rip when price pulls back to new support or resistance. Finally, we shift back to a momentum strategy when the resolution impulse sets into motion, buying high and selling higher or selling short low and covering lower.

2.6.4 Confluence

Confluence occurs when multiple ideas or strategies are used together to form a single, coherent idea or strategy. We briefly introduce two types of confluence: confluence of technical indicators and confluence of lines. Thus, we are looking for areas where two or more analysis tools or levels are intersecting.

2.6.4.1 Confluence of technical indicators In technical analysis, confluence results from employing several indicators or signals to come up with entry and exit points or spot reversals in a trend. This is because traders require several indications of a buy and sell signal combined together before they can affirm a trade decision. For instance,

traders use a confluence of indicators to determine a signal at a potential reversal. A price approaching a particular resistance or support line has the potential to reverse or continue pushing through the trendline. Consequently, traders examine several indicators occurring simultaneously or within a short-term time frame to confirm the trend through confluence. We can also combine TIs with Order Book and Fundamental analysis. We provide a few examples below:

- We can combine a MACD with a Stochastic Indicator and check that they confirm each other (see Section (3.1)). The MACD indicator is a trend-following indicator that depicts the connection between a 26 and 12-day EMA from which the MACD and the signal line are calculated. On the other hand, the Stochastic Indicator primarily gauges whether an asset is overbought or oversold.
- We can combine the RSI (relative strength index) along with Moving Averages (see Section (3.1)). The mid-level of the RSI is 50, and a reading over 50 indicates bullish momentum. On the other hand, when it falls below 50, it suggests a bearish trend. Thus, when the MAs provide a signal, we can use the RSI to determine whether the momentum is strong enough to justify entering the trade.
- We can combine technical indicators with fundamental analysis. Fundamental analysis offers the reason why the market moves. For instance, markets consolidate before important economic data or before critical political events.
- We can combine technical indicators with the Level 2 Order Book. An order book is a collection of orders for a given market that exchange records to evaluate the market depth and presents buying and selling interests from all market participants. It is used to gauge market sentiment. Hence, we combine various indicators to validate the market sentiment. For example, if an order book shows several large buy orders, and the indicators show bullish momentum, we can go long knowing that the price will increase.

2.6.4.2 Confluence of lines We can also define confluence as the intersection of two or more lines (getting a confluence zone) such as trend lines, horizontal levels, and even channels, which are really just equidistant trend lines. This is called a confluence of support or resistance. Market participants are more likely to gather around an area that involves two or more factors rather than an area with just one. Hence, these intersections provide some of the best areas to watch for buying and selling opportunities. Some examples include

- The crossing of a trend line and a horizontal level.
- The crossing of two trend lines or channels, one ascending and the other descending.
- Combining intersections: horizontal level, trend line, and channel.
- Combining a trend line, a horizontal level, and a Fibonacci. Fibonacci levels are used as confirmation tools.

The confluence zone exists on any time frame. It can be divided into two parts:

1. Confluence buying zone: Place where we have two or more buy signals near the price level. Examples include stock price above or near a trend line support, horizontal support, demand zone, Fibonacci level (50%, 61.8%), moving average line, bullish candles formation near support line/zone, price rejection at a key support level.
2. Confluence selling zone: Similarly, we can find a confluence zone that provides a sell signal or prevents us from buying stocks in a selling zone. Examples include stock price below or near a trend line resistance, horizontal resistance, supply zone, Fibonacci level (50%, 61.8%), moving average line, bearish candles formation near resistance line/zone, price rejection at key resistance level.

This is illustrated in Figure (10). In this example, the (decreasing) trend line intersects with a horizontal level and an (increasing) channel resistance, providing an area with three intersecting levels.



Figure 10: Confluence of support or resistance

These areas of confluence tend to attract more buyers or sellers resulting in a reversal or continuation of the current trend. In this example, one can also look at key levels in the channel resistance. A key level is one where we have several bounces on the support or resistance, intensifying the signal strength. That is, by simply watching price behaviour and its strength gives us some indication as whether price will hold resistance or break it.

2.7 Candlestick representation

Candlesticks are created by up and down movements in the price. While these price movements sometimes appear random, they often form patterns traders use for analysis or trading purposes. Traders use candlestick charts to make trading decisions based on irregularly occurring patterns that help forecast the short-term direction of the price.

2.7.1 Candlestick charts

In financial markets, we observe high frequency price action of specific underlying assets. Over a given fixed period of time (such as a day or less), they are sampled and expressed as open, high, low, and close (OHLC) prices. This leads to multivariate stock time series (four dimensions) illustrated in Figure (11).

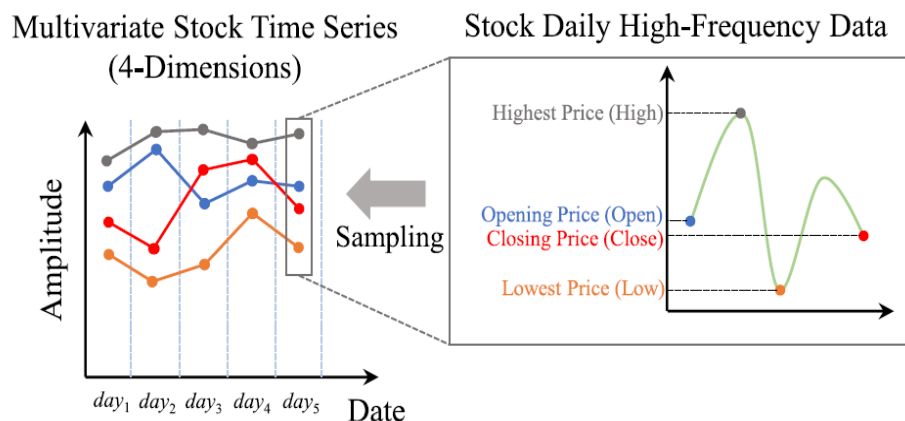


Figure 11: Multivariate stock time series.

A candlestick chart (also called K-line) is a graphical representation of stock price movements in chronological order, with time intervals ranging from one minute to one hour, day to month and prices such as open, high, low, and close. It is drawn by close, open, high, and low, for a given period, where the part between the close and open is called real body. If the asset closed higher than it opened, the body is filled with red colour in Chinese stock market while represented with white or green colour in European and American stock markets. If the close price is lower than open price, the body is painted with green colour in Chinese stock market while filled with black or red color in European and American stock markets. This is illustrated in Figure (12).

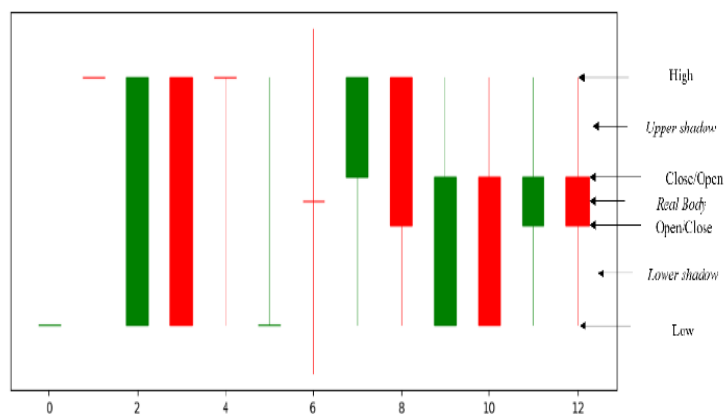


Figure 12: Candlestick representation.

A **shadow**, or a **wick**, is a line found on a candle in a candlestick chart that is used to indicate where the price of a stock has fluctuated relative to the opening and closing prices. They illustrate the highest and lowest prices at which a security has traded over a specific time period. The **price range** is the distance between the top of the upper shadow and the bottom of the lower shadow moved through during the time frame of the candlestick. The range is calculated by subtracting the low price from the high price.

The candlestick chart is considered as a special representation of a stock's multidimensional price series, and it is essentially a daily sampling of high-frequency price data. This is illustrated in Figure (13).

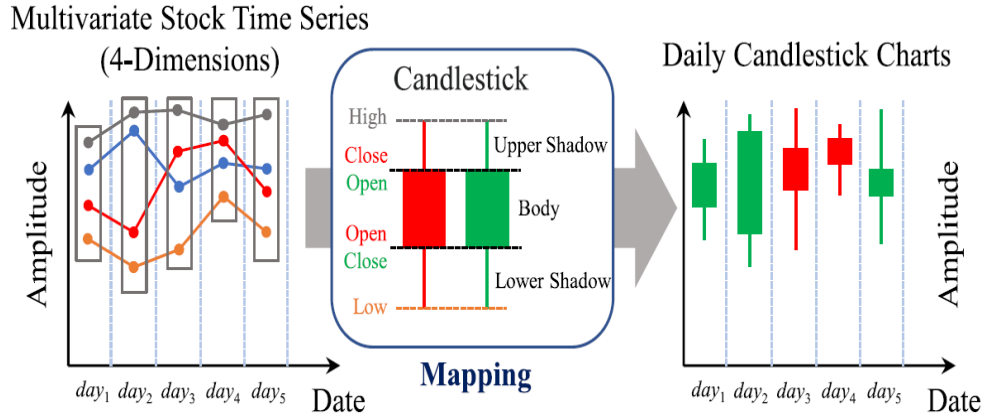


Figure 13: Daily candlestick charts.

In a high-volatility environment, price candles are usually larger and exhibit longer wicks. In a high-momentum environment, we only observe one colour of candles (very few candles moving against the trend), and smaller candle wicks against the trend direction.

2.7.2 Definition

We properly define candlesticks and present their relative positions. We use the following notation:

- o_t represents the open price at time t
- h_t represents the highest price at time t
- c_t represents the closing price at time t
- l_t represents the lowest price at time t

Definition 2.1 At time t , a candlestick $k_t = (o_t, h_t, l_t, c_t)$ is a tuple that consists of four basic prices of a stock. It is a basic element in identifying the candlestick pattern recognition. Additionally, $k_i = (o_i, h_i, l_i, c_i)$ denotes the i th candlestick for $i = 1, \dots, n$. A candlestick time series $T_n = \{k_1, \dots, k_n\}$ is a sequence of candlesticks of a stock, consisting of n candlesticks from day 1 to day n .

We consider 8 candlestick relative position (or trigrams) (see Lin et al. [2021] and Lin et al. [2021b]):

1. BullishHorn (BC_h): The oscillations of the day exceeded the previous cycle and reached a new high and a new low, reflecting the strong characteristics of an oscillation. It satisfies the following three conditions:
 - $h_t > h_{t-1}$
 - $l_t < l_{t-1}$
 - $c_t > c_{t-1}$
2. BearHorn (BC_l): The oscillations of the day exceeded the previous cycle and reached a new high and a new low, but the closing price was lower than the previous cycle, reflect the weak characteristics of an oscillation. It satisfies the following three conditions:
 - $h_t > h_{t-1}$

- $l_t < l_{t-1}$
 - $c_t < c_{t-1}$
3. BullishHigh (BH_h): It shows that the candlestick creates a new high price and the lowest price is higher than the previous period, reflecting a rising strong feature. It satisfies the following three conditions:
- $h_t > h_{t-1}$
 - $l_t > l_{t-1}$
 - $c_t > c_{t-1}$
4. BearHigh (BH_l): It reflects a weak rising feature. The candlestick creates a new high price while the closing price is lower than the previous period. It satisfies the following three conditions:
- $h_t > h_{t-1}$
 - $l_t > l_{t-1}$
 - $c_t < c_{t-1}$
5. BullishLow (BL_h): It contains the information that the stock is getting stronger. The stock hit a new low, but the closing price exceeds the previous cycle. It satisfies the following three conditions:
- $h_t < h_{t-1}$
 - $l_t < l_{t-1}$
 - $c_t > c_{t-1}$
6. BearLow (BL_l): It reflects the weak characteristics. The stock hit a new low and the closing price is lower than the previous periods. It satisfies the following three conditions:
- $h_t < h_{t-1}$
 - $l_t < l_{t-1}$
 - $c_t < c_{t-1}$
7. BullishHarami (BM_h): It shows the process of energy accumulation. The stock's amplitude is within the range of the previous period and the closing price is higher than the previous period. It satisfies the following three conditions:
- $h_t < h_{t-1}$
 - $l_t > l_{t-1}$
 - $c_t > c_{t-1}$
8. BearHarami (BM_l): It reflects another process of energy accumulation. The stock's amplitude is within the range of the previous period while the closing price is lower than the previous period. It satisfies the following three conditions:
- $h_t < h_{t-1}$
 - $l_t > l_{t-1}$
 - $c_t < c_{t-1}$

These patterns are displayed in Figure (14).

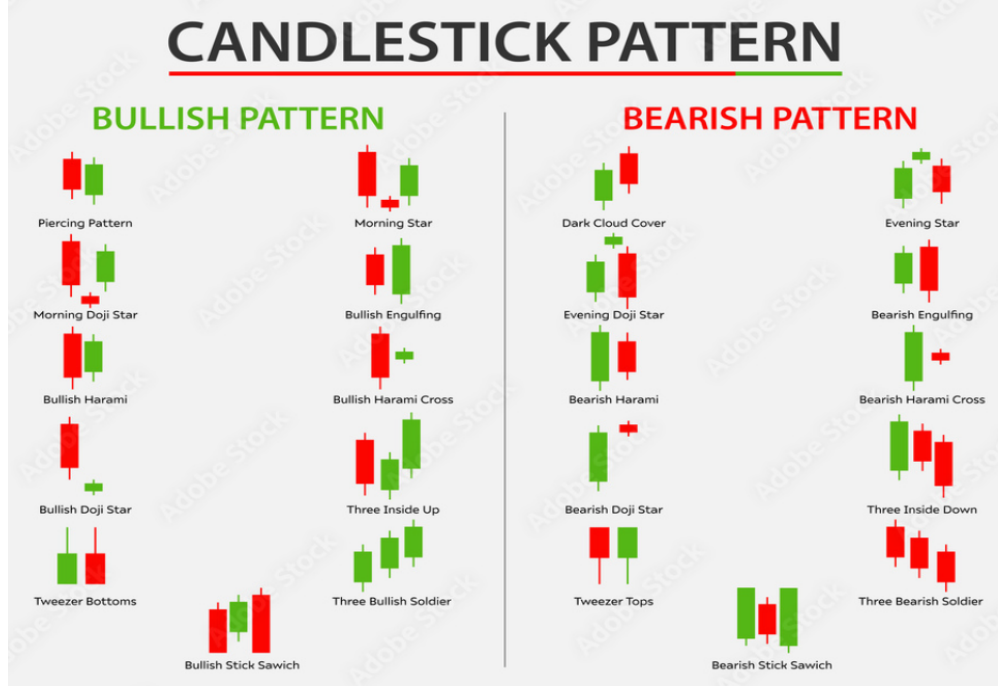


Figure 14: Candlestick patterns.

2.7.3 Some patterns

We denote loc_i the relative position of the i th candlestick. Then, $Loc_n = \{loc_1, \dots, loc_n\}$ is a sequence of relative position information of a stock, consisting of n relative position information from day 1 to day n . We can now define a candlestick pattern.

Definition 2.2 Candlestick pattern

A candlestick pattern (or K -line pattern) $p_j = \{T_j, Loc_j\}$ is a subsequence of consecutive candlesticks consisting of two parts: (1) a sequence of candlesticks, (2) a corresponding location sequence.

For example, a two-day pattern is defined as $p_2 = \{T_2, Loc_2\}$ where $T_2 = \{k_1, k_2\}$ and $Loc_2 = \{loc_1, loc_2\}$. According to the different values of open, high, low and close, one-day patterns can have 13 different forms. We display in Figure (15) the one-day pattern forms and two-day and three-day candlestick patterns.

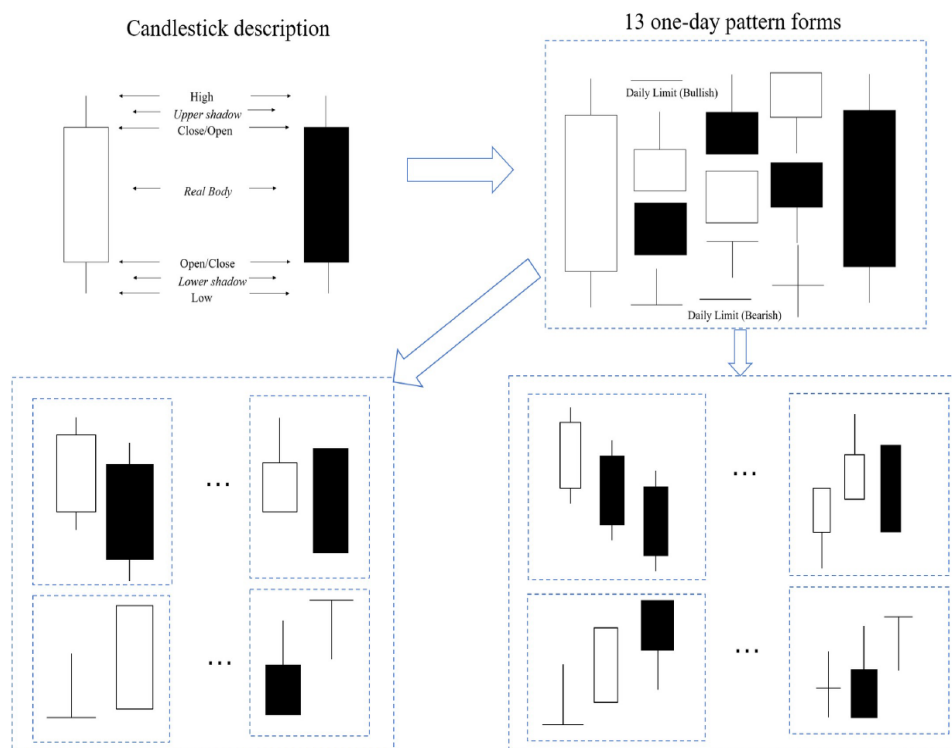


Figure 15: Candlestick representation two-day and three-day patterns.

Some candlestick patterns can be found in Bigalow [2014]. Examples of patterns are Bullish Engulfing Signal, Bearish Engulfing Signal, Bullish Harami, Bearish Harami, Hammer Signal, and Shooting Star.

2.8 Some chart patterns

Price action (PA) refers to the changes of a security's price over time. **Price action trading (PAT)** focuses almost solely on the movement of price over time. This movement is often analysed with respect to price changes in the recent past. Due to the repetitive nature of market participants and their reaction to global economic variables, the PA of a market tends to repeat itself in various patterns. Hence, price action analysis focuses on trends, chart patterns, and significant historical price events and movements. Patterns are recurring sequences found in OHLC candlestick charts (see Section (2.7)) which traders have historically used as buy and sell signals. Several studies found some correlation between patterns and future trends. These patterns are identified and analysed with the help of technical indicators, such as charts, trend lines, price bands, high and low swings, technical levels (of support, resistance and consolidation), etc. As a result, rather than trying to analyse a large number of economic variables each day, we can simply learn to trade price actions. When analysing the markets, traders look for an obvious price action pattern that has formed at a confluent point in the market (see Section (2.6.4)). We briefly present some of these chart patterns and will present some PAT in Section (4.1).

2.8.1 Head and shoulders

Head and shoulders is a chart pattern in which a large peak has a slightly smaller peak on either side of it. Traders look at head and shoulders patterns to predict a bullish-to-bearish reversal. In general, the first and third peak will be smaller than the second, but they will all fall back to the same level of support (known as the neckline). Once the

third peak has fallen back to the level of support, it can breakout into a bearish downtrend or a bullish uptrend. This is illustrated in Figure (16).

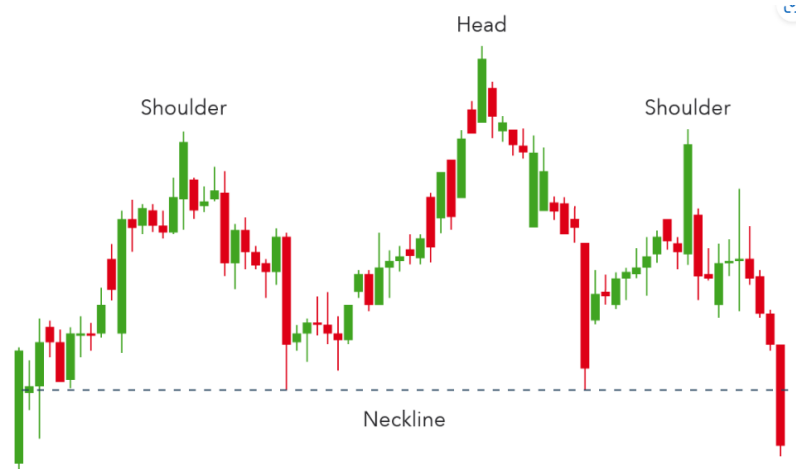


Figure 16: Head and shoulder

2.8.2 Double top

A double top is a pattern used to highlight trend reversals. Typically, an asset's price will experience a peak, before retracing back to a level of support. It will then climb up once more before reversing back more permanently against the prevailing trend. This is illustrated in Figure (17).



Figure 17: Double top

2.8.3 Double bottom

A double bottom indicates a period of selling, causing an asset's price to drop below a level of support. It will then rise to a level of resistance, before dropping again. Finally, the trend will reverse and begin an upward motion as the

market becomes more bullish. It is a bullish reversal pattern, because it signifies the end of a downtrend and a shift towards an uptrend. This is illustrated in Figure (18).

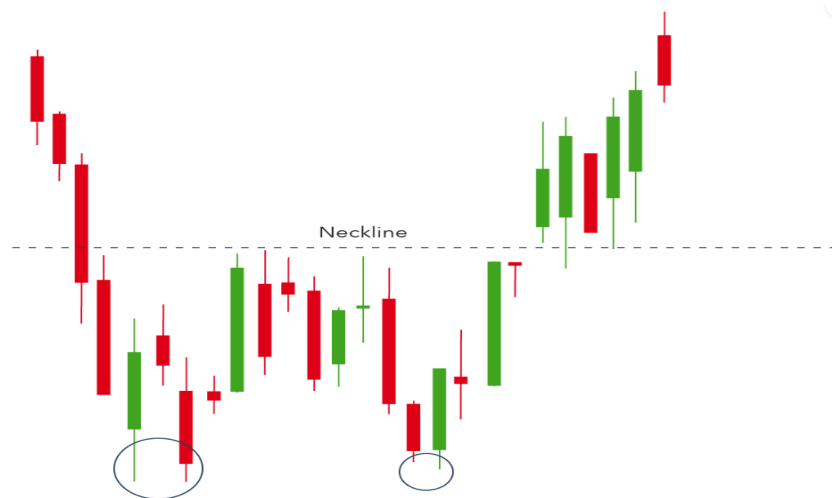


Figure 18: Double bottom

2.8.4 Rounding bottom

A rounding bottom chart pattern can signify either a continuation or a reversal. For instance, during an uptrend an asset's price may fall back slightly before rising once more. This would be a bullish continuation. For example, a bullish reversal rounding bottom is as follows: an asset's price is in a downward trend and a rounding bottom is formed before the trend reversed and entered a bullish uptrend. This is illustrated in Figure (19). Traders will seek to capitalise on this pattern by buying halfway around the bottom, at the low point, and capitalising on the continuation once it breaks above a level of resistance.

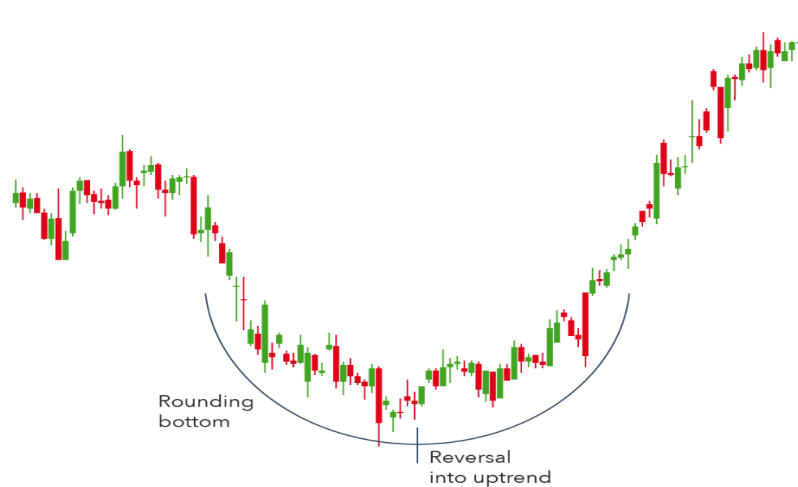


Figure 19: Rounding bottom

2.8.5 Cup and handle

The cup and handle pattern is a bullish continuation pattern used to show a period of bearish market sentiment before the overall trend finally continues in a bullish motion. The cup appears similar to a rounding bottom chart pattern, and the handle is similar to a wedge pattern. This is illustrated in Figure (20). Following the rounding bottom, the price of an asset will likely enter a temporary retracement, which is known as the handle because this retracement is confined to two parallel lines on the price graph. The asset will eventually reverse out of the handle and continue with the overall bullish trend.

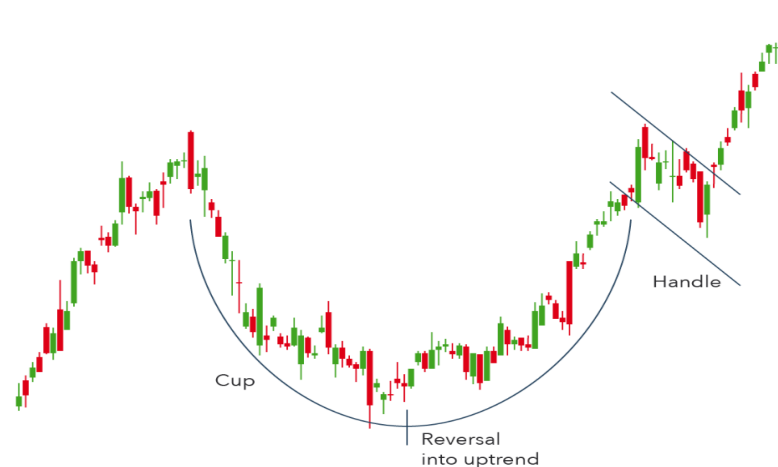


Figure 20: Cup and handle

2.8.6 Wedges

Wedges form as an asset's price movements tighten between two sloping trend lines. Both trendlines are moving in the same direction, either up or down. There are two types of wedge:

1. Rising wedge: it is represented by a trend line caught between two upwardly slanted lines of support and resistance. In this case the line of support is steeper than the resistance line. This pattern generally signals that an asset's price will eventually decline more permanently, which is demonstrated when it breaks through the support level. This is illustrated in Figure (21).
2. Falling wedge: it occurs between two downwardly sloping levels. In this case the line of resistance is steeper than the support. It is usually indicative that an asset's price will rise and break through the level of resistance. This is illustrated in Figure (22).

Both rising and falling wedges are reversal patterns, with rising wedges representing a bearish market and falling wedges being more typical of a bullish market.



Figure 21: Rising wedge



Figure 22: Falling wedge

2.8.7 Pennant, rectangle and flags

We often observe pauses in a trend (consolidation) in which the price action moves sideways, bound between parallel support and resistance lines. This is called a rectangle (or a trading range) and can last for short periods or many years.

Flags are a pause in the trend, where the price becomes confined in a small price range between parallel lines giving them the appearance of a flag (see Figure (2)). They are generally short in duration, lasting several bars, and do not contain price swings back and forth as a trading range or trend channel would. They may be parallel or upward or downward sloping. Flag patterns have five main characteristics: (1) the preceding trend, (2) the consolidation channel, (3) the volume pattern, (4) a breakout, (5) a confirmation where price moves in the same direction as the breakout. Bullish and bearish patterns have similar structures but differ in trend direction.

Pennant patterns are created after an asset experiences a period of upward movement, followed by a consolidation. Generally, there will be a significant increase during the early stages of the trend, before it enters into a series of smaller upward and downward movements. Pennants can be either bullish or bearish, and they can represent a continuation or a reversal. For example, a bullish continuation is illustrated in Figure (23). Pennants can be a form of bilateral pattern because they show either continuations or reversals. Wedges differ from pennants because a wedge is always ascending or descending, while a pennant is always horizontal.

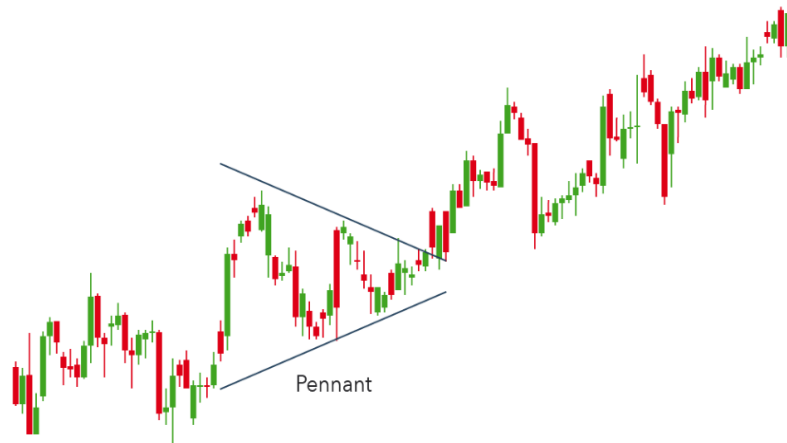


Figure 23: Pennant

2.8.8 The triangles

The triangles are defined as a converging of the price range, with higher lows and lower highs, such that the converging price action creates a triangle formation. While they vary in their duration, they have at least two swing highs in price and two swings lows in price. As price continues to converge, the price will eventually reach the apex of the triangle. The closer it gets to the apex, the tighter and tighter the price action becomes, thus making a breakout more imminent. There are three basic types of triangles: ascending, descending, and symmetrical.

2.8.8.1 Ascending triangle The ascending triangle is a bullish continuation pattern signifying the continuation of an uptrend. They can be drawn onto charts by placing a horizontal line along the swing highs (the resistance), and then drawing an ascending trend line along the swing lows (the support). They often have two or more identical peak highs which allow for the horizontal line to be drawn. The trend line signifies the overall uptrend of the pattern, while the horizontal line indicates the historic level of resistance for that particular asset. This is illustrated in Figure (24).



Figure 24: Ascending triangle

2.8.8.2 Descending triangle A descending triangle signifies a bearish continuation of a downtrend. Typically, a trader will enter a short position during a descending triangle (possibly with CFDs) in an attempt to profit from a falling market. They generally shift lower and break through the support because they are indicative of a market dominated by sellers, meaning that successively lower peaks are likely to be prevalent and unlikely to reverse. They can be identified from a horizontal line of support and a downward-sloping line of resistance. Eventually, the trend will break through the support and the downtrend will continue. This is illustrated in Figure (25).



Figure 25: Descending triangle

2.8.8.3 Symmetrical triangle The symmetrical triangle pattern can be either bullish or bearish, depending on the market. In either case, it is normally a continuation pattern, which means the market will usually continue in the same direction as the overall trend once the pattern has formed. They form when the price converges with a series of lower peaks and higher troughs. This is illustrated in Figure (26). In this example, the overall trend is bearish, but the symmetrical triangle shows us that there has been a brief period of upward reversals.

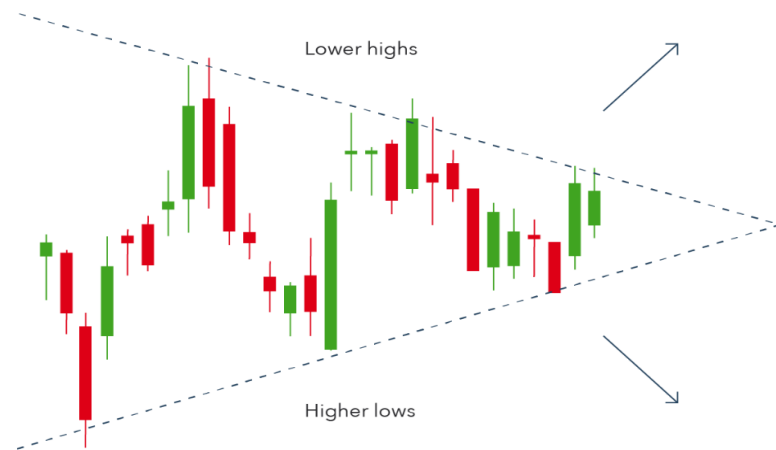


Figure 26: Symmetrical triangle

Note, if there is no clear trend before the triangle pattern forms, the market could break out in either direction. This makes symmetrical triangles a bilateral pattern, meaning they are best used in volatile markets where there is no clear indication of which way an asset's price might move. An example of a bilateral symmetrical triangle is illustrated in Figure (27).

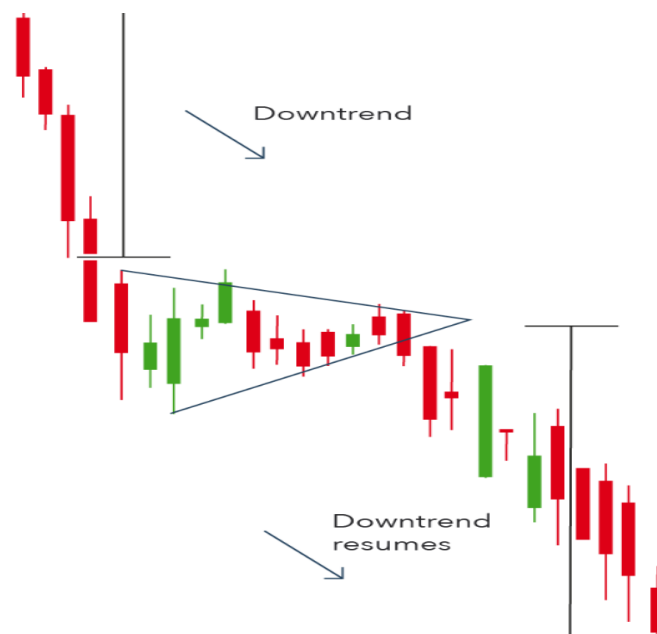


Figure 27: Bilateral symmetrical triangle

2.9 Some strategies using patterns

We present a few trading strategies using patterns in price action trading (PAT) (see Section (4)).

2.9.1 Channels charting

A channel occurs when the price of an asset oscillates between two parallel trendlines. There are three types of channels: horizontal (equal high, equal low), ascending (higher high, higher low), and descending (lower high, lower low) channels (the last two are also called trend channels). Channels that are angled up are called ascending channels (or bullish movement). Channels that are angled down are called descending channels (or bearish movement). This is illustrated in Figure (28). They provide traders with precise points for entering and exiting trades inside these ranges until prices breakout or breakdown. We can determine the likelihood of success of a trade through confirmations⁵: [1, 2] weak channel, [3, 4] adequate channel, [5, 6] strong channel. However, this strategy takes a long time to be validated as it requires at least two lows to connect, as well as two highs. Horizontal channels (also called price range or sideways trend) are trend lines that connect variable pivot highs and lows to show the price contained between the upper line of resistance and the lower line of support. This is illustrated in Figure (29). In that setting, buying and selling pressure is equal, and the prevailing price direction is sideways. In trend channels, the upper trendline connects the swing highs in price, while the lower trendline connects the swing lows in price.

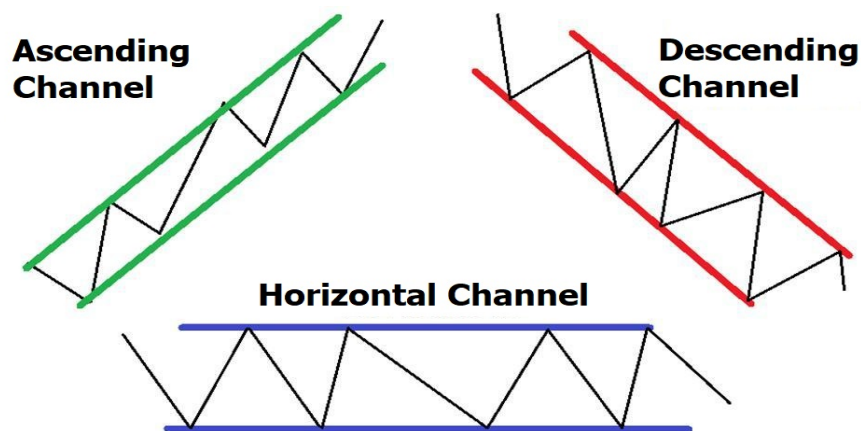


Figure 28: Channels charting



Figure 29: Horizontal channel

⁵Confirmations represent the number of times the price has rebounded from the top or bottom of the channel.

One possible trading strategy is as follows:

- When the price hits the top of the channel, sell your existing long position or take a short position.
- When the price is in the middle of the channel, do nothing if you have no trades open or hold onto your current trades.
- When the price hits the bottom of the channel, cover your existing short position or take a long position.

This strategy stops when the price breaks through the top or bottom of the channel. During a rising channel, focus on buying near the bottom of the channel and exiting near the top. During a descending channel, focus on shorting near the top of the channel and exiting near the bottom. Some other tools to use while channel trading include:

- The MACD is often near zero during horizontal channels. The MACD line crossing the signal line can also point out potential long trades near the bottom of a channel or short trades near the top of the channel.
- A stochastic crossover may signal a buying opportunity near the bottom of the channel or a selling opportunity near the top.
- Volume can aid in trading channels. Volume is often lower in channels, especially near the middle of the channel. Breakouts are often associated with high volume. If the volume isn't rising on a breakout, there is a greater likelihood the channel will continue.

2.9.2 Pullbacks and throwbacks

In general, a long-term trend is defined by a long-term moving average (such as a 200-day). We go long if the trend is positive and we go short if it is negative. Pullbacks and throwbacks exit both in positive and negative trends. Throwbacks are mirror images of the pullbacks occurring in the opposite direction. They are both a very common feature after a breakout ⁶. Volume on the breakout is a good indicator that a pullback/throwback will follow.

2.9.2.1 Pullbacks A pullback is usually applied to pricing drops that are relatively short in duration. The term means the upward movement or retracement which occurs following the downward thrust in price. It is a temporary reversal in the upside/downside price action of an asset or security. While pullbacks and reversals both involve a security moving off its highs/lows, pullbacks are temporary and reversals are longer-term. However, pullbacks could be the start of true reversals. A longer pause before the uptrend/downtrend resumes is generally referred to as consolidation. Pullbacks can provide an entry point for traders looking to enter a position when other technical indicators remain bullish/bearish. For instance, pullbacks involve a security's price moving down to an area of technical support, such as a moving average, pivot point, or Fibonacci retracement level, and bouncing back before resuming the uptrend/downtrend. This is illustrated in Figure (30).

⁶A breakout is when the price moves above a resistance level or moves below a support level.

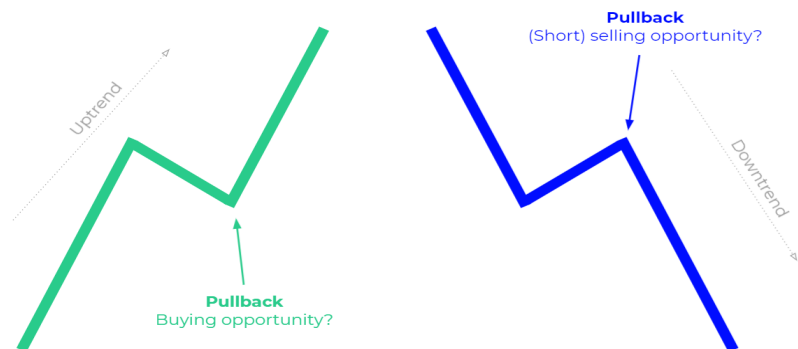


Figure 30: Pullback in positive and negative trend

Breakout pullbacks commonly happen at market turning points. They occur after a breakout: the price breaks through support, but then rebounds back up to that support level before resuming its direction (which depends on the long-term trend).

- Assuming a positive long-term trend, the price breaks through support (or just reach support), and then rebounds back up to that support level (or previous level) before resuming its upward trend. That is, up-down-up action. This is illustrated in Figure (31). Hence, a pullback trading strategy involves selling the stock at the top (when it starts decreasing) and buying it after it has experienced a decline in price (at or below the support level). The rationale behind this strategy being that the stock is likely to rebound from its recent decline and return to its previous price level or even higher.
- Assuming a negative long-term trend, the price breaks through support, and then rebounds back up to that support level (which becomes the resistance) before resuming its downward trend. That is, down-down-up-down action. This is illustrated in Figure (32). Hence, a pullback trading strategy involves buying the stock at the bottom (below support when it starts increasing) and selling it after it has experienced an increase in price (above the support level). The rationale behind this strategy being that the stock is likely to rebound from its recent increase and return to its previous price level or even lower.



Figure 31: Pullback in positive trend

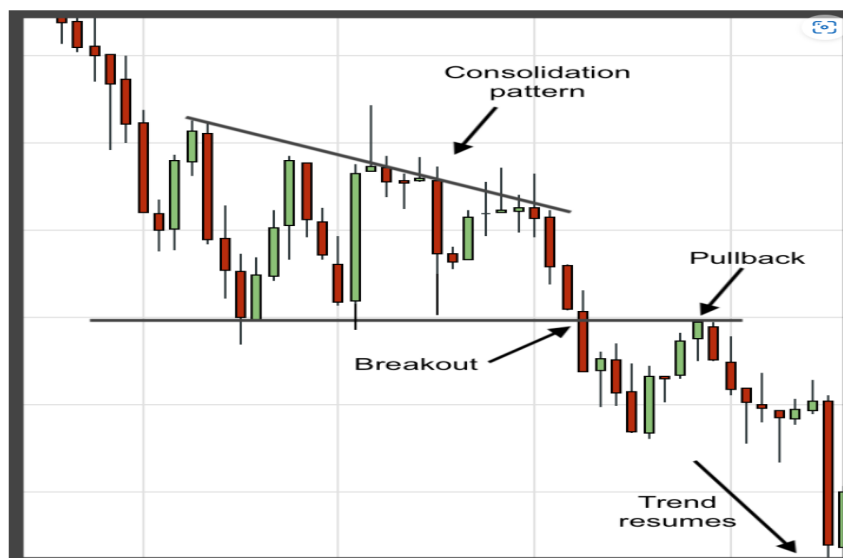


Figure 32: Pullback in negative trend

2.9.2.2 Throwbacks A throwback is usually applied to pricing rises that are relatively short in duration. The term means the downward movement or retracement which occurs following the upward thrust in price. It is a temporary reversal in the upside/downside price action of an asset or security. However, throwbacks could be the start of true reversals. Throwbacks can provide an entry point for traders looking to enter a position when other technical indicators remain bullish/bearish. For instance, throwbacks involve a security's price moving up to an area of technical resistance, such as a moving average, pivot point, or Fibonacci retracement level, and bouncing back before resuming the uptrend/downtrend.

Breakout throwbacks commonly happen at market turning points. They occur after a breakout: the price breaks

through resistance, but then rebounds back down to that resistance level before resuming its direction (which depends on the long-term trend).

- Assuming a positive long-term trend, the price breaks through resistance, and then rebounds back down to that resistance level (which becomes the support) before resuming its upward trend. That is, up-up-down-up action. This is illustrated in Figure (33). Hence, a throwback trading strategy involves selling the stock at the top (above resistance when it starts decreasing) and buying it after it has experienced a decline in price (below the resistance level). The rationale behind this strategy being that the stock is likely to rebound from its recent decline and return to its previous price level or even higher.
- Assuming a negative long-term trend, the price breaks through resistance (or just reach resistance), and then rebounds back down to that resistance level (or previous level) before resuming its downward trend. That is, down-up-down action. This is illustrated in Figure (34) with resistance represented by trade line. Hence, a throwback trading strategy involves buying the stock at the bottom (when it starts increasing) and selling it after it has experienced a recent increase in price (above the resistance level). The rationale behind this strategy being that the stock is likely to rebound from its recent increase and return to its previous price level or even lower.



Figure 33: Throwback in positive trend



Figure 34: Throwback in negative trend with resistance represented by trade line.

2.9.3 Multiple pullbacks and drawbacks

Multiple pullbacks and drawbacks are series of several pullbacks and drawbacks on strong long-term trends. It is characterised by price waves where the markets alternate between bullish (rising) and bearish (falling) trend waves. During an uptrend, the dominant trend waves moved higher. This is illustrated in Figure (35) in the case of pullbacks. The correction waves represent moves against the ongoing trend direction. In that setting, we enter each trade at the lowest possible price when the price start to pull back during the trend.

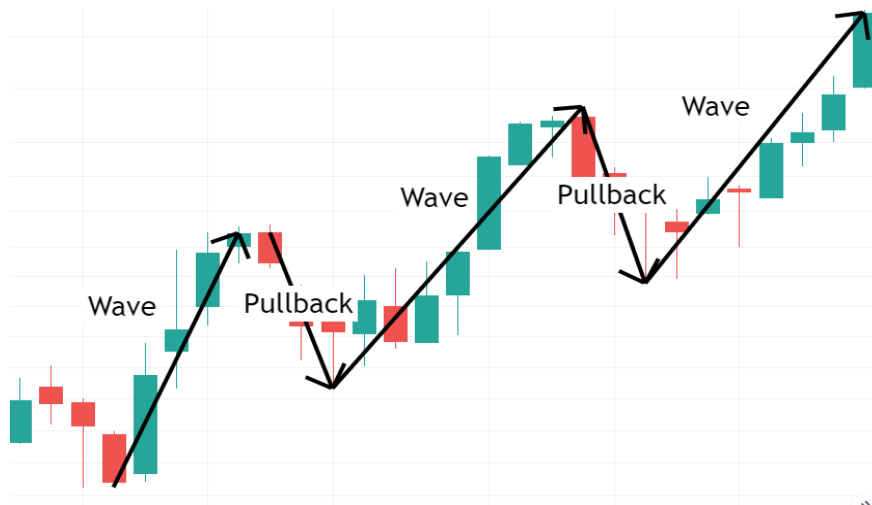


Figure 35: Pullback in positive wave trend

During ongoing trending phases, the price often present horizontal stepping patterns (or horizontal channels) leading to breakouts followed by pullbacks and/or throwbacks. This is illustrated in Figure (36) in the case of throwbacks.



Figure 36: Multiple throwbacks in positive trend

We can also use trend lines to characterise pullbacks and throwbacks (these are the trend channels). This is illustrated in Figure (37) in the case of throwbacks. However, the trendlines take longer to be validated as it requires several contact points (minimum two) to get validated. In general, we get a trendline at the third contact point so that pullbacks and throwbacks can only be traded at the third, fourth or fifth contact point.



Figure 37: Throwbacks with trendlines in negative trend

3 Trading with moving averages

Moving averages (MAs) (see Appendix (11.1)) are used to smooth out the short-term price fluctuations of an asset, providing a clearer picture of the underlying trend. MAs are lagging indicators because they are computed by using historical data. A faster moving average (short term or short lookback period) has less lag when compared to a slower moving average (long term or long lookback period). This is illustrated in Figure (38). There are different types of moving averages, including simple moving averages (SMA) (see Appendix (11.1.1)), exponential moving averages (EMA) (see Appendix (11.1.4)), and weighted moving average (WMA) (see Appendix (11.1.2)), each with its own method of calculating the average. The significant difference between the different MAs is the weight assigned to data points in the moving average period. For instance, SMA (see Equation (13.86)) give equal weight to all data points in the chosen period, EMA (see Equation (13.87)) give more weight to recent data points, making them more responsive to recent price changes, WMA also gives greater weight to recent prices, but each price point is assigned a weight based on its position in the period, with the most recent prices given the highest weights. It is a compromise between the SMA and EMA, providing a balance between smoothness and responsiveness. There are other types of MAs such as Smoothed Moving Average (SMMA), the Triangular Moving Average (TMA), and the Volume Weighted Moving Average (VWMA). The TMA applies more weight to data in the middle of the moving average period.

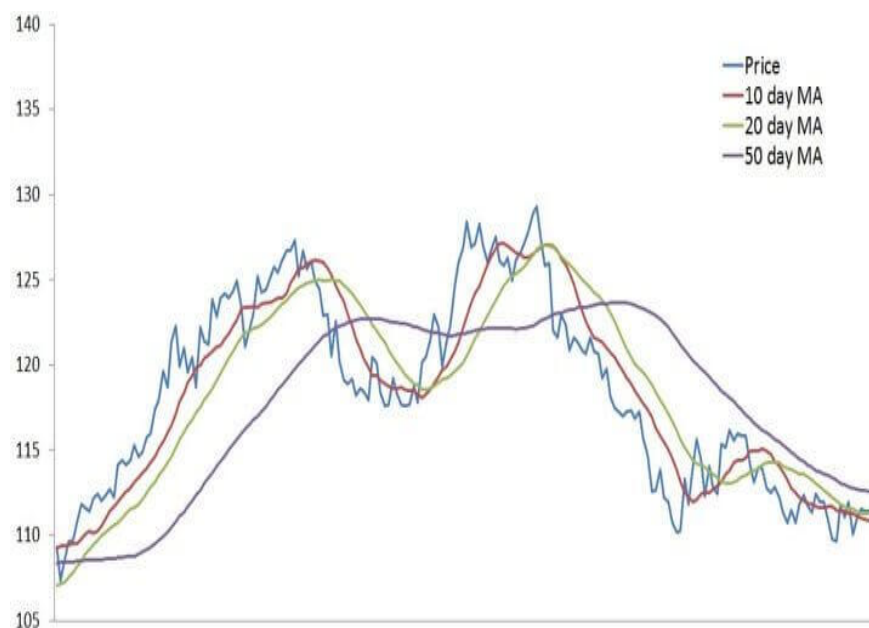


Figure 38: SMA with different speeds.

In moving average trading, each moving average indicator has its own pros and cons. The SMA moves much slower than other averages, keeping trades longer when there are short-lived price movements or price fluctuations. The EMA reacts faster when the price is changing direction. For example, if the price retraces lower, the EMA will start turning down to indicate a change in the trading signal.

3.1 Some strategies

3.1.1 Description

Moving averages (MAs) are used for identifying potential entry and exit points in the market. In general, we combine the moving averages to obtain moving average crossover (MAC) strategies (See Faith [2007]). A moving average

with a short time period will react much faster to price changes than a moving average with a long time period. They are powerful tools for identifying trends and market reversals. As a result, traders can gain insight into the direction of the market and the strength of the trend, making informed decisions in the market. Some of the most common MA strategies include:

- **Double Moving Average Crossover (DMAC):** This strategy uses two moving averages (MA) of different lengths. A buy signal is generated when the shorter moving average crosses (from below) above the longer moving average (a Golden Cross), and a sell signal is generated when the shorter moving average crosses (from above) below the longer moving average (a Death Cross).
- **Exponential Moving Average (EMA) Crossover:** This strategy uses exponential moving averages that give more importance to recent price data. When a shorter-term EMA crosses (from below) above a longer-term EMA, it generates a buy signal, and when it crosses (from above) below, it generates a sell signal.
- **Triple Moving Average Crossover (TMAC):** This strategy uses three moving averages of different lengths (fast, medium, and slow). A buy signal is generated when the fast moving average crosses (from below) above both the medium and slow moving averages, and a sell signal is generated when the fast moving average crosses (from above) below both the medium and slow moving averages. It is better equipped at dealing with false trading signals than the dual moving average crossover system. The buy signal is generated early in the development of a trend and a sell signal is generated early when a trend ends. When the security begins an uptrend, faster MA (short term) will begin rising much earlier than the slower MAs (medium and long term). The third MA is used in combination with the other two MAs to confirm or deny the signals they generate, reducing the probability of acting on false signals. For instance, the signal to buy is confirmed when the medium MA crosses (from below) above the slow MA. Similarly, the signal to sell is confirmed when the medium MA crosses (from above) below the slow MA. When the fast moving average goes above the medium moving average, the system exits its position. The system is out of the market when the relationship between the slow and medium moving averages do not match that between the medium and fast moving averages.
- **Moving Average Ribbon:** This strategy involves placing a large number of moving averages onto the same chart, creating a ribbon effect. Trading signals are generated in a similar manner to the triple MAC system, one must decide the number of crossovers to trigger a buy or sell signal. We buy when the faster MAs cross (from below) above the slower MAs and sell when the faster MAs cross (from above) below the slower MAs. The ribbon can help traders identify trend strength, potential entry and exit points, and trend reversals. When the ribbon is wide and moving averages are parallel, it indicates a strong trend, while a narrowing ribbon and converging moving averages signal a potential trend reversal. These MAC strategies can be powerful tools for traders to identify trend changes and potential entry and exit points in the market. However, they should be used in conjunction with other technical indicators to improve their effectiveness.
- **Moving Average Convergence Divergence (MACD):** See details in Appendix (13.1.2). The MACD is illustrated in Figure (39). The system buys when the MACD line crosses the signal line from below and sells when the MACD line crosses the signal line from above. It is understood that the trend in the underlying security is about to escalate in the direction of the crossover. One can also consider zero crossover when the slow and fast moving averages of the price curve crossover each other, or when the MACD series changes sign. A change from positive to negative is considered to be a bearish sign, while a change from negative to positive is considered as a bullish sign. The zero crossover provides confirmation about a change in trend but it is less reliable in triggering signals than the signal crossover. Alternatively, we can monitor the divergence between the MACD line and the signal line, which can be observed through the histogram. When the histogram starts falling (moves towards the zero line), it indicates that the trend is weakening, this happens when the MACD and signal lines are converging. In any case, when the signal line and MACD line are diverging, or the histogram is rising (moves away from the zero line), it is an indication that the trend is growing stronger.

- The Chande momentum oscillator (CMO) takes values in $[-100, 100]$ and measures the market trend strength. It can be used as a confirmation signal when it crosses above or below the zero line. It can also be used to spot bullish and bearish price divergence between the indicator and underlying security. A bearish divergence occurs if the underlying security is trending upward and the CMO is moving downwards. Conversely, a bullish divergence occurs if the price is declining but the oscillator is rising. This is illustrated in Figure (40). In this example, Apple made a high in late August and another one in late September. On the other hand, the CMO made a lower high in late September, confirming a bearish divergence. Hence, a trader deciding to sell short can place a stop-loss order above the September swing high and take profits when the oscillator crosses below -50 .
- The Bollinger bands (BB) can be used for determining overbought and oversold levels, as a trend following tool, and for monitoring for breakouts. In general, the outer bands coincide with Support and Resistance. This is illustrated in Figure (41). Some of the strategies include
 - Overbought and oversold strategy: When the price of an asset is trading near the upper BB (TopBand), it is considered overbought and may indicate that the price is likely to fall. Conversely, when the price is trading near the lower BB (BotBand), it is considered oversold and may indicate that the price is likely to rise. Hence, a typical system buys (+1) when price reaches the bottom (BotBand) and liquidates (−1) as the price moves up past the MA. Note, if the MACD shows a bullish crossover when the price bounces off the lower Bollinger Band, it could confirm a buy signal. The sell side is simply the opposite. BB identifies asset prices that have deviated from the mean. It is assumed that when a price goes beyond two standard deviations it should revert to the MA. Note, we can adjust your Bollinger Bands settings to 3 standard deviations (or higher) to identify even more overbought/oversold levels to trade-off. At last, some practitioners revert the logic and sell (−1) rather than buy when prices reach the lower band (BotBand), and vice versa with the upper band (TopBand). This is typical of range-bound markets where mean reversion strategies work well, as prices travel between the two bands like a bouncing ball.
 - Trend following: First look at the overall direction of price and only then take trade signals that align us with the trend. For example, if the trend is down, only take short positions when the upper band is tagged (or touched). The lower band can still be used as an exit if desired, but a new long position is not opened since that would mean going against the trend.
 - Squeeze strategy: A squeeze occurs when the price has been moving aggressively then starts moving sideways in a tight consolidation. In that case, the upper and lower bands get closer together, meaning that the volatility of the asset has decreased. This is called volatility contraction. After a period of consolidation, the price often makes a larger move in either direction (price break), ideally on high volume. When the price breaks through the upper or lower band, the trader buys or sells the asset, respectively.
- In the Bollinger double bands (BDB), whenever price holds between the upper TopBand1 and TopBand2 away from mean, the trend is up. We call it the buy zone. Conversely, whenever price holds between the lower BotBand1 and BotBand2 away from mean, the trend is down. We call it the sell zone. If price meanders between the upper TopBand1 and the lower BotBand1, we are in a neutral state (or zone). The bands adapt dynamically to price expanding and contracting as volatility increases and decreases. Therefore, the bands naturally widen and narrow in sync with price action, creating a very accurate trending envelope. Some of the strategies include
 - Trend following: Take a long position once the price enter the buy zone (crossing the upper TopBand1 from below). Close the long position if the candle on the candlestick charts turn red and more than 75% of its body is below the buy zone.
 - Trend exhaustion: Identify the turn in price (counter-trend). When prices fall out of the trend channel, short the next tag of the upper band.
 - Stop-loss points (above and below): Use the volatility of the market to help set a stop-loss level. Measure the width of the neutral zone and add it to the upper band to get the SL-points.

- The Relative Strength Index (RSI): The RSI can be used to predict the price behaviour of a security, validate trends and trend reversals, identify overbought and oversold securities, provide buy and sell signals. This is illustrated in Figure (42). Some of the strategies include
 - Trend following: During trends, the RSI readings may fall into a band or range. In an uptrend, the RSI tends to stay above 30 and should frequently hit 70. In a downtrend, the indicator frequently hits 30 or below and hardly exceed 70. Hence, we can create a horizontal trendline between the levels of 30 and 70 when a strong trend is in place to better identify the overall trend and extremes.
 - Overbought or oversold: When the RSI indicator crosses 30 on the RSI chart, we anticipate a bullish market, and when it crosses 70, we anticipate a bearish market. That is, values of 70 or above indicate that a security is becoming overbought or overvalued. It may be primed (ready) for a trend reversal or corrective price pullback. Conversely, a RSI reading of 30 or below indicates an oversold or undervalued condition.
 - Predicting trend lines: The RSI is a leading indicator so that we can use trendlines on the RSI chart and trade its break. When the RSI indicator is rising, an upward trendline is drawn by connecting two or more lows and projecting the line into the future. Similarly, when the RSI indicator is falling, a downward trendline is drawn by connecting two or more highs and projecting the line into the future. A break of an RSI trendline precedes an actual price reversal or continuation in the market. For instance, if the asset price breaks above a downward trendline, it is a signal that the price is about to edge upwards, either as a continuation of an uptrend or as a reversal of an existing downtrend in the market (a bullish divergence). Conversely, if the asset price breaks below an upward trendline, it is a signal that the price is about to edge downwards, either as a continuation of a downtrend or as a reversal of an existing uptrend in the market (a bearish divergence). This is illustrated in Figure (42).
 - RSI divergences: An RSI divergence occurs when price moves in the opposite direction of the RSI. That is, a chart might display a change in momentum before a corresponding change in price. A bullish divergence occurs when the RSI displays an oversold reading followed by a higher low that appears with lower lows in the price. It may indicate rising bullish momentum, and a break above oversold territory could be used to trigger a new long position. Conversely, a bearish divergence occurs when the RSI creates an overbought reading followed by a lower high that appears with higher highs on the price. This is illustrated in Figure (42).
 - Positive-negative RSI reversals: A positive RSI reversal may take place once the RSI indicator reaches a low that is lower than its previous low at the same time that a security's price reaches a low that is higher than its previous low price. We can see this formation as a bullish sign and a buy signal. Conversely, a negative RSI reversal may take place once the RSI indicator reaches a high that is higher than its previous high at the same time that a security's price reaches a lower high. This formation would be a bearish sign and a sell signal.
 - RSI swing rejections: The bullish swing rejection has four parts: (1) the RSI falls into oversold territory, (2) the RSI crosses back above 30, (3) the RSI forms another dip without crossing back into oversold territory, (4) the RSI then breaks its most recent high. Conversely, the bearish swing rejection also has four parts: (1) the RSI rises into overbought territory, (2) the RSI crosses back below 70, (3) the RSI forms another high without crossing back into overbought territory, (4) the RSI then breaks its most recent low.
- Combining RSI with MACD: The MACD measures the relationship between two EMAs, while the RSI measures price change momentum in relation to recent price highs and lows. Since the MACD uses moving averages in its computation, it is largely a lagging indicator and can be used to qualify signals generated by the RSI indicator, which is a leading indicator. For instance, if the RSI displays a reading of above 70, which implies overbought conditions in the market, a sell position can be initiated when the MACD series shifts from positive to negative (crossing below the 0 centreline). Conversely, if the RSI displays a reading of below 30, which

implies oversold conditions in the market, a buy position can be initiated when the MACD series shifts from negative to positive (crossing above the 0 centreline).

- Combining RSI with Bollinger bands: BB is a volatility indicator whose bands squeeze when there is low volatility but diverge when there is high volatility. Further, periods of consolidation are usually followed by massive breakouts, which makes any Bollinger Bands squeeze a vital time to watch the markets. We also know that a breakout signal will be delivered when there is an RSI overbought or oversold failure swing. For instance, a bearish signal happens when the RSI line rises above 70 and then falls below 70 again. Hence, we can open a sell position when the prices touch the lower Bollinger band BotBand in Equation (13.102). Conversely, a bullish signal happens when the RSI line falls below 30 and then rises above 30 again. In that case, we can open a buy position when the prices touch the upper Bollinger band TopBand in Equation (13.100).



Figure 39: MACD lines and its histogram.



Figure 40: CMO divergence.



Figure 41: Bollinger bands.



Figure 42: RSI oscillator.

3.1.2 Trading styles

A holding period is the amount of time the investment is held by an investor, or the period between the purchase and sale of a security. It is the time between the day after a stock is bought till the day the stock is sold. There are several types of a holding periods:

- Day Trading: Minutes to Hours
- Swing Trading: Hours to Days
- Position Trading: Days to Weeks
- Investment Timing: Weeks to Months

As a result, we must consider a particular trading style in order to determine the optimal time frame for MAC and other MA strategies. For examples:

- Day trading: For day traders, who hold positions for a few hours or less, shorter time frames are more suitable. A combination of 5, 8, and 13-bar simple moving averages (SMAs) can be effective for day trading strategies.
- Swing trading: Swing traders, who hold positions for a few days to a few weeks, can use moving average crossovers to enter trades. Commonly used time frames for swing trading include 20-day, 50-day, and 200-day SMAs.
- Long-term investing: For long-term investors, longer time frames are more appropriate. Commonly used periods for creating moving average lines include 50-day, 100-day, and 200-day moving averages.

Day trading require continuous feedback on short-term price action to make very fast buy and sell decisions. One approach is to use intraday bars wrapped in multiple moving averages, allowing quick analysis that highlights current risks (as well as the most advantageous entries and exits). The 5 – 8 – 13-bar simple moving averages (SMAs)

offer relatively strong inputs for day trading strategies. To improve results, one should also look at the slopes of the MAs which reflect subtle shifts in short-term momentum. Increases in observed momentum (acceleration) offer buying opportunities for day traders, while decreases in momentum (deceleration) may signal timely exits. Trend-less markets and periods of high volatility will force the 5 – 8 – 13-bar SMAs into large-scale whipsaws (oscillations), with horizontal orientation and frequent crossovers telling observant traders to stand aside. In that case, the slope tends to near zero, indicating that intraday trending is weak and profitable opportunities are limited. In fact, trading ranges expand in volatile markets and contract in trend-less markets.

When performing day trading with RSI, we choose between 6 and 9 periods, so that more overbought and oversold signals are generated. Ideally, these levels should correspond with support and resistance levels. Thus, in a ranging market, buy orders will be placed off a support area when the RSI reading is below 30, whereas sell orders will be placed off a resistance area when the RSI reading is above 70.

In trending markets, the RSI setting can be even lower (2 – 6) periods, so that we can identify the best value areas to enter the market in the direction of the dominant trend. For instance, in an uptrend, we will look to enter buy trades after a pullback that will be indicated by an oversold RSI reading. Similarly, in a downtrend, sell trades will be placed after a retracement that will be highlighted by an overbought RSI reading.

Since a lower setting will generate numerous overbought and oversold signals, it is important to combine RSI and other technical indicators, such as moving averages (MAs), in trending markets so as to filter out the best quality signals when day trading. MAs will confirm the trend and also provide possible retracement target levels where RSI overbought and oversold signals can confirm solid trading opportunities.

3.1.3 Some advantages and drawbacks

Moving average trading indicators help us trace the price changes and fluctuations so that we can take the right decisions regarding to the trading position. Using MA indicators, we can identify an uptrend (the price moves above the moving average) or a downtrend (the price moves below the moving average). They are used to predict the price change and the change in the trend of the financial market. Considering other factors such as the length of the trading period, moving average crossover, etc., we can find out the trading positions, the entry points when the prices are strongly trending. Further, MAs can be used as support/resistance points. The MAs indicator are responsive to new and updated information which means better predictions.

Some drawback of moving averages are:

- Lagging indicators: They provide signals after the trend has already begun, which can result in missed opportunities and false signals.
- Whipsaw markets: In markets with choppy price action and no clear trend, MAs can generate a large number of false signals, leading to losses.
- Slow-moving: MAs are slow-moving indicators, which means they may not respond quickly enough to sudden changes in market conditions.
- Not suitable for all market conditions: MAC strategies may work well in trending markets but may be less effective in range-bound or volatile markets (generating multiple trend reversals or trade signals). Adjusting the time frame can help with this problem.

To maximise the effectiveness of MACs, traders should use them in conjunction with other technical indicators and fundamental analysis. This way, they can confirm signals and make informed trading decisions.

3.2 Defining the trading rules

3.2.1 Mathematical formulation

Recall, we can use any specific moving averages when trading with MAs. For simplicity, we consider the SMA (see Equation (13.86)) at time t given by

$$MA_t(N) = \frac{1}{N} \sum_{i=0}^{N-1} P(t-i) = \frac{1}{N} \sum_{i=1}^N P(t-(i-1))$$

where $P(t)$ the price at time t . We could as well used the EMA in Equation (13.87).

3.2.1.1 The double moving average crossover In the case of a Double Moving Average Crossover (DMAC), we let N_S be the number of observations in the short period, N_L be the number of observations in the long period, X_b a percentage band. The MA rule is (N_S, N_L, X_b) and some popular ones are $(1, 50, 0)$, $(1, 100, 0)$. At time t , the buy signal is given by

$$MA_t(N_S) > MA_t(N_L) + X_b \rightarrow \text{buy at time } t$$

and the sell signal is reversed, as follows:

$$MA_t(N_S) < MA_t(N_L) - X_b \rightarrow \text{sell at time } t$$

3.2.1.2 The triple moving average crossover In the case of a Triple Moving Average Crossover (TMAC), we let N_S be the short period, N_M the medium period, N_L the long period, X_b a percentage band, and $P(t)$ the price at time t . At time t , the long (buy) signal is given by

$$MA_t(N_S) > MA_t(N_M) + X_b \text{ and } MA_t(N_M) > MA_t(N_L) + X_b \rightarrow \text{buy at time } t \quad (3.8)$$

and we unwind the long signal (we sell) when

$$MA_t(N_S) < MA_t(N_M) - X_b \quad (3.9)$$

The short (sell) signal is reversed, as follows:

$$MA_t(N_S) < MA_t(N_M) - X_b \text{ and } MA_t(N_M) < MA_t(N_L) - X_b \rightarrow \text{sell at time } t \quad (3.10)$$

and we unwind the short signal (we buy) when

$$MA_t(N_S) > MA_t(N_M) + X_b \quad (3.11)$$

3.2.1.3 The moving average convergence divergence The MACD line at time t , denoted $DIFF_t$, is given in Equation (13.95) and the signal line at time t , denoted $MACD(n; n_S, n_L)_t$, is given in Equation (13.97). Practically, we can use the MACD's histogram the same way that we use the MACD itself. The MACD is illustrated in Figure (43). Hence, we consider the MACD histogram (see Equation (13.98)) given by

$$Hist_t = (DIFF_t - MACD(n; n_S, n_L)_t) \quad (3.12)$$

It represents the distance between the line MACD and its signal line.



Figure 43: MACD histogram.

Some possible use of the histogram are as follows:

- Positive or negative crossovers: If MACD line is above the signal line, the histogram will be above the MACD's baseline (or zero line), otherwise if it is below, the histogram will be below the MACD's baseline. Zeroes in the histogram occur when the MACD line crosses the signal line above or below.
- Identify strength: The magnitude of height is directly proportional to the spread between the MACD and signal lines. When the spread between the two lines increases, the histogram becomes taller or deeper, depending on its direction. When the opposite scenario occurs (when the two lines draw closer), the histogram becomes shorter.
- Identify bullish (when MACD rises above the signal line) or bearish (when MACD falls below the signal line) momentum, and possibly identify overbought/oversold signals. MACD functions on a relative basis so that we should focus on the level and direction of the MACD/signal lines compared with preceding price movements in the security at hand. Peaks and troughs in the histogram indicate when a burst of bearish or bullish momentum is losing strength, and the curve is likely to return to its mean.
- Divergence: When MACD forms highs or lows that exceed the corresponding highs and lows on the price, it is called a divergence. A bullish divergence appears when MACD forms two rising lows that correspond with two falling lows on the price. This is a valid bullish signal when the long-term trend is still positive. Similarly, when MACD forms a series of two falling highs that correspond with two rising highs on the price, a bearish divergence has been formed. A bearish divergence appearing during a long-term bearish trend is considered confirmation that the trend is likely to continue.
- Rapid rises or falls: When MACD rises or falls rapidly (the shorter-term moving average pulls away from the longer-term moving average), it is a signal that the security is overbought or oversold and will soon return to normal levels. This analysis can be combined with the RSI or other TIs to verify overbought or oversold conditions.
- Predict change in trend: We can use MACD histogram to predict when a change in trend will occur. In that case, it makes MACD a leading indicator. This is done by computing the slope of the histogram. From the slope

of the histogram, one can get the most common signal. For example, when the current bar is higher than the preceding bar, the slope is up and one can go long. A slant divergence is a slope that is capable of creating a divergence without well-defined peaks/troughs.

For simplicity, we consider the case where we buy (long) the security when the MACD line crosses above the signal line, and sell (short) the security when the MACD line crosses below the signal line. That is, given some ϵ ,

- Go long when the MACD histogram turns up below zero, $Hist_t > \epsilon$.
- Go short when MACD histogram turns down above zero, $Hist_t < \epsilon$.
- Otherwise we don't trade.

3.2.1.4 The Bollinger double bands Given the Bollinger double bands (BDB) defined in Equations (13.105) and (13.106), we consider a mix of trend following and mean-reverting strategy. The buy zone is given by

$$\text{Buy Zone} = [\text{TopBand1}, \text{TopBand2}] \quad (3.13)$$

and sell zone is given by

$$\text{Sell Zone} = [\text{BotBand2}, \text{BotBand1}] \quad (3.14)$$

The BDB strategy becomes

- Go long when the price enters the buy zone. Unwind when the price exit (from above or below) the buy zone.
- Go short when the price enters the sell zone. Unwind when the price exit (from below or above) the sell zone.
- Otherwise we don't trade.

3.2.1.5 The RSI The RSI indicator is given in Equation (13.93). Since the main signal the RSI oscillator defines overbought and oversold price ranges, we consider the overbought or oversold strategy. When the RSI line goes above the level 70 or below the level 30, it signals that market is overbought/oversold, respectively. The rules for opening and closing positions based on the RSI signals are:

- If the RSI line crosses the level 30 from above (or first from above and then from below), a long position (buy) is opened. Exit on opposite indicator's signal. Unwind the position if the RSI line crosses the level 70 from below (or first from below and then from above).
- If the RSI line crosses the level 70 from below (or first from below and then from above), a short position (sell) is opened. Exit on opposite indicator's signal. Unwind the position if the RSI line crosses the level 30 from above (or first from above and then from below).

3.2.2 Trading rules

The Triple Moving Average (TMA) strategy has the following trading rules (See Faith [2007]):

1. Long position

- Enter long rules:
 - Open a long position when the $MA_t(N_S)$ crosses over the $MA_t(N_M)$, and
 - Both the $MA_t(N_S)$ and $MA_t(N_M)$ are above the $MA_t(N_L)$.
- Exit long rules:

- Sell the long position when the $MA_t(N_S)$ crosses below the $MA_t(N_M)$.

2. Short position

- Enter short rules:
 - Open a short position when the $MA_t(N_S)$ crosses under the $MA_t(N_M)$, and
 - Both the $MA_t(N_S)$ and $MA_t(N_M)$ are below the $MA_t(N_L)$.
- Exit short rules: Cover a short position when the $MA_t(N_S)$ crosses above the $MA_t(N_M)$.

Note, the TMA strategy only trades when two moving averages are on the same side of the $MA_t(N_L)$. Further, even though the TMA strategy lacks of stop-loss orders, there is an implicit stop. When prices move against our position, sooner or later the short MA crosses the middle MA and the strategy generates its exit signal. However, the exit time is random so that the price level is unknown. Hence, we must impose a global stop loss. At last, the TMAverage strategy underperforms significantly when prices move sideways. One solution is to filter out those low-probability trades (with specific technical indicators).

The template for the TMA strategy is:

1. Define strategy settings
2. Calculate strategy values
3. Determine long trading conditions
4. Code short trading conditions
5. Output strategy data
6. Submit entry orders
7. Submit exit orders

3.2.3 The entry and exit algorithms

We use the strategies defined in Section (3.2.1) to implement their entry and exit algorithms.

3.2.3.1 The triple moving average crossover

The entry and exit algorithms for the TMA strategy are:

Algorithm 1 Set entry rule for TMA.

Require: The current time t_j , the stock price S_{t_j} , the window size.

Require: The moving averages $MA_{t_j}(N_S)$, $MA_{t_j}(N_M)$, and $MA_{t_j}(N_L)$

- 1: **if** Equation (3.8) **then**
 - 2: Identify the trade as long: set $BoolLong = \text{True}$ and $BoolTrade = \text{True}$.
 - 3: **else if** Equation (3.10) **then**
 - 4: Identify the trade as short: set $BoolLong = \text{False}$ and $BoolTrade = \text{True}$.
 - 5: **else**
 - 6: Do not trade : set $BoolTrade = \text{False}$.
 - 7: **end if**
 - 8: **return** BoolTrade, BoolLong
-

Algorithm 2 Set exit rule for TMA.

Require: The current time t_j , the stock price S_{t_j} , the window size.**Require:** The moving averages $MA_{t_j}(N_S)$, $MA_{t_j}(N_M)$, and $MA_{t_j}(N_L)$ **Require:** BoolTrade, BoolLong

- 1: **if** $BoolLong$ And Equation (3.9) **then**
 - 2: Identify the exit time of a long position: set $BoolTrade = \text{False}$.
 - 3: **else if** $\neg BoolLong$ And Equation (3.11) **then**
 - 4: Identify the exit time of a short position: set $BoolTrade = \text{False}$.
 - 5: **end if**
 - 6: **return** BoolTrade
-

3.2.3.2 The moving average convergence divergence The entry and exit algorithms for the MACD strategy are:

Algorithm 3 Set entry rule for MACD.

Require: The current time t_j , the stock price S_{t_j} , the window size.**Require:** The MACD and signal lines $DIFF_{t_j}$ and $MACD(n; n_S, n_L)_{t_j}$ **Require:** The histogram $Hist_{t_j}$ given in Equation (3.12)

- 1: **if** $Hist_{t_j} > \epsilon$ **then**
 - 2: Identify the trade as long: set $BoolLong = \text{True}$ and $BoolTrade = \text{True}$.
 - 3: **else if** $Hist_{t_j} < \epsilon$ **then**
 - 4: Identify the trade as short: set $BoolLong = \text{False}$ and $BoolTrade = \text{True}$.
 - 5: **else**
 - 6: Do not trade : set $BoolTrade = \text{False}$.
 - 7: **end if**
 - 8: **return** BoolTrade, BoolLong
-

Algorithm 4 Set exit rule for MACD.

Require: The current time t_j , the stock price S_{t_j} , the window size.**Require:** The MACD and signal lines $DIFF_{t_j}$ and $MACD(n; n_S, n_L)_{t_j}$ **Require:** The histogram $Hist_{t_j}$ given in Equation (3.12)**Require:** BoolTrade, BoolLong

- 1: **if** $BoolLong$ And $Hist_{t_j} < \epsilon$ **then**
 - 2: Identify the exit time of a long position: set $BoolTrade = \text{False}$.
 - 3: **else if** $\neg BoolLong$ And $Hist_{t_j} > \epsilon$ **then**
 - 4: Identify the exit time of a short position: set $BoolTrade = \text{False}$.
 - 5: **end if**
 - 6: **return** BoolTrade
-

3.2.3.3 The Bollinger double bands The entry and exit algorithms for the BDB strategy are:

Algorithm 5 Set entry rule for BDB.

Require: The current time t_j , the stock price S_{t_j} , the window size.**Require:** The Buy Zone given in Equation (3.13) and the Sell Zone given in Equation (3.14)

- 1: **if** S_{t_j} enters the Buy Zone **then**
 - 2: Identify the trade as long: set $BoolLong = \text{True}$ and $BoolTrade = \text{True}$.
 - 3: **else if** S_{t_j} enters the Sell Zone **then**
 - 4: Identify the trade as short: set $BoolLong = \text{False}$ and $BoolTrade = \text{True}$.
 - 5: **else**
 - 6: Do not trade : set $BoolTrade = \text{False}$.
 - 7: **end if**
 - 8: **return** $BoolTrade, BoolLong$
-

Algorithm 6 Set exit rule for BDB.

Require: The current time t_j , the stock price S_{t_j} , the window size.**Require:** The Buy Zone given in Equation (3.13) and the Sell Zone given in Equation (3.14)**Require:** $BoolTrade, BoolLong$

- 1: **if** $BoolLong$ And S_{t_j} exit the Buy Zone **then**
 - 2: Identify the exit time of a long position: set $BoolTrade = \text{False}$.
 - 3: **else if** $\neg BoolLong$ And S_{t_j} exit the Sell Zone **then**
 - 4: Identify the exit time of a short position: set $BoolTrade = \text{False}$.
 - 5: **end if**
 - 6: **return** $BoolTrade$
-

3.2.3.4 The RSI The entry and exit algorithms for the RSI strategy are:

Algorithm 7 Set entry rule for RSI.

Require: The current time t_j , the stock price S_{t_j} , the window size.**Require:** The RSI indicator given in Equation (13.93)

- 1: **if** RSI line crosses the level 30 from above **then**
 - 2: Identify the trade as long: set $BoolLong = \text{True}$ and $BoolTrade = \text{True}$.
 - 3: **else if** RSI line crosses the level 70 from below **then**
 - 4: Identify the trade as short: set $BoolLong = \text{False}$ and $BoolTrade = \text{True}$.
 - 5: **else**
 - 6: Do not trade : set $BoolTrade = \text{False}$.
 - 7: **end if**
 - 8: **return** $BoolTrade, BoolLong$
-

Algorithm 8 Set exit rule for RSI.

Require: The current time t_j , the stock price S_{t_j} , the window size.**Require:** The RSI indicator given in Equation (13.93)**Require:** BoolTrade, BoolLong

- 1: **if** *BoolLong* And RSI line crosses the level 70 from below **then**
 - 2: Identify the exit time of a long position: set *BoolTrade* = False .
 - 3: **else if** !*BoolLong* And RSI line crosses the level 30 from above **then**
 - 4: Identify the exit time of a short position: set *BoolTrade* = False .
 - 5: **end if**
 - 6: **return** BoolTrade
-

4 Price action trading

When performing Price Action Trading (PAT), we need to identify trading patterns (see Section (2.8)), entry and exit levels, stop-losses, and related observations. That is, we need to identify trading opportunities and devise appropriate strategies. For instance, decide whether a stock will form a double top to go higher, or whether it will drop further following a mean reversion. When a defined breakout scenario is met, trading opportunity exists in terms of breakout continuation. We rely on technical indicators (TIs) on charts to identify patterns that can help predict how a security will behave in the future and to time entry and exit points of trades. Since Price Action reflects all variables affecting that market for any given period of time, there is no need of using lagging price indicators such as Stochastics, MACD, RSI. Price movements and their ranges (see Figure (14)) are the only useful information when performing PAT. As a result, volatility indicators developed to measure such price ranges.

Volatility is a statistical measure of the dispersion of returns around its mean for a given security or market index and a given period of time. Historical volatility is based on historical prices and represents the degree of variability in the returns of an asset. This number is without a unit and is expressed as a percentage. While variance captures the dispersion of returns around the mean of an asset, volatility is a measure of that variance bounded by a specific period of time. That is, we can report daily volatility, weekly, monthly, or annualised volatility. Volatility is the rate at which the price changes relative to the average, whereas momentum refers to trend strength in a particular direction. Hence, volatile markets have wide price ranges, while less volatile markets have narrow price ranges. As a result, practitioners developed volatility indicators (VIs) by devising formulae to measure price ranges. In general, when the line of a VI is higher, it implies that the volatility of the underlying asset is increasing, while when it is lower, it implies that the volatility of the underlying asset is decreasing. This is a misinterpretation and it should read: when VI is higher the price range is increasing, while when it is lower, the price range is decreasing. It establishes how far an asset's price can go within a given period of time. This is illustrated with the ATR in Figure (44). Popular volatility indicators include the ATR, Bollinger Bands, the Keltner Channels, the WVF (see Appendix (13.1)). We briefly describe a few strategies for trading with such volatility indicators.



Figure 44: ATR indicator.

4.1 Trading with volatility

4.1.1 Trading with the ATR

While the average true range (ATR) (see Appendix (13.1.1)) is presented as a volatility indicator, the true range is a measure of range between a high and a low, and the ATR measures the average range prices swing for an investment over a specified period. In fact, the ATR is roughly an average measure of the range/size of the past n candlesticks. When the candlesticks increase in size, the ATR also increases. It is described as measuring volatility caused by gaps and limit up or down moves. It is an indicator of the price volatility of an asset expressed in cash (dollar). It is used by practitioners to enter and exit trades. In the latter, the chandelier exit places a trailing stop under the highest high the stock has reached since we entered the trade. The distance between the highest high and the stop level is defined as some multiple multiplied by the ATR:

$$SL_D = HH_t(n) - \alpha \times ATR_t(n)$$

where $HH_t(n)$ is the highest high at time t , n is the number of discrete events in the look back period, and $\alpha \in [1, 3]$. We can also use the entry price instead of the highest high. This is illustrated in Figure (45). This trailing stop rapidly moves upward in response to the market action. The ATR changes based on the characteristics of the stock being traded. As the trading range expands or contracts, the distance between the stop and the closing price automatically adjusts and moves to an appropriate level.



Figure 45: ATR trailing stop indicator.

When the breakout occurs, the stock is likely to experience a sharp move resulting in higher volatility. ATR breakout systems can be used by strategies of any time frame. If our volatility indicator suddenly raise we can deduce that such a price move is happening. We can therefore profit from that volatility cycle: whenever price closes more than an ATR above the most recent close, a change in volatility has occurred. Thus, the ATR can be added or subtracted to the closing price. Traders can buy whenever the next day's price trades above that value and unwind their position when the price trades below the closing price minus the ATR (the stock is likely to enter a trading range or reverse direction at this point). We define an entry and exit indicator as follows:

$$ATR_t^{En} = C_t + ATR_t(n) \text{ and } ATR_t^{Ex} = C_t - ATR_t(n)$$

Since the ATR is only a volatility measure, it can be combined with other indicators such as trend indicators. Some examples include:

- Traders can overlay a moving average (MA) (or an exponential moving average) on the ATR that will act as a signal line. As an example, we can add a 20-period SMA over the ATR and watch out for crosses. This is illustrated in Figure (46). When prices are trending higher, an ATR cross above the signal line will confirm an uptrend, while when prices are drifting lower, an ATR cross below the signal line will confirm a downtrend. Note, when the ATR is above the MA line we do not know the direction of the trend. Further, when the ATR and the EMA are on top of each other, clustering together, the price is in a narrow sideways period.
- We can also use Stochastics (it is a pure trend strength indicator) because they deliver overbought and oversold signals. Low ATR values confirm ranging markets and buy/sell signals can be provided by Stochastics crossovers in overbought and oversold zones. This is illustrated in Figure (47). In a strong bullish trend, the Stochastic is above the 80 level while the ATR is at a low value, showing a low volatility and high momentum trending market. In a strong bearish trend strength, the Stochastic is below the 20 level while the ATR is at a high value, showing a high volatility and downward market.



Figure 46: ATR (orange) and moving average (blue) indicator.



Figure 47: ATR (black) and Stochastic (blue) indicator.

In a bearish market, as the downward trend comes to an end, the ATR also reached its peak. Then, as the ATR declined, the trend also stopped going lower. Even though the ATR is not a trend-following tool, changes in volatility can point to changes in market behaviour.

4.1.2 Trading with KC

Keltner Channels (KC) (see Appendix (13.1.2)) are volatility-based bands placed on either side of an asset's price that aid in determining the direction of a trend. The channels can contract and expand with volatility (price range). Since they are based on the average true range (ATR), they are smoother in appearance compared to other indicators. However, they don't have the clear squeezes and bounces which show areas of high and low volatility. Moves outside the channel can signal trend changes or an acceleration of the trend. The direction of the channel (up, down, or

sideways) can also aid in identifying the trend direction of the asset. This is illustrated in Figure (48). Some of the information gained by the KC are:

- The angle of the channel helps identifying trend direction: A rising channel means the price has been rising, while a falling or sideways channel indicates the price has been falling or moving sideways, respectively. In an uptrend, the price action tends to be confined in the upper half of the channel, between the middle line as support and the top line as resistance. This is illustrated in Figure (49). In a downtrend, the price action tends to be confined in the lower half of the channel, between the bottom line as support and the middle line as resistance. This is illustrated in Figure (50).
- A price move above the upper band shows price strength: It is an indication that an uptrend is in play, especially if the channel is angled upwards. This is illustrated in Figure (51).
- A drop below the lower band shows price weakness: It is evidence of a downtrend, especially if the channel is angled downward. This is illustrated in Figure (52).
- A signal that an uptrend is losing momentum: If the price is continually hitting the upper band but not the lower, when the price does finally reach the lower band, it could be a sign that the uptrend is losing momentum.
- A signal that a downtrend is near its end: If the price is constantly hitting the lower band but not the upper, when the price does finally reach the upper band, it could be a signal that the downtrend is near an end.
- A buying and selling indicator: The price may oscillate between the upper and lower bands. In that case, we can use the bands as support and resistance: Buy when the price reaches the lower band and then starts to move higher again, and sell or short after the price starts to fall again after reaching the upper band.
- A price breakout indicator: After a sideways period, if the price breaks above or below the channel and the channel starts to angle the same way, it may signal that a new trend is underway in that breakout direction.

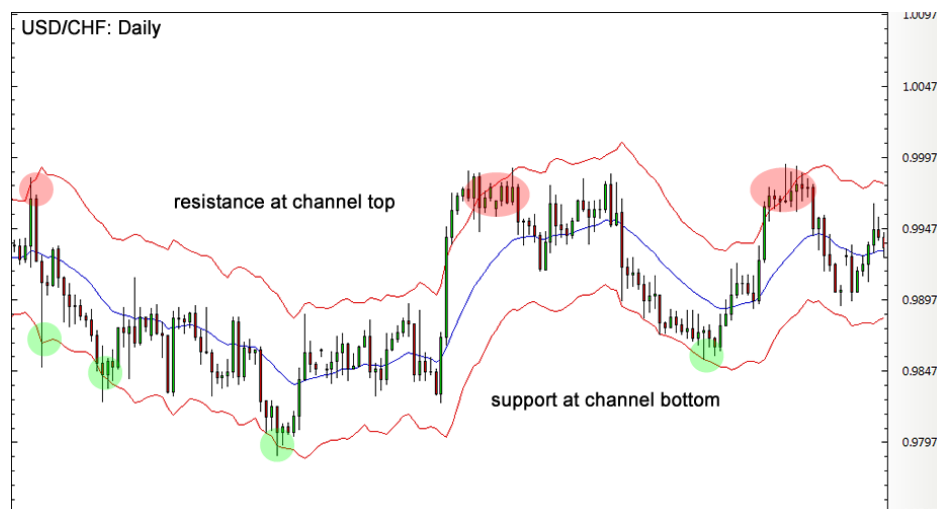


Figure 48: KC acting as support and resistance.



Figure 49: KC uptrend.

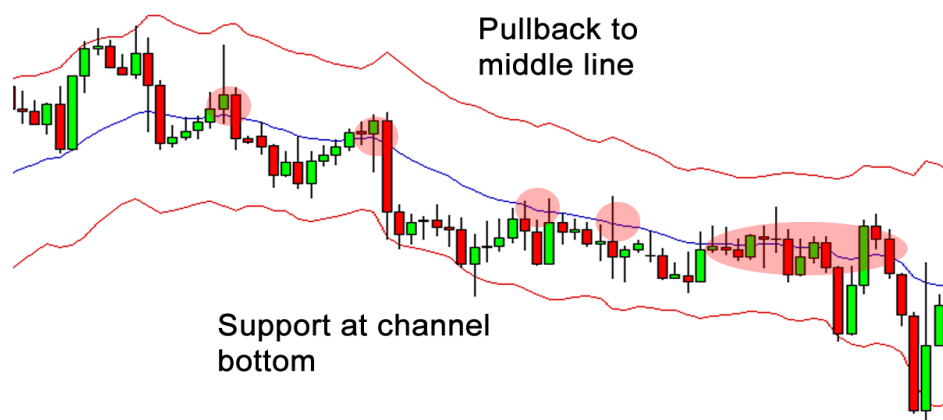


Figure 50: KC downtrend.

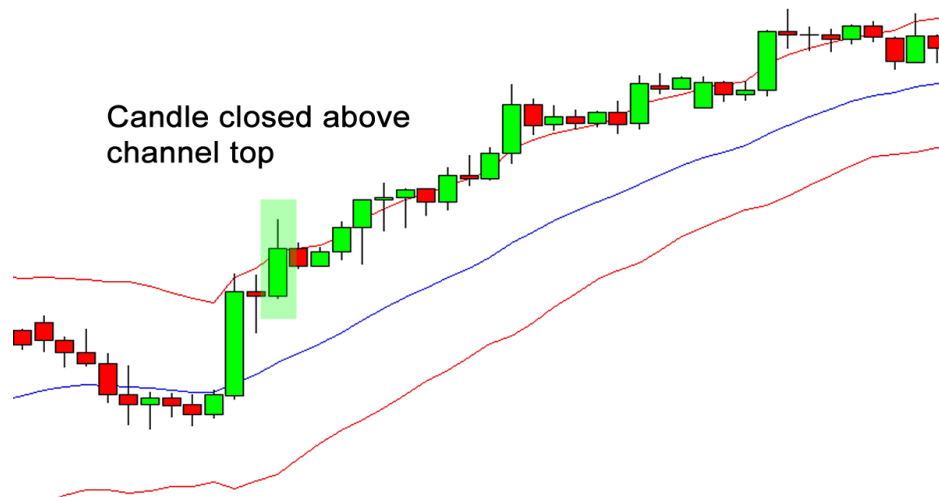


Figure 51: KC break above top band.



Figure 52: KC break below bottom band.

4.1.3 Trading with the WVF

4.1.3.1 The VIX Volatility can be seen as a technical indicator which rises in a market downturn. For example, the VIX is derived from the implied volatility (IV) ⁷ of stock index options on the Chicago Board Of Trade (CBOE). It was introduced in 1993 and was intended to represent the 'fear' or 'complacency' of the market (see Whaley [1993]). The calculation of VIX requires using market prices of options contracts to derive the size of the expected price move over the next 30 days. The expected volatility is computed by averaging the weighted prices of out-of-the-money puts and calls. The formula is

⁷The IV is the market's expectation of future volatility.

$$VIX = \sqrt{\frac{2e^{r\tau}}{\tau} \left(\int_0^F \frac{P(K, \tau)}{K^2} dK + \int_F^\infty \frac{C(K, \tau)}{K^2} dK \right)}$$

where τ is the number of average days in a month, r is the risk-free rate, F is the 30-day forward price, and $P(K, \tau)$ and $C(K, \tau)$ are prices for puts and calls with strike K and maturity τ , respectively. The steps are as follows:

- Select the options to be included in VIX calculation: a range of call and put strikes in two consecutive expirations around the target 30-day mark.
- Calculate each option's contribution to the total variance of its expiration.
- Sum up the contributions to get total variances of the first and second expiration.
- Calculate 30-day variance by interpolating the two variances, depending on the time to expiration of each.
- Take the square root to get volatility as standard deviation.
- Multiply the volatility (standard deviation) by 100. We get the VIX index value.

However, the VIX is only calculated for the *SP500*, Nasdaq, and the Dow Jones 30. The VIX for the *SP500* from 2014 to 2021 is displayed in Figure (53). In general, volatility is associated with negative stock market performance. Implied volatility (IV) is one of the major components in the price of an option: the higher the implied volatility, the more expensive the premium. Hence, it is a measure of risk: the more volatility in the markets, the more we need to pay for insurance. The VIX oscillates up and down: a high reading means investor sentiment is one of increased fear, while low readings are associated with low-volatility conditions (and market tops). Thus, the VIX has a negative correlation to the *SP500*. A high VIX normally means the market has fallen, at least in the short term, and the risk premium for owning stocks increases. As a result, we get buying opportunities as the VIX spikes. However, the time between VIX peaks and market lows is variable and unpredictable in real time.

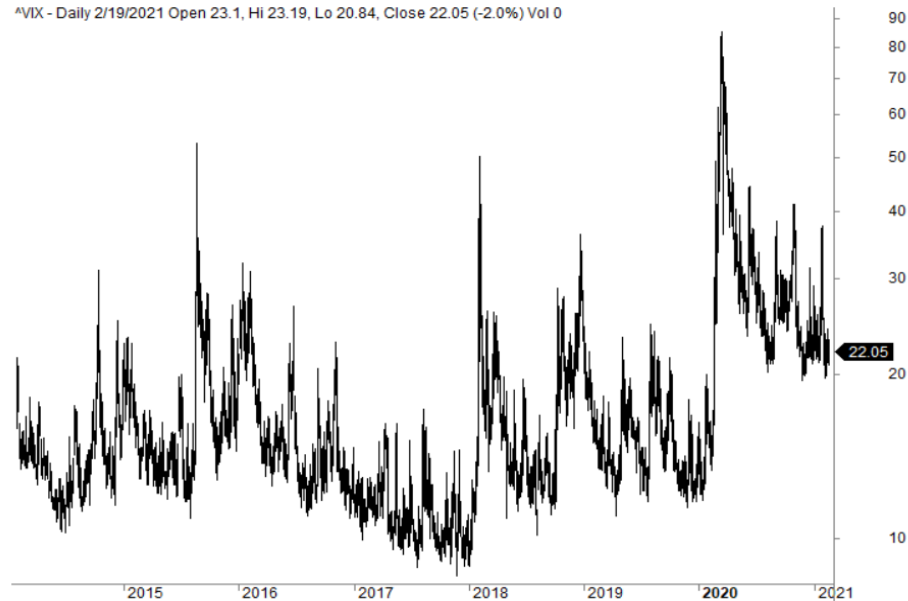


Figure 53: Daily VIX from 2014 to 2021.

4.1.3.2 The WVF The Williams Vix Fix (WVF) indicator (see Williams [2007]) is a way of creating a synthetic VIX for any security. The formula is as follows:

$$WVF_t = \frac{HC_t(m) - L_t}{HC_t(m)} \times 100 \quad (4.15)$$

where $HC_t(m)$ is the highest close at time t (or bar on any frequency) over m periods. It is multiplied by 100 to take values in $[0, 100]$. In general, we set $m = 22$, the normal number of trading days in a month. Hence, it measures the price volatility of the last 22 trading days. Practically, it measures by how much the current low price L_t statistically deviates from the maximum $HC_t(m)$ within a given look-back period. Since market prices jumps downward, the lower the current low the higher the WVF. Hence, it is more a measure of price jump than one of volatility (in the sense of quadratic variation). This is the reason why it is associated with identifying market bottoms. We can easily check how the WVF compares with the VIX. This is displayed in Figure (54).

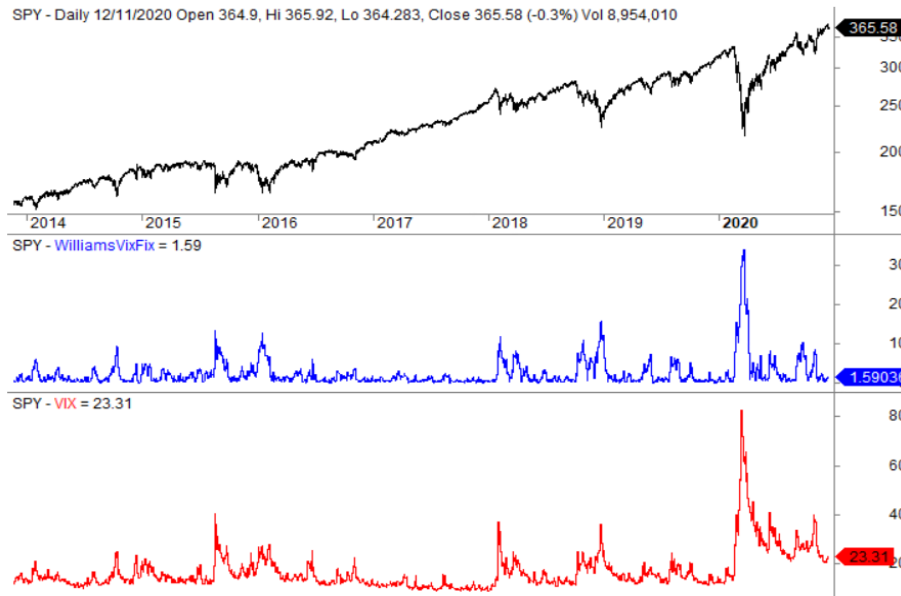


Figure 54: Daily WVF (blue line) versus VIX (red line).

We observe that both the WVF and VIX have similar characteristics but different order of magnitude. The levels, swings, magnitude, and timing are pretty similar. This is because the IV is mostly a result of short-term volatility which measure negative stock market performance. Thus, the WVF can be used on all securities. As a result, it can be used to identify volatile stocks in any market environment. We can also apply the WVF to benefit from price moves in the options market. For example, if volatility is higher than average, by selling options we should generate significant gains as volatility returns to a normal level and the options price declines. More importantly, it is a measure of market bottoms.

4.1.4 Some strategies

Most analysts consider high levels of VIX to be associated with market bottoms. Further, since markets tend to find the lowest prices during times of highest volatility, the WVF can help in finding market bottoms (see Figure (54)). Similarly, its inverse (how much the current high value statistically deviates from the minimum within a given look-back period) may indicate market tops. Nonetheless, investor sentiment is more magnified at market bottoms

than at tops. However, since the WVF is not an oscillator, its values are unbounded. That is, it is always possible that the WVF move higher. Thus, in order to compare current volatility with past ones, we must combine the WVF with another indicator that averages it over the past. Williams [2007] suggested to plot Bollinger Bands (BB) (see Appendix (13.1.2)) to help identify potential over extended zones without resorting to fixed values. This is displayed in Figure (55). We observe that the BB is an envelop to the WVF. He also proposed to construct a 14-day stochastic of the WVF. Most of the market lows come when the stochastic of the WVF posts a high reading (above 80%). Later, Chris Moody's indicator (2014) combined the WVF with Bollinger Bands (BB) to generate bottom signals, the Hestla strategy (see Hestla-Barnhart [2015]) used a 20-SMA, and the MOPOI strategy (see Odink [2016]) used the upper level of the 20-Bollinger Band.

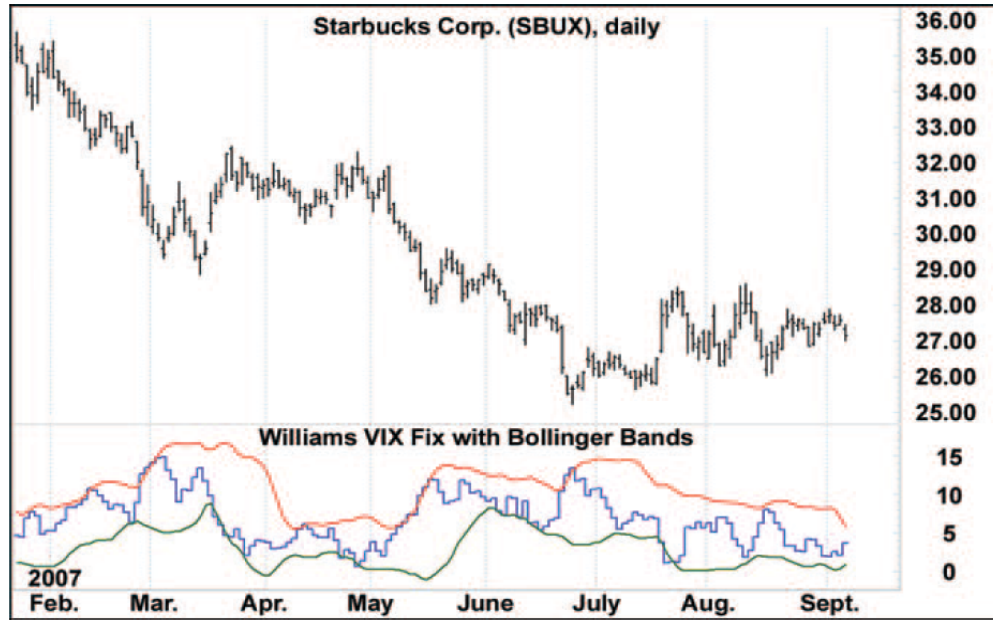


Figure 55: Starbucks WVF (blue line) versus Bollinger Bands (red line).

4.1.4.1 The Hestla strategy In order to test the effectiveness of the WVF, Hestla-Barnhart [2015] applied a 20-day MA to daily data which she tested on all stocks of the *SP500* from 1999 to October 2014. Assuming that volatility is mean reverting, a buy is triggered when the WVF falls below the MA and a sell is generated when the WVF rises above the MA. The results showed that buying stocks after a peak in volatility was rewarding.

4.1.4.2 The MOPOI strategy Odink [2016] proposed to calculate and assigns a volatility level of downward volatility (the WVF) on the underlying asset to each trading day. He computed the moving average and a standard deviation on these volatility levels. Put another way, he applied Bollinger Bands (BB) directly on the WVF. Hence, we assume that the WVF are Gaussian $N(\mu, \sigma)$ with values seen at time t for the maturity $T = t + \Delta t$ given by

$$WVF(t, T) = \mu \Delta t \pm \alpha \times \sigma \sqrt{\Delta t}$$

where α is the number of standard deviation. That is, we replace the Typical Price in the BB with the WVF and compute the SMA and standard deviation

$$SM_t(n) = \frac{1}{n} \sum_{i=1}^n WVF_{t-i+1} \text{ and } SD_t(n) = \sqrt{\frac{1}{n} \sum_{i=1}^n (WVF_{t-i+1} - SM_t(n))^2} \quad (4.16)$$

Again, the bands (see Equations (13.100) (13.102)) are

$$\text{TopBand} = SM_t(n) + \alpha \times SD_t(n), \text{MidBand} = SM_t(n), \text{BotBand} = SM_t(n) - \alpha \times SD_t(n) \quad (4.17)$$

where $\alpha = 2$. The best result was obtained by using a length of around $n = 10$ days. The author called these bands the MOPOI levels and the TopBand the MOPOI line. The MOPOI algorithm is an algorithm that aims to identify entry moments in which a reversal follows up after an overreaction (see Brown et al. [1988]). Overreaction occurs on the underlying asset when the WVF level of a given trading day is higher than the TopBand. A reversal movement is expected subsequent to this overreaction. The author chose to open a long position at the moment when overreaction is no longer assumed on the underlying asset. That is, enter a long position at time t when

$$WVF_{t-2} > \text{TopBand}_{t-2} \text{ and } WVF_{t-1} < \text{TopBand}_{t-1}$$

This algorithm was implemented and tested in Tradingview. This is illustrated in Figure (56) for the SP500 from August 2014 to December 2014. Note, all the WVF values and the MOPOI line have been multiplied by -1 for a more intuitive illustration.

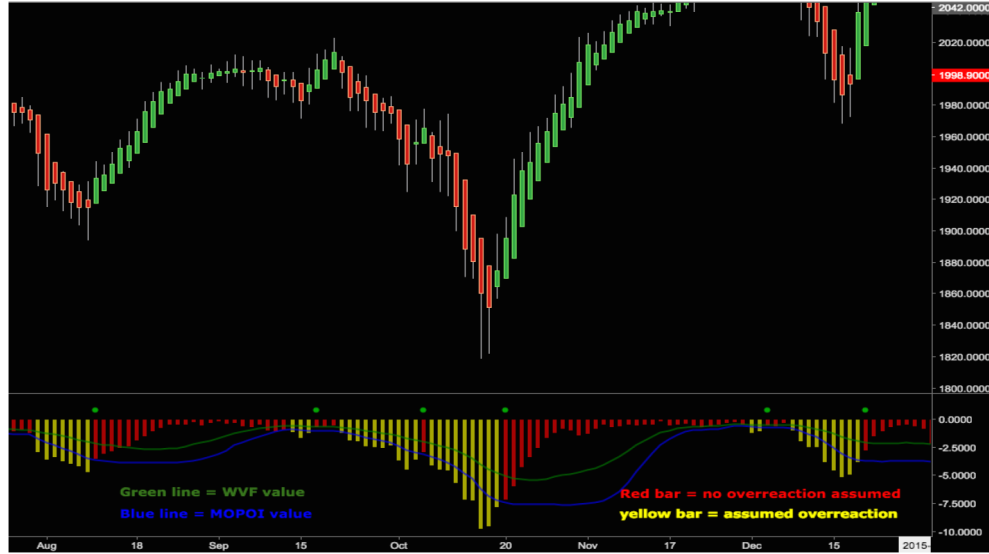


Figure 56: MOPOI algorithm from August 2014 to December 2014.

The length of the red and yellow bars illustrates the WVF values in Equation (4.15) for that given trading day. The green line represents a moving average of these WVF values in Equation (4.16), and the blue line is the MOPOI line in Equation (4.17) representing the threshold level with which overreaction of the underlying asset is assumed. When the bar (of the WVF) exceeds the blue line ($WVF > MOPOI$) then overreaction is assumed on the underlying asset, and the bar turns yellow. A long position will be entered at the open of the trading day subsequent to the first red bar that followed up a yellow bar. Note, potential errors have been identified in times of long-lasting upward movements and times of flat movements of the underlying asset subsequent to a downward movement.

4.1.4.3 The Percent Rank strategy The percentile rank (PR) of a given score is the percentage of scores in its frequency distribution that are less than that score. It is given by

$$PR = \frac{CF - \frac{1}{2}F}{N} \times 100$$

where the cumulative frequency CF is the count of all scores less than or equal to the score of interest, F is the frequency for the score of interest, and N is the number of scores in the distribution. The Percent Rank function returns the rank of a value in a data set as a percentage of the data set.

By using a longer lookback periods, we can consider Percent Rank as a variable. The Percent Rank function generates a ranged value in the form of a percentage between 0 and 100. Hence, we use the Percent Rank function to rank the current WVF against the last n values. We get $WVFrank(n)$, that is, WVF_t is ranked against the past n values in our data set. We multiply this number by 100 and drop the fraction to get an integer in the range $[0, 100]$. As an example, percentile rank will show a reading of $[98, 100]$ range if there is a sharp spike in the WCF readings. We can then devise the following strategy:

- Go long if the WVF closed high on the n -day Percent Rank. We can use values in the range $[95\%, 98\%]$ as the threshold and entry on the close.
- The exit is when today's close is higher than yesterday's close. Exit on the close.

The strategy works for a lookback period of up to 50 days. Note, we can go short if it closes in the lower 2% of the range, use a longer lookback period of 50 days, and exit when the close is below the 5-day moving average. This indicator is very good at picking the low points in price.

4.1.4.4 High range and low range Tradingview allows to compute the high range and low range of the WVF. The formulae are

$$rangeHigh_t = \alpha_H \times HWVF_t(n) \text{ and } rangeLow_t = \alpha_L \times LWVF_t(n)$$

where $HWVF_t(n)$ is the highest WVF at time t for n periods and $LWVF_t(n)$ is the lowest WVF at time t for n periods, α_H is the highest percentile and α_L is the lowest percentile. The highest percentile option makes a difference similar to the stand deviation and Bollinger Band. The higher the percentile, the less likely it will get triggered but the more likely it is to be correct.

4.2 CM-WVF indicators

Chris Moody (2014) proposed an indicator (the CM-WVF) that uses the Williams Vix Fix (WVF) indicator (see Williams [2007]) to identify bottoms. A threshold is calculated using Bollinger Bands and based on percentile of LookBack period Percentile High number of candles. The time frame is $\{5, 15, 30\}$ minutes. The data values are displayed in the form of a Histogram Chart. The Bars are plotted above the zero line (or below when multiplied by -1). There are two colours: Grey and Lime. That is, when it is Lime colour it is a bottom, otherwise it is not. We enter the trade after a Lime turns Grey. However, they can alternate and be false positive (FP). Hence, Chris Moody (2015) improved the indicator to filter out FP and get a better entry point. He devised different filtering criteria to increase the probability of success. This indicator was further improved to identify tops.

4.2.1 Formulae, settings, and illustration

4.2.1.1 The formulae We saw in Section (4.1.3) that the Williams Vix Fix (WVF) indicator was a way of creating a synthetic VIX for any security. The formula is given in Equation (4.15), which was used with the Bollinger Bands to identify bottoms (see Equation (4.17)).

We can modify this indicator to identify tops by flipping the indicator and adjusting colours.

$$WVFT_t = \frac{LC_t(m) - H_t}{LC_t(m)} \times 100 \quad (4.18)$$

where $LC_t(m)$ is the lowest close at time t (or bar on any frequency) over m periods. Then, we replace the WVF with the WVFT and compute the SMA and standard deviation

$$SM_t(n) = \frac{1}{n} \sum_{i=1}^n WVFT_{t-i+1} \text{ and } SD_t(n) = \sqrt{\frac{1}{n} \sum_{i=1}^n (WVFT_{t-i+1} - SM_t(n))^2} \quad (4.19)$$

Again, the bands (see Equation (13.102)) are

$$\text{MidBandTop} = SM_t(n), \text{ BotBandTop} = SM_t(n) - \alpha \times SD_t(n) \quad (4.20)$$

where $\alpha = 2$.

Further, we have

$$\text{rangeLowTop}_t = \alpha_L \times LWVFT_t(n)$$

where $LWVFT_t(n)$ is the lowest WVFT at time t for n periods, and α_L is the lowest percentile.

4.2.1.2 Settings in Tradingview The settings for the Support Indicator in Tradingview are:

- $pd = 22$, LookBack Period Standard Deviation High
- $bbl = 20$, Bollinger Band Length
- $mult = \alpha = 2.0$, $minval = 1$, $maxval = 5$, Bollinger Band Standard Deviation Up
- $lb = 50$, LookBack Period Percentile High
- $ph = \alpha_H = 0.85$, Highest Percentile (0.90, 0.95, 0.99)
- $pl = \alpha_L = 1.01$, Lowest Percentile (1.10 = 90%, 1.05 = 95%, 1.01 = 99%)
- $hp = False$, Show High Range - Based on Percentile and LookBack Period
- $sd = False$, Show Standard Deviation Line
- $new = False$, Highlight Bars below use original criteria
- $sbc = True$, Show Highlight Bars if WVF was True and is now False
- $sbcc = False$, Show Highlight Bars if WVF is True
- $new2 = False$, Highlight Bars below use filtered criteria
- $sbcFilt = True$, Show Highlight Bars for filtered entry
- $sbcAggr = False$, Show Highlight Bars for aggressive filtered entry
- $new3 = False$, Check below to turn all bars Grey, then check the boxes above, and you will have some colours as WVF
- $sgb = False$, Check box to turn Bars Grey
- $ltLBB = 40$ bars ago, $minval = 25$, $maxval = 99$, long-term LookBack current bar has to close below this value or $medium - term - default = 40$
- $mtLBB = 14$ bars ago, $minval = 10$, $maxval = 20$, medium-term LookBack current bar has to close below this value or $long - term - default = 14$

- $ltLBA = 40$ bars ago, $minval = 25$, $maxval = 99$, long-term LookBack current bar has to close above this value or $medium - term - default = 40$
- $mtLBA = 14$ bars ago, $minval = 10$, $maxval = 20$, medium-term LookBack current bar has to close above this value or $long - term - default = 14$
- $str = 3$ bars ago, $minval = 1$, $maxval = 9$, entry price action Strength (short-term reversal), $C > X$ bars back - $default = 3$
- $new4 = False$, Turn on/off alerts below
- $new5 = False$, To activate alerts you have to check the boxes below for any alert criteria you want

4.2.1.3 Illustration

- Lime bar = **bottom assumed**, Grey bar = **no bottom**.
- Entry rule: go long after first Grey bar following a Lime bar.
- False positives may occur, hence filtering was introduced.



Figure 57: CM-WVF indicator: Lime (bottom) and Grey (neutral) bars.

4.2.2 Charts representation

We define a few building blocks that we use to characterise some chart patterns.

4.2.2.1 Building blocks We present a few building blocks expressed as open, high, low, and close (OHLC) prices that we use to define filters and alerts. These blocks are filtered bar criteria defining price action. They are as follows:

- $UpRange_t = L_t > L_{t-1}$ and $C_t > H_{t-1}$
- $DownRange_t = H_t < H_{t-1}$ and $C_t < L_{t-1}$

- $Aggr_UpRange_t = C_t > C_{t-1}$ and $C_t > O_{t-1}$
- $Aggr_DownRange_t = C_t < C_{t-1}$ and $C_t < O_{t-1}$
- $UpMove_t = C_t > C_{str}$
- $DownMove_t = C_t < C_{str}$

where $C_{name} = C_{t-name}$ and $name$ is found in the above settings. Then, we can define some up and down trends as well as some aggressive up and down trends as follows:

- Short-term up trend: $ST_UpTrend_t = [UpRange_t \text{ and } UpMove_t]$, current up trend
- Short-term down trend: $ST_DownTrend_t = [DownRange_t \text{ and } DownMove_t]$, current down trend
- Short-term aggressive up trend: $ST_AggrUpTrend_t = [Aggr_UpRange_t \text{ and } UpMove_t]$
- Short-term aggressive down trend: $ST_AggrDownTrend_t = [Aggr_DownRange_t \text{ and } DownMove_t]$
- Long-term up trend: $LT_UpTrend_t = [C_t > C_{ltLBA} \text{ or } C_t > C_{mtLBA}]$
- Long-term down trend: $LT_DownTrend_t = [C_t < C_{ltLBB} \text{ or } C_t < C_{mtLBB}]$

4.2.2.2 Defining patterns Using these building blocks, we can characterise some chart patterns. We start by considering the pullbacks and throwbacks defined in Section (2.9.2). They are implemented as follows:

- $PullBack_t = [LT_DownTrend_t \text{ and } ST_UpTrend_t]$, long-term down trend with current up trend
- $ThrowBack_t = [LT_UpTrend_t \text{ and } ST_DownTrend_t]$, long-term up trend with current down trend
- $AggrPullBack_t = [LT_DownTrend_t \text{ and } ST_AggrUpTrend_t]$, long-term down trend with aggressive current up trend
- $AggrThrowBack_t = [LT_UpTrend_t \text{ and } ST_AggrDownTrend_t]$, long-term up trend with aggressive current down trend

4.2.3 Filters and alerts

The filters use the WVF to identify the bottoms and tops defining the entry moments, while the alerts warn us that such bottoms or tops are currently unrolling.

4.2.3.1 Finding bottoms The filtered criteria for price action are:

- $Filtered_Bot_t = [WVF_t \geq TopBand_t \text{ or } WVF_t \geq rangeHigh_t]$
- $Not_Filtered_Bot_t = [WVF_t < TopBand_t \text{ and } WVF_t < rangeHigh_t]$
- $Mod_Filtered_Bot_t = [Filtered_Bot_{t-1} \text{ and } Not_Filtered_Bot_t]$
- $AGGR_Mod_Filtered_Bot_t = [Filtered_Bot_{t-1} \text{ and not } Not_Filtered_Bot_t]$

Recall, the $Filtered_Bot_t$ above is an overreaction (at the bottom) possibly leading to reversal movement. In the MOPOI algorithm, we open a long position at the moment when overreaction is no longer assumed on the underlying asset. That is, when $Mod_Filtered_Bot_t$ is True. When $AGGR_Mod_Filtered_Bot_t$ is True, we don't wait the end of the overreaction to open a long position.

We let $I_{(i)} = \{0, 1\}$ be a Boolean indicator returning 1 if the i th event is True. We get the following Alert criteria:

- $I_{(AB1)} = [\text{if } Filtered_Bot_t]$
- $I_{(AB2)} = [\text{if } Mod_Filtered_Bot_t]$
- $I_{(AB3)} = [PullBack_t]$ and $[Mod_Filtered_Bot_t]$, pullback
- $I_{(AB4)} = [AggrPullBack_t]$ and $[AGG_Mod_Filtered_Bot_t]$, aggressive pullback

4.2.3.2 Finding tops The filtered criteria for price action are:

- $Filtered_Top_t = [WVFT_t \leq BotBandTop_t \text{ or } WVFT_t \leq rangeLowTop_t]$
- $Not_Filtered_Top_t = [WVFT_t > BotBandTop_t \text{ and } WVFT_t > rangeLowTop_t]$
- $Mod_Filtered_Top_t = [Filtered_Top_{t-1} \text{ and } Not_Filtered_Top_t]$
- $AGGR_Mod_Filtered_Top_t = [Filtered_Top_{t-1} \text{ and not } Not_Filtered_Top_t]$

Recall, the $Filtered_Top_t$ above is an overreaction (at the top) possibly leading to reversal movement. Hence, we open a short position at the moment when overreaction is no longer assumed on the underlying asset. That is, when $Mod_Filtered_Top_t$ is True. When $AGGR_Mod_Filtered_Top_t$ is True, we don't wait the end of the overreaction to open a short position.

We let $I_{(i)} = \{0, 1\}$ be a Boolean indicator returning 1 if the i th event is True. We get the following Alert criteria:

- $I_{(AT1)} = [\text{if } Filtered_Top_t]$
- $I_{(AT2)} = [\text{if } Mod_Filtered_Top_t]$
- $I_{(AT3)} = [ThrowBack_t]$ and $[Mod_Filtered_Top_t]$, throwback
- $I_{(AT4)} = [AggrThrowBack_t]$ and $[AGG_Mod_Filtered_Top_t]$, aggressive throwback

4.2.4 Assigning colours

We let $I_{(i)} = \{0, 1\}$ be a Boolean indicator returning 1 if the i th event is True. We denote C_t^j be the j th colour at time t . Then, we assign colours as follows:

4.2.4.1 Finding bottoms Highlight bar criteria:

- $C_t^0_Bot = \text{if } [sgb \text{ and } C_t] \text{ then bar colour is Grey}$
- $C_t^1_Bot = \text{if } [sbcc \text{ and } I_{(AB1)}] \text{ then bar colour is Lime else Grey}$
- $C_t^2_Bot = \text{if } [sbc \text{ and } I_{(AB2)}] \text{ then bar colour is Aqua else Grey}$
- $C_t^3_Bot = \text{if } [sbcFilt \text{ and } I_{(AB3)}] \text{ then bar colour is Fuchsia else Grey}$
- $C_t^4_Bot = \text{if } [sbcAggr \text{ and } I_{(AB4)}] \text{ then bar colour is Orange else Grey}$

Colouring criteria of WVF:

$$C_t^1_Bot = \text{if } [I_{(AB1)}] \text{ then bar colour is Lime else Grey}$$

4.2.4.2 Finding tops Highlight bar criteria:

- $C_t^0_Top =$ if $[sgb \text{ and } C_t]$ then bar colour is Grey
- $C_t^1_Top =$ if $[sbcc \text{ and } I_{(AT1)}]$ then bar colour is Raspberry else Grey
- $C_t^2_Top =$ if $[sbc \text{ and } I_{(AT2)}]$ then bar colour is Azur else Grey
- $C_t^3_Top =$ if $[sbcFilt \text{ and } I_{(AT3)}]$ then bar colour is Pink else Grey
- $C_t^4_Top =$ if $[sbcAggr \text{ and } I_{(AT4)}]$ then bar colour is Yellow else Grey

Colouring criteria of WVFT:

$$C_t^1_Top = \text{if } [I_{(AT1)}] \text{ then bar colour is Raspberry else Grey}$$

4.3 Squeeze momentum indicators

Market squeeze describes a variety of situations that involve some sort of market pressure such as short-sellers purchasing stock to cover losses or investors selling long positions to realise capital gains. For example, a **long squeeze** occurs in a strong financial market when there are sharp price decreases and investors who are long a stock sell a portion of their position, pressuring more long holders of the stock into selling their shares to protect against a dramatic loss. Conversely, a **short squeeze** occurs in the equities market where a stock's price increases and its purchase volume spikes because short sellers are exiting their positions and cutting their losses. A **Volatility Squeeze** occurs when the volatility of a stock falls below its recent levels, meaning that the stock is in a period of consolidation and trending in a narrow range. That is, prices consolidate into a tighter and tighter range, indicating that there will be an explosive price movement either to the upside or downside. The tighter the volatility, the more violent and explosive the move will be.

Market equities alternate between periods of low volatility and high volatility. Since Bollinger Bands (BBs) identify points of high and low volatility, we can use these bands to identify volatility squeeze via the BB squeeze formula given in Equation (13.103). To determine breakout direction, Bollinger suggested to look at the divergence between the BBs and other indicators such as the RSI. For example, if the indicators are heading upward while price is heading down, it is a bullish sign. Conversely, if price is moving higher but the indicators are showing negative divergence, we expect a downside breakout, especially if there have been increasing volume spikes on down days. Further, in expansion, when a powerful trend is born, the resulting explosive volatility increase is often so great that the lower band will turn downward in an upside break, or the upper band will turn higher in a downside breakout.

Carter [2018] proposed the TTM Squeeze volatility indicator which measures the relationship between Bollinger Bands (BB) and Keltner Channels (KC) to help identify consolidations and signal when prices are likely to break out (whether up or down). The main idea being that as BB compress within KC, price volatility decreases, giving way for a potential explosive price movement up or down. Practically, the Squeeze Indicator finds sections of the BB which fall inside the KC, where the market is said to be in a squeeze (indicator turns off (strong squeeze), displayed with grey diamond shapes). The tighter the band, the stronger the squeeze and the more violent the price explosion will be, leading to a BreakOut or BreakDown condition. When the volatility declines, the distance between the bands decreases, conversely, when the volatility increases that distance increases, and in such cases the squeeze is said to be released (indicator turns on (weak squeeze), displayed with triangle up (green) or triangle down (red) shapes). This indicator was improved by adding a momentum indicator (Squeeze Momentum Indicator), and it was further improved with the Squeeze PRO Arrow and Squeeze Momentum Deluxe. To produce Buy/Sell signals, the Squeeze Indicator is plotted along with Momentum Oscillator. When the indicator is on and the Momentum Oscillator Histogram is blue (or green), it is considered a Buy signal (this signal is supposed to be correct until two blue (or green) bars in a row).

When the indicator is on and the Momentum Oscillator Histogram is red, it is considered a Sell signal (this signal is supposed to be correct until two red bars in a row). When the market finishes a move, the indicator turns off (Grey) and no trade is recommended. This is illustrated in Figure (59).

4.3.1 The indicators

The TTM Squeeze indicator has both volatility and momentum components: (1) the Squeeze dots signal when volatility conditions are right to buy such as potential breakouts after periods of low volatility. (2) the momentum histogram indicates the direction (long or short) after the breakout in which to trade.

4.3.1.1 Squeeze indicator The Squeeze values are displayed in the form of a Scatter Chart. We use two volatility indicators: (1) the Bollinger Band (see Equations (13.100) (13.102)), and (2) the Keltner Channels (see Equation (13.99)). Both BB and KC are constructed upon a moving average with the addition of Standard Deviation (Stdev) and Average True Range (ATR) respectively. A squeeze occurs when the Stdev is lower than the ATR. Here, the KC is made of a triple band to account for different range levels (Squeeze PRO Arrow) acting as thresholds on the BB. The Pro version helps differentiate between levels of squeeze (compression) as the BB moves through the KC. That is, the greater the compression, the more potential for explosive moves, while less compression means more squeezing.

The scatter dots are plotted on the Zero Line with four levels of squeeze: No Squeeze S^{No} (Green), Low Squeeze S^{Low} (Black), Mid Squeeze S^{Mid} (Red), and High Squeeze S^{High} (Orange). The settings are:

- Bollinger Band: $n = 14$, $\alpha_{bb} = 2$
- Keltner Channels: $n = 14$, $\alpha_{kc_high} = 1$, $\alpha_{kc_mid} = 1.5$, $\alpha_{kc_low} = 2$. Note, α_{kc_high} corresponds to most points centred around the mean, and α_{kc_low} corresponds to extreme points of the distribution. α_{kc_mid} is the coefficient used in Tradingview.

There is a sub-indicator function that calculates the Squeeze Conditions, such as No Squeeze, Low Squeeze, Mid Squeeze and High Squeeze. The calculation is as follows:

- $S^{Low} = BB_BotBand \geq KC_BotBand_Low$ and/or $BB_TopBand \leq KC_TopBand_Low$. Low compression, high band spread: the Squeeze is on and the indicator is off.
- $S^{Mid} = BB_BotBand \geq KC_BotBand_Mid$ and/or $BB_TopBand \leq KC_TopBand_Mid$. Mid compression, mid band spread: the Squeeze is on and the indicator is off.
- $S^{High} = BB_BotBand \geq KC_BotBand_High$ and/or $BB_TopBand \leq KC_TopBand_High$. High compression, small band spread: the Squeeze is on and the indicator is off.
- $S^{Off} = BB_BotBand < KC_BotBand_Low$ and/or $BB_TopBand > KC_TopBand_Low$. No compression, largest band spread: the Squeeze is off (squeeze release) and the indicator is on.
- $S^{No} = [\text{if } S^{Off} = \text{False and } S^{Low} = \text{False}], \text{ no Squeeze.}$

Even though the middle band in KC can be computed either with SMA or EMA on the Typical Price, in the former it coincides with that of BB (see Equation (13.99) and Equation (13.101)). As a result, we can express the squeeze in terms of the difference between the standard deviation and the ATR. That is, the squeeze is On or Off when

$$S^{On} = [\text{if } \alpha_{bb} \times SD_t(n) \leq \alpha_{kc} \times ATR_t(n)] \text{ and } S^{Off} = [\text{if } \alpha_{bb} \times SD_t(n) > \alpha_{kc} \times ATR_t(n)] \quad (4.21)$$

where $\alpha_{bb} = 2$ and $\alpha_{kc} = \{1, 1.5, 2\}$. We get the pairs $\{(\alpha_{kc_low}, S^{Low}), (\alpha_{kc_mid}, S^{Mid}), (\alpha_{kc_high}, S^{High})\}$. Here, $\alpha_{kc} \times ATR_t(n)$ acts as a dynamic threshold on the standard deviation where the dynamics is given by $ATR_t(n)$.

4.3.1.2 Momentum indicator Next, we compute a Momentum Indicator with a linear regression-based momentum calculation. At time t , given a window with n periods, compute the SMA of close prices $SC_t(n) = \frac{1}{n} \sum_{i=1}^n C_{t-i+1}$ and the Donchian midline, that is, the average between highest high and lowest low $\overline{HL}_t(n) = \frac{1}{2} (HH(n)_t + LL(n)_t)$. Then, take their averages to get a kind of average price $\overline{P}_t(n) = \frac{1}{2} (\overline{HL}_t(n) + SC_t(n))$. At each time k in that window we compute the difference (or delta) between the close price and this average price, getting $\Delta_k = C_k - \overline{P}_t(n)$. Note, Δ_k is a kind of dispersion value⁸ around an average price (expressed in dollar) which can be positive or negative. Over that window, we get the time series $diff_t(n) = \{\Delta_{t-n+1}, \dots, \Delta_{t-1}, \Delta_t\}$. Then, this time series is smoothed by linearly regressing it, getting some kind of momentum indicator

$$M_t(n) = f_{LRC}(diff_t(n), n, 0) \quad (4.22)$$

where f_{LRC} is a linear regression curve⁹. At time t , we get the point $M_t(n)$ on that regression line. This indicator produces both positive and negative values, with positive values signifying an upward price movement and increasing momentum, while negative values indicate a downward price movement and decreasing momentum.

We can make the values of the Momentum Indicator become comparable throughout time and across securities by normalising the calculation of delta. This way, we get a nuanced assessment of Bullish and Bearish momentum. That is, we define the delta as

$$\Delta_k = \frac{C_k - \overline{P}_t(n)}{ATR_t(n)} \times 100 \quad (4.23)$$

where the ATR is a measure of volatility (range) over a given window (see Equation (13.90)). Note, the ATR could be computed on a shorter window size. Here, Δ_k takes value in $[-100, 100]$.

Traders may still experience difficulties in identifying the direction in which the price would move after a breakout. One solution is to devise an indicator for identifying the trade direction. Thus, we define the average momentum $\overline{M}_t(m) = \frac{1}{m} \sum_{i=1}^m M_{t-i+1}(n)$ and the direction is given by

$$dir_t = M_t(n) - \overline{M}_t(m) \quad (4.24)$$

Note, for the direction to be positive or negative, the current momentum $M_t(n)$ has to be significantly different from its average $\overline{M}_t(m)$. This equation leads to

- A crossover if $[dir_t > 0 \text{ and } dir_{t-1} < 0]$
- A crossunder if $[dir_t < 0 \text{ and } dir_{t-1} > 0]$

4.3.2 Assigning colours

The Histogram shows price momentum whereas the coloured dots (along the zero line) show where the BB are in relation to the KC.

4.3.2.1 The setting The on/off state of the squeeze is shown with small dots on the zero line of the indicator. The colours on the Squeeze are as follows:

- Orange Dots: High compression / large squeeze (One or both of the BB is inside the 1st (1.0 ATR) KC).
- Red Dots: Medium Squeeze (One or both of the BB is inside the 2nd (1.5 ATR) KC).

⁸Statistical dispersion means the extent to which numerical data is likely to vary about an average value. It helps understand the distribution of the data.

⁹Linear regression $y = a + bx$, where a is the intercept and b is the slope.

- Black Dots: Low compression / wide squeeze (One or both of the BB is inside the 3rd (2.0 ATR) KC).
- Green (or Grey) Dots: No Squeeze / Squeeze fired (One or both of the BB is outside of the 3rd (2.0 ATR) KC).

Sometimes, Grey crosses signify that the market is in Squeeze (on) and Green/Red triangles signify Squeeze release (off).

The colours on the Histogram are as follows:

- Transition from Light Aqua (Green) to Dark Blue (Green): Indicates a potential end to the bullish momentum.
- Shift from Dark Blue (Green) to Light Aqua (Green): Suggests an increase in bullish momentum. It's an indicator for traders to consider bullish positions.
- Transition from Yellow (Light Red) to Red (Dark Red): Signals that the bearish momentum might be fading.
- Change from Red (Dark Red) to Yellow (Light Red): Implies that bearish momentum is picking up. Traders might want to explore bearish strategies under this condition.
- Rapid Change from Red (Light Red) to Aqua (Light Green): This swift shift indicates a quick transition from bearish to bullish sentiment. It is a strong signal for traders to consider switching to bullish positions.
- Quick Shift from Aqua (Light Green) to Red (Light Red): Demonstrates a speedy change from bullish to bearish momentum. It suggests that traders might want to adjust their strategies to align with the emerging bearish trend.

A bar that is lower than the previous bar will be a darker color (dark blue or dark red), and a bar that is higher than the previous bar will be a lighter colour (light blue (aqua) or light red (yellow)).

4.3.2.2 Strategy In an ideal scenario, the ticker enters the Squeeze, black dots warn of the beginning of a low compression squeeze. As the BB continue to constrict within the KC, red dots highlight a medium compression. As the price action and momentum continues to compress, an orange dot shows warning of high compression. As price action leaves the squeeze, the colouring would reverse, going from orange to red to black and to green. Any compression squeeze is considered fired at the first green dot that appears. In practice, any type of squeeze (and colour sequence) may appear at any time, so that we should focus primarily on the green dots after any type of compression. Carter recommended entering a position after at least 5 black dots or wait for 1st green dot. Exit occurs on second blue or yellow bar or, alternatively, remain in the position after confirming a continuing trend through a separate indicator.

4.3.2.3 Squeeze The colours for the squeeze are:

- $C_t^{High} = \begin{cases} \text{if } S^{High} \end{cases}$ the colour is Orange, else it is Green
- $C_t^{Mid} = \begin{cases} \text{if } S^{Mid} \end{cases}$ the colour is Red, else it is Green
- $C_t^{Lowh} = \begin{cases} \text{if } S^{Low} \end{cases}$ the colour is Black, else it is Green

When there is no squeeze, the colour is green.

4.3.2.4 Momentum

The colours of the momentum histogram are

- $UpMom_t = \left[\text{if } [M_t > 0] \text{ then if } M_t > M_{t-1}, \text{ the colour is Aqua, else it is Blue.} \right]$
- $DownMom_t = \left[\text{if } [M_t < 0] \text{ then if } M_t < M_{t-1}, \text{ the colour is Red, else it is Yellow.} \right]$

We can define the Bull and Bear Squeeze as

- $Bull_Squeeze_t = \left[\text{if } S^{No} \text{ and } UpMom_t \right] \text{ or } \left[\text{if } UpMom_t \text{ and } DownMom_{t-1} \right]$
- $Bear_Squeeze_t = \left[\text{if } S^{No} \text{ and } DownMom_t \right] \text{ or } \left[\text{if } DownMom_t \text{ and } UpMom_{t-1} \right]$

The Bull and Bear watch out conditions are

- $Buy_Conf = \left[\text{if } UpMom_t \text{ and } DownMom_{t-1} = \text{False} \right] \text{ and not } Bull_Squeeze_t$
- $Sell_Conf = \left[\text{if } DownMom_t \text{ and } UpMom_{t-1} = \text{False} \right] \text{ and not } Bear_Squeeze_t$



Figure 58: Squeeze momentum indicator.

4.3.2.5 Direction

Alternatively, we can use the direction indicator devised in Equation (5.25) to define the squeeze and its release. In that case we get

- In Squeeze (low or mid compression): $\left[\text{if } S^{Low} \text{ or } S^{Mid} \right]$ then the shape is diamond and the colour is Grey (sometimes black). It signifies low volatility, market preparing itself for an explosive move (up or down).
- Squeeze release up (Bull): $\left[\text{if } S^{Off} \text{ and } dir_t \geq 0 \right]$ then the shape is triangle up and the colour is Green (sometimes grey).
- Squeeze release down (Bear): $\left[\text{if } S^{Off} \text{ and } dir_t < 0 \right]$ then the shape is triangle down and the colour is Red (sometimes grey).

Carter suggested to wait till the first green/red (grey cross) triangle after a grey cross (black cross), and take a position in the direction of the momentum. If the histogram is above zero, open a long position, while if it is below zero, open a short one. Exit the position when the momentum changes: increase or decrease (signified by a colour change). We can also use exit signals such as Fibonacci numbers, certain range expansion, or significant support/resistance levels.



Figure 59: Squeeze momentum indicator.

4.3.3 Some analysis

We present the Squeeze Momentum Deluxe which improves the previous squeeze indicators by adding reversion and trend-following analysis. It makes use of the normalised Momentum Indicator with normalised delta given in Equation (4.23) as well as the direction in Equation (5.25). This is illustrated in Figure (60). Further, Directional Flux, Confluence Gauges, and Divergences are added to improve trading with the indicator.



Figure 60: Normalised Squeeze momentum indicator.

The Directional Flux measures the overall direction of price volatility, offering insights into trend sentiment. It is

a slight modification of the Chande momentum oscillator (CMO) given in Equation (13.91) that identifies relative strength or weakness in a market. At time t , given a window with n periods, we set $Up_H_t = \max(H_t - H_{t-1}, 0)$ and $Down_L_t = \min(L_t - L_{t-1}, 0)$ (rather than using the close) and compute the up and down flux direction:

$$U_t(n) = \frac{EMA(Up_H_t, n)}{ATR_t(n)} \text{ and } D_t(n) = \frac{EMA(Down_L_t, n)}{ATR_t(n)}$$

where $EMA(x, n)$ is the exponential moving average over n periods (see Equation (13.88)). In principle, the Flux at time t should be

$$F_t^o = \frac{U_t(n) - D_t(n)}{U_t(n) + D_t(n)} \times 100$$

which takes value in $[-100, 100]$. However, in the Deluxe version it is further smoothed to get

$$F_t^o = EMA\left(\frac{U_t(n) - D_t(n)}{U_t(n) + D_t(n)}, \left\lfloor \frac{n}{2} \right\rfloor\right) \times 100$$

Then, define F_t^s as

$$F_t^s \text{ is not defined if } F_t^o \in [-25, 25], \left[\text{if } F_t^o > 25 \text{ do } F_t^s = F_t^o - 25 \right], \left[\text{if } F_t^o < -25 \text{ do } F_t^s = F_t^o + 25 \right]$$

The crossover and crossunder between X and Y are given by

$$\begin{aligned} CO(X_t, Y_t) &= \left[\text{if } X_t > Y_t \text{ and } X_{t-1} \leq Y_{t-1} \right] \text{ from bottom to top} \\ CU(X_t, Y_t) &= \left[\text{if } X_t < Y_t \text{ and } X_{t-1} \geq Y_{t-1} \right] \text{ from top to bottom} \end{aligned}$$

The flux alerts are defined as

- Flux Midline Crossover: $[CO(F_t^o, \mathbf{0})]$ where $\mathbf{0}$ is a vector of zeros
- Flux Midline Crossunder: $[CU(F_t^o, \mathbf{0})]$

The Confluence Gauges, placed at the top and bottom of the indicator, measure confluence in the relationship between the Momentum Indicator and Directional Flux. They provide interpretable visual aid for detecting market sentiment. Reversal doritos displayed alongside them contribute to mean reversion analysis. The strong and weak confluence gauges are:

- Strong gauge up: $SG_Up_t(0) = [F_t^o > \mathbf{0} \text{ and } M_t > 0]$
- Weak gauge up: $WG_Up_t(0) = [F_t^o > \mathbf{0} \text{ or } M_t > 0]$
- Strong gauge down: $SG_Down_t(0) = [F_t^o < \mathbf{0} \text{ and } M_t < 0]$
- Weak gauge down: $WG_Down_t(0) = [F_t^o < \mathbf{0} \text{ or } M_t < 0]$

where (0) applies to the momentum indicator M_t . The colours on the gauges are:

$$\begin{aligned} G_Up_t &= \left[\text{if } SG_Up_t(0) \text{ then } Cgp \right] \text{ Else if } [WG_Up_t(0) \text{ get light } Cgp] \\ G_Down_t &= \left[\text{if } SG_Down_t(0) \text{ then } Cgn \right] \text{ Else if } [WG_Down_t(0) \text{ get light } Cgn] \end{aligned}$$

where Cgp is a colour for up gauge and Cgn is a colour for down gauge. The confluence alerts are defined as

- Weak Bullish Confluence: it is when we go from no colour to light Cgp.
- Strong Bullish Confluence: it is when we go from no colour to Cgp.
- Weak Bearish Confluence: it is when we go from no colour to light Cgn.
- Strong Bearish Confluence: it is when we go from no colour to Cgn.

This is illustrated in Figure (61).



Figure 61: Confluence gauge.

The divergences are

- The divergence down: $Div_Dw_t = CO(M_t, \overline{M}_t)$, that is, $[dir_t > 0 \text{ and } dir_{t-1} < 0]$.
- The divergence up: $Div_Up_t = CU(M_t, \overline{M}_t)$, that is, $[dir_t < 0 \text{ and } dir_{t-1} > 0]$.

The confluence buy and sell conditions are

- The confluence buy: $[Div_Dw_t \text{ and } SG_Down_t(-40)]$
- The confluence sell: $[Div_Up_t \text{ and } SG_Up_t(40)]$

These indicators detect real-time divergences between price and the oscillator. This is illustrated in Figure (62).

an investor with power utility had preference for odd number moments of any order and was averse to even number moments of any order. Gormsen et al. [2019] found that the skewness related moments were negatively correlated with the kurtosis related moments. Further, there are periods where the return distribution becomes both more left-skewed and more fat tailed. As a result, we propose to revisit some of the main Technical Indicators by accounting for skewness and kurtosis. However, there are numerous ways for combining the first four moments to approximate non-Gaussian distributions (see Appendix (11.6)). We are going to present a few approaches to account for higher moments.

5.1.1 Skew and kurtosis indicators

In financial market skewness hints at the probable direction of asset price movements, while kurtosis alerts to dramatic market shifts. Hence, these higher moments can help manage risk. That is, skewness can help identify the direction in which the data is tending, and kurtosis can be used to measure the strength or decisiveness of that trend's direction. As such, we can directly use sample skew $\hat{S}$ and sample kurtosis $\hat{K}$ defined in Appendix (11.4.3) as technical indicators. In this section, we let the skew be $\mathcal{S}_t = \beta \times \hat{S}$ (it isn't scaled by the standard error of skewness) and the excess kurtosis be $\mathcal{K}_t = \gamma \times (\hat{K} - 3)$ (it isn't scaled by the standard error of kurtosis).

5.1.1.1 Skew indicator In a Gaussian distribution, most of the data is centred around its mean within $\pm\alpha \cdot \sigma$, for some volatility σ and constant α . Following the same approach, we use the notion of bands as a threshold on the skew values. It acts as an alert for abnormal markets by measuring spikes of volatility. Here, the middle band is set to zero: $\text{MidBand} = 0$. In the first set of bands, the top and bottom bands are

$$\text{BotBand1} = -\alpha_1 \text{ and } \text{TopBand1} = \alpha_1$$

In the second set of bands, the top and bottom bands are

$$\text{BotBand2} = -\alpha_2 \text{ and } \text{TopBand2} = \alpha_2$$

That is, check when the skew is above α_1 or below $-\alpha_1$ to alert for a deviation from a normal regime. We can also check when the skew is above α_2 or below $-\alpha_2$ for $\alpha_2 > \alpha_1$. In general, we set $\alpha_1 = \frac{1}{2}$ and $\alpha_2 = 1$ (see Bulmer [1979]). For a skewness in the range $[-\alpha_1, \alpha_1]$, the distribution is considered approximately symmetric. Beyond $\pm\alpha_2$ the distribution is highly skewed. Below $-\alpha_1$ the distribution is left-skewed (there are more outliers distributed on left side), and above α_1 it is right-skewed (there are more outliers distributed on right side). When the distribution is right-skewed, it means that the potential trend is increasing because increasingly price returns tends to be high. Conversely, the potential trend will be decreasing when the distribution tends to be left-skewed. In a sliding window, once we have detected a skewed distribution, spike of volatility becomes the new normal and the skew goes back to around zero, indicating that volatility has been consumed. That is, after a change of regime, as soon as prices slightly change direction (negative/positive), the skew decreases/increases going below/above zero.

5.1.1.2 Kurtosis indicator We can also use the notion of bands as a threshold on the excess kurtosis values. It acts as an alert for abnormal markets by measuring extreme returns and peakness (high apex). Positive kurtosis measures the increase of the variance due to low frequent outliers. Here, the middle band is set to zero to represent the normal distribution: $\text{MidBand} = 0$. In the first set of bands, the top and bottom bands are

$$\text{BotBand1} = -\alpha_1 \text{ and } \text{TopBand1} = \alpha_1$$

In the second set of bands, the top and bottom bands are

$$\text{BotBand2} = -\alpha_2 \text{ and } \text{TopBand2} = \alpha_2$$

That is, check when the excess kurtosis is in the range $[-\alpha_1, \alpha_1]$ to alert for excess kurtosis. We can also check for extreme kurtosis when the excess kurtosis is in between the two bands for $\alpha_2 > \alpha_1$. In general, we set $\alpha_1 = 1$

and $\alpha_2 = 5$. An excess kurtosis in the range $[-\alpha_1, \alpha_1]$ has moderate kurtosis. A high negative kurtosis implies a distribution having shorter and thinner tails, and often its central peak is lower and broader. If a distribution has low apex, the data is dispersed. That is, a negative excess kurtosis means that the majority of the prices are uniformly spread across the distribution. A high positive kurtosis implies that an investment will yield occasional extreme returns. It indicates either large positive returns or extreme negative returns. If a distribution has high apex, the data has a concentrated trend. That is, a positive excess kurtosis means that the majority of the prices are around the average and just a few of them are outside the standard deviation. It describes a sideways market with possible breakout. On the other hand, a low kurtosis value indicates a more stable and predictable return profile, hence lower risk. It is characteristic of a trending market.

5.1.1.3 Trading with skew and kurtosis We can define the trading logic as follows:

1. Entry:

- Long position: When the kurtosis is negative (trend) and the skewness is larger than α_1 (increasing trend), then buy.
- Short position: When the kurtosis is negative (trend) and the skewness is smaller than $-\alpha_1$ (decreasing trend), then sell.

2. Exit:

- When the loss reach at some level, then payoff.
- When skewness changes direction, then payoff.

Momentum: want to know the potential trend and the strength of the trend.

5.1.2 Bands indicators

Since Gaussian distributions are defined by the first two moments, we can use Band Indicators (BIs) (such as the Bollinger Bands) to infer information on non-normal distributions. That is, BIs are attempts by practitioners at characterising higher moments by only relying on the first two moments. We propose to improve these indicators by accounting for higher moments. Since there are numerous ways for combining the first four moments to approximate non-Gaussian distributions, we are going to discuss a few methods to account for the standard deviation and the skew.

At time t , given a window with n periods, we use the Typical Price M_t to compute the sample mean $\hat{\mu}_n$ and the sample standard deviation $\hat{\sigma}_n$ given in Equation (11.65). When necessary, we let μ_n be the sample mode¹⁰. We also compute the sample skew $\hat{S}$ and the sample kurtosis $\hat{K}$ defined in Appendix (11.4.3). In this section, we let the skew be $\mathcal{S}_t = \beta \times \frac{\hat{S}}{6}$ (it is scaled by the standard error of skewness) and the excess kurtosis be $\mathcal{K}_t = \gamma \times \frac{\hat{K}-3}{24}$ (it is scaled by the standard error of kurtosis).

5.1.2.1 Method 1: signed deviation Following the spirit of the two-piece normal distribution (see Appendix (11.6.2)), we consider the scale parameters σ_1 and σ_2 in Equation (11.71). However, instead of using their estimated values given in Equation (11.74), we only make use of the negative and positive deviation from the mean s_1 and s_2 , respectively (see Equation (11.73)).

For the i th observation in our sample, set $R_i(n) = X_i - \mu_n$ and compute

$$R_i^+ = \mathcal{H}^+ R_i^2(n) \text{ and } R_i^- = \mathcal{H}^- R_i^2(n)$$

where $\mathcal{H}^+ = I_{\{R_i(n)>0\}}$ and $\mathcal{H}^- = I_{\{R_i(n)<0\}}$. We obtain the positive and negative deviation to the mean as

¹⁰The mode of a sample is the element that occurs most often in the collection. The numerical value of the mode is the same as that of the mean and median in a normal distribution, but it is be very different in highly skewed distributions.

$$s_2 = \frac{1}{|R_n^+|} \sum_{i \in |R_n^+|} R_i^+ \text{ and } s_1 = \frac{1}{|R_n^-|} \sum_{i \in |R_n^-|} R_i^-$$

Then, $\sigma_1 = \sqrt{s_1}$ controls the scale (spread) of the distribution on the left-hand side of μ_n , while $\sigma_2 = \sqrt{s_2}$ controls the scale on the right-hand side of μ_n . Next, we follow a double band approach: In the first set of bands, the top and bottom bands are

$$\text{BotBand1} = \mu_n - \alpha \times \sigma_1 \text{ and } \text{TopBand1} = \mu_n + \alpha \times \sigma_2$$

Note, if the distribution is symmetric (and n is large), then $\hat{\mu}_n = \mu_n$, $|R_n^+| = |R_n^-| = \frac{n}{2}$, and $\sigma_1 = \sigma_2$ is the standard deviation. Hence, we recover existing bands such as the Bollinger Bands.

5.1.2.2 Method 2: two-piece normal distribution We can also directly follow the two-piece normal distribution (see Appendix (11.6.2)) with estimates for σ_1 and σ_2 given in Equation (11.74). The mean and variance of that distribution (see Equation (11.72)) are:

$$\mu^{TPD} = \mu_n + \sqrt{\frac{2}{\pi}}(\sigma_2 - \sigma_1) \text{ and } V^{TPD} = (1 - \frac{2}{\pi})(\sigma_2 - \sigma_1)^2 + \sigma_1 \sigma_2$$

Next, we follow a double band approach: In the first set of bands, the top and bottom bands are

$$\text{BotBand1} = \mu^{TPD} - \alpha \times \sigma^{TPD} \text{ and } \text{TopBand1} = \mu^{TPD} + \alpha \times \sigma^{TPD}$$

where $\sigma^{TPD} = \sqrt{V^{TPD}}$. Note, if the distribution is symmetric (and n is large), then $\hat{\mu}_n = \mu_n$, and $\sigma_1 = \sigma_2$. Hence, we recover existing bands such as the Bollinger Bands.

5.1.2.3 Method 3: skew First, we define the direction of a data sample and the strength of its deviation from the normal distribution as

$$Dir_t = (\mathcal{H}^+ - \mathcal{H}^-)\mathcal{S}_t = |\mathcal{S}_t|$$

where $\mathcal{H}^+ = I_{\{\hat{S} > 0\}}$ and $\mathcal{H}^- = I_{\{\hat{S} < 0\}}$. Following our study of the impacts of skewness and kurtosis on distributions in Appendix (11.4.4), we modify the standard deviation (SD) to account for the skew. We consider a left-skewed SD and a right-skewed SD given by

$$\sigma_1 = \hat{\sigma}_n - \mathcal{H}^- \mathcal{S}_t \text{ and } \sigma_2 = \hat{\sigma}_n + \mathcal{H}^+ \mathcal{S}_t$$

Next, we follow a double band approach: In the first set of bands, the top and bottom bands are

$$\text{BotBand1} = \hat{\mu}_n - \sigma_1 \text{ and } \text{TopBand1} = \hat{\mu}_n + \sigma_2$$

Note, when the skew is zero, then $\sigma_1 = \sigma_2$ is the standard deviation. Hence, we recover existing bands such as the Bollinger Bands.

5.1.2.4 Second part: kurtosis In the second set of bands, the top and bottom bands are

$$\text{BotBand2} = \text{BotBand1} - \mathcal{K}_t \text{ and } \text{TopBand2} = \text{TopBand1} + \mathcal{K}_t$$

Note, when the excess kurtosis is positive, TopBand2 is above TopBand1 and BotBand2 below BotBand1. Conversely, when the excess kurtosis is negative, TopBand2 is below TopBand1 and BotBand2 above BotBand1. When the excess kurtosis is zero, the double bands collapse to the single ones. We will see later that the excess kurtosis has a major impact on the Squeeze Indicator.

5.1.2.5 Applications We can use these higher moments bands in the same way as we use the Bollinger Bands. We can also apply it to the WVF.

5.1.3 Momentum indicator

Given a sample of market returns, a well known performance measure is the Sharpe ratio which measures the excess return per unit of deviation in an investment asset (see Appendix (12.2)). It can be used as a momentum indicator. However, it assumes market returns to be normally distributed and suffer from the above drawbacks. To account for this drawback, several modified Sharpe ratios were developed (see Kazemi et al. [2003]). For instance, Pezier [2004] and Pezier et al. [2006] proposed the adjusted Sharpe ratio (ASR) which explicitly rewards positive skewness and low kurtosis in its calculation. We follow this approach to devise the Adjusted Momentum Indicator (AMI).

At time t , given a window with n periods, compute the log-returns of the Typical Price M_t , getting $LR_t = \log \frac{M_t}{M_{t-1}}$. We write the sample Sharpe ratio (see Equation (12.81)) as

$$M_t^{SR}(n) = \frac{\hat{\mu}_n - R_b}{\hat{\sigma}_n}$$

where $\hat{\mu}_n$ and $\hat{\sigma}_n$ are the sample mean and sample standard deviation, respectively, given in Equation (11.65). When volatility is low and returns are high the sample SR is high, conversely when returns are low the sample SR is low. We can now incorporate tail risk in the sample SR by considering the ASR given in Equation (12.82), which is derived from a Taylor series expansion of expected exponential utility function (see Appendix (12.3)). We are using this performance measure to devise the adjusted momentum indicator for long and short position. That is,

$$M_t^{AMI}(n) = M_t^{SR}(n) \left[1 + \frac{\hat{S}}{6} M_t^{SR}(n) - \frac{(\hat{K} - 3)}{24} (M_t^{SR}(n))^2 \right]$$

where $\hat{S}$ and $\hat{K}$ are the Skew and Kurtosis of the return distribution, respectively, defined in Appendix (11.4.3). Clearly, in a long position with positive rates, negative skewness ($\hat{S} < 0$) and positive excess kurtosis ($\hat{K} > 0$) penalise the Momentum. Conversely, in a short position with negative rates, positive skewness ($\hat{S} > 0$) and positive excess kurtosis ($\hat{K} > 0$) penalise the Momentum.

In order to identify the direction in which the price would move after a breakout, we devise an indicator for identifying the trade direction. Thus, we define the average momentum $\bar{M}_t^{AMI}(m) = \frac{1}{m} \sum_{i=1}^m M_{t-i+1}^{AMI}(n)$, with sample size m , and the direction is given by

$$dir_t^{AMI} = M_t^{AMI}(n) - \bar{M}_t^{AMI}(m) \quad (5.25)$$

Note, for the direction to be positive or negative, the current momentum $M_t^{AMI}(n)$ has to be significantly different from its average $\bar{M}_t^{AMI}(m)$. This equation leads to

- A crossover if $[dir_t^{AMI} > 0 \text{ and } dir_{t-1}^{AMI} < 0]$
- A crossunder if $[dir_t^{AMI} < 0 \text{ and } dir_{t-1}^{AMI} > 0]$

We can then use the conditions and alerts defined both in Section (4.3.2) and in Section (4.3.3) by replacing the momentum indicator $M_t(n)$ with $M_t^{AMI}(n)$ and the direction dir_t with dir_t^{AMI} .

5.1.4 Squeeze indicators

We saw in Equation (4.21) that the Squeeze was a measure of compression of the probability distribution measured with respect to the standard deviation. That is, it assumes that returns are Gaussian and that standard deviation is the risk measure. More generally, we define the Squeeze as a measure of compression of the distribution measured with respect to Risk. In non-Gaussian distributions, Risk is defined with respect to higher moments. Hence, the Squeeze has to be measured with respect to the left tail and the right tail of the distribution.

As discussed in Appendix (11.4.4), distributions with positive excess kurtosis have returns concentrated on the mode with a lower than normal probability of a medium-sized move, but a higher than normal probability of both a large move and small move. Conversely, distributions with negative excess kurtosis have returns (uniformly) spread across the distribution, but with a lower than normal probability of a large move. Hence, practitioners focusing on standard deviation alone as a measure of risk are not correct. When we squeeze somewhere on the distribution, it goes somewhere else (see Figure (72)). We must therefore decide on the definition of Squeeze we are interested in. We give two possible meanings:

1. Negative kurtosis: Having low standard deviation and a lower than normal probability of outliers, but returns spread across the distribution.
2. Positive kurtosis: Having low standard deviation and lower than normal probability of medium-sized move but higher than normal probability of outliers.

In general, practitioners are after that moment of burst when the market gets large moves (outliers), which has a higher probability of occurrence with positive excess kurtosis. We devise two Squeeze Indicators to account for the above two characteristics.

5.1.4.1 Indicator 1 We consider the spread between the first set of bands and the second set of bands: $Diff_1 = \text{TopBand1} - \text{BotBand1}$ and $Diff_2 = \text{TopBand2} - \text{BotBand2}$. In the first set of bands, we get

- Method 1: $Diff_1 = \alpha \times (\sigma_1 + \sigma_2)$. From the two piece distributions (see Appendix 11.6.2), we get $Diff_1 = \alpha \times \frac{2}{w}$ where w is the weight between the two densities.
- Method 2: $Diff_1 = \alpha \times 2 \times \sigma^{TPD} = \alpha \times 2 \times \sqrt{(1 - \frac{2}{\pi})(\sigma_2 - \sigma_1)^2 + \sigma_1\sigma_2}$.
- Method 3: $Diff_1 = \sigma_1 + \sigma_2 = 2\hat{\sigma}_n + Dir_t$.

In the second set of bands, we get

$$Diff_2 = Diff_1 + 2\mathcal{K}_t$$

which gives

- Method 1: $Diff_2 = \alpha \times (\sigma_1 + \sigma_2) + 2\mathcal{K}_t$.
- Method 2: $Diff_2 = \alpha \times 2 \times \sigma^{TPD} + 2\mathcal{K}_t$.
- Method 3: $Diff_2 = (\sigma_1 + \sigma_2) + 2\mathcal{K}_t = 2\hat{\sigma}_n + Dir_t + 2\mathcal{K}_t$.

The Squeeze On and Off is now given by

$$S_t^{On} = Diff_2 < \epsilon_t^{on} \text{ and } S_t^{Off} = Diff_2 > \epsilon_t^{off}$$

for some thresholds ϵ_t^{on} and ϵ_t^{off} . In the special case where both the skew and kurtosis are null, we recover the Squeeze in Equation (4.21). Clearly the Squeeze is On when σ is small, $\hat{S}_t = 0$ and $\hat{K}_t - 3 < 0$, conversely it is Off when σ is large, $\hat{S}_t \neq 0$ and $\hat{K}_t - 3 > 0$.

5.1.4.2 Indicator 2 Arguably, the ratio $\frac{1}{\hat{\sigma}_n}$ is a measure of squeeze: it is high for low volatility and low for high volatility. Further, a squeeze is characterised by no skewness and positive kurtosis. As a result, we can easily modify the above measure of momentum (M_t^{AMI}) to get a measure of squeeze. That is, given the volatility ratio $M_t^{VR}(n) = \frac{1}{\hat{\sigma}_n}$, we propose the Adjusted Squeeze Indicator

$$M_t^{ASI}(n) = M_t^{VR}(n) \left[1 - \frac{|\hat{S}|}{6} M_t^{VR}(n) + \frac{(\hat{K} - 3)}{24} (M_t^{VR}(n))^2 \right]$$

This measure gets larger with lower volatility, no skew, and positive excess kurtosis. On the other hand, it is penalised with high volatility, skewness, and negative excess kurtosis. Note, we can always change the sign on the kurtosis if we want to satisfy the first definition of a squeeze.

5.2 Using signatures

6 Trading with options

We saw in Section (3) how to use technical indicators (TIs) to identify patterns and predict future price movements. However, TIs are data driven and do not represent expectation about future events. It is well known that Statistical Models (such as moving averages) fit the past and hope it will repeat, while option prices estimates future events on average. As a result, we can use Option Theory to infer the distribution of future market returns and devise quantitative strategies. That is,

- Use option payoffs to identify opportunities and protect capital.
- Use option payoffs to infer market return behaviour.
- Use option theory to devise trading rules.
- Dynamically maximise Risk/Return.

The benefits of using options to forecast the expected move of the stock are as follows:

- Help to create a trade with a higher probability of success
- Set reasonable profit targets
- Choose our profit stops and stop loss for the trade setup

Thus, we are now going to explain how we can use option prices to devise quantitative strategies. We will first consider a diffusion model to represent the dynamics of stock prices (or returns), and derive their first two expected moments. We will then relate these moments to traded contracts and use them for risk management. At last, we will perform a risk/reward analysis with option prices.

6.1 Modelling returns

We let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space where $(\mathcal{F}_t)$ is a right-continuous filtration including all $\mathbb{P}$ negligible sets in $\mathcal{F}$. We just assume that the stock price $(S_t)_{t \geq 0}$ is a semimartingale with a unique representation. For simplicity, we let the stock price $\{S_u; u \in [t, T]\}$, over the life of the option price, be a diffusion process.

6.1.1 A diffusion model

We write the dynamics of the stock price, under the historical measure $\mathbb{P}$, as

$$\frac{dS_t}{S_t} = b_t dt + dM_{P,S}(t) \quad (6.26)$$

with drift and martingale

$$b_t = r_t - q_t + \sigma_t \lambda_t \text{ and } dM_{P,S}(t) = \sigma_t d\widehat{W}_S(t), \sigma_t = \sqrt{V_t}$$

where $\widehat{W}_S$ is a Brownian motion in the $\mathbb{P}$ -measure, V_t is a stochastic variance process, and λ_t is a market price of risk (MPR) process. The stock price S_t is given by

$$S_t = x e^{\int_0^t (r_s - q_s + \sigma_s \lambda_s) ds} \mathcal{E}(\sigma \widehat{W}_S)(t), S_0 = x$$

where the Doleans-Dade martingale is

$$\mathcal{E}(\sigma \widehat{W})(t) = e^{-\frac{1}{2} \int_0^t \sigma_s^2 ds + \int_0^t \sigma_s d\widehat{W}_S(s)}$$

For $u \geq t$, the first two moments of the infinitesimal returns are

$$E\left[\frac{dS_u}{S_u} | \mathcal{F}_t\right] = E[r_u + \sigma_u \lambda_u | \mathcal{F}_t] du \text{ and } Var\left(\frac{dS_u}{S_u} | \mathcal{F}_t\right) = E[V_u | \mathcal{F}_t] du$$

When $u = t$, we get

$$\frac{1}{dt} E\left[\frac{dS_t}{S_t} | \mathcal{F}_t\right] = r_t + \sigma_t \lambda_t \text{ and } \frac{1}{dt} Var\left(\frac{dS_t}{S_t} | \mathcal{F}_t\right) = V_t$$

We define the cumulative mean drift, under the historical measure $\mathbb{P}$, as

$$\begin{aligned} H_P(t, T) &= \frac{1}{T-t} \int_t^T (r_s - q_s + \sigma_s \lambda_s) ds \\ &= \frac{1}{T-t} \int_t^T r_s ds - \frac{1}{T-t} \int_t^T q_s ds + \frac{1}{T-t} \int_t^T \sigma_s \lambda_s ds = H^r(t, T) + H^q(t, T) + H^\lambda(t, T) \end{aligned}$$

The expected cumulative mean drift, under the historical measure $\mathbb{P}$, is

$$H_{P,0}(t, T) = E[H_P(t, T) | \mathcal{F}_t]$$

We define the mean variance as

$$I^2(t, T) = \frac{1}{T-t} Y_t \text{ for } Y_t = \int_t^T V(X_s) ds$$

where the cumulative variance process Y_t is not an adapted process. We denote $\tilde{I}(t, T) = (T-t)I^2(t, T)$ the total variance over the period $[t, T]$. The expected mean variance, under the historical measure, is

$$I_0^2(t, T) = E[I^2(t, T) | \mathcal{F}_t]$$

From Equation (6.26), we observe that the cumulative return, under the historical measure $\mathbb{P}$, is

$$\begin{aligned}
R_P(t, T) &= \int_t^T \frac{dS_u}{S_u} = (T-t)H_P(t, T) + \int_t^T dM_{P,S}(u) \\
&= (T-t)H^r(t, T) + (T-t)H^q(t, T) + (T-t)H^\lambda(t, T) + \int_t^T dM_{P,S}(u)
\end{aligned}$$

Taking conditional expectation, we get the expected cumulative return

$$\begin{aligned}
R_{P,0}(t, T) &= E[R_P(t, T)|\mathcal{F}_t] = (T-t)H_{P,0}(t, T) \\
&= (T-t)H_{Q,0}(t, T) + (T-t)H_0^\lambda(t, T)
\end{aligned} \tag{6.27}$$

We are interested in the moments of the cumulative returns over the life of the option, that is,

$$\begin{aligned}
E\left[\int_t^T \frac{dS_u}{S_u} du | \mathcal{F}_t\right] &= E\left[\int_t^T (r_u - q_u + \sigma_u \lambda_u) du | \mathcal{F}_t\right] = (T-t)H_{P,0}(t, T) \\
Var\left(\int_t^T \frac{dS_u}{S_u} du | \mathcal{F}_t\right) &= E\left[\int_t^T V_u du | \mathcal{F}_t\right] = (T-t)I_0^2(t, T)
\end{aligned}$$

Thus, in a diffusion model, returns are characterised by

1. The expected mean drift $H_{P,0}(t, T)$
2. The expected mean variance $I_0^2(t, T)$

6.1.2 Traded contracts

6.1.2.1 The forward price Given the dynamics of the stock price in Equation (6.26), the spot price at time T seen at time t is

$$S_{P,T} = S_t C_P(t, T) \mathcal{E}_P^\sigma(t, T)$$

where

$$C_P(t, T) = e^{(T-t)H_P(t, T)} \text{ and } \mathcal{E}_P^\sigma(t, T) = e^{-\frac{1}{2}(T-t)I^2(t, T) + \int_t^T \sigma_s dW(s)}$$

Taking expectation, we get

$$C_{P,0}(t, T) = E[C_P(t, T)|\mathcal{F}_t], \quad E[\mathcal{E}_P^\sigma(t, T)|\mathcal{F}_t] = 1$$

where $C_{P,0}(t, T)$ is the normalised forward price seen at time t given by

$$\begin{aligned}
C_{P,0}(t, T) &= E[e^{(T-t)H_P(t, T)} | \mathcal{F}_t] = E[e^{(T-t)H^r(t, T) + (T-t)H^q(t, T) + (T-t)H^\lambda(t, T)} | \mathcal{F}_t] \\
&= E[e^{(T-t)H^r(t, T)} | \mathcal{F}_t] E[e^{(T-t)H^q(t, T)} | \mathcal{F}_t] E[e^{(T-t)H^\lambda(t, T)} | \mathcal{F}_t] \Psi(t, T) \Psi^\lambda(t, T) \\
&= \frac{D_q(t, T)}{P(t, T)} \Lambda^\sigma(t, T) \Psi(t, T) \Psi^\lambda(t, T)
\end{aligned}$$

and

- $P(t, T) = E[e^{-(T-t)H^r(t, T)} | \mathcal{F}_t]$ is the discount factor

- $D_q(t, T) = E[e^{(T-t)H^q(t, T)} | \mathcal{F}_t]$ is the repo factor
- $\Lambda^\sigma(t, T) = E[e^{(T-t)H^\lambda(t, T)} | \mathcal{F}_t]$ is the market price of risk factor.

Note, $\Psi(t, T)$ is an adjustment term accounting for the joint variation between the returns of all the stochastic processes in the drift and S , except for the market price of risk λ . It is a direct application of the semimartingale exponential. The quadratic variation can be composed of several covariances depending on the number of stochastic processes. Further, we let $\Psi^\lambda(t, T)$ denote the specific adjustment term accounting for the covariance between the market price of risk, λ , and the spot price S . We denote $\Lambda^{\Psi^\lambda}(t, T)$ the modified MRP factor

$$\Lambda^{\Psi^\lambda}(t, T) = E^{S^{\tilde{S}}} [e^{(T-t)H^\lambda(t, T)} | \mathcal{F}_t] = \Lambda^\sigma(t, T) \Psi^\lambda(t, T)$$

which corresponds to the MRP factor, under the $P^{\tilde{S}}$ measure. We can then rewrite the normalised forward price as

$$C_{P,0}(t, T) = \frac{D_q(t, T)}{P(t, T)} \Lambda^{\Psi^\lambda}(t, T) \Psi(t, T)$$

6.1.2.2 The yield curve Note, we can express the cumulative mean drift as

$$(T - t)H_P(t, T) = \log C_P(t, T)$$

Taking expectation, we get the expected cumulative mean drift

$$E[(T - t)H_P(t, T)] = (T - t)H_{P,0}(t, T) = E[\log C_P(t, T)]$$

We can apply Taylor series on the logarithm ¹¹, getting (at the first order)

$$\log C_P(t, T) \approx C_P(t, T) - 1$$

Hence, we can rewrite the above equation as

$$(T - t)H_{P,0}(t, T) \approx E[C_P(t, T) - 1] = C_{P,0}(t, T) - 1 \quad (6.28)$$

We let $y_P(t, T)$ be the yield curve of the stock price, under the $\mathbb{P}$ -measure, defined as

$$C_{P,0}(t, T) = e^{y_P(t, T)(T-t)}$$

Taking the logarithm, we get

$$\begin{aligned} y_P(t, T) &= \frac{1}{T-t} \log C_{P,0}(t, T) \\ &= y^r(t, T) + y^q(t, T) + y^\lambda(t, T) + y^\psi(t, T) \end{aligned} \quad (6.29)$$

where

- $y^r(t, T) = -\frac{1}{T-t} \log P(t, T)$ is the yield for the spot rate
- $y^q(t, T) = \frac{1}{T-t} \log D_q(t, T)$ is the yield for the repo
- $y^\lambda(t, T) = \frac{1}{T-t} \log \Lambda^{\Psi^\lambda}(t, T)$ is the yield for the market price of risk.

¹¹For $x \in \mathbb{R}$ satisfying $0 < x < 2$, $f(x) = \ln(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1)^n$

For $|y_P(t, T)(T - t)| < 1$, we can apply Taylor series on the exponential¹², getting the normalised forward price

$$C_{P,0}(t, T) \approx 1 + y_P(t, T)(T - t)$$

Hence, we get the relation

$$y_P(t, T; S_t) \approx H_{P,0}(t, T)$$

That is, the yield curve of the stock price is approximately equal to the expected cumulative mean drift.

6.1.2.3 A link to option prices The forward price, under the $\mathbb{P}$ -measure, is

$$\begin{aligned} F_P(t, T; S_t) &= E[S_{P,T} | S_t] = S_t E[C_P(t, T) | S_t] = S_t C_{P,0}(t, T) \\ &= S_t \frac{D_q(t, T)}{P(t, T)} \Lambda^{\Psi^\lambda}(t, T) \Psi(t, T) \end{aligned} \quad (6.30)$$

so that the normalised forward price (cumDrift) is

$$C_{P,0}(t, T; S_t) = \frac{F_P(t, T; S_t)}{S_t}$$

From the put-call parity in Equation (15.115), we get

$$C_{P,0}(t, T; S_t) = \frac{\bar{C}_P(t, S_t; K, T) - \bar{P}_P(t, S_t; K, T) + K}{S_t} \quad (6.31)$$

For simplicity, we can choose the at-the-money strike $K^* = S_t$, and get

$$C_{P,0}(t, T; S_t) = \frac{\bar{C}_P(t, S_t; K^*, T) - \bar{P}_P(t, S_t; K^*, T)}{K^*} + 1$$

in which case

$$C_{P,0}(t, T; S_t) - 1 = \frac{\bar{C}_P(t, S_t; K^*, T) - \bar{P}_P(t, S_t; K^*, T)}{K^*}$$

Further, the yield curve in Equation (6.29) becomes

$$\begin{aligned} y_P(t, T; S_t) &= \frac{1}{T - t} \log C_{P,0}(t, T) \\ &= \frac{1}{T - t} \log \frac{F_P(t, T; S_t)}{S_t} \\ &= \frac{1}{T - t} \log \left(\frac{\bar{C}_P(t, S_t; K^*, T) - \bar{P}_P(t, S_t; K^*, T)}{K^*} + 1 \right) \end{aligned} \quad (6.32)$$

For $\left| \frac{\bar{C}_P(t, S_t; K^*, T) - \bar{P}_P(t, S_t; K^*, T)}{K^*} + 1 \right| < 1$, we can apply Taylor series on the logarithm¹³, getting (at the first order)

¹² $e^x = 1 + x + \dots$ for $|x| < 1$.

¹³ $\ln(1 + x) = x - \frac{x^2}{2} + \dots$ for $|x| < 1$.

$$\begin{aligned}
y_P(t, T; S_t) &\approx \frac{\overline{C}(t, S_t; K^*, T) - \overline{P}(t, S_t; K^*, T)}{K^*(T-t)} \\
&= \frac{C_{P,0}(t, T; S_t) - 1}{(T-t)} \\
&= \frac{1}{(T-t)} \left(\frac{F_P(t, T; S_t)}{S_t} - 1 \right)
\end{aligned} \tag{6.33}$$

6.2 Risk management

We estimate the expected simple and logarithm returns (see Appendix (12.1)) and link them to obtain the expected mean variance. Then, we present a few measures of risk under the historical measure $\mathbb{P}$.

6.2.1 Estimating the first two moments

6.2.1.1 Defining returns The returns over the period $[t, T]$ are estimated as follows:

- Simple return

$$R_{t,T} = \frac{S_{P,T}}{S_t} - 1 = C_P(t, T) \mathcal{E}_P^\sigma(t, T) - 1$$

with expected value

$$\begin{aligned}
E[R_{t,T} | \mathcal{F}_t] &= C_{P,0}(t, T) - 1 = \frac{F_P(t, T; S_t)}{S_t} - 1 \\
&\approx (T-t)y_P(t, T) \\
&\approx (T-t)H_{P,0}(t, T)
\end{aligned}$$

Hence, we get

$$E[R_{t,T} | \mathcal{F}_t] \approx E[R_P(t, T) | \mathcal{F}_t] = R_{P,0}(t, T)$$

- Log-return

$$\begin{aligned}
r_L(t, T) = \log \frac{S_{P,T}}{S_t} &= \log C_P(t, T) + \log \mathcal{E}_P^\sigma(t, T) \\
&= (T-t)H_P(t, T) - \frac{1}{2}(T-t)I^2(t, T) + \int_t^T \sigma_s dW(s)
\end{aligned}$$

with expected value

$$\begin{aligned}
E[r_L(t, T) | \mathcal{F}_t] &= (T-t)H_{P,0}(t, T) - \frac{1}{2}(T-t)I_0^2(t, T) \\
&\approx (C_{P,0}(t, T) - 1) - \frac{1}{2}(T-t)I_0^2(t, T) \\
&= E[R_{t,T} | \mathcal{F}_t] - \frac{1}{2}(T-t)I_0^2(t, T)
\end{aligned} \tag{6.34}$$

Thus, we can deduce that

$$(T-t)I_0^2(t, T) \approx 2 \left\{ E[R_{t,T} | \mathcal{F}_t] - E[r_L(t, T) | \mathcal{F}_t] \right\}$$

6.2.1.2 The log-contract Taking the logarithm of the spot price at maturity, we get

$$\log \frac{S_{P,T}}{S_t} = \log C_P(t, T) + \log \mathcal{E}_P^\sigma(t, T) = r_L(t, T)$$

which is the log-return defined above. Taking the expectation, we get the log-contract

$$E[\log \frac{S_{P,T}}{S_t} | \mathcal{F}_t] \approx (C_{P,0}(t, T) - 1) - \frac{1}{2}(T - t)I_0^2(t, T)$$

From Equation (6.33), the log-contract becomes

$$\begin{aligned} E[\log \frac{S_{P,T}}{S_t} | \mathcal{F}_t] &\approx \left(\frac{F_P(t, T; S_t)}{S_t} - 1 \right) - \frac{1}{2}(T - t)I_0^2(t, T) \\ &= \frac{\bar{C}(t, S_t; K^*, T) - \bar{P}(t, S_t; K^*, T)}{K^*} - \frac{1}{2}(T - t)I_0^2(t, T) \end{aligned}$$

Thus, the expected mean variance is approximated by

$$\begin{aligned} I_0^2(t, T) &\approx \frac{2}{T - t} \left[\left(\frac{F_P(t, T; S_t)}{S_t} - 1 \right) - E[\log \frac{S_{P,T}}{S_t} | \mathcal{F}_t] \right] \\ &= \frac{2}{T - t} \left\{ \frac{\bar{C}(t, S_t; K^*, T) - \bar{P}(t, S_t; K^*, T)}{K^*} - E[\log \frac{S_{P,T}}{S_t} | \mathcal{F}_t] \right\} \end{aligned} \quad (6.35)$$

6.2.2 Measure of risk

From Equation (6.26), we observe that

$$\int_t^T \frac{dS_u}{S_u} du - \int_t^T \sigma_s dW(s) = (T - t)H_P(t, T)$$

Taking the expectation, we get

$$H_{P,0}(t, T) = \frac{1}{T - t} \int_t^T E[\frac{dS_u}{S_u} | \mathcal{F}_t] du$$

Thus, decomposing $H_{P,0}$, we get

$$\int_t^T E[\sigma_s \lambda_s | \mathcal{F}_t] ds = \int_t^T E[\frac{dS_u}{S_u} | \mathcal{F}_t] du - \int_t^T E[r_s | \mathcal{F}_t] ds$$

which is

$$(T - t)H_{P,0}^\lambda(t, T) = \int_t^T E[\frac{dS_u}{S_u} | \mathcal{F}_t] du - (T - t)H_{P,0}^r(t, T)$$

A natural measure of risk over the life of the option (see Equation (6.34)) is

$$M_R^{(1)}(t, T) = \frac{(T - t)H_{P,0}(t, T)}{\sqrt{(T - t)I_0^2(t, T)}} = \sqrt{(T - t)} \frac{H_{P,0}(t, T)}{I_0(t, T)}$$

From Equation (6.28), it can be approximated by

$$M_R^{(1)}(t, T) \approx \frac{C_{P,0}(t, T) - 1}{\sqrt{(T - t)I_0^2(t, T)}}$$

Alternatively, splitting the interest rate from the market price of risk, we get

$$M_R^{(2)}(t, T) = \frac{(T-t)H_0^\lambda(t, T)}{\sqrt{(T-t)I_0^2(t, T)}} = \frac{\int_t^T E\left[\frac{dS_u}{S_u} | \mathcal{F}_t\right] du - (T-t)H_0^r(t, T)}{\sqrt{(T-t)I_0^2(t, T)}}$$

Differentiating $M_R^{(2)}$ with respect to time t , we get

$$M_R^{(3)} = \frac{(r_t + \sigma_t \lambda_t) - r_t}{\sigma_t} = \frac{\frac{1}{dt} E\left[\frac{dS_t}{S_t} | \mathcal{F}_t\right] - r_t}{\sqrt{\frac{1}{dt} \text{Var}\left(\frac{dS_t}{S_t}\right)}}$$

which is an infinitesimal measure of risk when the model parameters are stochastic. In the special case where r_t, σ_t , and λ_t are constant, we get $M_R^{(3)} = \lambda$, and we recover the Sharpe ratio in Equation (15.112).

6.2.3 Trading with the first two moments

From Equation (6.33), the yield curve of the stock price is

$$\begin{aligned} y_P(t, T) &= \frac{1}{T-t} \log C_{P,0}(t, T) \\ &= \frac{1}{T-t} \log \frac{F_P(t, T; S_t)}{S_t} \\ &= \frac{1}{T-t} \log \left(\frac{\bar{C}(t, S_t; K^*, T) - \bar{P}(t, S_t; K^*, T)}{K^*} + 1 \right) \end{aligned}$$

We can compute the yield curve for different maturities and decide to trade based on the shape of that curve. From Equation (6.35), we get the cumulative returns statistics

$$\begin{aligned} C_{P,0}(t, T) - 1 &\approx (T-t)H_{P,0}(t, T) = \frac{F_P(t, T; S_t)}{S_t} - 1 \\ (T-t)I_0^2(t, T) &\approx 2 \left\{ \left(\frac{F_P(t, T; S_t)}{S_t} - 1 \right) - E\left[\log \frac{S_{P,T}}{S_t} | \mathcal{F}_t\right] \right\} \end{aligned} \quad (6.36)$$

and the $M_R^{(1)}(t, T)$ risk measure is

$$M_R^{(1)}(t, T) = \frac{\frac{F_P(t, T; S_t)}{S_t} - 1}{\sqrt{2 \left[\left(\frac{F_P(t, T; S_t)}{S_t} - 1 \right) - E\left[\log \frac{S_{P,T}}{S_t} | \mathcal{F}_t\right] \right]^{\frac{1}{2}}}}$$

We want to enter a trade if either the absolute value of the expected return or the absolute value of the expected log-return are larger than a threshold:

- $|E[R_{t,T} | \mathcal{F}_t]| > \epsilon_L$: this is given by $|C_{P,0}(t, T) - 1| > \epsilon_L$
- $|E[r_L(t, T) | \mathcal{F}_t]| > \epsilon_L$: this is given by $\left| (C_{P,0}(t, T) - 1) - \frac{1}{2}(T-t)I_0^2(t, T) \right| > \epsilon_L$

Then, the direction of the trade is given by the sign of the returns:

- if $E[R_{t,T} | \mathcal{F}_t] > 0$ we go long, else we go short
- if $E[r_L(t, T) | \mathcal{F}_t] > 0$ we go long, else we go short

6.2.4 Example: the GBM

So far we have presented results based on a general diffusion model. We are now going to specify the approach by considering the simple case of a GBM. The dynamics of the stock price, under the $\mathbb{P}$ -measure, are

$$\frac{dS_t}{S_t} = (r - q + \sigma\lambda)dt + \sigma dW_t$$

where $\sigma > 0$ is the volatility and λ is a constant MPR. Integrating that dynamics in the range $[t, T]$ the stock price is given by

$$S_{P,T} = S_t C_P(t, T) \mathcal{E}_P^\sigma(t, T) \text{ where } C_P(t, T) = e^{(r-q+\sigma\lambda)(T-t)}, \mathcal{E}_P^\sigma(t, T) = e^{-\frac{1}{2}\sigma^2(T-t) + \sigma W_{T-t}}$$

Thus,

$$(T - t)H_P(t, T) = (r - q + \sigma\lambda)(T - t)$$

Taking the expectation, we get

$$(T - t)H_{P,0}(t, T) = (r - q + \sigma\lambda)(T - t) \text{ and } \mathcal{E}_{P,0}^\sigma(t, T) = 1$$

so that $H_P(t, T) = H_{P,0}(t, T)$. Similarly,

$$C_{P,0}(t, T) = E[e^{(r-q+\sigma\lambda)(T-t)} | \mathcal{F}_t] = e^{(r-q+\sigma\lambda)(T-t)}$$

so that

$$C_P(t, T) = C_{P,0}(t, T) \approx 1 + (r - q + \sigma\lambda)(T - t) = 1 + (T - t)H_{P,0}(t, T)$$

It can be computed with option prices as

$$C_{P,0}(t, T) - 1 = \frac{\overline{C}(t; K^*, T) - \overline{P}(t; K^*, T)}{K^*}$$

where $K^* = S_t$. The forward price becomes

$$F_P(t, T; S_t) = S_t C_{P,0}(t, T)$$

The yield curve of the stock price is

$$y_P(t, T) = \frac{1}{T - t} \log C_{P,0}(t, T) = H_{P,0}(t, T) = r - q + \sigma\lambda$$

Similarly, the logarithm of the stock price $S_{P,T}$ is given by

$$\log \frac{S_{P,T}}{S_t} = (r - q + \sigma\lambda)(T - t) + \log \mathcal{E}_P^\sigma(t, T)$$

and taking the expectation, we get

$$E[\log \frac{S_{P,T}}{S_t} | \mathcal{F}_t] = (r - q + \sigma\lambda - \frac{1}{2}\sigma^2)(T - t) = (T - t)H_{P,0}(t, T) - \frac{1}{2}I_0(t, T)(T - t)$$

Hence, the cumulative returns statistics defined in Equation (6.36) become

$$\begin{aligned}
 (T-t)H_{P,0}(t, T) &= (r-q+\sigma\lambda)(T-t) \approx C_{P,0}(t, T) - 1 \\
 (T-t)I_0^2(t, T) &= 2\left\{(r-q+\sigma\lambda)(T-t) - \left(r-q+\sigma\lambda - \frac{1}{2}\sigma^2\right)(T-t)\right\} \\
 &= \sigma^2(T-t)
 \end{aligned}$$

The measure of risk $M_R^{(1)}(t, T)$ becomes

$$M_R^{(1)}(t, T) = \frac{(r-q+\sigma\lambda)(T-t)}{\sqrt{T-t}\sigma} \approx \frac{C_{P,0}(t, T) - 1}{\sqrt{(T-t)I_0^2(t, T)}}$$

The measure of risk $M_R^{(2)}(t, T)$ becomes

$$M_R^{(2)}(t, T) = \frac{(-q+\sigma\lambda)(T-t)}{\sqrt{T-t}\sigma}$$

In that setting, the upper and lower bounds of the stock price are given by

$$S_{P,T}^{\pm} = S_t e^{(r-q+\sigma\lambda-\frac{1}{2}\sigma^2)(T-t) \pm \alpha\sigma\sqrt{T-t}} = S_t C_{P,0}(t, T) e^{-\frac{1}{2}(T-t)I_0^2(t, T) \pm \alpha\sqrt{T-t}I_0(t, T)}$$

where $\alpha \in [0.5, 2]$ to capture the probability distribution function. We can express these bonds with respect to log-returns as follows:

$$\begin{aligned}
 \ln \frac{S_{P,T}^{\pm}}{S_t} &= (r-q+\sigma\lambda-\frac{1}{2}\sigma^2)(T-t) \pm \alpha\sigma\sqrt{T-t} \\
 &= (T-t)H_{P,0}(t, T) - \frac{1}{2}(T-t)I_0^2(t, T) \pm \alpha\sqrt{T-t}I_0(t, T)
 \end{aligned}$$

We can also express these bonds in terms of returns as follows:

$$\begin{aligned}
 R_{t,T}^+ &= \frac{S_{P,T}^+ - S_t}{S_t} = C_{P,0}(t, T) e^{-\frac{1}{2}\sigma^2(T-t) + \alpha\sigma\sqrt{T-t}} - 1 \\
 R_{t,T}^- &= \frac{S_t - S_{P,T}^-}{S_t} = C_{P,0}(t, T) e^{-\frac{1}{2}\sigma^2(T-t) - \alpha\sigma\sqrt{T-t}} - 1
 \end{aligned}$$

We set the distance as

$$\begin{aligned}
 D_{t,T} = S_{P,T}^+ - S_{P,T}^- &= C_{P,0}(t, T) e^{-\frac{1}{2}\sigma^2(T-t)} (e^{+\alpha\sigma\sqrt{T-t}} - e^{-\alpha\sigma\sqrt{T-t}}) \\
 &\approx C_{P,0}(t, T) 2\alpha\sigma\sqrt{T-t}
 \end{aligned}$$

where we have ignored the term $e^{-\frac{1}{2}\sigma^2(T-t)}$. We define the stop-loss as follows:

$$SL^{\pm} = S_{P,T}^{\pm} \mp \epsilon D_{t,T} \quad (6.37)$$

where SL^+ is the stop-loss when we are short and SL^- is the stop-loss when we are long.

6.3 Risk/reward analysis

We apply the Option Theory to the risk/reward analysis presented in Section (2.4). First, we use option prices to identify the entry and direction of a trading strategy. Then, we use option prices to estimate the reward/risk target $S_{t,T}^\pm$, the expected reward/risk $\Delta_{t,T}^\pm$, and the expected return/risk $R_{t,T}^\pm$.

6.3.1 Estimating the entry and direction

We consider the dynamics of the spot price, under the historical measure, given in Equation (6.26). The forward price $F(t, T)$ for the time-to-maturity $\tau = T - t$ is given in Equation (6.30). For simplicity, we denote the MPR $\Lambda^{\Psi^\lambda}(t, T)\Psi(t, T)$ as $\Lambda(t, T)$. Hence, the forward price becomes

$$F(t, T) = S_t C_{P,0}(t, T), \quad C_{P,0}(t, T) = \frac{D(t, T)}{P(t, T)} \Lambda(t, T)$$

where $C_{P,0}(t, T)$ is the normalised forward price.

Given the forward price and the put-call parity in Equation (15.115), setting $K^* = S_t \frac{D(t, T)}{P(t, T)}$, we get

$$\overline{C}(t, S_t; K^*, T) - \overline{P}(t, S_t; K^*, T) = K^*(\Lambda(t, T) - 1)$$

so that the modified MPR is

$$\Lambda^*(t, T) = \Lambda(t, T) - 1 = \frac{\overline{C}(t, S_t; K^*, T) - \overline{P}(t, S_t; K^*, T)}{K^*}$$

Note, for τ small, or for $(T - t)H^\lambda(t, T) < 1$, we get

$$\Lambda^*(t, T) \approx E\left[\int_t^T \sigma_s \lambda_s ds\right] = (T - t)H_{P,0}^\lambda(t, T) \approx (T - t)y^\lambda(t, T)$$

That is, the modified MPR corresponds to the yield for the market price of risk. Thus, we choose to enter the trade if $|\Lambda^*(t, T)| > \epsilon_L$, where $\epsilon_L > 0$ is a positive constant. Further, we go long if $\Lambda^*(t, T) > 0$ and short if $\Lambda^*(t, T) < 0$.

Alternatively, if we cannot directly estimate $D(t, T)$ and $P(t, T)$, we set $K^* = S_t$ and get

$$\overline{C}(t, S_t; K^*, T) - \overline{P}(t, S_t; K^*, T) = K^*(C_{P,0}(t, T) - 1)$$

so that the modified normalised forward price is

$$\begin{aligned} C_{P,0}^*(t, T) = C_{P,0}(t, T) - 1 &= \frac{F(t, T)}{S_t} - 1 \\ &= \frac{\overline{C}(t, S_t; K^*, T) - \overline{P}(t, S_t; K^*, T)}{K^*} \end{aligned} \quad (6.38)$$

Note, for τ small, or for $(T - t)H^r(t, T) + (T - t)H^\lambda(t, T) < 1$, we get (see Equation (6.28))

$$C_{P,0}^*(t, T) \approx E\left[\int_t^T (r_s + \sigma_s \lambda_s) ds\right] = (T - t)H_{P,0}(t, T) \approx (T - t)y_P(t, T; S_t)$$

That is, the modified normalised forward price corresponds to the yield curve of the stock price. The trading rules are as follows:

- Entry: We choose to enter the trade if $|C_{P,0}^*(t, T)| > \epsilon_L$, where $\epsilon_L > 0$ is a positive constant.
- Direction: We go long if $C_{P,0}^*(t, T) > 0$ and short if $C_{P,0}^*(t, T) < 0$.

6.3.2 Estimating risk/reward targets

We consider the risk/reward analysis presented in Section (2.4) and want to estimate the risk/reward targets $S_T^\pm$. We described in Section (2.4.4) how to use the technical indicators (TIs) for marking the risk/reward targets by finding the next resistance level likely to come into play within the time constraints of the holding period. However, TIs do not represent expectation about future events. Thus, an alternative solution is to use the Option Theory.

6.3.2.1 The mean-variance approach First, we assume that the stock price is normally distributed. Then, the expected stock price (the forward price) and its variance seen at time t for the maturity T are

$$F(t, T) = E[S_T | \mathcal{F}_t] \text{ and } V(t, T) = \text{Var}(S_T | \mathcal{F}_t) = E[S_T^2 | \mathcal{F}_t] - F^2(t, T)$$

We let $SD(t, T) = \sqrt{V(t, T)}$ be the standard deviation of the stock price. Hence, we can estimate the risk and reward targets as follows:

$$S_{t,T}^+ = F(t, T) + \alpha SD(t, T) \text{ and } S_{t,T}^- = F(t, T) - \alpha SD(t, T)$$

for some positive constant α . Then, the distance between the risk and reward targets becomes

$$\Delta^\pm = S_{t,T}^+ - S_{t,T}^- = 2\alpha SD(t, T)$$

This approach is similar to the Bollinger bands (BB) (see Equations (13.100) and (13.102)) except that it is looking forward when the BB is looking backward.

6.3.2.2 The straddle strategy A better solution is to estimate the reward $\Delta_{t,T}^+$ and the risk $\Delta_{t,T}^-$ given in Equation (2.1) with ATM call and put options as it is done in Appendix (15.6.1). That is, given the strike $K^* = S_t$, we get

$$\Delta_{t,T}^+ = C(t, S_t; T, K^*; \sigma_{BS}) \text{ and } \Delta_{t,T}^- = P(t, S_t; T, K^*; \sigma_{BS})$$

where $C(t, S_t; T, K^*; \sigma_{BS})$ and $P(t, S_t; T, K^*; \sigma_{BS})$ are the ATM call and put option prices, respectively, with volatility σ_{BS} . As a result, the target reward and target risk become

$$S_{t,T}^+ = S_t + C(t, S_t; T, K^*; \sigma_{BS}) \text{ and } S_{t,T}^- = S_t - P(t, S_t; T, K^*; \sigma_{BS}) \quad (6.39)$$

The distance between the target risk and target reward in Equation (2.2) corresponds to the straddle strategy presented in Appendix (15.6.1). That is,

$$D_{t,T}^\pm = S_{t,T}^+ - S_{t,T}^- = C(t, S_t; T, K^*; \sigma_{BS}) + P(t, S_t; T, K^*; \sigma_{BS})$$

which is Equation (15.130) with $L = 0$.

6.3.2.3 Link to digital prices Note, we can now use the theory of digital prices (see Section (15.1.4)) and rewrite the risk/reward targets in terms of Digital Bound and Digital Stock. From Equation (15.119), the reward target becomes

$$\begin{aligned} S_T^+ &= S(S_t, t, T; \xi_C^*) - K^* D(S_t, t, T; \xi_C^*) + S_t \\ &= S(S_t, t, T; \xi_C^*) - K^* (1 - D(S_t, t, T; \xi_P^*)) + S_t \\ &= S(S_t, t, T; \xi_C^*) + K^* D(S_t, t, T; \xi_P^*) \end{aligned}$$

where $\xi_C^* = \{S_T > K^*\}$ and $\xi_P^* = \{K^* > S_T\}$. Similarly, from Equation (15.120), the risk target becomes

$$\begin{aligned}
S_T^- &= S_t - K^* D(S_t, t, T; \xi_P^*) + S(S_t, t, T; \xi_P^*) \\
&= S_t - K^* (1 - D(S_t, t, T; \xi_C^*)) + S(S_t, t, T; \xi_P^*) \\
&= K^* D(S_t, t, T; \xi_C^*) + S(S_t, t, T; \xi_P^*)
\end{aligned}$$

When computing the reward target S_T^+ , we can safely assume that $P(\xi_P^*)$ is null. Similarly, when computing the risk target S_T^- , we can safely assume that $P(\xi_C^*)$ is null. Thus, the risk and reward targets become

$$S_{t,T}^+ = S(S_t, t, T; \xi_C^*) \text{ and } S_{t,T}^- = S(S_t, t, T; \xi_P^*)$$

which are expressed in terms of Digital Stock.

In conclusion, given the ATM Digital Stock with strike K^* , we get an idea of the distribution of the stock price around the strike K^* and for the maturity T .

6.3.2.4 The strangle strategy An alternative solution is to estimate the reward $\Delta_{t,T}^+$ and the risk $\Delta_{t,T}^-$ defined in Equation (2.1) with a linear combination of call and put options around the ATM strike K^* . We follow the strategy presented in Appendix (15.6.1) and assume the existence of a strike chain $\mathcal{K} = \{K_1, \dots, K_n\}$ uniformly spaced, where the change in strike is denoted ΔK . Given the distances $d_j = |S_t - K_j|$, for all j , choose d^* such that the distance is minimum. Denote the resulting strike K_i . Thus, $K_i \approx S_t$. If $K_i = S_t$, then we get the same results as above. Given $K_{i \pm L} \in \mathcal{K}$, where L is the total number of options considered, we get

$$\Delta_{t,T}^+ = \sum_{l=0}^L \omega_l C(t, S_t; T, K_{i+l}; \sigma_{BS,i+l}) \text{ and } \Delta_{t,T}^- = \sum_{l=0}^L \omega_l P(t, S_t; T, K_{i+l}; \sigma_{BS,i+l})$$

where $C(t, S_t; T, K_i; \sigma_{BS,i})$ and $P(t, S_t; T, K_i; \sigma_{BS,i})$ are call and put options with strike K_i and volatility $\sigma_{BS,i}$. As a result, the risk and reward targets become

$$S_{t,T}^+ = \sum_{l=0}^L \omega_l C(t, S_t; T, K_{i+l}; \sigma_{BS,i+l}) + S_t \text{ and } S_{t,T}^- = S_t - \sum_{l=0}^L \omega_l P(t, S_t; T, K_{i+l}; \sigma_{BS,i+l})$$

The distance between the target risk and the target reward given in Equation (2.2) corresponds to the straddle strategy presented in Appendix (15.6.1). That is,

$$D_{t,T}^\pm = S_T^+ - S_T^- = \sum_{l=0}^L \omega_l C(t, S_t; T, K_{i+l}; \sigma_{BS,i+l}) + \sum_{l=0}^L \omega_l P(t, S_t; T, K_{i+l}; \sigma_{BS,i+l})$$

which is Equation (15.130).

6.4 Defining the trading rules

6.4.1 Pricing options

In order to apply Option Theory to define the trading rules for a particular strategy, we need to compute option prices on the underlying asset in the historical measure $\mathbb{P}$. There are several ways for computing option prices on real market data. One solution is to assume that the stock price follows a Geometric Brownian motion (GBM) and use the BS-formula (see Equation (15.113)) to price European options. From the symmetry of Brownian motions, the GBM model is also symmetric. However, when adding the market price of risk (MPR), stochastic volatility and jumps to this process, this is no-longer the case. The market distribution is asymmetric and we need models to capture this behaviour. Another solution is to use statistics to compute the sample mean and variance and use Monte Carlo simulation to calculate the option prices (see Section (8.2.1)). A better solution is to use Machine Learning to directly

compute the conditional expectation from the market. To make sure that the ML modes are accurate at pricing options on the market we need to test these models on

- well known continuous stochastic processes: Black-Scholes Model, Merton Model, Heston Model, Bates Model
- well known payoffs: European options, barrier options, digital options, density (butterfly)

We can then use these models on real market data to compute options in the historical measure.

6.4.2 Strategies with options

6.4.2.1 Overview We saw in Section (6.3) that we could develop quantitative strategies using option payoffs such as European options, barrier options, digital options. For example, we can compute option metrics to devise strategies and use results to improve PnL. We can compute option indicators:

- Straddle: Determine the magnitude of the move
- Barriers: Determine the direction of the move
- Density: Probabilities to the expected move

We can then use these option indicators to devise simple strategies. For example,

- Buy stock when probability of hitting an up barrier is higher than $X\%$
- Sell stock when probability of hitting a lower barrier is higher than $X\%$
- Buy/sell stock when expected return is higher than $X\%$

Practically, we use the risk/reward targets for trading as follows:

- Use Call/Put options to estimate the Risk/Reward targets
- Compute the Distance between R/R targets (Straddle strategy)
- Compute the Risk/Reward ratio
- Use Call/Put options to estimate the market distribution
- Use barrier options to estimate the Speed of reaching a target level
- Compute the Greeks to get sensitivities

We can then implement a profit protection strategy as follows:

- Enter trade if R/R ratio greater than 1 : 3
- Use the Distance and Speed to determine stop-loss strategy
- Use Distance and Speed to determine optimum portfolio allocation
- In a trend market: place a trailing stop to protect gains
- Use the Speed to determine optimum exit point
- Repeat process to identify new entry/exit points

6.4.2.2 A simple strategy For simplicity, we consider day trading. We use the modified normalised forward price given in Equation (6.38) to define the entry point and the trade direction.

- Entry: We choose to enter the trade if $|C_{P,0}^*(t, T)| > \epsilon_L$, where $\epsilon_L > 0$ is a positive constant.
- Direction: We go long if $C_{P,0}^*(t, T) > 0$ and short if $C_{P,0}^*(t, T) < 0$.

Then, we follow the straddle strategy, where the target reward and target risk are computed from Equation (6.39). Assuming a long position, we set the profit taking to the reward target and the local stop loss (LSL) to the risk target. Every time the stock price hit the LSL we exit the position. We can enter and exit a position several times during the day. This is illustrated in Figure (63) in the case of a GBM with stochastic MPR (see Section (9.4)).



Figure 63: Dynamic Stop Loss PnL.

The blue curve is the stock price over 10 days. The red curve corresponds to the local stop loss and the green curve corresponds to the profit taking. The blue dot is an entry point and the red cross corresponds to an exit point either because it is the end of the day or because the stock price hit the LSL.

7 Optimising strategies

7.1 Mathematical formulation

7.1.1 Formulating the trading problem

Let $(X_t)_{t \geq 0}$ be the logarithm of stock process in a given market. Since other market participants transact that security, we let its mark-to-market (MtM) at time t be denoted P_t . Given the set of opportunities $\mathcal{I} = (1, \dots, I)$ over a fixed

trading period T (a discrete sequence of individual trades), consider an investment strategy S ¹⁴ that invests in I opportunities¹⁵. For each opportunity $i \in \mathcal{I}$, the strategy S takes a position of $w_i \in (-\infty, \infty)$ units of the log-security. The transaction entered by such an opportunity is priced at a value $w_i P_{i,0}$, where $P_{i,0}$ is the average price per unit at which the w_i securities were transacted. Here, $P_{i,0}$ is the entry level of opportunity i . The mark-to-market (MtM) value of that opportunity i after t observed transactions is $w_i P_{i,t}$, which represents the value of opportunity i if it was liquidated at the price observed in the market after t transactions. Hence, the MtM profit/loss (PnL) of opportunity i after t transactions is

$$\pi_{i,t} = \pm w_i (P_{i,t} - P_{i,0} \mp c) \quad (7.40)$$

where c is a transaction cost per unit. Here, $+$ represents a long position and $-$ represents a short position. For simplicity we will only consider long position. The trading rules provide the logic for entering and exiting opportunity i at time $t_0 = 0$ and $t = T_i$, $i \in \mathcal{I}$, respectively. They should also include the thresholds defining taking profit and stop-loss orders. In that case, one of the following two conditions is verified:

1. $\pi_{i,T_i} \geq \bar{\pi}_i$, where $\bar{\pi}_i$ is the profit-taking threshold for opportunity i .
2. $\pi_{i,T_i} \leq \underline{\pi}_i$, where $\underline{\pi}_i$ is the stop-loss threshold for opportunity i .

Note, $(\bar{\pi}_i, \underline{\pi}_i)$ can themselves be obtained with some parametric functions. In a long position we have $\underline{\pi}_i < \bar{\pi}_i$, leading to the existence of a single exit condition. As an example, in the approach described in Section (7.4.1), the (periodic) trading strategy is defined as the act of entering a trade when $X_t = a$ and exiting the trade at $X_t = m$, where a and m are fixed. We then need to wait a random time for the process to get back to level a before initiating a new trade. In that setting, $P_{i,0} = a$ and $P_{i,T_i} = m$, where T_i is a first-hitting time of an OU process. Here, P_{i,T_i} is a random variable observed at time T_i for opportunity i .

Given an existing trade ($P_{i,0}$ is known) and, assuming that opportunity i can be exited at T_i , its final PnL is π_{i,T_i} . The objective of the trading problem is to realise the expected profit

$$E_0[\pi_{i,T_i}] = w_i (E_0[P_{i,T_i}] - P_{i,0} - c), \forall i \in \mathcal{I} \quad (7.41)$$

where $E_0[P_{i,T_i}]$ is the expected price of opportunity i seen at time $t = 0$ for the maturity T_i . If $E_0[\pi_{i,T_i}] > 0$, we are making an expected profit, else we are making an expected loss. This is the alpha calculated from the historical price data. It corresponds to the level targeted by the trader for opportunity i . Note, the value $E_0[P_{i,T_i}]$ is called the forward log-price, which is traded in the market.

Quantitative trading is about estimating the expected profit $E_0[\pi_{i,T_i}]$ to identify trading opportunities and mapping these opportunities to desired positions. We can use (1) data-driven methods or (2) process-driven methods to estimate the expected PnL. In case (1), we use chart analysis to infer information on the direction of $E_0[P_{i,T_i}] - P_{i,0}$ from past data. In case (2), we model the logarithm of the underlying security $(X_t)_{t \geq 0}$ with a stochastic process and use it to compute $E_0[P_{i,T_i}]$. For example, if we let the stock price be a $GBM(\mu, \sigma)$, we have $X_{i,T_i} = X_{i,0} + \mu T_i - \frac{1}{2} \sigma^2 T_i + \sigma W_{T_i}$, and the forward log-price is given by $E_0[P_{i,T_i}] = P_{i,0} + \mu T_i - \frac{1}{2} \sigma^2 T_i$. Hence, the expected profit seen at $t = 0$ for the maturity T_i becomes

$$E_0[\pi_{i,T_i}] = w_i \left[\left(\mu - \frac{1}{2} \sigma^2 \right) T_i - c \right]$$

It leads to the well know mean-variance quantity $\mu - \frac{1}{2} \sigma^2 > 0$ (see Equation (6.34)).

Note, we can modify the objective to guarantee a profit and no-loss. Set $K = P_{i,0} + c$ and compute the European call option

$$E_0[\pi_{i,T_i}^+] = w_i E_0[(P_{i,T_i} - K)^+], \forall i \in \mathcal{I}$$

¹⁴An investment is an asset acquired with the goal of generating income or appreciation. An investment strategy is a set of principles that guide investment decisions.

¹⁵An investment opportunity is any situation where you have the option of purchasing something that has a chance to gain value in the future.

7.1.2 Optimum timing problems

One of the main goal of quantitative trading is to develop highly automated trading strategies that map an alpha to a desired position via trading rules. We saw in Section (2.3) that these trading rules provide a set of decisions to enter and exit a position based on conditions defined through reaching values and thresholds (possibly for some parametric functions) for profit-taking and stop-losses. In addition, these strategies can also rely on mathematical models to be successful. These models are parametric and must be calibrated along with the trading rules to maximise profit. Hence, it leads to two trading problems:

1. Optimum entry and exit timing: Find the optimum timing strategy to enter a trade and then liquidate the position. This is related to determining the optimal trading rule (OTR).
2. Optimally exiting an existing position: Given an existing trade, find the best timing strategy to liquidate a position.

In all cases, we must determine the best entry and exit thresholds that maximise future returns for a particular underlying instrument. This gave rise to the interesting and challenging problem of finding the optimum timing strategy to enter a position in the first place and then liquidate that position. Mathematically, we can recognise these problems as stochastic timing problems with constraints in the sense that it installs a path-dependent random time (as opposed to buying and selling at a fixed time) into the entry and liquidation problem, rendering the problem significantly more difficult to analyse and solve. These optimum timing problems can be solved with

1. Data-driven methods: Compute averages along the realised trajectories and calibrate the model parameters via historical simulations in a backtesting engine.
2. Process-driven methods: Assuming a stochastic process, compute expectations and first hitting time.

In the problem (1), one approach is to use chart analysis (CA) and technical indicators (TIs) to time the buying and selling of a stock to capitalise on any breakout or trend reversals. They identify the entry and exit timing with the help of moving averages reaching values and thresholds, or crossing each others (see details in Section (3)). These model parameters are calibrated via historical simulations in a backtesting engine. However, this practice leads to the problem of backtest overfitting since these parameters target specific observations in-sample. It leads to investment strategies that are so attached to the past that they become unfit for the future. Bailey et al. [2013] [2014] discussed the problem of backtest overfitting, and provided methods to determine to what extent a simulated performance may be inflated due the overfitting (see Section (7.2)).

Another approach to the problem (1) is to construct a stationary, mean reverting synthetic asset as a linear combination of traded securities. One example is the method of pairs trading (see Vidyamurthy [2004]) which allows for the construction of a trading strategy where trades are entered when the process reaches an extreme value and exited when the process reverts to some equilibrium value. Bertram [2009] provided a mathematical framework for the construction and analysis of statistical arbitrage methods in the case of continuous time trading strategies. He showed that when considering statistical based strategies, the time between trades (trading frequency) becomes an important quantity to consider. In such an environment there is a need for analytic and approximate solutions. One solution is to assume that the traded security is described by an Ornstein-Uhlenbeck process. The resulting analysis provides a mathematical model for calculating optimal statistical arbitrage strategies. As a result, we can construct a continuous trading strategy and express the trade length and return of the strategy in terms of the first-passage time of the process. Optimal trading strategies can be found by constructing objective functions that are expressed in terms of the expected value and variance of the return. One can then derive pseudo-analytic solutions for the strategy maximising expected return. In general, the solution satisfies a real valued integral equation.

In continuous trading, another solution to avoid backtest overfitting consists in modelling the underlying variable by generating pseudorandom scenarios and measuring the performance of the resulting investment strategy for those

scenarios. It relies on generating a distribution of outcomes, rather than relying on a single OOS performance estimate. Assuming security price follows an Ornstein-Uhlenbeck process, Bertram [2009] and Zeng et al. [2014] examined the problem of choosing an optimal strategy by expressing it in terms of the first-passage time of the process. Carr et al. [2014] also let the security price follows an OU-process and proposed a procedure for determining the optimal trading rule (OTR) without running alternative model configurations through a backtest engine.

In the problem (2), trailing stops are popular trade orders widely used by proprietary traders and retail investors to provide downside protection (see Section (2.5)). In contrast to a stop-loss exit, which closes a position at a fixed price, a trailing stop is characterised by a stochastic floor that moves based on the running maximum of the asset price. This provides a dynamic downside protection as the stochastic floor is automatically adjusted upward whenever the asset price moves to a new high. Practically, a trailing stop is triggered when the prevailing price of an asset falls below the stochastic floor. The asset price at the trailing stop may be higher than its starting price. Hence, it allows an investor to specify a limit on the maximum possible loss without limiting the maximum possible gain. This is particularly relevant in common trend-following strategies, and trailing stop provides an automatic trigger to exit when prices start to trend downward due to regime switching (or for other reasons). In addition to setting a trailing stop order, the investor can also use a limit order to sell at a certain price target. If the price is sufficiently high, the investor may prefer to take profit immediately, rather than waiting further with the possibility of setting off the trailing stop. As a result, the investor's position will be liquidated by either order, whichever comes first.

Yin et al. [2010] developed stochastic approximation algorithms to estimate the optimal trailing stop percentage. They proposed a stochastic optimisation approach to recursively estimate the desired trailing stop percentage. They developed a modification using projection to ensure that the approximation sequence constructed stays in a reasonable range. Convergence of the algorithm was obtained and interval estimates were constructed. Simulation examples were presented to compare the algorithm with Monte Carlo methods. They used real market data to demonstrate the algorithms. Leung et al. [2019], Zhang [2019] showed that it is optimal to use a limit sell order at a sufficiently high price. That is, once the investor enters the market, he/she can immediately set the optimal limit sell order together with the trailing stop order, and wait for either one to be automatically executed. Incorporating trailing stops may lead to nontrivial and possibly quite consequential effects to a trading system.

7.2 Backtesting strategies

7.2.1 Overview

Given an underlying security X and a fixed trading period T , an investment strategy S invests in I opportunities (a discrete sequence of individual trades), where for each opportunity, $i \in \mathcal{I}$, a position is taken in the underlying security. The trading rules provide the logic for entering and exiting each opportunity based on conditions defined through reaching values and thresholds for profit-taking and stop-losses. We can then compute the profit and loss (PnL) for every position and use a particular metric to compute the strategy performance over that period of time.

Backtests are defined as simulations of performance based on historical market data. They allow practitioners to evaluate the risk/reward profile of an investment strategy before committing funds. Financial models to detect patterns are parametric and add up to the parameters from the trading rules that need to be estimated. The number of combinations to be tested over all underlyings is very large (to the billions). For each of those backtests, we can compute a performance metric (such as the Sharpe ratio) to identify the best combination of parameters. In general, practitioners estimate the parameters of an investment strategy by maximising its performance. We say that the parameters are calibrated via historical simulations (backtesting). However, since we are applying the test on the same data multiple times, this approach is prone to false positives, leading to biased conclusion. In fact, the probability of finding false positives increases with the number of tests conducted on the same data (see Miller [1981]). In the case of low signal-to-noise ratio (like financial data), a test of hypothesis must be conducted on a large sample as a way of assessing the existence of a phenomenon. Otherwise, the parameters estimated via calibration are chosen to profit from past noise rather than future signal.

In general, this approach leads to the problem of backtest overfitting, where the parameters target specific observations and become unfit for the future. Hence, practitioners prevent overfitting by withholding a portion of the available data sample for separate testing and validation as out-of-sample (OOS) performance. If the IS and OOS performance levels are congruent, the investor might decide to reject the hypothesis that the backtest is overfit. In finance, the IS dataset is usually too short to fit, and the OOS dataset too short to conclude anything with sufficient confidence. One need at least 1,000 observations in the hold-out sample (see Weiss et al. [1990]). More importantly, as long as the researcher tries more than one strategy configuration, overfitting is always present.

Bailey et al. [2013] developed computational techniques to control for the increased probability of false positives as the number of trials increases. They defined overfitting as occurring when the strategy configuration that delivers maximum performance in-sample (IS) systematically underperform the remaining configurations out-of-sample (OOS). It can be modelled by using an abstract probability space in which the sample space consists of pairs of IS and OOS test results. For a given strategy, the probability of backtest overfitting (PBO) (even though backtest overfitting is a deterministic event) is evaluated as the conditional probability that this strategy underperforms the median OOS while remaining optimal IS. The authors proposed combinatorially symmetric cross-validation (CSCV) as a way of producing estimates of PBO for several useful examples.

Another approach consists in modelling the underlying variable by generating pseudorandom scenarios and measuring the performance of the resulting investment strategy for those scenarios (see Carr et al. [2014]). It relies on generating a distribution of outcomes, rather than relying on a single OOS performance estimate. However, the model generating random series of the underlying variable may also be overfit, or may not contain all relevant statistical features.

A backtest is realistic when the IS performance is consistent with the OOS performance. Given any financial series, it is relatively simple to overfit an investment strategy so that it performs well IS. Overfitting denotes the situation when a model targets particular observations rather than a general structure. In general, a pseudo-mathematical argument is spuriously validated through a backtest. We cannot apply ML techniques for solving overfitting to investment problems.

7.2.2 Formulating the optimisation problems

We consider the trading problem presented in Section (7.1.1).

We let $\Phi(\mathcal{I}) = \{(\underline{\pi}_i, \bar{\pi}_i)\}$ be a set of parameters (trading rule or model configuration) associated to the opportunity set $\mathcal{I}$, $\mathcal{P}(\Phi)$ be the powerset, and $\Omega = \{\Phi(\mathcal{I})\}$ be the set of alternative values of $\Phi(\mathcal{I})$. For simplicity, we will denote Φ when the reference to the opportunity set $\mathcal{I}$ is obvious. Further, in the case of several set of parameters, we let $[\Phi]^k$ be the k th set of parameters.

Carr et al. [2014] defined a trading rule as follows:

Definition 7.1 *Trading rule*

A trading rule for strategy S is defined by the set of parameters $\Phi(\mathcal{I}) = \{(\underline{\pi}_i, \bar{\pi}_i)\}$, $i \in \mathcal{I}$.

7.2.2.1 Investment opportunities

We can use the following steps to calibrate the trading rule:

1. Define a set of alternative values of $\Phi(\mathcal{I})$, $\Omega = \{\Phi(\mathcal{I})\}$.
2. Simulate historically (also called backtest) the performance of strategy S under alternative values of $\Phi(\mathcal{I}) \in \Omega$. That is, for $[\Phi]^k$, $k = 1, \dots, \mathcal{K}$.
3. Select the optimal set $\Phi^*(\mathcal{I})$. That is,

$$\Phi^*(\mathcal{I}) = \arg \max_{\Phi(\mathcal{I}) \in \Omega} \Psi(P_{\mathcal{I}}; \Phi(\mathcal{I})) \quad (7.42)$$

where $\Psi(P_{\mathcal{I}}; \cdot)$ is a particular metric which depends on the prices $P_{\mathcal{I}} = \{P_{1,T_1}, \dots, P_{I,T_I}\}$.

As an example of a metric, we can consider the sample Sharpe Ratio given by

$$\hat{\Psi}_{SR} = \frac{\hat{E}[\pi_{\mathcal{I}} | \Phi(\mathcal{I})]}{\hat{\sigma}[\pi_{\mathcal{I}} | \Phi(\mathcal{I})]}, \pi_{\mathcal{I}} = \{\pi_{1,T_1}, \dots, \pi_{I,T_I}\} \quad (7.43)$$

where $\hat{E}[\cdot]$ and $\hat{\sigma}[\cdot]$ are respectively the sample expected value and sample standard deviation of $\pi_{\mathcal{I}}$, conditional on trading rule $\Phi(\mathcal{I})$. Thus, the above equation in Step [3] maximise the Sharpe Ratio of strategy S on I opportunities over the space of alternative trading rules Ω . Since we count with $2I$ variables to maximise Ψ_{SR} over a sample of size I , it is easy to overfit $\Phi(\mathcal{I})$. A trivial overfit occurs when each pair $(\pi_i, \bar{\pi}_i)$ targets the specific opportunity i (along the realised trajectory).

7.2.2.2 Generating pseudorandom scenarios We consider the case where we model the underlying variable by generating pseudorandom scenarios and measure the performance of the resulting investment strategy for those scenarios. We let the prices $(P_{i,t})_{t \geq 0}$ follow a stochastic process and simulate synthetic market returns under different scenarios. We focus on opportunity i with maturity T_i and we let, p , be the distribution of the price P_{i,T_i} . In that setting, we can compute the expected profit $\bar{\pi}_{i,T_i} = E_0[\pi_{i,T_i}]$. We can use the following steps to calibrate the trading rule:

1. Define a set of alternative values of Φ_i , $\Omega_i = \{\Phi_i\}$.
2. Simulate the performance of strategy S under alternative values of $\Phi_i \in \Omega_i$.
3. Select the optimal set Φ_i^* . That is,

$$\Phi_i^* = \arg \max_{\Phi_i \in \Omega_i} \Psi(P_{i,T_i}; \Phi_i)$$

where $\Psi(P_{i,T_i}; \cdot)$ is a particular metric which depends on the prices $P_{i,T_i} \sim p$.

4. Repeat steps [1] to [3] and obtain the distribution $\Phi_{\mathcal{I}}^* = \{\Phi_1^*, \dots, \Phi_I^*\}$.

Then, the Sharpe Ratio is given by

$$\Psi_{SR,i} = \frac{E_{P_{i,T_i} \sim p}[\pi_{i,T_i} | \Phi_i]}{\sigma_{P_{i,T_i} \sim p}[\pi_{i,T_i} | \Phi_i]} \quad (7.44)$$

where $E_{P_{i,T_i} \sim p}[\cdot]$ and $\sigma_{P_{i,T_i} \sim p}[\cdot]$ are respectively the expected value and standard deviation of π_{i,T_i} , conditional on trading rule Φ_i . Thus, the above equation in Step [3] maximise the Sharpe Ratio of strategy S on the i th opportunity over the space of alternative trading rules Ω_i . We obtain a distribution of Sharpe ratio for the set $\mathcal{I}$.

7.3 Probability of backtest overfitting

We saw in Section (7.1.1) that, given an underlying security X and a fixed trading period T , an investment strategy S invests in I opportunities (a discrete sequence of individual trades), where for each opportunity $i \in \mathcal{I}$ a position is taken in the underlying security. The trading rule provides the logic for entering and exiting each opportunity. We can then compute the profit and loss (PnL) for every position and use a particular metric to compute the strategy performance over that period of time. Further, we can test a large number of combinations in a backtesting engine and perform an optimisation to maximise profit (see Equation (7.42)). However, this approach leads to backtest overfitting. The solution proposed by Bailey et al. [2013] to avoid backtest overfitting only requires time series of backtested performance (no calibration of parameters). Due to the difficulty of creating a truly out-of-sample (OOS) dataset, and since it doesn't allow us to quantify a backtest overfit (it depends on the number of trials involved), the authors don't rely on preserving an OOS test-set, and they don't require knowledge of either the trading rule or forecasting equation. Instead, they compute the probability of backtest overfitting and obtain a distribution of OOS performance measure.

7.3.1 Description

The authors proposed the following backtest strategy selection process:

Using a given performance measure Ψ (such as the Sharpe ratio), select the best strategy ¹⁶ (seen as trading rules or model configurations) from the set $\mathcal{S} = (S_1, \dots, S_N)$ labelled with counting set $\mathcal{E} = (1, \dots, N)$. Use the random vectors $R = (R_1, \dots, R_N)$ and $\bar{R} = (\bar{R}_1, \dots, \bar{R}_N)$ to represent the IS and OOS performance (over a period of time) of the N strategies, respectively. Let $\mathcal{C}$ represents a sample space of pairs of IS and OOS samples. For a given sample $c \in \mathcal{C}$ (pair of IS and OOS samples), let R^c and $\bar{R}^c$ represent the performances of the N strategies on the IS and OOS pair given by c . In order to compare the ranking of the selected strategies IS and OOS, consider the ranking space Ω consisting of the $N!$ permutations of $\mathcal{E}$ indicating the ranking of the N strategies. Use the random vectors r and $\bar{r}$ to represent the ranking of the components of R and $\bar{R}$, respectively. For instance, for $N = 3$, considering the Sharpe ratio as a metric and given a particular sample $c \in \mathcal{C}$, we could have the vectors $R^c = (0.5, 1.1, 0.7)$ and $\bar{R}^c = (0.6, 0.7, 1.3)$, leading to $r^c = (1, 3, 2)$ and $\bar{r}^c = (1, 2, 3)$. We can deduce that the best performing strategy IS is $n = 2$ and the best performing strategy OOS is $n = 3$.

The null hypothesis assumes that backtest overfitting has taken place. Given some empirical evidence and priors, the authors infer the posterior probability that overfitting has taken place. In order to define backtest overfitting, the authors defined the following subset of Ω ,

$$\Omega_n^* = \{f \in \Omega \mid f_n = N\} \quad (7.45)$$

Note, this is an arbitrary partitioning of the ranking space Ω . The subset $\Omega_{n^*}^*$ is such that n^* is the best performing strategy IS. Following on the above example, we have

$$\Omega = \{r(1, 2, 3), r(1, 3, 2), r(2, 1, 3), r(2, 3, 1), r(3, 1, 2), r(3, 2, 1)\}$$

and from Equation (7.45), we have

$$\Omega_1^* = \{r(3, 1, 2), r(3, 2, 1)\}; \Omega_2^* = \{r(1, 3, 2), r(2, 3, 1)\}; \Omega_3^* = \{r(1, 2, 3), r(2, 1, 3)\}$$

For instance, given $r(2, 3, 1)$, we can deduce that $n^* = 2$ is the best performing strategy, and that this vector belongs to Ω_2^* .

Definition 7.2 Backtest Overfitting

We say that the backtest strategy selection process overfits if a strategy with optimal performance IS has an expected ranking below the median OOS. By the Bayesian formula and using the notation above that is

$$\sum_{n=1}^N E[\bar{r}_n \mid r \in \Omega_n^*] P(r \in \Omega_n^*) \leq \frac{N}{2}$$

Remark 7.1 It is not clear what an expected ranking means, given that the ranking set Ω is made of integers.

Note, $r \in \Omega_n^*$ means that $r_n = N$, that is, the n th strategy is always the best. Hence, we can rewrite the probability $P(r \in \Omega_n^*)$ as $P(r_n = N)$, and the above formula becomes

$$\sum_{n=1}^N E[\bar{r}_n \mid r_n = N] P(r_n = N) \leq \frac{N}{2}$$

This formula computes the expected value of the performance OOS rank, $\bar{r}_n$, given the optimal performance IS rank r_n (that is n^*), and weights the result by the probability $P(r_n = N)$. We then sum up this result, for all $n \in \mathcal{E}$. It must

¹⁶An investment is an asset acquired with the goal of generating income or appreciation. An investment strategy is a set of principles that guide investment decisions.

be less or equal to $\frac{N}{2}$ to get a backtest overfit.

In our example, we get $P(r_n = 3) = \frac{1}{3}$, for $n = 1, 2, 3$. Hence, we can write the above formula as

$$E[\bar{r}_1 | r_1 = 3]P(r_1 = 3) + E[\bar{r}_2 | r_2 = 3]P(r_2 = 3) + E[\bar{r}_3 | r_3 = 3]P(r_3 = 3)$$

Note, the formula in Definition (7.2) corresponds to part of the marginal mean $E[\bar{r}]$ of the OOS ranking vector ¹⁷, given by

$$\begin{aligned} E[\bar{r}] &= \sum_{m=1}^N \sum_{n=1}^N E[\bar{r}_m | r_n = N]P(r_n = N) \\ &= \sum_{n=1}^N E[\bar{r}_n | r_n = N]P(r_n = N) + \sum_{m=1}^N \sum_{n \neq m}^N E[\bar{r}_m | r_n = N]P(r_n = N) \end{aligned}$$

For the performance OOS rank to match the optimal performance IS rank, we must only consider the elements on the diagonal of the $N \times N$ matrix, where $m = n$. This is what the formula in Definition (7.2) does. In our example, the above formula becomes

$$\begin{aligned} &E[\bar{r}_1 | r_1 = 3]P(r_1 = 3) + E[\bar{r}_1 | r_2 = 3]P(r_2 = 3) + E[\bar{r}_1 | r_3 = 3]P(r_3 = 3) \\ + &E[\bar{r}_2 | r_1 = 3]P(r_1 = 3) + E[\bar{r}_2 | r_2 = 3]P(r_2 = 3) + E[\bar{r}_2 | r_3 = 3]P(r_3 = 3) \\ + &E[\bar{r}_3 | r_1 = 3]P(r_1 = 3) + E[\bar{r}_3 | r_2 = 3]P(r_2 = 3) + E[\bar{r}_3 | r_3 = 3]P(r_3 = 3) \end{aligned}$$

which gives $\sum_{m=1}^3 E[\bar{r}_m]$.

Definition 7.3 *Probability of Backtest Overfitting*

A strategy with optimal performance IS is not necessarily optimal OOS. Moreover, there is a non-null probability that this strategy with optimal performance IS ranks below the median OOS. This is what we define as the probability of backtest overfit (PBO). More precisely,

$$PBO = \sum_{n=1}^N P(\bar{r}_n < \frac{N}{2} | r \in \Omega_n^*)P(r \in \Omega_n^*)$$

which we rewrite as

$$\begin{aligned} PBO &= \sum_{n=1}^N P(\bar{r}_n < \frac{N}{2} | r_n = N)P(r_n = N) \\ &= \sum_{n=1}^N E[I_{\{\bar{r}_n < \frac{N}{2}\}} | r_n = N]P(r_n = N) \end{aligned}$$

¹⁷From the conditional expectation of Y given X , written $E[Y|X]$, we can compute the marginal mean $E[Y]$ as

$$\begin{aligned} E[Y] &= \sum_x E[Y|X = x]P(X = x) \\ &= \sum_x \sum_y y f_{Y|X}(y|x) f_X(x) \end{aligned} \tag{7.46}$$

This formula states that given the optimal performance IS rank r_n (that is n^*), we count one if $\bar{r}_n < \frac{N}{2}$ and weight the result by the probability $P(r_n = N)$. We then sum the result for all $n \in \mathcal{E}$ to get the probability of backtest overfitting. That is, a strategy selection process overfits if the expected performance of the strategies selected IS is less than the median performance rank OOS of all strategies. Here, IS corresponds to the subset of observations used to select the optimal strategy among the N alternatives. The authors refer to overfitting in relation to the strategy selection process, not a strategy's model calibration (such as in the context of regressions).

7.3.2 The CSCV algorithm

Estimating the probability (PBO) in a particular application relies on schemes for selecting samples of IS and OOS pairs. For instance, in a trend-following moving average (MA) strategy, the researcher might try alternative sample lengths on which the moving averages are computed, entry thresholds, exit thresholds, stop losses, holding periods, and sampling frequencies. Hence, we run N alternative model configurations (or trials) over the time set $\mathcal{T} = (t_1, \dots, t_T)$ labelled with the counting set $\mathcal{T} = (1, \dots, T)$ (synchronised), out of which one is chosen according to some performance evaluation criterion, such as the Sharpe ratio. That is, for each pair of model configuration $n \in \mathcal{E}$ and time $t \in \mathcal{T}$, denoted (n, t) , we get the performance $R_{n,t}$ computed from Equation (7.40). Running the N model configurations over each of these combinations allowed the authors to derive a relative ranking, expressed as a logit. To facilitate the evaluation of PBO, they proposed the combinatorially symmetric cross-validation (CSCV) algorithm. This estimate is generic, symmetric, model-free and non-parametric. This algorithm is described below:

1. Form a matrix $M \in \mathbb{R}^{T \times N}$ by collecting the performance series from the N trials over the time set $\mathcal{T}$. Each column $n \in \mathcal{E}$ represents a vector of profit and losses (PnL) over time $t \in \mathcal{T}$ observations associated with a particular model configuration. M must be a true matrix, and the performance evaluation metric can be estimated on subsamples of each column. That is,

$$M = \begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ R_1(T) & R_2(T) & \cdots & R_N(T) \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}$$

where $R_n(T) = (R_{n,1}, \dots, R_{n,T})$, $n \in \mathcal{E}$, are performance vectors of size T .

2. Partition M across rows, into an even number S of disjoint submatrices of equal dimensions, denoted $M_s \in \mathbb{R}^{\frac{T}{S} \times N}$ with $s = 1, \dots, S$. These submatrices correspond to performances over different periods of time. That is,

$$M_s = \begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ R_1(\frac{T}{S}) & R_2(\frac{T}{S}) & \cdots & R_N(\frac{T}{S}) \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}$$

and

$$M = \begin{pmatrix} M_1 \\ M_2 \\ \vdots \\ M_S \end{pmatrix}$$

That is, concatenating (or joining) all the submatrices M_s , for $s = 1, \dots, S$, we recover the matrix M . Further, concatenating the submatrices M_s , for $s = 1, \dots, \frac{S}{2}$, we get a matrix of size $\frac{T}{2} \times N$.

3. Form all combinations C_S of submatrices M_s ¹⁸ taken in groups of size $\frac{S}{2}$. The total number of combinations is given by

¹⁸Using C_r^n as a symbol for the number of combination of r objects taken from n different objects, we can write

$$\binom{S}{S/2} = \binom{S-1}{S/2-1} \frac{S}{S/2} = \cdots = \prod_{i=0}^{\frac{S}{2}-1} \frac{S-i}{\frac{S}{2}-i}$$

That is, $|\{C_S\}| = \binom{S}{S/2}$. Each combination $c \in C_S$ is composed of $\frac{S}{2}$ submatrices M_s . Concatenating (or joining) these submatrices gives a matrix of order $\frac{T}{2} \times N$.

4. For each combination $c \in C_S$

- (a) Form the training set J , by joining the $\frac{S}{2}$ submatrices M_s that constitute c in their original order. J is a matrix of order $\frac{T}{2} \times N$. That is,

$$J = \begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ R_1(\frac{T}{2}) & R_2(\frac{T}{2}) & \cdots & R_N(\frac{T}{2}) \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}$$

- (b) Form the testing set $\bar{J}$ as the complement of J in M . That is, join the remaining $\frac{S}{2}$ submatrices to get the matrix $\bar{J}$ of order $\frac{T}{2} \times N$.
- (c) Form a vector R^c of performance statistics of order N , where the n th component R_n^c reports the performance associated with the n th column of J (over time $\frac{T}{2}$). The vector r^c is the IS ranking of the N strategies.
- (d) Repeat [3] with J replaced by $\bar{J}$ to derive the vectors $\bar{R}^c$ and $\bar{r}^c$.
- (e) Determine the element n^* such that $r_{n^*}^c \in \Omega_{n^*}^*$. That is, n^* is the best performing strategy IS (for a given combination c).
- (f) Define the relative rank of $\bar{r}_{n^*}^c$ by $\bar{\omega}_c = \frac{\bar{r}_{n^*}^c}{N+1} \in (0, 1)$. This is the relative rank of the best performing strategy IS in the OOS. If the strategy optimisation procedure is not overfitting, we should observe that $\bar{r}_{n^*}^c$ systematically outperforms OOS, just as $r_{n^*}^c$ outperformed IS.
- (g) Define the logit $\lambda_c = \ln \frac{\bar{\omega}_c}{1-\bar{\omega}_c}$. High logit values imply a consistency between IS and OOS performances, which indicates a low level of backtest overfitting.

5. Compute the distribution of OOS ranks by collecting all the λ_c , for $c \in C_S$. Define the relative frequency at which λ occurred across all C_S by

$$f(\lambda) = \sum_{c \in C_S} \frac{I_\lambda(\lambda_c)}{|\{C_S\}|}$$

where $I_A(x) = I_{\{x \in A\}}$ is the indicator function, and $|\{C_S\}| = \sharp(C_S)$ is the number of elements in C_S . This is the frequency (PDF) of logits. Then,

$$PBO = \int_{-\infty}^0 f(\lambda) d\lambda$$

which represents the rate at which optimal IS strategies underperform the median of the OOS trials. Note, $\int_{-\infty}^{\infty} f(\lambda) d\lambda = 1$. Hence, when $PBO \gg 0$, it indicates high likelihood of overfitting. The authors proposed to reject models for which PBO was estimated to be greater than 0.05.

$$C_r^n = \frac{n!}{(n-r)!r!}$$

The combination symbol C_r^n is also called the binomial coefficient $\binom{n}{r}$.

For an implementation, see

<https://github.com/vnazariy/agentBasedSimulation>

For example, given $S = 4$, the number of combinations is $\binom{4}{2} = 6$. That is,

$$C_4 = \{(M_1, M_2), (M_1, M_3), (M_1, M_4), (M_2, M_3), (M_2, M_4), (M_3, M_4)\}$$

If $c = \{(M_1, M_2)\}$, then

$$J = \begin{pmatrix} M_1 \\ M_2 \end{pmatrix} \text{ and } \bar{J} = \begin{pmatrix} M_3 \\ M_4 \end{pmatrix}$$

Recall, for $N = 4$ and Ω_2^* , we have $\{f \in \Omega \mid f_2 = 4\} = \{(1, 4, 2, 3), (2, 4, 1, 3), (1, 4, 3, 2), (3, 4, 1, 2), \dots\}$. Similarly, for Ω_3^* , we have $\{f \in \Omega \mid f_3 = 4\} = \{(1, 2, 4, 3), (2, 1, 4, 3), (1, 3, 4, 2), (3, 1, 4, 2), \dots\}$. Thus, if we are given $r^c = (1, 4, 2, 3)$, then the best performing strategy is at index 2, and $n^* = 2$. That is, the second strategy is always the best and we consider the subspace Ω_2^* .

7.3.3 Some statistics

In addition to the Probability of Backtest Overfitting (PBO), the authors also presented three additional statistics.

1. Performance degradation: For each combination $c \in C_S$, while $R_{n^*}^c$ is the maximum among the components of R^c , $\bar{R}_{n^*}^c$ is not necessarily the maximum among the components of $\bar{R}^c$. Since we are trying every combination of M_s taken in groups of size $\frac{S}{2}$, we expect to get

$$\bar{R}_{n^*}^c < \max(\bar{R}^c) \approx \max(R^c) = R_{n^*}^c$$

Hence, performance degradation determines the extent to which greater performance IS leads to lower performance OOS by considering the regression

$$\bar{R}_{n^*}^c = \alpha + \beta R_{n^*}^c + \epsilon^c$$

for some constants α and β . Note, β will be negative in most practical cases due to compensation effects.

2. Probability of loss: We can then compute the probability that the model selected as optimal IS will deliver a loss OOS

$$P(\bar{R}_{n^*}^c < 0)$$

3. Stochastic dominance: It allows us to rank gambles or lotteries without having to make strong assumptions regarding an individual's utility function. Here, it determines if the distribution of $\bar{R}_{n^*}$ across all $c \in C_S$ stochastically dominates over the distribution of all $\bar{R}$. That is if the procedure used to select a strategy IS is preferable to randomly choosing one model configurations among the N alternatives.

The first order stochastic dominance occurs if

$$P(\bar{R}_{n^*} \geq x) \geq P(\text{Mean}(\bar{R}) \geq x), \forall x$$

and

$$P(\bar{R}_{n^*} \geq x) > P(\text{Mean}(\bar{R}) \geq x) \text{ for some } x$$

One can visually check that the cumulative distribution function of $\bar{R}_{n*}$ is not above that of R for all possible outcomes. A less demanding criterion is the second-order stochastic dominance (SD2), which requires that:

$$SD2(x) = \int_{-\infty}^x (P(\text{Mean}(\bar{R}) \leq x) - P(\bar{R}_{n*} \leq x)) dx \geq 0, \forall x$$

and that

$$SD2(x) > 0 \text{ for some } x$$

7.3.4 Some problems

The method of division in CSCV to get the samples IS and OOS is symmetric. However, if the performance measure as a time series has a strong autocorrelation, then such a division may obscure the characterisation, especially for large S . For some backtests other methods, such as K-FCV, may well be better suited. Further, since we do not know the distribution of the backtest performance measure, the CSCV estimate of the probability measure assumes all the sample statistics carries the same weight.

This procedure only takes into account structural breaks as long as they are present in the dataset $\mathcal{T}$ of length T . If a structural break occurs outside of the dataset, the strategy may be overfit to a particular data regime. Hence, the dataset used for any backtest is assumed to be representative of future states of the modelled financial variable. One shouldn't apply CSCV to guide the search for an optimal strategy. Strathern [1997] stated that when a measure becomes a target, it ceases to be a good measure. Thus, PBO should not be the objective function on which strategy selection relies.

7.4 Determining the entry and exit thresholds

In the case of continuous trading, letting the security price follows an Ornstein-Uhlenbeck (OU) process, Bertram [2009] and Zeng et al. [2014] provided analytic formulas to calculate the trade length's mean and variance by framing the problem in terms of the first-passage time of that process. They examined the problem of choosing an optimal strategy under two different objective functions: the expected return and the Sharpe ratio. They used this result to calculate the expected return and the variance per unit of time of a mean-reverting trading strategy. The solution satisfies a real valued integral equation which is approximated via a Taylor series expansion. This approach allows us to reduce the problem of finding the optimal trading thresholds to a simple maximisation problem.

7.4.1 The approach by Bertram

Bertram [2009] derived analytic formulae for statistical arbitrage trading where the price of an asset follows an exponential Ornstein-Uhlenbeck process. By framing the problem in terms of the first-passage time of the process, he first derived the expressions for the mean and the variance of the trade length. Then, he derived analytic formulae for the expected return and the variance of the return per unit of time in terms of the trade entry, exit levels, and transaction costs. Hence, a solution to the problem of choosing optimal trading thresholds was proposed by maximising the expected return and the Sharpe ratio.

7.4.1.1 The trading strategy We assume a long position and let the continuous (mean-reverting) underlying security (log-price) be denoted $(X_t)_{t \geq 0}$. The continuous trading strategy is defined as the act of entering a trade when $X_t = a$ and exiting the trade at $X_t = m$, where a and m are fixed and such that $a < m$. This is illustrated in Figure (64).

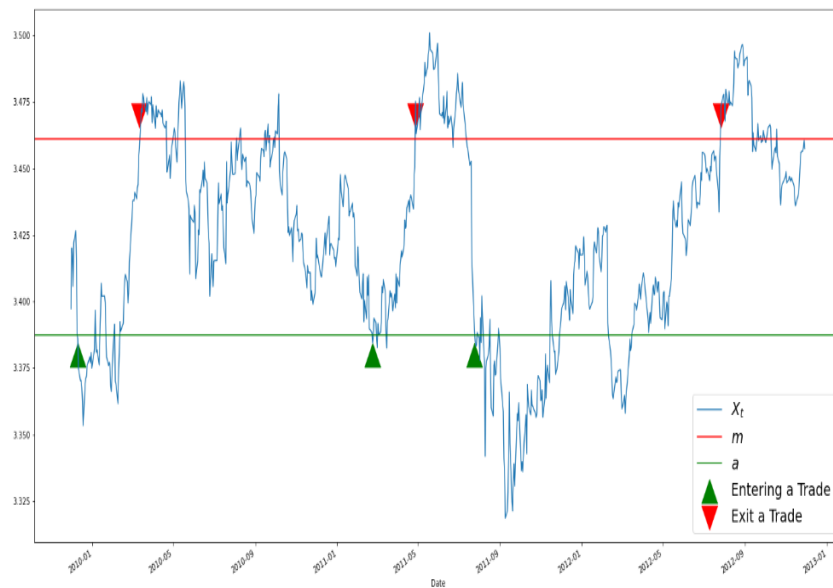


Figure 64: Entry and exit trade.

Note, since a and m are fixed, it takes a certain time for the (mean-reverting) process X_t to get back to a (where a new trade will start) after reaching m . Hence, the trading cycle is completed when the trajectory X_t starting at a reaches m and gets back to a . Such a strategy can be thought of as periodic, since the actions are repeated between trade entry points. This is illustrated in Figure (65).

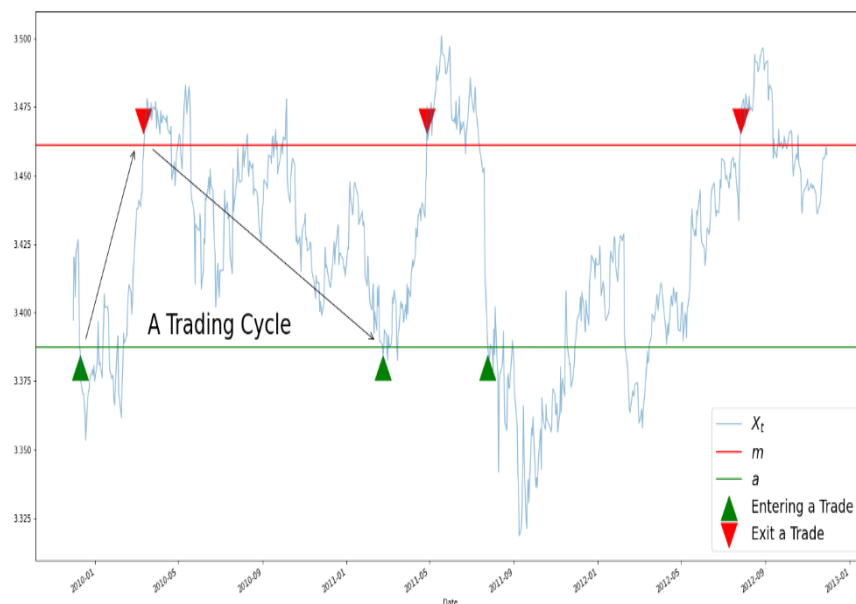


Figure 65: Trading cycle.

Since X_t is a stochastic process, the time taken to complete the trade cycle is a random variable. The total trade length L_T is defined as the time needed to complete a trading cycle (start in a , move to m , and get back to a). It can be decomposed into two subintervals:

$$L_T = L_{T,1} + L_{T,2}$$

where $L_{T,1}$ is the time taken for the process to travel from a to m and $L_{T,2}$ is the time taken from m back to a . They corresponds to a first-passage time for the process X_t . In addition, when X_t is a Markov process, $L_{T,1}$ and $L_{T,2}$ are independent. Depending on the properties of X_t and L_T a trade may take a long time before reaching the exit level and experience a significant deviation away from the exit level during that time.

7.4.1.2 Description Let the price of the traded security at time t be

$$P_t = e^{X_t}, X_{t_0} = x_0 \quad (7.47)$$

where X_t follows an Ornstein-Uhlenbeck (OU) process with dynamics given by

$$dX_t = \varphi(b_{\infty} - X_t)dt + \sigma dW_t$$

where b_{∞} is the long-term mean, $\varphi > 0$ is speed at which the values will converge to b_{∞} , and $\sigma > 0$ is the volatility of the OU process. Note, the original paper assumes that the long-term mean b_{∞} is null. The process Y_{τ} is a dimensionless series transformed from the original time series X_t (see Ricciardi et al. [1988]) as follows:

$$\begin{cases} \tau = \varphi t \\ Y_t = \frac{\sqrt{2\varphi}}{\sigma}(X_t - b_{\infty}) \end{cases}$$

with dynamics

$$dY_{\tau} = -Y_{\tau}d\tau + \sqrt{2}dW_{\tau}$$

In that setting, the entry level, the exit level, and the transaction cost rescale to

$$\bar{m} = \frac{(m - b_{\infty})\sqrt{2\varphi}}{\sigma}, \bar{a} = \frac{(a - b_{\infty})\sqrt{2\varphi}}{\sigma}, \bar{c} = \frac{(c - b_{\infty})\sqrt{2\varphi}}{\sigma}$$

where $\theta = \frac{\sigma}{\sqrt{2\varphi}}$ is the steady state standard deviation. Further, the trade length rescales to $\bar{L}_T = \varphi L_T$. After some calculation, the mean of the total trade length L_T is

$$E[L_T] = \frac{\pi}{\varphi} (Erfi(\frac{\bar{m}}{\sqrt{2}}) - Erfi(\frac{\bar{a}}{\sqrt{2}}))$$

where $Erfi(x) = iErf(ix)$ is the imaginary error function. The variance of the total trade length is

$$Var(L_T) = \frac{1}{\varphi^2} [w_1(\bar{m}) - w_1(\bar{a}) - w_2(\bar{m}) + w_2(\bar{a})]$$

where

$$\begin{aligned} w_1(z) &= \left(\frac{1}{2} \sum_{k=1}^{\infty} \Gamma\left(\frac{k}{2}\right) \frac{(\sqrt{2}z)^k}{k!} \right)^2 - \left(\frac{1}{2} \sum_{k=1}^{\infty} (-1)^k \Gamma\left(\frac{k}{2}\right) \frac{(\sqrt{2}z)^k}{k!} \right)^2 \\ w_2(z) &= \sum_{k=1}^{\infty} \Gamma\left(\frac{2k-1}{2}\right) \Psi\left(\frac{2k-1}{2}\right) \frac{(\sqrt{2}z)^{2k-1}}{(2k-1)!} \end{aligned}$$

where $\Psi(x) = \psi(x)\psi(1)$ and $\psi(x)$ is the digamma function.

Recall, the trading strategy is periodic since the actions are repeated between trade entry points. As a result, there is no maturity to the strategy. Instead, it is characterised by the behaviour of L_T . Hence, the return on the strategy must be expressed in terms of unit of time. A trading cycle is completed as follows: we start in a , move to m , and get back to a after a random time. The trading frequency is specified by the number of times the strategy trades per unit of time. This value is dependent on how long it takes in total to move from one trade entry point to the next, passing though the exit point along the way. Here, the variable $\frac{1}{L_T}$ represents the number of trades per unit time (or trade frequency). As a result, expressions for the mean and the variance of the return per unit of time of the strategy can be formulated in terms of the trading frequency. Since the value of L_T is independent for each trade cycle in the Markov case, the trade frequency is a renewal process and we can make use of the following results from renewal theory

$$E\left[\frac{1}{L_T}\right] \sim \frac{1}{E[L_T]} \text{ and } Var\left(\frac{1}{L_T}\right) \sim \frac{Var(L_T)}{E^3[L_T]}$$

Let $r(a, m, c)$ be the return per trade as a function of entry a , exit levels m , and transaction cost c . By multiplying the return per trade by the above expressions for the frequency, the mean and variance of the trading strategy return per unit of time are

$$\mu_s(a, m, c) = \frac{r(a, m, c)}{E[L_T]} \text{ and } \sigma_s^2(a, m, c) = \frac{r^2(a, m, c)Var(L_T)}{E^3[L_T]}$$

The return for a single trade can be expressed as a function of the exit and entry values, minus the total transaction costs associated with that trade. Since the variable X_t represents the log-price, the function

$$r(a, m, c) = (m - a - c)$$

gives the continuously compound rate of return for a single trade and accounts for the transaction cost. We can then get an analytic form of the expected return and expected variance for the strategy, allowing us to determine the properties of the strategy in terms of the trade entry and exit levels and transaction costs.

7.4.1.3 Optimisation The authors estimated the optimal entry and exit thresholds a and m , respectively, by maximising the expected return or the Sharpe ratio (depending on our goal) per unit of time for a given transaction cost/risk-free rate. In the former, we want to find the values of a and m that maximise the expected return $\mu_s(a, m, c)$ defined above. That is, differentiating $\mu_s(a, m, c)$ with respect to a and m , and setting the derivatives equal to zero. They showed that $\frac{\partial \mu_s}{\partial a} = \frac{\partial \mu_s}{\partial m} = 0$ required that $(m - b_{\mathcal{O}})^2 = (a - b_{\mathcal{O}})^2$. They found a similar result in the case of the Sharpe ratio. Thus, the maximum expected return/Sharpe ratio occurs when $(m - b_{\mathcal{O}})^2 = (a - b_{\mathcal{O}})^2$. Since $a < m$, in the case where $b_{\mathcal{O}} = 0$, we get $a < 0$ and $m = -a$. Otherwise, it implies that $m = 2b_{\mathcal{O}}a$. Therefore, for a given transaction cost/risk-free rate, the following equation can be maximised to find optimal a and m :

$$\mu_s^*(a, c) = \frac{r(a, 2b_{\mathcal{O}}a, c)}{E[L_T]} \text{ and } \Psi_{SR}^*(a, c, r_f) = \frac{\mu_s(a, 2b_{\mathcal{O}}a, c) - r^*}{\sigma_s(a, 2b_{\mathcal{O}}a, c)}$$

where $r^* = \frac{r_f}{E[L_T]}$ and r_f is the risk-free rate. Note, r^* corresponds to the risk-free rate over the same time period as the return produced by the trading strategy. While the root of the first equation can be found numerically, the author constructed an approximate solution via Taylor series.

7.4.2 The approach by Zeng

Zeng et al. [2014] extended the previous results by allowing for short positions. They derived a polynomial expression for the expectation of the first-passage time of an OU process with a two-sided boundary. Then, they simplified the problem of optimising the expected return per unit of time to a problem of solving an equation. Recall, X_t is an OU process and X_t is the underlying security.

7.4.2.1 The trading strategy The trading strategy is defined as follows:

$$\begin{cases} \text{Open a short trade when } Y_\tau = a_d \text{ and close the existing short trade at } Y_\tau = b_d \\ \text{Open a long trade when } Y_\tau = -a_d \text{ and close the existing long trade at } Y_\tau = -b_d \end{cases}$$

where a_d and b_d are the entry and exit thresholds in the dimensionless system, respectively. Recall, the process Y_τ is a dimensionless series transformed from the original time series X_t as follows:

$$\begin{cases} \tau = \varphi t \\ Y_t = \frac{\sqrt{2\varphi}}{\sigma} (X_t - b_\infty) \end{cases}$$

The trading strategy is illustrated in Figure (66).



Figure 66: Entry and exit long-short trade.

7.4.2.2 Description The mean of the trade length L_T is

$$E[L_T] = \frac{1}{2\varphi} \sum_{k=0}^{\infty} \Gamma\left(\frac{2k+1}{2}\right) \frac{1}{(2k+1)!} ((\sqrt{2}a_d)^{2k+1} - (\sqrt{2}b_d)^{2k+1})$$

The variance of the trade length is

$$\text{Var}(L_T) = \frac{1}{\varphi^2} (\text{Var}(L_T(a_d, b_d)) + \text{Var}(L_T(-a_d, a_d, b_d)))$$

where $\text{Var}(L_T(a_d, b_d))$ is the variance of the time taken for the OU process to travel from a_d to b_d , and $\text{Var}(L_T(-a_d, a_d, b_d))$ is the variance of the time taken for the O-U process to travel from b_d to a_d or $-a_d$. That is,

$$\text{Var}(L_T(a_d, b_d)) = w_1(a_d) - w_1(b_d) - w_2(a_d) + w_1(b_d)$$

where $w_i, i = 1, 2$ are defined above, and

$$\text{Var}(L_T(-a_d, a_d, b_d)) = E[L_T^2(-a_d, a_d, b_d)] - E^2[L_T(-a_d, a_d, b_d)]$$

where

$$\begin{aligned} E[L_T(-a_d, a_d, b_d)] &= \frac{1}{2} \sum_{k=1}^{\infty} \Gamma(k) \frac{1}{(2k)!} ((\sqrt{2}a_d)^{2k} - (\sqrt{2}b_d)^{2k}) \\ E[L_T^2(-a_d, a_d, b_d)] &= e^{\frac{1}{4}(b_d^2 - a_d^2)} [g_1(a_d, b_d) - g_2(a_d, b_d)] \end{aligned}$$

where

$$\begin{aligned} g_1(a_d, b_d) &= \frac{1}{m^2(\lambda, a_d)} [m''(\lambda, b_d)m(\lambda, a_d) - m'(\lambda, a_d)m'(\lambda, b_d)] \Big|_{\lambda=0} \\ g_2(a_d, b_d) &= \left[\frac{1}{m^2(\lambda, a_d)} [m''(\lambda, a_d)m(\lambda, b_d) + m'(\lambda, a_d)m'(\lambda, b_d)] - \frac{1}{m^3(\lambda, a_d)} 2(m'(\lambda, a_d))^2 m(\lambda, b_d) \right] \Big|_{\lambda=0} \end{aligned}$$

where

$$m(\lambda, x) = D_{-\lambda}(x) + D_{-\lambda}(x)$$

and

$$D_{-\lambda}(x) = \sqrt{\frac{2}{\pi}} e^{\frac{x^2}{4}} \int_0^{\infty} t^{-\lambda} e^{-\frac{t}{2}} \cos(xt + \frac{\lambda\pi}{2}) dt$$

The mean and variance of the trading strategy return per unit of time are as in the previous section with m replaced by b . We get the optimal thresholds by maximising the expected return. The authors presented the following two rules:

1. Conventional optimal rule: When $0 \leq b_d \leq a_d$, the maximum expected return occurs when $b_d = 0$. Therefore, for a given transaction cost, the following equation can be solved to find optimal a_d

$$\frac{1}{2} \sum_{k=0}^{\infty} \Gamma\left(\frac{2k+1}{2}\right) \frac{(\sqrt{2}a_d)^{2k+1}}{(2k+1)!} = (a - c) \frac{\sqrt{2}}{2} \sum_{k=0}^{\infty} \Gamma\left(\frac{2k}{2}\right) \frac{(\sqrt{2}a_d)^{2k}}{(2k+1)!}$$

2. New optimal rule: When $-a_d \leq b_d \leq 0$, the maximum expected return occurs when $b_d = -a_d$. Therefore, for a given transaction cost, the following equation can be solved to find optimal a_d

$$\frac{1}{2} \sum_{k=0}^{\infty} \Gamma\left(\frac{2k+1}{2}\right) \frac{(\sqrt{2}a_d)^{2k+1}}{(2k+1)!} = (a - \frac{c}{2}) \frac{\sqrt{2}}{2} \sum_{k=0}^{\infty} \Gamma\left(\frac{2k}{2}\right) \frac{(\sqrt{2}a_d)^{2k}}{(2k+1)!}$$

After calculating optimal thresholds in the dimensionless system, we need to use the following formula to transform them back to the original system

$$k = k_d \frac{\sigma}{\sqrt{2}\varphi} + b_{\mathcal{O}}$$

where

$$\begin{cases} k_d = a_d, b_d, -a_d, -b_d \\ k = a_s, b_s, a_l, b_l \end{cases}$$

where a_s, b_s denotes the entry and exit thresholds for a short position, and a_l, b_l denotes the entry and exit thresholds for a long position.

Practically, we estimate statistically the model parameters $\varphi, b_{\sigma}, \sigma$ for the OU process on the training data (see Leung et al. [2015]). Then, we set the transaction costs to some value (such as $c = 0.02$), and use the new optimal rule to get the trading thresholds a_s, b_s, a_l, b_l . Here, the model parameters are redetermined every 6-months with a 1-year history. Note, when using real-world data, we often encounter problems related to the OU process not being well fitted or that cannot be fitted to a given dataset.

7.5 Calibrating trading rules

We saw that the parameters from trading rules were calibrated via historical simulations (backtesting) to maximise profit. Carr et al. [2014] proposed an approach to avoid the risk of overfitting in the context of calibrating trading rules. That is, they proposed a procedure for determining the optimal trading rule (OTR) without running a backtest. In theory it can be accomplished by estimating the optimal parameters for the trading rules directly from data generated with a calibrated stochastic process rather than using historical simulations. They considered the case of market prices following a discrete Ornstein-Uhlenbeck process, and focused on the problem of maximising performance by finding the optimal exit value of an existing position. They calibrated the OU process to the entire historical sample, which they used to generate the observed stream of returns. Then, they derived the optimal values for the trading rule's parameters without requiring a historical simulations.

7.5.1 The problem

Given the trading problem defined in Section (7.1.1), we consider the case where we are in an existing trade and search for the exit threshold that will maximise profit. The authors modified the definition of backtest overfitting given in Definition (7.2) to apply it to the problem of trading rule overfitting. They used the sample Sharpe ratio given in Equation (7.43) to define Overfit Trading Rule as follows:

Definition 7.4 *Overfit Trading Rule*

The optimal set of parameters (trading rule or model configuration) Φ^* is overfit if

$$\frac{\hat{E}[\pi_{\mathcal{J}} | \Phi^*(\mathcal{I})]}{\hat{\sigma}[\pi_{\mathcal{J}} | \Phi^*(\mathcal{I})]} < Me_{\Omega} \left\{ E_{\Omega} \left[\frac{\hat{E}[\pi_{\mathcal{J}} | \Phi(\mathcal{J})]}{\hat{\sigma}[\pi_{\mathcal{J}} | \Phi(\mathcal{J})]} \right] \right\}, \mathcal{J} = (I + 1, \dots, J)$$

for some constant M and some alternative trading rules $\Phi(\mathcal{J}) \in \Omega$.

That is, an optimal in-sample (IS) trading rule $\Phi^*(\mathcal{I})$ is overfit when it is expected to underperform the median of alternative trading rules $\Phi(\mathcal{J}) \in \Omega$ out-of-sample (OOS). Bailey et al. [2014] showed that it was hard not to overfit a backtest, particularly when there were free variables able to target specific observations IS, or the number of elements in Ω was large. The outcome is that the backtest profits from random noise IS, making Φ^* unfit for OOS opportunities. They showed that overfitting leads to negative performance OOS when $\Delta\pi_{i,t}$ exhibits serial dependence.

We can also consider the case where we model the underlying variable by generating pseudorandom scenarios and measure the performance of the resulting investment strategy for those scenarios. Using the Sharpe ratio given in Equation (7.44), we can define Overfit Trading Rule as follows:

Definition 7.5 *Overfit Trading Rule (pseudorandom scenarios)*

The optimal set of parameters (trading rule or model configuration) Φ^* , is overfit if, for $P_{j,T_j} \sim p, j \in \mathcal{J}$, we have

$$\frac{E_{P_{j,T_j} \sim p}[\pi_{j,T_j} | \Phi^*(\mathcal{I})]}{\sigma_{P_{j,T_j} \sim p}[\pi_{j,T_j} | \Phi^*(\mathcal{I})]} < Me_{\Omega} \left\{ E_{\Omega} \left[\frac{E_{P_{j,T_j} \sim p}[\pi_{j,T_j} | \Phi_j(\mathcal{J})]}{\sigma_{P_{j,T_j} \sim p}[\pi_{j,T_j} | \Phi_j(\mathcal{J})]} \right] \right\}, j \in \mathcal{J} = (I + 1, \dots, J)$$

for some constant M and some alternative trading rules $\Phi(\mathcal{J}) \in \Omega$.

7.5.2 An alternative approach

The authors characterised a stochastic process for generating the returns (and the observations $\pi_{i,t}$) of strategy S by assuming that prices at time t were following a discrete Ornstein-Uhlenbeck (OU) process

$$P_{i,t} = (1 - \varphi)\bar{X} + \varphi P_{i,t-1} + \sigma \epsilon_{i,t}, P_{i,0} \quad (7.48)$$

where $\epsilon_{i,t} \sim N(0, 1)$. It corresponds to the $AR(1)$ model given in Equation (16.134) with speed of mean-reversion $K_1 = \varphi$ and σ should be $\bar{\sigma} = \sigma\sqrt{\Delta t}$. Assuming stationarity, given the maturity T_i , the unconditional mean is $E_0[P_{i,T_i}] = \bar{X}$ (see Equation (16.136)). Note, the authors reversed this relation by setting $\bar{X} = E_0[P_{i,T_i}]$. Hence, $\{P_{i,0}, E_0[P_{i,T_i}]\}$ are the initial conditions, and φ determines the speed at which $P_{i,0}$ converges towards $E_0[P_{i,T_i}]$.

Remark 7.2 Starting the price process $P_{i,0}$ at the entry level and setting the long-term mean $\bar{X}$ equal to the forecasted price of opportunity i seen at $t = 0$, $E_0[P_{i,T_i}]$, is a way of internalising the entry and exit pair (a, m) in the dynamics of the price process.

Remark 7.3 Here, the price P_{i,T_i} is a random variable observed at time T_i . The initial conditions $\{P_{i,0}, E_0[P_{i,T_i}]\}$ are given. We want to find the thresholds $(\bar{\pi}_i, \underline{\pi}_i)$ that maximise the expected profit for the trade. The entry and exit pair (a, m) are internalised in the dynamics of the price process. In the approach described in Section (7.4.1), $P_{i,T_i} = m$ is known, but T_i is a random variable. The mean and variance of the trading strategy return per unit of time are expressed in terms of entry and exit thresholds a and m . These thresholds are optimally estimated to maximise the expected return or the Sharpe ratio.

Given the definition of the PnL, $\pi_{i,t}$, in Equation (7.40), the above equation implies that the performance of opportunity i is characterised by the process

$$\frac{1}{w_i} \pi_{i,t} = (1 - \varphi)E_0[P_{i,T_i}] - P_{i,0} + \varphi P_{i,t-1} + \sigma \epsilon_{i,t}$$

Remark 7.4 The PnL, $\pi_{i,t}$, is now related to the entry and exit pair (a, m) .

From Equation (16.135), it can be shown that the distribution of the process specified in the above equation has a closed-form given by

$$\pi_{i,t} \sim N\left(w_i((1 - \varphi)E_0[P_{i,T_i}] - P_{i,0}), w_i^2 \sigma^2 \sum_{j=0}^{t-1} \varphi^{2j}\right)$$

where $\varphi \in (-1, 1)$ for the process to be stationary. Given a set of input parameters $\{\sigma, \varphi\}$ and initial conditions $\{P_{i,0}, E_0[P_{i,T_i}]\}$ associated with opportunity i , we want to find out if there is an OTR $\Phi^* = \{(\underline{\pi}_i, \bar{\pi}_i)\}$. Further, if there is a strategy S predicting a profit target $\bar{\pi}_i$, can we compute the optimal stop-loss $\underline{\pi}_i$ given the input values $\{\sigma, \varphi\}$? In the case of an affirmative answer, no backtest would be needed to determine Φ_i^* . As a result, this approach avoids the problem of overfitting the trading rule.

7.5.2.1 The algorithm The authors presented a procedure to derive numerically the OTR for any specification in general, and the OU specification in particular.

1. Estimate the input parameters $\{\sigma, \varphi\}$ by linearising Equation (7.48) as follows:

$$P_{i,t} = E_0[P_{i,T_i}] + \varphi(P_{i,t-1} - E_0[P_{i,T_i}]) + \xi_t$$

We can then form vectors X and Y by sequencing opportunities:

$$X = \begin{bmatrix} P_{0,0} - E_0[P_{0,T_0}] \\ P_{0,1} - E_0[P_{0,T_0}] \\ \dots \\ P_{0,T-1} - E_0[P_{0,T_0}] \\ \dots \\ P_{I,0} - E_0[P_{I,T_I}] \\ \dots \\ P_{I,T-1} - E_0[P_{I,T_I}] \end{bmatrix}, Y = \begin{bmatrix} P_{0,1} \\ P_{0,2} \\ \dots \\ P_{0,T} \\ \dots \\ P_{I,1} \\ \dots \\ P_{I,T} \end{bmatrix} \text{ and } Z = \begin{bmatrix} E_0[P_{0,T_0}] \\ E_0[P_{0,T_0}] \\ \dots \\ E_0[P_{0,T_0}] \\ \dots \\ E_0[P_{I,T_I}] \\ \dots \\ E_0[P_{I,T_I}] \end{bmatrix}$$

such that

$$Y = Z + \varphi X + \xi_t$$

Applying OLS on this linearised equation, we can estimate the original OU parameters as

$$\hat{\varphi} = \frac{Cov(X, Y)}{Cov(X, X)}, \hat{\xi}_t = Y - Z - \hat{\varphi} X, \hat{\sigma} = \sqrt{Cov(\hat{\xi}_t, \hat{\xi}_t)}$$

2. Construct a mesh of stop-loss and profit-taking pairs $(\underline{\pi}_i, \bar{\pi}_i)$. For example, a Cartesian product of $\underline{\pi}_i = \{0, -\frac{1}{2}\sigma, -\sigma, \dots, -10\sigma\}$ and $\bar{\pi}_i = \{0, \frac{1}{2}\sigma, \sigma, \dots, 10\sigma\}$ gives 21×21 nodes, each constituting an alternative trading rule Φ_i .
3. Generate a large number of paths (such as 100,000) for $\pi_{i,t}$, applying the estimates $\{\hat{\sigma}, \hat{\varphi}\}$ on $P_{i,t}$. Use the observed initial conditions $\{P_{i,0}, E_0[P_{i,T_i}]\}$ associated with an opportunity i as seed values. Impose a maximum holding period (such as 100 observations) at which point the position is exited even though $\underline{\pi}_i \leq \pi_{i,100} \leq \bar{\pi}_i$.
4. Apply the 100,000 paths generated in Step [3] on each node of the 21×21 mesh $(\underline{\pi}_i, \bar{\pi}_i)$ generated in Step [2]. For each node, we apply the stop-loss and profit-taking logic, giving 100,000 values of π_{i,T_i} . For each node we compute the Sharpe ratio Ψ_{SR} associated with that trading rule.
5. The result in Step [4] can be used in two different ways (Steps 5a, Step 5b and 5c):
 - (a) Determine the pair $(\underline{\pi}_i, \bar{\pi}_i)$ within the mesh of trading rules that is optimal, given the input parameters $\{\hat{\sigma}, \hat{\varphi}\}$ and the observed initial conditions $\{P_{i,0}, E_0[P_{i,T_i}]\}$.
 - (b) If strategy S provides a profit target $\bar{\pi}_i$ for a particular opportunity i , we can use that information in conjunction with the results in Step [4] to determine the optimal stop-loss $\underline{\pi}_i$.
 - (c) If the trader has a maximum stop-loss $\underline{\pi}_i$ imposed by the fund's management, we can use that information in conjunction with the results in Step [4] to determine the optimal profit taking $\bar{\pi}_i$ within the range of stop-losses $[0, \underline{\pi}_i]$.

7.6 Modelling trailing stops

If we let an investor be long the risky asset, the problem of finding the right time to sell stocks is an optimal stopping problem (see Appendix (11.7)). That is, under no constraint, it is optimal to sell the asset when its price is sufficiently high. However, this is no longer the case when we add constraints. Both stop-loss and trailing stop trading strategies provide downside protection for an existing position by determining the optimal trading regions. A stop-loss exit closes a position at a fixed price, while a trailing stop consist of a stochastic floor moving based on the running maximum of the asset price. In these cases, the exit is triggered when the asset price crosses a fixed or random barrier level, respectively. Glynn et al. [1995] studied the expected discounted reward at a trailing stop under a discrete-time random walk or a geometric Brownian motion (GBM) model, and found that it would be optimal to never use the trailing stop if the stock followed a GBM with a positive drift. Warburton et al. [2006] performed a

probabilistic analysis of a variant of trailing stop in a random walk model. Yin et al. [2010] implemented a stochastic approximation scheme to determine the optimal percentage trailing stop level that maximises the expected discounted simple return from liquidation. Imkeller et al. [2014] compared the performance of a number of trading rules with fixed and trailing stops under an arithmetic Brownian motion model. Considering a linear diffusion framework, Leung et al. [2019] studied the problem of optimal early liquidation of a risky asset, given a pre-specified trailing stop mandatory liquidation order. They expressed the solution as an optimal double stopping problem with a random path-dependent maturity. They solved an optimal liquidation problem with a trailing stop and derived the optimal timing to buy the asset.

Details on linear diffusion can be found in Borodin et al. [2002].

7.6.1 Linear diffusion

We let the risky asset $X = \{X_t; t \geq 0\}$ be a linear diffusion on $I \equiv (l, r) \subset \mathbb{R}$ with operator

$$\mathcal{L} = \frac{1}{2}\sigma^2(x)\frac{\partial^2}{\partial x^2} + \mu(x)\frac{\partial}{\partial x}, \forall x \in I$$

where $(\mu(\cdot), \sigma(\cdot))$ is a pair of real-valued functions on I . For any $q > 0$, the Sturm-Liouville equation

$$(\mathcal{L} - q)u(x) = 0$$

has a positive increasing solution ϕ_q^+ and a positive decreasing solution ϕ_q^- . For an arbitrary fixed $\kappa \in I$, the solutions can be expressed as

$$\phi_q^+ = \begin{cases} E_x[e^{-q\tau_X^+(\kappa)}] & \text{if } x \leq \kappa \\ \frac{1}{E_\kappa[e^{-q\tau_X^+(\kappa)}]} & \text{if } x > \kappa \end{cases} \quad \text{and} \quad \phi_q^- = \begin{cases} \frac{1}{E_\kappa[e^{-q\tau_X^-(\kappa)}]} & \text{if } x \leq \kappa \\ E_x[e^{-q\tau_X^-(\kappa)}] & \text{if } x > \kappa \end{cases}$$

where $\tau_X^+(y)$ and $\tau_X^-(y)$ are the first passage times of X to level $y \in I$ from below and above, respectively. That is,

$$\tau_X^\pm(y) = \inf\{t > 0; X_t \geq y\}, \forall y \in I$$

The functions $\phi_q^\pm$ are also closely related to two-sided exit problems of X (see Lehoczy [1977]). Specifically, we have:

Lemma 7.1 Suppose that $l < y \leq x \leq z < r$, then for $q > 0$, we have

$$\begin{aligned} E_x[e^{-q\tau_X^-(y)} I_{\{\tau_X^-(y) < \tau_X^+(z)\}}] &= \frac{\phi_q^-(x)\psi_q(z) - \psi_q(x)}{\phi_q^-(y)\psi_q(z) - \psi_q(y)} \\ E_x[e^{-q\tau_X^+(z)} I_{\{\tau_X^-(y) > \tau_X^+(z)\}}] &= \frac{\phi_q^-(x)\psi_q(x) - \psi_q(y)}{\phi_q^-(z)\psi_q(z) - \psi_q(y)} \end{aligned}$$

where $\psi_q : I \rightarrow \mathbb{R}^+$ is a strictly increasing function defined as

$$\psi_q(x) = \frac{\phi_q^+(x)}{\phi_q^-(x)}, \forall x \in I$$

By the boundary behaviour of X , we have $\phi_q^-(l+) = \phi_q^+(r-) = \infty$, so that $\psi_q(I) = (0, \infty)$.

For simplicity, we assume the investor holds long one unit of the risky asset X . For any $x_M \in I$, the running maximum of X is

$$M_t = x_M \vee \sup_{s \in [0, t]} X_s, t \geq 0$$

where $\vee$ is the maximum. We can model liquidation time by a stopping time τ of the underlying process X , and denote the reward to be realised upon liquidation by $h(X_\tau)$, where h is assumed to be a real-valued increasing function on I . Here, h represents the proceeds from selling the risky asset.

Assumption 1 *The reward function h is increasing, twice differentiable on I , and there is an x_0 in the interior of I such that*

$$(\mathcal{L} - q)h(x) \geq 0 \text{ iff } x \leq x_0$$

Moreover, we have

$$\lim_{x \rightarrow r} \frac{h(x)}{\phi_q^+(x)} \leq \sup_{x \geq x_0} \frac{h(x)}{\phi_q^+(x)} \leq \infty$$

Lemma 7.2 *Given Assumption (1), there is an $x^* \in [x_0, r)$ such that*

$$\sup_{\tau \in \mathcal{T}} E_x[e^{-r\tau} h(X_\tau) I_{\{\tau < \infty\}}] = E_x[e^{-r\tau_X^+(x^*)} h(X_{\tau_X^+(x^*)}) I_{\{\tau_X^+(x^*) < \infty\}}]$$

This shows that Assumption (1) is sufficient for the optimality of upcrossing strategy $T_{x^*}^+$ in the optimal stopping problem (see Appendix (11.7))

$$\sup_{\tau \in \mathcal{T}} E_x[e^{-r\tau} h(X_\tau) I_{\{\tau < \infty\}}]$$

where x^* is a constant in (x_0, r) . That is, under no constraint (no trailing stop), it is optimal to sell the asset when its price is sufficiently high. However, this is no longer the case when we add constraints. Since the investor is long the risky asset, he must protect his downside risk with a stop-loss exit (selling the asset).

7.6.2 Fixed stop-loss exit

First, let's consider the optimal stopping problem where the investor uses a fixed stop-loss exit. Arbitrarily fix $y \in I$ and consider the following class of problems

$$V_y(x) = \sup_{\tau \in \mathcal{T}_y^S} E_x[e^{-q\tau} h(X_\tau) I_{\{\tau < \infty\}}]$$

where $q > 0$ is a subjective discounting rate, and $\mathcal{T}_y^S$ is the set of all stopping times of X that stops no later than the first passage time to level y . Since the investor is long the risky asset, we get

$$\tau_X^-(y) = \inf\{t > 0; X_t < y\}$$

This problem puts a mandatory liquidation constraint upon hitting the fixed stop-loss level y from above. Thus, we call this problem the optimal liquidation problem. We let $c \in [0, \sup_{x \in I} [V_y(x) - h(x)]]$ be a transaction fee for asset acquisition. The special case where the reward function $h(x) = x - c$ is driven by the OU and CIR processes was studied by Cartea et al. [2015], Leung et al. [2014] [2015a] [2015b].

Proposition 1 *Under Assumption (1), for any fixed $y \in (l, x_0)$ there is a finite threshold $b(y) \in (x_0, r)$ such that*

$$V_y(x) = E_x[e^{-q[\tau_X^+(b(y)) \wedge \tau_X^-(y)]} h(X_{[\tau_X^+(b(y)) \wedge \tau_X^-(y)]})], \forall x \in I$$

where $\wedge$ is the minimum.

This proposition shows how we can specify a limit on the maximum possible loss (with a fixed stop-loss exit) without limiting the maximum possible gain.

7.6.3 Trailing stop exit

The trailing stop can be viewed as a random maturity or stopping time constraint in the optimal stopping problem. The trailing stop ρ_f is defined as the first time the asset value X reaches the stochastic floor $f(M)$ from above. That is,

$$\rho_f = \inf\{t > 0; X_t < f(M_t)\} \quad (7.49)$$

This is equivalent to a barrier option with a stochastic barrier level being a function of the running maximum.

Assumption 2 Let f be continuous, strictly increasing on I . For all $x \in I$, $f(x) \in I$, $f(x) < x$.

One way of modelling a trailing stop is to consider a stochastic floor that moves based on the running maximum of the asset price. Here, the stochastic floor is automatically adjusted upward whenever the asset price moves to a new high. A trailing stop can be defined as the first time when the asset price X drops below $f(M)$, where M is the running maximum of X ¹⁹, and f is an increasing function such that $f(x) < x$ for all x in the support of X . This stopping time is related to the drawdown process and its first passage time. There exists several ways to model the floor function f , including,

- If $I \in \mathbb{R}$, we let $f(x) = x - a$, for some $a > 0$, gives the absolute drawdown floor, and ρ_f is the first time X falls from its maximum M by a units.
- If $I \in \mathbb{R}^+$, then $f(x) = (1 - \alpha)x$, for some $\alpha \in (0, 1)$, gives the percentage drawdown, and ρ_f is the first time X falls from its maximum M by $(100 \times \alpha)\%$

We let h be a real-valued increasing function on I such that $\{x \in I; h(x) > 0\} \neq \emptyset$. The optimal stopping problem is

$$v_f(x, x_M) = \sup_{\tau \in \mathcal{T}_f^T} E_{x, x_M} [e^{-q\tau} h(X_\tau) I_{\{\tau < \infty\}}]$$

where $\mathcal{T}_f^T$ is the set of all stopping times of X that stop no later than the trailing stop ρ_f . This value function yields the optimal trading strategy that involves buying a risky asset and selling it later while being protected by a trailing stop.

Remark 7.5 Compute the value function

$$v_f(t_i, x_i) = E_{x_i} [h(X_{t_i})], \forall (t_i, x_i)$$

and then take the supremum.

Let $g_f(x, x_M)$ be the expected discounted reward from waiting to sell at the trailing stop given by

$$g_f(x, x_M) = E_{x, x_M} [e^{-q\rho_f} h(X_{\rho_f}) I_{\{\rho_f < \infty\}}]$$

We can then quantify the gain in terms of expected discounted reward from liquidating earlier than the trailing stop time ρ_f . The early liquidation premium is defined by

$$p_f(x, x_M) = v_f(x, x_M) - g_f(x, x_M)$$

We have $p_f(x, x_M) \geq 0$ for all $x, x_M \in I$ with $x \leq x_M$. The concepts of early/delayed exercise/purchase premium have been studied in the pricing American options (see Carr et al. [1992]).

¹⁹ $M(t, T) = \max\{X_u \mid u \in [t, T]\}$

Proposition 2 *If floor functions f_1 and f_2 satisfy Assumption (2), and $f_1(x) \leq f_2(x)$ for all $x \in I$, then for every fixed $x \in I$ we have the inequalities:*

$$h(x) \leq v_{f_2}(x, x_M) \leq v_{f_1}(x, x_M) \leq \sup_{\tau \in \mathcal{T}} E_x[e^{-q\tau} h(X_\tau) I_{\{\tau < \infty\}}]$$

where $\mathcal{T}$ is the set of all stopping times of X .

We let b_f^* be a candidate threshold type strategy for liquidation before the trailing stop ρ_f . We assume a unique $b_f^* \geq x_0$ such that $b[f(x_M)] > x_M$ if and only if $x_M < b_f^*$.

Theorem 7.1 *Given Assumption (1), for $x, x_M \in I$ with $x \leq x_M$ and $x_M \geq b_f^*$, we have*

$$v_f(x, x_M) \equiv V_{f(x_M)}(x)$$

So the optimal stopping time is $\rho_f \wedge \tau_X^+(b[f(x_M)])$.

At last, in order to establish the optimality of the stopping rule $\rho_f \wedge \tau_X^+(b_f^*)$, we need to consider the remaining case $l < x \leq x_M < b_f^*$. It leads to two expectations that can be computed with standard calculation using excursion theory.

8 Generating synthetic data

8.1 Introduction

8.1.1 Overview

We saw in Section (7.2) that parametric models (and strategies) calibrated to a single market trajectory (or a few trajectories) were not robust to market changes, performing poorly on out of sample data. One solution is to calibrate these models on synthetic data covering a wide range market scenarios, hoping they will be more robust to market changes and perform more consistently. In order to use the Futuretesting engine, we need to generate two types of market data: (1) consistent synthetic market data closely related to the original time series with predictive capability, (2) scenarios for stress testing and risk management. In case (1), the simulated data must be statistically indistinguishable from a given original dataset. In case (2), we do not want to infer the true law of the historical price process, but simply generate market data with similar statistical properties. We can then stress this synthetic data to generate a large range of scenarios spanning the whole time-space domain.

Synthetic data generation can roughly be categorised into two distinct classes:

1. **Process-driven methods:** It derives synthetic data from a computational or mathematical model of an underlying psychical process. Define a stochastic model of the underlying asset processes (see Bakshi [1997], Bates [1996] [2000]), calibrate it to the observed option prices, and use it to simulate many trajectories. The main problem is that the chosen model may not be representative of the known characteristics of financial assets, such as long memory and GARCH effects. One can also add complexity and further parameters to the models, taking their weighted averages with weights calibrated to the current market conditions (see Duembgen et al. [2014]).
2. **Data-driven methods:** It has historically relied on modelling multivariate probability models such as Gaussian Mixture Models (GMM), Gaussian copulas, and Bayesian networks (see Russell et al. [2009]). In the case of deep learning, several model architectures have been investigated, with selforganising maps (SOMs), Variational Autoencoders (VAs) (see Kingma et al. [2014]), and Generative Adversarial Networks (GANs) (see Goodfellow et al. [2014]) being the most promising research trajectories. These methods are nonparametric, nonlinear estimators of latent variables. VAs learn how to represent high-dimensional observations in a low-dimensional space by maximising the likelihood of observing the given (original) data samples under the generated samples. SOMs apply competitive learning and use a neighbourhood function to preserve the topological properties of

the input space. GANs train two competing neural networks, where one network (the generator) is tasked with generating simulated observations from a distribution function, and the other network (the discriminator) is tasked with predicting the probability that the simulated observations are false given the true observed data.

Process-driven methods are widely used in banks because stochastic models calibrated to the observed prices tend to reflect forward-looking views of the market. However, no matter how complex the models are, they cannot generate realistic time series. On the other end, the strength of data-driven methods using Machine Learning (ML) is that it can uncover the hidden variables involved in complex systems such as the actual time series, and better reproduce their characteristics. As an example, Andreasen [2023] used Autoencoders to identify risk-neutral stochastic dynamics of interest rates yield curves in terms of factors. One can then use augmented data (via Signatures) to generate arbitrage-free dynamics for interest rates and fit the autoencoder on this synthetic data.

8.1.2 Generative models

The generation of synthetic data heavily depends on Generative Models. Hence, we give a very light description of some of these models.

Autoencoders (AEs) are neural networks used to learn efficient codings of unlabelled data in unsupervised learning (see Kramer [1991], Hinton et al. [1994]). An autoencoder has two main parts: (1) an encoder that maps the input into the code, and (2) a decoder that maps the code to a reconstruction of the input. The encoding is validated and refined by attempting to regenerate the input from the encoding. The search for the optimal autoencoder can be accomplished either with any mathematical optimisation technique, or by gradient descent. Traditionally applied to dimensionality reduction, or feature learning, the concept became widely used for learning generative models of data (see Kingma et al. [2014]). Modern variations of AEs have been applied to other tasks such as anomaly detection, image processing, drug discovery, machine translation, among others.

Generative models (GMs) are statistical models of the joint probability distribution function (pdf) of two observed random variables. In general, they focus on providing a parametric specification of a pdf and can be trained by maximising the log-likelihood. However, in these models, the pdf is generally intractable, requiring numerical approximation to compute the gradient. Recently, deep generative models (DGMs) were developed through the combination of generative models and deep neural networks by approximating the pdf via deep neural networks from learned samples. They represent the log-likelihood implicitly while generating samples from the desired distribution. For instance, generative stochastic networks can be trained with exact backpropagation, but rely on Markov chains for computation. Kingma et al. [2014] and Rezende et al. [2014] developed more general stochastic backpropagation rules, allowing one to backpropagate through Gaussian distributions with finite variance, and to backpropagate to the covariance parameter as well as the mean. They trained variational autoencoders (VAEs), which combines a differentiable generator network with a second neural network. The main idea being that given one random variable, z , with one distribution, we can create another random variable $x = g(z)$ with very different distribution. The deterministic function, g , is learned from the data through the function approximation capacity of neural networks. Let the joint probability distribution be the generative model we are sampling from. VAEs rely on the maximum likelihood by adjusting the generative process via backpropagation to maximise the (unknown) probability of observing the given training samples. The authors relied on the inherent property of neural networks as functional approximators and chose the initial distribution as a d -dimensional Gaussian. We optimise the model by using stochastic gradient descent to update the network parameters. Thus, VAEs require differentiation through the hidden units.

Schmidhuber [1992] proposed a predictability minimisation approach where two neural networks compete each others. For each representational unit, there is an adaptive predictor which tries to predict the unit from the remaining units. In turn, each unit tries to react to the environment such that it minimises its predictability. Following a similar approach, Goodfellow et al. [2014] proposed GAN models which eliminate Markov chains. GANs require differentiation through the visible units, and thus cannot model discrete data, while VAEs require differentiation through the

hidden units, and thus cannot have discrete latent variables. In GANs, one network produces a rich, high dimensional vector that is used as the input to another network, and attempts to choose an input that the other network does not know how to process. Here, the competition between the networks is the only training criterion. It is based on a minimax game with a value function that one agent seeks to maximise and the other seeks to minimise. The game terminates at a saddle point corresponding to a minimum for one player and a maximum for the other player. GANs are implicit generative models, which means that they do not explicitly model the likelihood function nor provide means for finding the latent variable corresponding to a given sample (see Mohamed et al. [2016]).

8.2 The two main methods

8.2.1 Process-driven methods

We saw in Section (8.1.1) that, depending on the task at hand, we could generate two types of synthetic data. In all cases, we start by specifying a particular stochastic differential equation (SDE), such as the diffusion process given in Equation (6.26), and then we select the model parameters. We can then simulate as many trajectories as required with a Monte Carlo engine.

- In the case (1), where the data must be statistically indistinguishable from the original dataset, we estimate statistically the model parameters on the original dataset (see Bakshi [1997], Bates [1996] [2000]), which we can then perturbate.
- In the case (2), where we need to generate scenarios, we assume a (discrete) range for the model parameters and consider all the combinations which we simulate in turn with a Monte Carlo engine.

Focussing on case (1), the simplest approach is to assume that returns are normally distributed, fit the model parameters on the original dataset, and perturbate these parameters. Given a realisation of the stock price $S = (S_1, \dots, S_N)$, where $T = N\Delta t$, we estimate statistically its first two sample moments (see Appendix (11.4)). We denote $\hat{\mu}$ and $\hat{\sigma}$ the sample mean and sample standard deviation, respectively.

First, we let the returns be Gaussian $N(\hat{\mu}, \hat{\sigma})$ with dynamics given by

$$\frac{dS_t}{S_t} = \hat{\mu}dt + \hat{\sigma}dW_t \quad (8.50)$$

In order to generate scenarios for the stock price, we perturbate the model parameters by assuming they follow a Gaussian distribution centred on their values. We modify the sample mean $\bar{\mu} \sim N(\hat{\mu}, \sigma^*)$ and the sample standard deviation $\bar{\sigma} \sim N(\hat{\sigma}, \sigma^*)$, where σ^* is a chosen volatility. The dynamics of returns become

$$\frac{dS_t}{S_t} = \bar{\mu}dt + \bar{\sigma}dW_t$$

We can also complexify the dynamics of returns by adding jumps and stochastic volatility (see Bates [2000]). That is,

$$\frac{dS_t}{S_{t-}} = b_t dt + dM_{P,S}(t)$$

with drift and martingale

$$b_t = \hat{\mu} + \sigma_t \lambda_t \text{ and } dM_{P,S}(t) = \sigma_t d\widehat{W}_S(t) + dM_S^{J,P^\gamma}(t)$$

where the drift is centred around the sample mean $\hat{\mu}$, $W_S^{P^\gamma}$ is a $\mathbb{P}^\gamma$ -Brownian motion and $M_S^{J,P^\gamma}(t)$ is an adjusted random measure for jump processes. As an example of a jump process, we can follow the approach by Merton (see Merton [1976]) and consider a Poisson process with positive jump size log-normally distributed. In addition, $\sigma_t = \sqrt{V_t}$ is a stochastic volatility and λ_t is a stochastic market price of risk (MPR). We can further complexify the

dynamics of returns by replacing the Brownian motions with fractal (or multi-fractal) Brownian motions. Let x_t be a OU-process with dynamics

$$dx_t = a_x(b_x - x_t)dt + \sigma_x dW_x(t), x_0 = x \quad (8.51)$$

where a_x is the speed of mean reversion, b_x is the infinite value of x , and $\sigma_x > 0$ is the instantaneous volatility. Let v_t be a square-root process with dynamics

$$dv_t = (b_v - a_v v_t)dt + \sigma_v \sqrt{v_t} dW_v(t), v_0 = v$$

where b_v, σ_v are positive and $a_v \in \mathbb{R}$. The infinite variance is $\bar{v} = \frac{b_v}{a_v}$. For $b_v > \frac{1}{2}\sigma_v$ the process v_t never reaches zero. In our setting, x can represent the spot rate and the market price of risk, that is, $x = r, \lambda$.

As an example, we can let the MPR be an Ornstein-Uhlenbeck (OU)-process and the variance be a square-root process. Hence, we get

$$d\lambda_t = a_\lambda(b_\lambda - \lambda_t)dt + \sigma_\lambda dW_\lambda(t), \lambda_0 = \lambda$$

We can then define a (discrete) range for the model parameters: $a_\lambda, b_\lambda, \sigma_\lambda, \lambda, a_v, b_v, \sigma_v$, and v . At last, we simulate the trajectories with a Monte Carlo engine.

8.2.2 Data-driven methods

We now consider the case (1) where we want to generate consistent synthetic market data closely related to the original time series. The simulated data must be statistically indistinguishable from a given original dataset. That is, the generated sequences of returns must be representative of the known characteristics of financial assets, such as serial autocorrelation, cross correlation, and volatility clustering. Further, a good model for generating time-series data should preserve temporal dynamics, in the sense that new sequences respect the original relationships between variables across time. However, the problem of defining metrics that can accurately describe the performance of financial data synthesis is still an open issue. Synthesised samples are usually validated by a stock market prediction algorithm (see Li et al. [2019], Sidi [2020]). Dogariu et al. [2018] tackled the problem of evaluating the quality of synthetic financial time-series by introducing qualitative and quantitative metrics, along with a portfolio trend prediction framework. When performing option pricing and risk management, we are interested in determining appropriate similarity metrics on the level of stochastic processes. Buehler et al. [2020a] discussed the generalisations of similarity metrics in the static case (marginal distributions) to the dynamic one (metrics on path space). Brigo et al. [2000] and Brigo [2019] gave examples of diffusion models which are statistically indistinguishable using samples on a fixed time (coarse) grid yet which have arbitrarily different European option prices on a finer grid. This suggest that you can not use a statistical approach to distinguish stochastic process, but that option prices will do the job (specially with path-dependent payoffs).

Data-driven methods can be used to generate synthetic data for (1) prediction, (2) data augmentation. In both cases, there are several possible approaches:

1. Prediction. Predict the next sample(s) in a sequence based on the available recent history. It can be modelled as a regression problem. Kim et al. [2019] considered financial time series prediction with stacked LSTM, attention networks, and weighted attention networks. A more general approach is to use Generative Models to generate the sequence of returns in such a way that it is representative of the known characteristics of financial assets, such as serial autocorrelation, cross correlation, and volatility clustering (see Yoon et al. [2019], Wiese [2019], Dogariu et al. [2018] [2021], Ni et al. [2020], Alonso et al. [2023], Kinlay [2023]). Eckerli et al. [2021] tested three known GAN architectures on financial time series, and the generated data was evaluated on its statistical properties, yielding solid results. One can also combine generative models, image representations of time series and data augmentation (see Hellermann et al. [2022]).

2. Time series augmentation. The goal is to generate a fixed number of consecutive samples at a time, in a dataset augmentation paradigm. One solution is to use some filter methods to decompose the time series and re-compose it with more or less noise. However, it is challenging to produce a synthetic series that looks substantially different from the original. This method can easily be combined with resampling trades techniques on the sub-samples. For instance, one can combine the bootstrap approach with shuffling time series which have been generated by Fourier transform to simulate new time series (see Schreiber et al. [2000]). One can also rely on transformation and bootstrapping methods (see Bergmeir et al. [2016]). There exists multiple time series augmentation methods such as Markov Chain Monte Carlo, local-global-trend algorithm, exponential smoothing (see Smyl et al. [2016]). Kondratyev et al. [2019] proposed a Restricted Boltzmann Machine (RBM) to build a powerful generator of synthetic market data that can replicate the probability distribution of the original market data. In the case of nonparametric Monte Carlo experiments, one can consider variational autoencoders (VAs), selforganising maps (SOMs), or generative adversarial networks (GANs). Note, when augmenting financial time series, GANs can only be trained on a very low number of data samples. Buehler et al. [2020a] [2020b] proposed to use the signature of a rough paths combined with Variational Autoencoder framework. They could generate new samples from a particularly small observable dataset to a high precision. Ni et al. [2021] proposed the SigWGAN to generate high fidelity samples. They used Rough Paths Theory to reduce the min-max formulation of GANs to an optimisation problem. Carreira [2023] introduced core signatures, which are subsets of the original signatures with the same size as log signatures and a faster and easier calculation algorithm. This method aims at reproducing enough of the original core signature to improve interpretability.
3. Aggregate all the above methods.

The main advantage of data-driven methods is that we can decide which types of scenarios best suit the application at hand (portfolio trading or risk management) and sample the data accordingly (see Kinlay [2023]). This can be done in a variety of ways such as setting a minimum required correlation between the synthetic and real price series, or designing a system of stratified sampling in which the desired outcomes are sampled according to a stipulated frequency distribution. That is, we decide on the level of closeness of the generated distribution of returns against that of the real return process.

8.3 Evaluating synthetic data

The problem of defining metrics that can accurately describe the performance of financial data synthesis is still an open issue. A good generative model for time-series data should preserve temporal dynamics (including the temporal correlations). One can approach the evaluation at three levels: (1) qualitatively, (2) quantitatively, and (3) via a predictive accuracy test (see Dogariu et al. [2018]).

8.3.1 Qualitative analysis

We can consider three criteria:

1. Diversity: Samples should be distributed to cover the real data. As a visual test, apply t-SNE and PCA analyses on both the original and synthetic datasets (flattening the temporal dimension).
2. Fidelity: Samples should be indistinguishable from the real data. Train a post-hoc time-series classification model to distinguish between sequences from the original and generated datasets. Report the classification error on the held-out test set.
3. Usefulness: Samples should be just as useful as the real data when used for the same predictive purposes. It should be able to capture conditional distributions over time. Using the synthetic dataset, we train a post-hoc sequence-prediction model to predict next-step temporal vectors over each input sequence. Then, we evaluate the trained model on the original dataset. Performance is measured in terms of the mean absolute error (MAE).

One approach is to target a number of essential stylised facts of the data in the training and in the evaluation of the generative procedure. To capture as much information as possible, we randomly select one batch of real data and one of synthesised data during each epoch and we compare several metrics. Some evaluation scores checking for some of the most relevant stylised facts include (but are not limited to) the following:

- **Central moments:** One way to determine whether two distributions are similar is to examine their statistical central moments. If the samples generated from two distributions have similar behaviour in terms of mean, variance, skew and kurtosis, it is a strong indicator that the two distributions might be similar.
- **Autocorrelation:** a well-known property of financial time series is that they do not possess a linear predictability, meaning that the autocorrelation of the returns is a diminishing function.
- **Correlation-, and cross-dependence scores:** To detect serial autocorrelation in the time-series, and analogously for multidimensional time-series, cross-correlation scores.
- **Heavy-tailed distribution:** Financial time series are known to exhibit a heavy-tailed behaviour, that is, their distribution presents a higher probability than a normal distribution of sampling very high and very low values. This translates in having a taller peak than normal distributions and thinner middle. When the two distributions do not overlap, it is a clear indicator that the model does not behave well. The properties of the underlying distribution that control the tail behaviour are higher order moments such as skewness and kurtosis:

$$\frac{1}{N_x} \sum_{j=1}^{N_x} \|Skew(x_j) - Skew([\hat{x}_j^{(1)}, \dots, \hat{x}_j^{(m)}])\| \text{ and } \frac{1}{N_x} \sum_{j=1}^{N_x} \|Kurt(x_j) - Kurt([\hat{x}_j^{(1)}, \dots, \hat{x}_j^{(m)}])\|$$

- **Autocorrelation:** Obtain box plots over an entire batch for both real and synthetic samples. Display an example of box plot of the autocorrelation function for different time lags. Financial time-series should have diminishing values of the autocorrelation starting with lags greater than 1 and the spread should be similar for real and synthetic cases.
- **Volatility clustering:** It means that changes (either high or low) come grouped in clusters. In other words, small changes tend to be followed by small changes and high changes tend to be followed by high changes, respectively. This can be assessed by examining the autocorrelation plot of non-linear transformations of returns such as squared returns or absolute returns where no significant autocorrelation can be observed. The plots for assessing whether the cluster volatility characteristic is met or not are similar to the ones for the autocorrelation.
- **Cumulative sum:** plotting the cumulative sum of the returns of any company should yield a non-monotonous curve with varying shapes. The cumulative sum for any company, at time t is defined as $c_t = \sum_{k=0}^t r_k$, where r_t the log return. The cumulative sum is plotted under the form of a line plot. Real samples display distinct forms of these lines for each company. Inflexion points usually occur at different times for different companies and only companies that have direct links (either they are owned by the same mother company or have very similar activity) can behave similarly but, even then, only for short periods of time (5 – 10 consecutive samples).
- **Trend ratios:** we define the trend ratio as

$$\alpha_t = \frac{C_t}{n_t}$$

where

- $C_t = \frac{c_t}{c_t - L_t} - 1$ is the stock's trend
- $n_t = \sum_{k=t-L_t}^t \left| \frac{c_k}{c_{t-1}} - 1 \right|$ is the noise corresponding to the same stock

and L_t is the trend lookback window (set to 20). Trend ratios are displayed under the form of line plots and behave similarly to the cumulative sum.

- conduct t-SNE and PCA analyses to visualise how well the generated distributions resemble the original distributions.

8.3.2 Quantitative analysis

Classical metrics on distributions are known as integral probability metrics (see Appendix (18)). For instance, distributional metrics target the cumulative distribution functions, in order to ensure that the distributions of the generated samples closely match the historical ones. The well known two-sample tests provide a flexible framework to compare the empirical distributions of two given data samples. Thus, we can adapt metrics that are generally used in the information theory field (see Appendix (17)) with metrics designed for generative models, but in other fields, such as image and speech processing. Some examples include:

- Kullback-Leibler divergence: It is a measure of how one probability distribution is different from another (see Kullback et al. [1951]). Since it is not a symmetric result, we set the real data distribution as reference and compute the metric for all synthetic data distributions. This measures the capability of the models to assign high probability to most realistic points. For continuous probabilities it is defined as:

$$D_{KL}(P||Q) = \int_{-\infty}^{\infty} p(x) \log\left(\frac{p(x)}{q(x)}\right) dx$$

where P and Q are distributions of two continuous random variables, and p and q denote the probability densities of P and Q , respectively. We are interested in the lowest values that we can attain.

- Jensen-Shanon divergence: It is symmetric, and it is a measure of similarity between two probability distributions (see Lin [1991]). It is defined as:

$$D_{JSD}(P||Q) = \frac{1}{2} D_{KL}(P||M) + \frac{1}{2} D_{KL}(Q||M)$$

where $M = \frac{1}{2}(P + Q)$ is the average of the two distributions, P and Q . Again, we are interested in the lowest values.

- Kolmogorov-Smirnov test statistics: The K-S statistic quantifies a distance between the empirical distribution function of the sample and the cumulative distribution function of the reference distribution, or between the empirical distribution functions of two samples (see Massey [1951]). Let F_n be the empirical distribution function for n i.i.d. ordered observations x_i . The K-S statistic for a given cumulative distribution function $F(x)$ is

$$D_n = \sup_x \|F_n(x) - F(x)\|$$

The statistic takes the largest absolute difference between the two distribution functions across all x values. By the Glivenko-Cantelli theorem, if the sample comes from distribution $F(x)$, then D_n converges to 0 almost surely in the limit when n goes to infinity. We can also test whether two underlying one-dimensional probability distributions differ. In this case, the K-S statistic becomes

$$D_{n,m} = \sup_x \|F_{1,n}(x) - F_{2,m}(x)\|$$

where $F_{1,n}(x)$ and $F_{2,m}(x)$ are the empirical distribution functions of the first and the second sample respectively. We run this on two randomly selected batches, one of real data and one of synthesized data and report

the K-S statistics. A low value means that we cannot reject the hypothesis that the two instances are from the same data distribution. With the increase in the number of samples for each instance we expect a more precise statistic. For large samples, the null hypothesis is rejected at level α if

$$D_{n,m} > c(\alpha) \sqrt{\frac{n+m}{n \cdot m}} \text{ where } c(\alpha) = \sqrt{-\ln \frac{\alpha}{2} \cdot \frac{1}{2}}$$

- **Earth Mover's distance:** It is the first Wasserstein distance between two 1D distributions (see Kantorovich [1960]). It can be seen as the minimum amount of work required to transform distribution u into distribution v , where work is measured as the amount of distribution weight that must be moved, multiplied by the distance it has to be moved. It is defined as:

$$l_1(u, v) = \inf_{\pi \in \Gamma(u, v)} \int_{\mathbb{R} \times \mathbb{R}} |x - y| d\pi(x, y)$$

where $\Gamma(u, v)$ is the set of (probability) distributions on $\mathbb{R} \times \mathbb{R}$ whose marginals are u and v on the first and second factors respectively.

8.4 Similarity metrics on path space

Generative models (GMs) approximate the underlying distribution implicitly, by drawing samples from the desired distribution and comparing their similarity to the original dataset based on some similarity metrics. In finance, the generation of synthetic data was used for the purpose of time series prediction. Hence, metrics were developed around accurately describing the prediction performance of financial data (see Section (8.3)). When computing the joint probability distribution function (pdf), the conditional density is related to the one-step forecast problem. Thus, when generating synthetic market data, we want to get full information on the distributions of the generated paths. However, when generating synthetic data for quantitative trading strategies and option pricing, we need to define similarity metrics on the level of stochastic processes. It is not enough to consider the distributional properties of the synthetic time series, and we must also account for the forward transition probabilities. To illustrate this statement, we are going to consider the problem of options pricing with machine learning. We will start by characterising the problem of defining similarity metrics in the path space and we will present a few solutions. We will then use forward starting options to introduce the notion of future density, and we will use the Kolmogorov-Compatibility condition to derive a probabilistic metric to identify similarity between two samples of stochastic processes.

8.4.1 Problems and solutions

8.4.1.1 Challenges for similarity metrics We presented in Section (8.3) some qualitative and quantitative static methods. However, when considering the path space, it is a challenge to determine whether two sets of sample paths originate from the same underlying distribution. Generalisations of the static case (similarity of marginal distributions) to the dynamic case (similarity metrics on path space) are not straightforward. Some of these challenges include (see Buehler et al. [2020a]):

- The potential non-universality of features to control for in the generated time-series. If the chosen set of optimisation objectives is bespoke to one application, the generated time series may not carry over easily to other applications.
- The underlying distribution of the data generating model is often not known explicitly in generative models and only controlled implicitly through the generated data samples and their similarity to the original data. Therefore, two-sample tests may be better suited as performance evaluation metrics than distributional metrics and divergences (such as KL-divergence, Fisher Information metric, Wasserstein metric). The latter may only be applicable after inferring from the available data sample to the underlying distribution.

- Non-continuous observation of the original data. Usually only discrete time observations of sample paths are available. While for classical models this is less problematic, for generative models an appropriate feature set needs to be specified. Modelling sample paths as a learning problem of vector-valued data is problematic for several reasons: the number, n , and position of observation time points might change from sample to sample. Also, in some applications (for example high-frequency data), n , can get very large. Finally, from a pricing and hedging perspective it is delicate (and often not sufficient) to match marginal distributions on a finite number of observation dates only (see Brigo et al. [2000] for discrete models). One can construct examples that are indistinguishable from one another on a finite set of marginals in the physical measure, but lead to arbitrarily different hedging strategies and option prices (see Brigo [2019]).
- Non-stationarity of the target distribution. Financial time-series are typically non-stationary and underlying distributions change with market conditions. It is beneficial to design a conditional version of the market generator, which allows to produce samples that are conditioned on specific market states.

8.4.1.2 Some performance metrics If the distributions of the original data generating process and the generated series were known explicitly, we could easily compute the quantitative metrics presented in Section (8.3.2). However, in practice, the underlying distribution of the original data is (often) only available through the given samples and not known explicitly. As an alternative, Gretton et al. [2012] determined if two samples were drawn from different distributions p and q based on the largest distance in expectations over predefined set of functions in the unit ball of a Reproducing Kernel Hilbert Space (RKHS), denoted $\mathcal{H}$. The maximum mean discrepancy (MMD) (see Appendix (18.1)) is

$$MMD[\mathcal{H}, p, q] = \sup_{f \in \mathcal{H}} (E_{X \sim p}[f(X)] - E_{Y \sim q}[f(Y)]) \quad (8.52)$$

where the space $\mathcal{H}$ is rich enough to distinguish between metrics p and q . This approach became popular in generative modelling. However, the data samples must be sufficiently large to accurately compute the expected values $E[f(\cdot)]$ with respect to the distributions p and q .

Chevyrev et al. [2018] developed a Maximum Mean Discrepancy based on two-sample that relies on a feature map from stochastic analysis called the signature of a path. This test is based on a notion that is reminiscent of the role of moment generating functions on path space and hence characterises the distribution of the stochastic process uniquely. Signatures also provide the right framework to match hedging objectives (see Lyons [1998], Armstrong et al. [2018]). The Ito-Lyons map (see Lyons [1998]) states that the mapping from the driving path controlled differential equation (CDE) to its solution is continuous (in fact Lipschitz) in a suitable rough path norm. Hence, if the signatures of two driving paths are close, then so are the solutions of the corresponding CDEs. Since the performance of a hedging strategy can be written in form of a CDE (the performance is an Ito integral of the hedging strategy against the price process) then if two price paths have similar signatures, they will have similar performance under the same hedging strategy.

8.4.2 A probabilistic metric

Since we cannot estimate the full signatures of stochastic process, we must truncate them to a finite dimension. As a result, the method which consists in combining Generative Models with Signatures struggles at capturing the temporal dependence of joint probability distributions induced by time-series data. In practices, we need to find the right truncation dimension that is large enough to capture the characteristics of the path, but small enough to avoid overfitting. Instead of using signatures to define similarity metrics in the path space, we propose a probabilistic approach which consists in computing conditional expectations in the time-space domain on a fixed grid. To illustrate our approach, we will use the pricing of options with machine learning (ML) in the entire time-space domain. In ML, they can be seen as value functions or action-value functions, depending on the chosen payoff.

When pricing path-dependent derivatives, we are concerned with capturing the properties of the true process in the entire time-space domain. For instance, when pricing forward starting options (see Appendix (15.2)), different mathematical models for the dynamics of the underlying assets having the same one-step marginal densities will have different forward densities. Hence, the probability distribution function (pdf) is not sufficient to fully characterise the true process (for path-dependent derivatives). We show that by augmenting the information given by the pdf with forward conditional densities we can characterise a stochastic process on a fixed time-space grid. Hence, we propose to use the prices of all European options and all forward starting options on that grid to fully characterise the true process.

8.4.2.1 The setting Given the time period $[0, T]$, assume that there exist a true continuous process $\{X_t; t \in [0, T]\}$. We discretise that process on a fixed time grid $\mathcal{T} = \{t_0 = 0, t_1, \dots, t_n = T\}$, where n is a given integer defining the time step $T = n \times \Delta t$. We let the time set starting at time t_i be denoted $\mathcal{T}_i = \{t_i, t_{i+1}, \dots, t_n = T\}$, where $i \in I$ and $I = \{0, \dots, n\}$. Let $X = (X_1, X_2, \dots, X_n)$ be an n -dimensional vector of random variables. We want to test the sample path $Y = (Y_1, Y_2, \dots, Y_n)$ to see if it is similar to X . To do so, we propose to compute all the options in the time set $\mathcal{T}$ using X as underlying asset. We then repeat this step using Y as underlying asset. Given a distance metric, we can then measure the closeness of the options priced with Y against the ones priced with X . Obviously, this similarity metric is only valid on the discrete time set $\mathcal{T}$. We can refine similarity by increasing the integer n .

We define the option price at time t_i for the maturity T_j , $j > i$, and strike K as

$$C(t_i, T_j; K) = E[h(X_j) | X_{\leq i}], \forall i < j \text{ and } j \in \mathcal{T}$$

where h is the payoff function. For simplicity, we don't distinguish all the strikes in space and denote $C(t_i, T_j) = C(t_i, T_j; K)$. We can obtain a probability by letting the payoff function be $h = I_{\{A\}}$ for the event A . For $k \in I$, we consider the trading time t_k and let A_k be a one dimensional event involving the random variable X_k . We need to compute all the option prices in Table (1).

#	t_k	t_{k+1}	t_{k+2}		t_{n-1}
T_{k+1}	$C(t_k, T_{k+1})$	$\dots$			
T_{k+2}	$C(t_k, T_{k+2})$	$C(t_{k+1}, T_{k+2})$	$\dots$		
T_{k+3}	$C(t_k, T_{k+3})$	$C(t_{k+1}, T_{k+3})$	$C(t_{k+2}, T_{k+3})$	$\dots$	
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
T_n	$C(t_k, T_n)$	$C(t_{k+1}, T_n)$	$C(t_{k+2}, T_n)$	$\dots$	$C(t_{n-1}, T_n)$

Table 1: All the option prices

We let $C_X(t, T)$ and $C_Y(t, T)$ be vectors of option prices of size n on the X -path and the Y -path, respectively. That is, for $j > i$, $C_X(t_i, T_j)$ is an option price on the X -path and $C_Y(t_i, T_j)$ is an option price on the Y -path.

8.4.2.2 The problem Recall, the conditional probability mass function (pmf) of $(X_{k+1}, X_{k+2}, \dots, X_n)$ given $(X_1, X_2, \dots, X_k)$ is

$$f_{X_{k+1}, \dots, X_n | X_{\leq k}}(x_{k+1}, \dots, x_n | x_{\leq k}) = \prod_{i=k}^{n-1} f_{X_{i+1} | X_{\leq i}}(x_{i+1} | x_{\leq i})$$

Note, the random variables X_i , for $i = 1, \dots, n$, are drawn one at a time in a sequence. Thus, the conditional density $f_{X_{i+1} | X_{\leq i}}$ is related to the one-step forecast problem. For instance, in the Autoencoder, the one-step forecast is given by

$$\hat{X}_{k+1} = \psi(\phi(X_1, X_2, \dots, X_k; M); W)$$

where ϕ is the encoder and ψ is the decoder with weight matrices M and W , respectively. For $h(X_j) = \mathcal{H}(X_j - K)$, where $\mathcal{H}(\cdot)$ is the heaviside function, we get $C(t_i, T_j) = P(X_j | X_{\leq i})$, for $j > i$ and $j \in \mathcal{T}_k$. Therefore, we can express the conditional density $f_{X_{i+1} | X_{\leq i}}$ in terms of derivatives as $C(t_i, T_{i+1}) = f_{X_{i+1} | X_{\leq i}}(x_{i+1} | x_{\leq i})$. We can therefore rewrite the conditional pmf as

$$f_{X_{k+1}, \dots, X_n | X_{\leq k}}(x_{k+1}, \dots, x_n | x_{\leq k}) = \prod_{i=k}^{n-1} C(t_i, T_{i+1})$$

We see that we just need the derivatives on the diagonal of Table (1) to compute the conditional pmf. It corresponds to the one-step forecast problem. However, in finance, we need to make p -step forecasts when computing forward start options. Different mathematical models for the dynamics of $(X_t; t \in [0, T])$ having the same one-step marginal densities will have different forward densities. For instance, the local volatility model and the Heston model do not lead to the same dynamics of the IVS. Thus, in this problem, the conditional pmf is not sufficient to fully characterise the true process X_t .

8.4.2.3 Characterising the true process On way of characterising the process $(X_t)_{t \in [0, T]}$ is to price all European options and all forward starting options in $\mathcal{T}$. From the Kolmogorov-Compatibility condition (see Definition (15.1)), there is an infinity of solutions for each forward probability density function (pdf) $P(X_j | X_i)$ for all $j > i$ and $j \in \mathcal{T}$. Given the option prices in Table (1), we let $\Theta(X_k; K, T_{k+p} - t_k; r, \sigma(\cdot))$ be the density operator for the state X_k at time t_k to go to the state K at maturity T_{k+p} .

In that setting, the probability of going from state X_k at time t_k to state K at maturity T_{k+2} is

$$\Theta(X_k; K, T_{k+2} - t_k; r, \sigma(\cdot)) = \int \Theta(X_k; K, t_{k+1} - t_k; r, \sigma(\cdot)) \Theta(x_1; K, T_{k+2} - t_{k+1}; r, \sigma(\cdot)) d\lambda(x_1)$$

where $\Theta(X_k; K, t_{k+1} - t_k; r, \sigma(\cdot))$ is given and $\Theta(x_1; K, T_{k+2} - t_{k+1}; r, \sigma(\cdot))$ is unknown. Similarly, the probability of going from state X_k at time t_k to state K at maturity T_{k+3} is

$$\Theta(X_k; K, T_{k+3} - t_k; r, \sigma(\cdot)) = \int \Theta(X_k; K, t_{k+2} - t_k; r, \sigma(\cdot)) \Theta(x_2; K, T_{k+3} - t_{k+2}; r, \sigma(\cdot)) d\lambda(x_2)$$

where $\Theta(x_2; K, T_{k+3} - t_{k+2}; r, \sigma(\cdot))$ is unknown. By recursion, the probability of going from state X_k at time t_k to state K at maturity T_n is

$$\Theta(X_k; K, T_n - t_k; r, \sigma(\cdot)) = \int \Theta(X_k; K, t_{n-1} - t_k; r, \sigma(\cdot)) \Theta(x_{n-1}; K, T_n - t_{n-1}; r, \sigma(\cdot)) d\lambda(x_{n-1})$$

which is

$$\Theta(X_k; K, T_n - t_k; r, \sigma(\cdot)) = \int \int \dots \left(\int \Theta(X_k; K, t_{k+1} - t_k; r, \sigma(\cdot)) \Theta(x_1; K, T_{k+2} - t_{k+1}; r, \sigma(\cdot)) d\lambda(x_1) \right) \Theta(x_{n-1}; K, T_n - t_{n-1}; r, \sigma(\cdot)) d\lambda(x_{n-1})$$

We see that we can characterise the true process X_t on the time grid $\mathcal{T}$ by inferring the conditional densities

$$\Theta(X_k; K, t_{k+1} - t_k; r, \sigma(\cdot)), \Theta(x_1; K, T_{k+2} - t_{k+1}; r, \sigma(\cdot)), \dots, \Theta(x_{n-1}; K, T_n - t_{n-1}; r, \sigma(\cdot))$$

from the data sample. In our setting, we do it by computing all the options $C(t_i, T_j)$, for all i and all $j \in \mathcal{T}_k$ in Table (1) with Machine Learning. That is, we need to price all the European options plus all the forward start options seen from time t_k , $k \in I$.

8.4.2.4 The minimum price discrepancy In the spirit of the maximum mean discrepancy (MMD) given in Equation (8.52), we propose the Minimum Price Discrepancy (MPD) as a probabilistic similarity metric on path space. It is define as

$$MPD(X, Y) = \|C_X(t, T) - C_Y(t, T)\|_2^2 = \sum_{i=0}^{n-1} \sum_{j=i+1}^n (C_X(t_i, T_j) - C_Y(t_i, T_j))^2$$

where $C_X(t, T)$ and $C_Y(t, T)$ are vectors of option prices on the X -path and the Y -path, respectively. We get similarity on path space when $MPD(X, Y) < \epsilon$, for some positive constant ϵ . We can use the MPD to generate synthetic data in an Autoencoder as follows:

1. Let the encoder generate data $Y \in \mathcal{Y}$ and minimise the MPD in the loss function of the decoder.
2. Keep generating new data in $[1]$ until the MPD is minimised.

8.5 Some limitations

In order to generate synthetic data that is statistically indistinguishable from the real data, we tested several popular approaches. First, we briefly explain the problems related to using Signatures as features in neural networks (NNs) and give a general view of some challenges that still need to be solved. Then, we present some of the models tested and describe the problems encountered.

8.5.1 The problems explained

The development of Rough Path Theory (see Lyons [1998], [2007]) provided a framework for understanding differential systems driven by irregular input signals. By appropriately compensating Riemann sums, the pathwise convergence can be achieved consistently across all possible sequences of approximations. Further, the assumption on the regularity of the stock price can be weakened. Rough Path Theory achieves the stated convergence by deploying higher-order expansions of integrators and integrands. The compensation term is the technical ingredient that distinguishes the new integrals from the classical Ito integrals. If the compensation is not incorporated, then the Riemann sums cannot converge pathwise along arbitrary sequences of partitions with vanishing mesh-size.

When generating synthetic data with Generative Models (GMs), we need an efficient encoding (feature map) of financial time series (FTS) to enhance the training of the models. Rough paths and signatures provide a means of efficiently encoding FTS. In the VA and GAN models used to generate synthetic data with Signatures as input features, the discriminator compute the signatures and take their expected values. However, this can be very slow and often one cannot compute the signatures to a high level. One must therefore come up with a way of reducing the rough paths while preserving convergence of the integral. Several approaches have been proposed. Since the full signature (even truncated) is still redundant, it is usual to calculate the log signature (a more compressed representation of the signatures, but harder to calculate), which has a lower size. An asset-price process arising from a diffusion model may be associated a rough-path. In diffusion models, the quadratic variation is a well-defined pathwise notion. Armstrong et al. [2021] proposed the **enhanced price path** as a reduced rough path which is obtained by enhancing a continuous pricing signal with a specification for its rough bracket (the enhancer). This rough bracket provides the appropriate analogue of quadratic variation for the asset price. It is a pathwise measure, as opposed to the historical volatility which is probabilistic. Carreira [2023] introduced **core signatures**, which are subsets of the original signatures with the same size as log signatures and a faster and easier calculation algorithm. This method aims at reproducing enough of the original core signature to improve interpretability. He proposed an inversion algorithm using Chen's Identity to find which concatenation of two (or three) signatures of line segments can generate the same core signature (up to a truncation level).

8.5.2 Empirical results

We present results on a few Generative Models. Some of the models tested are:

- Buehler et al. [2020a] proposed a data-driven market simulator for small data environments. They used the signature of a rough paths combined with VA. Following this approach we used Conditional Autoencoder (CAE) to generate synthetic signatures from real signatures. However, reconstructing the original path from the signature is tricky. We tested Genetic Algorithm (GA) and Neural Network (NN) to find the paths that have the closest signatures to the generated signatures. Both methods were not very successful. We found that the NN overfit the data and the GA was not very stable.
- Ni et al. [2020] [2021] combined the Logsig-RNN models with the signature W_1 metric to generate high fidelity samples. The generator was not particularly stable and quite sensitive to the choice of training data such as length of the paths, volatility of the paths, etc.

In general, these models struggle at capturing the temporal dependence of joint probability distributions induced by time-series data. Some general challenges that still need to be solved are:

1. Validation of generated data: Since it is a challenge to check the statistical characteristics of generated data against realised data (see Section (8.4)), we used the Geometric Brownian Motion (GBM) as our original dataset. We then used GMs to generate synthetic data and tested the quality of the generated data against the GBM process. Tests like Shapiro-Wilk for normality, chi-square test for independence were used to validate the generated data.
2. Finding the right depth of the signature: Theoretically, the signature of a path is infinite dimensional. However, in practice we need to truncate the signature to a finite dimension. Further, the truncated signature should converge to the original signature as the truncation dimension goes to infinity. In practice, we need to find the right truncation dimension that is large enough to capture the characteristics of the path, but small enough to avoid overfitting.
3. Length of the training dataset: The model trained on paths that exceed 100 points (such as 100 days) often produce worse results than paths that are shorter than 100 days. Further, the volatility of the training data also affects the performance of the model.

9 Futuretesting strategies

9.1 Introduction

9.1.1 Overview

Given an investment strategy S that invests in I opportunities over a fixed period of time, the trading rules provide the logic for entering and exiting each opportunity based on conditions defined through reaching values and thresholds for profit-taking and stop-losses. In addition, these strategies can also rely on mathematical models to be successful. These models are parametric and must be calibrated along with the trading rules to maximise profit. It leads to the optimum trading problems (OTPs) introduced in Section (7.1.2). If we choose to build quantitative strategies with a Backtesting engine (see Section (7.2)), we compute moving averages (MAs) along the realised trajectories, devise some trading rules, and calibrate the model parameters via historical simulations. That is, we compute the profit and loss (PnL) for every position along the market realisation and use a particular metric to compute the strategy performance over that period of time. We can then run a backtest for each combination of parameters and find the best parameters for strategy S by maximising its performance (the calibration). In general, the number of parameters for the trading rules and financial models is very large, leading to a huge number of combinations to be tested. Further, since we are applying the test on the same data multiple times, this approach is prone to false positives, leading to

biased conclusion. This approach leads to the problem of backtest overfitting, where the parameters target specific observations and become unfit for the future.

Bailey et al. [2013] proposed an approach to perform a backtest on a single market trajectory that tries to avoid backtesting overfit (see Section (7.2)). It only requires time series of backtested performance (no calibration of parameters) and constructs an abstract probability space in which the sample space consists of pairs of in-sample (IS) and out-of-sample (OOS) test results. For a given strategy, the probability of backtest overfitting (PBO) is evaluated as the conditional probability that this strategy underperforms the median OOS while remaining optimal IS. In continuous trading, another approach consists in modelling the underlying variable by generating pseudorandom scenarios and measuring the performance of the resulting investment strategy for those scenarios. It relies on generating a distribution of outcomes, rather than relying on a single OOS performance estimate. For instance, assuming security prices follow an Ornstein-Uhlenbeck process, Bertram [2009] and Zeng et al. [2014] examined the problem of choosing an optimal strategy by expressing it in terms of the first-passage time of the process (see Section (7.4)). Carr et al. [2014] also let the security prices follow an OU-process and proposed a procedure for determining the optimal trading rule (OTR) without running alternative model configurations through a backtest engine (see Section (7.5)). However, the models generating random series of the underlying variable may also be overfit, or may not contain all relevant statistical features.

Following the continuous trading approach, we want to compute probabilistic Risk/Return profiles for a set of trading strategies everywhere in the Time-Space domain. To do so, we need to test different quantitative strategies and trading rules under different market scenarios in a Futuretesting engine on realistic theoretical data. We choose to perform multiple futuretests on different datasets and under different scenarios to obtain a distribution of performance metrics. Given some accuracy measures, such as the mean-variance criterion, we can then find the optimum strategy maximising returns and minimising risk. Finally, we test this optimum strategy on real market data on different securities and over different periods of time. In Section (8), we discussed two main approaches for generating realistic synthetic data: (1) Simulate synthetic returns under different mathematical models. To make sure that the generated returns contain all relevant statistical features, we consider factor models with stochastic drift, stochastic volatility, and jumps. Some well known price models include Black-Scholes-Merton (BSM) model, Heston model, Merton model, Bates model. Since the returns must be generated in the historical measure, we need to introduce the market price of risk in the drift. These price models are parametric with model parameters to be estimated. We can try to statistically estimate them on real market data, or alternatively consider a family of sets of model parameters. However, the models must be very complex to generate realistic time series. (2) Use Machine Learning (ML) models to simulate synthetic returns, since they can uncover the hidden variables involved in complex systems, such as the actual time series, and better reproduce their characteristics.

In this section, we present our probabilistic framework, detail the creation of market scenarios, present the tools used for measuring the performance of the strategies, and provide quantitative trading algorithms. To conclude, we consider a stochastic model to generate market returns and perform a Futuretesting on a few quantitative strategies and display the results.

9.1.2 High level description

Some of the tasks we can perform with the Futuretesting engine include:

- Devise and calibrate quantitative strategies (QS) by maximising their performance in a realistic environment
- Test quantitative trading strategies in a stressed environment
- Risk management of trading strategies: mean-variance analysis, VaR, etc.
- Portfolio allocation for trading strategies

- Risk management of derivatives: market and credit risk

We should consider two types of data (see Section (8)):

1. Real market data: history of realised time series.
2. Simulate synthetic market returns under different scenarios. We model the underlying variable by generating pseudorandom scenarios and measure the performance of the resulting investment strategy for those scenarios. There are several approaches for simulating synthetic returns:
 - Statistical models: ARMA, FARIMA, GARCH, E-Garch, etc. Calibrate model parameters on real market data. See Appendix (14).
 - Stochastic processes: GBM, Heston, Merton-Heston, etc. Use different combinations of model parameters. See Section (8.2.1)
 - Generative models: VAE, GAN, etc. See Section (8.2.2).

The process is as follows: Define measures of performance (see Section (9.2)) and perform multiple backtests on different datasets and under different scenarios (see Section (9.3)). Obtain distributions of performance. We can then validate good strategies and discard the bad ones. The benefits are as follows:

- Walk-forward, stress testing
- Reduce risk of: overfitting, outliers, storytelling

The steps needed to perform multiple backtests on different datasets and under different scenarios are:

- Generate synthetic data: (1) process-driven methods, (2) data-driven methods. See Section (8.2).
- Derivation of quantitative strategies: see Sections (3) and (6).
- Implementation and testing of the system to generate synthetic returns
- Implementation and testing of quantitative strategies. See Section (9.3).
- Calibrating parameters to maximise profit and minimise risk. See Section (9.4).
- Optimise portfolio allocation

9.2 Measuring performance

9.2.1 The setting

9.2.1.1 The trajectories We let $x(t, \omega)$ be a real random function (r.r.f.), where $\omega \in \Omega$ and $t \in T$. By fixing a particular ω , we can visualise that random function, getting its trajectory. On the other hand, for each $t \in T$ fixed, $x_t = x(t, \cdot)$ is a random variable defined on Ω . We assume that we have a set of m (possibly independent) trajectories denoted $\mathcal{X} = (x_{0,p} = x, x_{1,p}, x_{2,p}, \dots, x_{n,p})_{p=1}^m$, with $(n+1)$ observations, all starting from the initial state x and ending at time $T = n\Delta t$. The p -th trajectory (or history) is denoted $e_p = \{x_{0,p}, x_{1,p}, x_{2,p}, \dots, x_{n,p}\}$. At each time t (or observation), we observe the values $\{x_{t,1}, x_{t,2}, \dots, x_{t,m}\}$ in Ω .

We let the underlying asset $\{x_t; t \in T\}$ be a real-valued stochastic process, and define the discrete time as $\{t_1, \dots, t_n\} \subset T$, where $t_1 < t_2 < \dots < t_n$. The corresponding one-period simple net return (see Appendix (12.1)) seen at time t_k is given by

$$r(t_k, t_{k+1}) = \frac{x_{t_{k+1}}}{x_{t_k}} - 1$$

The logarithmic return is

$$r_L(t_k, t_{k+1}) = \ln(1 + r(t_k, t_{k+1})) = \ln \frac{x_{t_{k+1}}}{x_{t_k}}$$

We assume that the process x takes values in the range $[x_{min}, x_{max}]$. At time t_k , we let $B_{i,k}$, for $i = 1, \dots, N - 1$, be the bins spanning the range $[x_{min}, x_{max}]$. That is, when in state i (or bin $B_{i,k}$), the process x_k is in the range

$$B_{i,k} = [x_{min} + (i - \frac{1}{2})\Delta, x_{min} + (i + \frac{1}{2})\Delta)$$

for $i = 1, \dots, N - 1$.

9.2.1.2 The returns We are given a family of investment strategies $\mathcal{S} = \{S_1, \dots, S_I\}$ consisting of a family of set of trading model parameters $\mathcal{M} = \{M_1, \dots, M_I\}$, where M_i is a set of trading model parameters, and a family of set of market parameters $\mathcal{P} = \{P_1, \dots, P_J\}$, where P_j is a set of market parameters. For each pair (i, j) of trading model parameters and market parameters, we generate m trajectories from $\mathcal{X}$, and denote the associated set $\mathcal{X}_{i,j}$. In that set, for each trajectory e_p , we compute the returns of the strategy. Following Section (7.1.1), given the set of opportunities $\mathcal{I}_p = (1, \dots, I_p)$ over a fixed trading period T (a discrete sequence of individual trades), consider an investment strategy S_i that invests in I_p opportunities. From Equation (7.40), the MtM profit/loss (PnL) of opportunity i_p after t transactions is

$$\pi_{i_p,t}^p = \pm w_{i_p} (P_{i_p,t}^p - P_{i_p,0}^p \mp c)$$

where w_{i_p} is the number of units of the underlying asset, c is a transaction cost per unit, and $P_{i_p,t}^p$ is the market price at time t . Summing up all the PnL on the p -th trajectory, we get the return $r_{i,j}^p$. We can then compute the mean return on the set $\mathcal{X}_{i,j}$ (over the m -trajectories), denoted $\bar{r}_{i,j}$ and its associated variance $\sigma_{i,j}^2$.

9.2.2 The performance metrics

Given a family of set of trading model parameters $\mathcal{M}$ and a family of set of market parameters $\mathcal{P}$, we need to select a performance metric (see Appendix (??)). We can then obtain a distribution of performance metric for every fixed set of model parameters.

For a fixed set of model parameters M_i , we compute the mean returns, variance, and downside variance of all the sets of market parameters, getting $\bar{r}_i = \{\bar{r}_{i,1}, \bar{r}_{i,2}, \dots, \bar{r}_{i,J}\}$, $\sigma_i^2 = \{\sigma_{i,1}^2, \sigma_{i,2}^2, \dots, \sigma_{i,J}^2\}$, $\sigma_i^{2,-} = \{\sigma_{i,1}^{2,-}, \sigma_{i,2}^{2,-}, \dots, \sigma_{i,J}^{2,-}\}$, respectively.

For a fixed set of model parameters M_i , we can generate a histogram (distribution) of

- the mean returns $\bar{r}_{i,j}$, $j = 1, \dots, J$, and compute its mean $M_{\bar{r}_i}$ and its dispersion $\Sigma_{\bar{r}_i}$. We get

$$M_{\bar{r}_i} = \frac{1}{J} \sum_{j=1}^J \bar{r}_{i,j} \text{ and } \Sigma_{\bar{r}_i}^2 = \frac{1}{J} \sum_{j=1}^J (M_{\bar{r}_i} - \bar{r}_{i,j})^2$$

We can compute a Sharpe ratio of the mean returns $\frac{M_{\bar{r}_i}}{\Sigma_{\bar{r}_i}}$.

- the variance $\sigma_{i,j}^2$, $j = 1, \dots, J$, and compute its mean $M_{\sigma_i^2}$ and its dispersion $\Sigma_{\sigma_i^2}$. We get

$$M_{\sigma_i^2} = \frac{1}{J} \sum_{j=1}^J \sigma_{i,j}^2 \text{ and } \Sigma_{\sigma_i^2}^2 = \frac{1}{J} \sum_{j=1}^J (M_{\sigma_i^2} - \sigma_{i,j}^2)^2$$

- the downside variance $\sigma_{i,j}^{2,-}$, $j = 1, \dots, J$, and compute its mean $M_{\sigma_i^{2,-}}$ and its dispersion $\Sigma_{\sigma_i^{2,-}}$. We get

$$M_{\sigma_i^{2,-}} = \frac{1}{J} \sum_{j=1}^J \sigma_{i,j}^{2,-} \text{ and } \Sigma_{\sigma_i^{2,-}} = \frac{1}{J} \sum_{j=1}^J (M_{\sigma_i^{2,-}} - \sigma_{i,j}^{2,-})^2$$

For a fixed set of model parameters M_i , we can also compute some performance metrics by using these moments:

- The Sharpe ratio $\{M_{i,1}^{SR}, M_{i,2}^{SR}, \dots, M_{i,J}^{SR}\}$, where $M_{i,j}^{SR} = \frac{\bar{r}_{i,j}}{\sigma_{i,j}}$. We can generate a histogram of Sharpe ratio and compute the average $\bar{M}_i^{SR}$ for the i -th index and its dispersion $\Sigma_{\bar{M}_i^{SR}}$. We get

$$\bar{M}_i^{SR} = \frac{1}{J} \sum_{j=1}^J M_{i,j}^{SR} \text{ and } \Sigma_{\bar{M}_i^{SR}} = \frac{1}{J} \sum_{j=1}^J (\bar{M}_i^{SR} - M_{i,j}^{SR})^2$$

- The Sortino ratio $\{M_{i,1}^S, M_{i,2}^S, \dots, M_{i,J}^S\}$, where $M_{i,j}^S = \frac{\bar{r}_{i,j}}{\sigma_{i,j}^-}$. We can generate a histogram of Sortino ratio and compute the average $\bar{M}_i^S$ for the i -th index and its dispersion $\Sigma_{\bar{M}_i^S}$. We get

$$\bar{M}_i^S = \frac{1}{J} \sum_{j=1}^J M_{i,j}^S \text{ and } \Sigma_{\bar{M}_i^S} = \frac{1}{J} \sum_{j=1}^J (\bar{M}_i^S - M_{i,j}^S)^2$$

We repeat this process for all the model parameters M_i , $i = 1, \dots, I$, and select the set of model parameters M_i with the highest average performance metric $\bar{M}_i$ and the lowest dispersion $\Sigma_{\bar{M}_i}$. That is, we can compute a performance metric of the performance metrics $\frac{\bar{M}_i}{\Sigma_{\bar{M}_i}}$ for all i .

For a fixed set of model parameters M_i , we want to adjust the mean returns with their risk level. We call it the risk adjusted mean return, denoted $\bar{r}_{i,j}^*$. We can normalise the mean returns as follows:

- In the case of the Sharpe ratio: we get $\bar{M}_i^{SR} = \frac{1}{J} \sum_{j=1}^J \frac{\bar{r}_{i,j}}{\sigma_{i,j}}$. This is equivalent to $\bar{M}_i^{SR} = \frac{1}{J} \sum_{j=1}^J w_j \bar{r}_{i,j}$, where $w_j = \frac{1}{\sigma_{i,j}}$. For $\bar{w} = \sum_{j=1}^J w_j$, we get the normalised weight $w_j^* = \frac{w_j}{\bar{w}} = \frac{\sum_{j=1}^J \sigma_{i,j}}{\sigma_{i,j}}$. Thus, we get

$$\bar{M}_i^{SR,*} = \frac{1}{J} \sum_{j=1}^J w_j^* \bar{r}_{i,j} \text{ and } \Sigma_{\bar{M}_i^{SR,*}} = \frac{1}{J} \sum_{j=1}^J (\bar{M}_i^{SR,*} - w_j^* \bar{r}_{i,j})^2$$

- In the case of the Sortino ratio: we get $\bar{M}_i^{SR} = \frac{1}{J} \sum_{j=1}^J \frac{\bar{r}_{i,j}}{\sigma_{i,j}^-}$. This is equivalent to $\bar{M}_i^S = \frac{1}{J} \sum_{j=1}^J w_j^- \bar{r}_{i,j}$, where $w_j^- = \frac{1}{\sigma_{i,j}^-}$. For $\bar{w} = \sum_{j=1}^J w_j^-$, we get the normalised weight $w_j^{-,*} = \frac{w_j^-}{\bar{w}} = \frac{\sum_{j=1}^J \sigma_{i,j}^-}{\sigma_{i,j}^-}$. Thus, we get

$$\bar{M}_i^{S,*} = \frac{1}{J} \sum_{j=1}^J w_j^{-,*} \bar{r}_{i,j} \text{ and } \Sigma_{\bar{M}_i^{S,*}} = \frac{1}{J} \sum_{j=1}^J (\bar{M}_i^{S,*} - w_j^{-,*} \bar{r}_{i,j})^2$$

More generally, we can define the weighted mean return $\bar{r}_{i,j}^* = w_j^* \bar{r}_{i,j}$ for some normalised weights w_j^* , $j = 1, \dots, J$, and compute its mean $M_{\bar{r}_i^*} = \frac{1}{J} \sum_{j=1}^J \bar{r}_{i,j}^*$ and its dispersion $\Sigma_{\bar{r}_i^*}$. We can then define a simple performance measure by taking their ratio

$$\frac{M_{\bar{r}_i^*}}{\Sigma_{\bar{r}_i^*}}$$

9.3 The QS algorithms

We briefly present two types of algorithms for quantitative strategies: (1) an algorithm using moving averages (see Section (3)), (2) an algorithm using option prices (see Section (6)).

9.3.1 Settings

We consider the case of day trading (see Section (3.1.2)), where buying and selling occur until the end of the trading day. Hence, we can enter the trade at the start of the day and must close trading positions every day at market close (MC). As a result, the unit of time for the backtest is the day, and the user must specify a start date D_S and an end date D_E . We let $\mathcal{D} = \{D_1, \dots, D_N\}$ be the set of trading days for the days $D_i, i = 1, \dots, N$, t_0 is the market opening time (MOT) and T_C is the market close time (MCT) for these days. Following Section (7.1.1), we let $\mathcal{I}_i = (1, \dots, I_i)$ be the set of opportunities over the i th day D_i (a discrete sequence of individual trades), and consider an investment strategy S that invests in I_i opportunities (or trades). For each opportunity $k \in \mathcal{I}_i$, the strategy S takes a position of $w_k \in (-\infty, \infty)$ units of the security.

We consider two approaches to devise Quantitative Strategies:

1. We use Technical Indicators (TI) presented in Section (3).
2. We use the results on option pricing presented in Section (6).

In the approach (2), at time t , given a stock price S_t we let $C(T, K)$ and $P(T, K)$ be a call price and put price, respectively, with strike K and maturity T . At maturity T , we assume the existence of a strike chain $\mathcal{K} = \{K_1, \dots, K_n\}$ uniformly spaced, where the change in strike is denoted ΔK . Every day $D_i, i = 1, \dots, N$, we consider the time interval $[t_0, T_O]$ for the option chain, where $T_O > t_0$ is a pre-defined fixed option maturity.

We now need to simulate trajectories of the underlying security by using a particular jump-diffusion price model. We need to define a trading frequency, that is, a unit bar of minutes, for the trading. We divide the day D_i in J observed values uniformly allocated in time with time difference δ . That is, $t_j = j\delta$ and $t_{j+1} = (j+1)\delta$, such that $\Delta t_j = t_{j+1} - t_j = \delta$ and $\Delta W_{t_j} = W(t_{j+1}) - W(t_j) = W(\delta)$. For simplicity, we let the stock price S_t follows a $GBM(\mu_t, \sigma_t)$ (see Equation (8.50)). In that setting, the stock price for the time $t_{j+1} = (j+1)\delta$, seen at time $t_j = j\delta$, is given by

$$S_{t_{j+1}}^{(i)} = S_{t_j}^{(i)} e^{(\mu_{t_j} - \frac{1}{2}\sigma_{t_j}^2)\Delta t_j - \sigma_{t_j}\Delta W_{t_j}}, j = 0, \dots, J-1$$

where $\mu_t = r_t - q_t + \sigma_t \lambda_t$ and λ_t is the market price of risk. Every day D_i , we let the market price of risk λ_i be drawn from a Gaussian distribution with mean μ_L and variance σ_L^2 , that is, $\lambda_i \sim N(\mu_L, \sigma_L^2)$. Then, we generate a trajectory in the range $[t_0^{(i)}, T_C^{(i)}]$.

Every day $D_i, i = 1, \dots, N$, at time $t_j \in [t_0, T_O]$, look at the option strike prices on the chain and identify the one that is closest to the current market value of our stock. Given the distances $d_l = |S_{t_j}^{(i)} - K_l|$, for all $l = 1, \dots, n$, choose d^* such that the distance is minimum. Denote the resulting strike K_l^* . Thus, $K_l^* \approx S_t$.

In that setting, there are two sets of parameters:

1. Market parameters $P_j \in \mathcal{P}$: $r, q, \sigma, \mu_L, \sigma_L, \delta$.
2. Strategy parameters $M_i \in \mathcal{M}$: $\alpha^\pm, \epsilon_L, \epsilon_{GSL^\pm}, \epsilon_{LSL^\pm}, T_O$ plus the parameters from the TIs.

9.3.1.1 High level algorithm Every day, we consider the time interval $[t_0, T]$, where t_0 is the market opening (MO) and T is a pre-defined fixed maturity between t_0 and the market close (MC). During trading day D_i , we consider an investment strategy S that invests in I_i opportunities (or trades). We use options to devise the following algorithms.

Given a set of market and model parameters, the high level algorithm is as follows:

1. For $i \geq 0$
 - (a) Draw the the market price of risk λ_i . Create the stock trajectory $S^{(i)}$, for $i = 1, \dots, N$.
 - (b) Set the global stop-loss
 - (c) For $j \geq 0$
 - Check entry condition
 - i. If use TIs: compute the indicators and check entry condition.
 - ii. If use options: compute the yield $y_P = y_P(t_0, T_O)$. If $y_P > 0$ go long, else go short.
 - Check unwind condition. If true, compute the return.
 - (d) Compute the daily return
 - (e) For each k , aggregate returns $\bar{R}_k$.
2. Repeat K -times. Each time modify the market parameters P_k . Aggregate returns $\bar{R}$.

9.3.2 Basic algorithms

In all cases, at day D_i , $i = 1, \dots, N$, draw the market price of risk λ_i , and generate a trajectory in the range $[t_0, T]$. The main algorithm is given in Algorithm (19).

9.3.2.1 Compute metrics We compute the metrics (from the TI analysis and Risk/Reward analysis):

Algorithm 9 Compute Measures: compute TIs and set entry rule.

Require: The current time t_j , the stock price S_{t_j} , the window size.

Require: BoolTrade

- 1: Compute the TI at time t_j
 - 2: **if** $BoolTrade = \text{False}$ **then**
 - 3: Identify if we trade or not (see Algorithm (1)): if we trade set $BoolTrade = \text{True}$
 - 4: Identify if we are long or short: if we are long set $BoolLong = \text{True}$ else set $BoolLong = \text{False}$.
 - 5: **end if**
 - 6: **return** BoolTrade, BoolLong
-

Algorithm 10 Compute Measures: compute TIs and check exit rule.

Require: The current time t_j , the stock price S_{t_j} , the window size.

Require: BoolTrade

- 1: Compute the TI at time t_j
 - 2: **if** $BoolTrade = \text{True}$ **then**
 - 3: Identify if we exit trade or not (see Algorithm (2)): if we exit trade set $BoolTrade = \text{False}$
 - 4: **end if**
 - 5: **return** BoolTrade, BoolLong
-

Algorithm 11 Compute Measures: Use Risk/Reward analysis.

Require: The current time t_j , the stock price S_{t_j} and the option maturity T_O .

- 1: Compute the reward Δ_{t_j, T_O}^+ and the risk Δ_{t_j, T_O}^- : Use statistical models or options.
 - 2: Compute the reward target S_{t_j, T_O}^+ and the risk target S_{t_j, T_O}^-
 - 3: Compute the distance $D^\pm(t_j, T_O)$
 - 4: Compute the Risk/Reward ratio $RR^\mp(t_j, T_O)$
 - 5: Set $t^* = t_j$
 - 6: **return** $S_{t_j, T_O}^+, S_{t_j, T_O}^-, D^\pm(t_j, T_O), RR^\mp(t_j, T_O)$
-

9.3.2.2 Compute the global SL We compute the Global Stop Loss:

Algorithm 12 Compute Global Stop-Loss.

Require: The current time t_j **Require:** Set $\epsilon_{GSL} \in (0, 1)$ to limit the strategy's drawdown and prevent big losses**Require:** BoolLong

- 1: **if** BoolLong **then**
 - 2: Set the global stop-loss $GSL^-(t_j) = S_{t^*} \times (1 - \epsilon_{GSL})$
 - 3: **else**
 - 4: Set the global stop-loss $GSL^+(t_j) = S_{t^*} \times (1 + \epsilon_{GSL})$
 - 5: **end if**
 - 6: **return** $GSL^-(t_j), GSL^+(t_j)$
-

Algorithm 13 Compute Global Stop-Loss: Use options.

Require: The current time t_j , the risk target S_{t^*, T_O}^- , reward target S_{t^*, T_O}^+ , the risk Δ_{t^*, T_O}^- , the reward Δ_{t^*, T_O}^+ , the global SL level ϵ_{GSL} **Require:** $\epsilon \in (0, 1)$ **Require:** BoolLong

- 1: **if** BoolLong **then**
 - 2: **if** $S_{t^*, T_O}^- < S_{t^*}$ **then**
 - 3: Set the global stop-loss $GSL^-(t_j) = S_{t^*, T_O}^- + \epsilon_{GSL} \Delta_{t^*, T_O}^-$
 - 4: **else**
 - 5: Set the global stop-loss $GSL^-(t_j) = S_{t^*} \times (1 - \epsilon)$
 - 6: **end if**
 - 7: **else**
 - 8: **if** $S_{t^*, T_O}^+ > S_{t^*}$ **then**
 - 9: Set the global stop-loss $GSL^+(t_j) = S_{t^*, T_O}^+ - \epsilon_{GSL} \Delta_{t^*, T_O}^+$
 - 10: **else**
 - 11: Set the global stop-loss $GSL^+(t_j) = S_{t^*} \times (1 + \epsilon)$
 - 12: **end if**
 - 13: **end if**
 - 14: **return** $GSL^-(t_j), GSL^+(t_j)$
-

9.3.2.3 The entry and unwind conditions The entry and unwind algorithms are as follows:

Algorithm 14 Entry condition.

Require: The current time t_j , the stock price $S_{t_j}^{(i)}$ **Require:** The global stop-loss $GSL^-(t_j)$ and $GSL^+(t_j)$ **Require:** BoolLong, BoolUnwind

- 1: **if** ($BoolLong$ And $S_{t_j}^{(i)} > GSL^-(t_{unwind})$) Or ($BoolLong$ And $BoolTrade = True$) Or ($!BoolLong$ And $S_{t_j}^{(i)} < GSL^+(t_{unwind})$) Or ($!BoolLong$ And $BoolTrade = True$) **then**
 - 2: Enter the position at time t_j . If long, buy the position. Else, if short, sell the position.
 - 3: Set $t_{entry} = t_j$
 - 4: Set BoolUnwind = False
 - 5: **end if**
-

Algorithm 15 Unwind condition.

Require: The current time t_j , the stock price $S_{t_j}^{(i)}$.**Require:** The local stop-loss $SL(t_j)$ and global stop-loss $GSL^-(t_j)$ and $GSL^+(t_j)$ **Require:** BoolTrade, BoolLong, BoolUnwind

- 1: **if** ($BoolLong$ And $S_{t_j}^{(i)} \leq GSL^-(t_j)$) Or ($BoolLong$ And $BoolTrade = False$) Or ($!BoolLong$ And $S_{t_j}^{(i)} \geq GSL^+(t_j)$) Or ($!BoolLong$ And $BoolTrade = False$) **then**
 - 2: Unwind the position at time t_j . If long, sell the position. Else, if short, buy the position.
 - 3: Set $t_{unwind} = t_j$
 - 4: Set BoolUnwind = True
 - 5: **end if**
-

9.3.2.4 Identify trades This algorithm identify trades:

Algorithm 16 Identify Trade for Benchmark Backtest: Use TIs.

Require: The current time t_j , the stock price $S_{t_j}^{(i)}$ **Require:** BoolLong

- 1: Compute metrics: set entry rule and call Algorithm (9).
 - 2: **if** $BoolTrade$ **then**
 - 3: Set the global stop-loss: call Algorithm (12).
 - 4: Set $t_{entry} = t_j$
 - 5: **end if**
-

Algorithm 17 Identify Trade for Benchmark Backtest: Use options.

Require: The current time t_j , the stock price $S_{t_j}^{(i)}$, and the option maturity T_O .

Require: BoolLong

- 1: Compute the Market Risk Factor $\Lambda^*(t_j, T_O)$ or $C_0^*(t_j, T_O)$.
 - 2: Compute the yield $Y = Y(t_j, T_O)$.
 - 3: Enter the trade and set $BoolTrade = \text{True}$
 - 4: **if** $Y > 0$ **then**
 - 5: Go long. Set $BoolLong = \text{True}$
 - 6: **else**
 - 7: Go short.
 - 8: **end if**
 - 9: Compute metrics: call Algorithm (11).
 - 10: Set the global stop-loss: call Algorithm (13).
 - 11: Set $t_{entry} = t_j$
-

Algorithm 18 Compute daily returns.

Require: The unwind time t_{unwind} , the entry time t_{entry} , the stock prices $S_{t_{unwind}}^{(i)}$ and $S_{t_{entry}}^{(i)}$

Require: Aggregate returns $\bar{R}_k$

Require: BoolLong

- 1: Compute the daily return $R^{(i)}(t_{unwind}) = \frac{S_{t_{unwind}}^{(i)} - S_{t_{entry}}^{(i)}}{S_{t_{entry}}^{(i)}}$.
 - 2: **if** $\neg BoolLong$ **then**
 - 3: $R^{(i)}(t_{unwind}) = -R^{(i)}(t_{unwind})$
 - 4: **end if**
 - 5: Aggregate returns $\bar{R}_k = \bar{R}_k + R^{(i)}(t_{unwind})$
-

9.3.2.5 The benchmark The benchmark algorithm is as follows:

Algorithm 19 Benchmark backtest.**Require:** The settings to generate the stock returns for the set $\mathcal{D}$.**Require:** The market parameters $P_k \in \mathcal{P}, k = 1, \dots, K$.

```

1:  $BoolTrade = \text{False}$ ,  $BoolLong = \text{False}$ ,  $BoolUnwind = \text{False}$ 
2:  $\bar{R}_k = 0, \bar{R} = 0$ 
3: Set the number of simulations  $K$ , the number of trajectories  $N$ , the number of time steps  $J$ 
4: for  $k = 1$  to  $k = K$  do
5:   Get the market parameters  $P_k$ .
6:   Draw the Brownian motions for the stock trajectories  $W^{(i)}, i = 1, \dots, N$ .
7:   for  $i = 1$  to  $i = N$  do
8:     Draw the the market price of risk  $\lambda_i$ . Create the stock trajectory  $S^{(i)}$ .
9:     for  $j = 0$  to  $j = J - 1$  do
10:      if  $BoolTrade = \text{False}$  then
11:        Identify trade: call Algorithm (16) or Algorithm (17).
12:      else
13:        Compute metrics: check exit rule and call Algorithm (10).
14:        Test the unwind condition: call Algorithm (15)
15:        if  $BoolUnwind = \text{True}$  then
16:          Compute the daily returns: call Algorithm (18).
17:          Test the entry condition: call Algorithm (14)
18:        end if
19:      end if
20:    end for
21:    if  $BoolUnwind = \text{False}$  then
22:      Set  $t_{unwind} = t_j$ 
23:      Compute the daily returns: call Algorithm (18).
24:    end if
25:  end for
26:  Aggregate returns  $\bar{R} = \bar{R} + \bar{R}_k$ 
27: end for

```

9.3.3 Accounting for the risk/reward ratio

In all cases, at day $D_i, i = 1, \dots, N$, draw the market price of risk λ_i , and generate a trajectory in the range $[t_0, T]$. The main algorithm is given in Algorithm (25).

9.3.3.1 Compute the local SL Using the results from the above section, the algorithm is as follows:

Algorithm 20 Set Local Stop-Loss Condition.

Require: The current time t_j , the risk target S_{t^*, T_O}^- , reward target S_{t^*, T_O}^+ , the risk Δ_{t^*, T_O}^- , the reward Δ_{t^*, T_O}^+ , the local SL level ϵ_{LSL}

Require: BoolLong

```

1: if BoolLong then
2:   Set the local stop-loss  $SLC^-(t_j) = S_{t^*} + \epsilon_{LSL} \Delta_{t^*, T_O}^+$ 
3: else
4:   Set the local stop-loss  $SLC^+(t_j) = S_{t^*} - \epsilon_{LSL} \Delta_{t^*, T_O}^-$ 
5: end if

```

Algorithm 21 Set Local Stop-Loss.

Require: The current time t_j , the stock price $S_{t_j}^{(i)}$, and the option maturity T_O .**Require:** The local stop-loss conditions $SLC^-(t^*)$ and $SLC^+(t^*)$ **Require:** BoolLong

- 1: **if** (*BoolLong* And $S_{t_j}^{(i)} > SLC^-(t^*)$) Or ($\neg$ *BoolLong* And $S_{t_j}^{(i)} < SLC^+(t^*)$) **then**
 - 2: Set the local stop-loss at $SL(t_j) = S_{t_j}^{(i)}$.
 - 3: Call Algorithm (11)
 - 4: Call Algorithm (13)
 - 5: **end if**
-

9.3.3.2 Identify trades This algorithm identify trades:

Algorithm 22 Identify Trade for Simple Backtest.

Require: The current time t_j , the stock price S_{t_j} and the option maturity T_O .**Require:** BoolLong

- 1: Compute the Market Risk Factor $\Lambda^*(t_j, T_O)$ or $C_0^*(t_j, T_O)$.
 - 2: Compute the yield $Y = Y(t_j, T_O)$.
 - 3: **if** $|Y| > \epsilon_L$ **then**
 - 4: Choose the level of risk λ that define the Risk/Return ratio.
 - 5: Enter the trade and set *BoolTrade* = True
 - 6: **if** $Y > 0$ **then**
 - 7: Go long. Set *BoolLong* = True
 - 8: **else**
 - 9: Go short.
 - 10: **end if**
 - 11: Compute metrics at t_j : Call Algorithm (11)
 - 12: Set the global stop-loss: Call Algorithm (12)
 - 13: Set Local Stop-Loss Condition: Call Algorithm (20)
 - 14: Set $t_{entry} = t_j$
 - 15: **end if**
-

9.3.3.3 The entry and unwind conditions We define the entry and unwind conditions:

Algorithm 23 Unwind condition.

Require: The current time t_j , the stock price $S_{t_j}^{(i)}$.**Require:** The local stop-loss $SL(t_j)$ and global stop-loss $GSL^-(t_j)$ and $GSL^+(t_j)$ **Require:** BoolLong, BoolUnwind

- 1: **if** (*BoolLong* And $S_{t_j}^{(i)} \leq GSL^-(t_j)$) Or (*BoolLong* And $S_{t_j}^{(i)} \leq SL(t_j)$) Or ($\neg$ *BoolLong* And $S_{t_j}^{(i)} \geq GSL^+(t_j)$) Or ($\neg$ *BoolLong* And $S_{t_j}^{(i)} \geq SL(t_j)$) **then**
 - 2: Unwind the position at time t_j . Break.
 - 3: Set $t_{unwind} = t_j$
 - 4: Set BoolUnwind = True
 - 5: **end if**
-

Algorithm 24 Entry condition.**Require:** The current time t_j , the stock price $S_{t_j}^{(i)}$ **Require:** The local stop-loss conditions $SLC^-(t_j)$ and $SLC^+(t_j)$ **Require:** BoolLong

- 1: **if** ($BoolLong$ And $S_{t_j}^{(i)} > SLC^-(t_{unwind})$) Or ($\neg BoolLong$ And $S_{t_j}^{(i)} < SLC^+(t_{unwind})$) **then**
 - 2: Enter the position at time t_{entry}
 - 3: Set $t_{entry} = t_j$
 - 4: Set BoolUnwind = False
 - 5: **end if**
-

9.3.3.4 The main algorithm The main algorithm is as follows:**Algorithm 25** Simple backtest.**Require:** The settings to generate the stock returns for the set $\mathcal{D}$.**Require:** The market parameters $P_k \in \mathcal{P}$, $k = 1, \dots, K$.

- 1: $BoolTrade = \text{False}$, $BoolLong = \text{False}$, $BoolUnwind = \text{False}$
 - 2: $\bar{R}_k = 0$, $\bar{R} = 0$
 - 3: **for** $k = 1$ to $k = K$ **do**
 - 4: Get the market parameters P_k .
 - 5: Draw the Brownian motions of the stock trajectories $W^{(i)}$, $i = 1, \dots, N$.
 - 6: **for** $i = 1$ to $i = N$ **do**
 - 7: Draw the the market price of risk λ_i . Create the stock trajectory $S^{(i)}$.
 - 8: **for** $j = 0$ to $j = J - 1$ **do**
 - 9: **if** $BoolTrade = \text{False}$ **then**
 - 10: Identify Trade: Call Algorithm (22)
 - 11: **else**
 - 12: Set Local Stop-Loss: Call Algorithm (21)
 - 13: Unwind condition: Call Algorithm (23)
 - 14: **if** BoolUnwind = True **then**
 - 15: Compute the daily returns: call Algorithm (18).
 - 16: Test the entry condition: Call Algorithm (24)
 - 17: **end if**
 - 18: **end if**
 - 19: **end for**
 - 20: **if** BoolUnwind = False **then**
 - 21: Set $t_{unwind} = t_j$
 - 22: Compute the daily returns: call Algorithm (18).
 - 23: **end if**
 - 24: **end for**
 - 25: Aggregate returns $\bar{R} = \bar{R} + \bar{R}_k$
 - 26: **end for**
-

9.4 Results

We consider the case of day trading, where buying and selling occur until the end of the trading day, since no positions are held overnight. Given the quantitative algorithms described in Section (9.3), we focus on three simple strategies:

1. Risk/Reward analysis with options: We consider the simple strategy presented in Section (6.4).
-

2. Double Moving Average Crossover: We consider the simple strategy presented in Section (3.2).
3. The Bollinger bands (BB): We consider the simple strategy presented in Section (3.2).

For simplicity, we consider the process-driven methods presented in Section (8.2.1), where we need to generate scenarios. To illustrate our approach, we let the stock prices follow a modified GBM with a stochastic market price of risk (MPR). The dynamics of the stock price, under the $\mathbb{P}$ -measure, are given by

$$\frac{dS_t}{S_t} = (r - q + \sigma_S \lambda_t)dt + \sigma_S d\widehat{W}_S(t) \quad (9.53)$$

where $\widehat{W}_S$ is a Brownian motion under the historical measure. Here, the repo rate q and the volatility σ_S are positive constants. We let the market price of risk, λ_t , follows an Ornstein-Uhlenbeck process (see Equation (8.51)). The Brownian motions are correlated as follows: $d\langle \widehat{W}_S, W_\lambda \rangle_t = \rho_{S\lambda} dt$.

9.4.1 R/R analysis with options

The settings for the model generating the returns, the option contracts, and the strategy are as follows:

- Market parameters taking values in $\mathcal{P}$:
 - GBM: $r = \{-0.1, 0, 0.1\}$, $q = \{0\}$, $\sigma_S = \{0.1, 0.2, 0.3\}$
 - MPR: $\lambda_0 = \{0.1\}$, $a_\lambda = \{0.1, 0.5\}$, $b_\lambda = \{0.1, 1.0\}$, $\sigma_\lambda = \{0.1, 0.5\}$
- Simulation parameters: sample frequency 15 minutes, time steps $N = 780$ which gives 10 days, number of simulation 1,000
- Option parameters taking value in $\mathcal{P}$: maturity $T_O = \{1, 5, 10\}$ days, strike $K = S_0$
- Strategy parameters taking value in $\mathcal{M}$:
 - Entry level: $\epsilon_L = \{0.02, 0.03, 0.04, 0.05\}$
 - Global distance level: $\epsilon_{GSL} = \{0.1, 0.3, 0.5, 0.7, 1.0\}$
 - Local distance level: $\epsilon_{LSL} = \{0.1, 0.3, 0.5, 0.7, 1.0\}$

For each combination of Strategy Parameters M_i , $i = 1, \dots, I$, test 72 Market Scenarios. Following the approach in Section (9.2), for each such combination, we compute a distribution of Sharpe Ratio (SR) and Sortino (S) measures. Given the entry level $\epsilon_L > 0.04$, the SR-distribution is illustrated in Figure (67) and the S-distribution is illustrated in Figure (68).

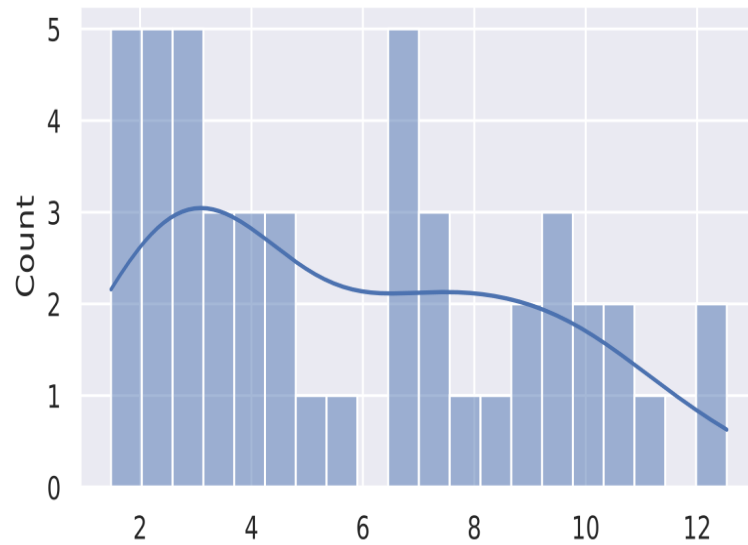


Figure 67: Entry level $\epsilon_L > 0.04$: Sharpe distribution with a mean of 5.65.

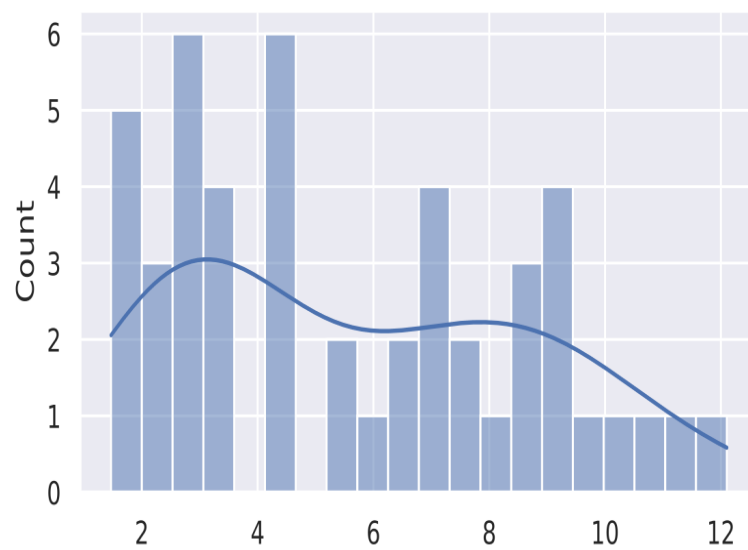


Figure 68: Entry level $\epsilon_L > 0.04$: Sortino distribution with a mean of 5.76.

Both the SR-distribution and the S-distribution are positive, with mean 5.65 and 5.76, respectively.

10 Appendix

11 Miscellaneous

11.1 The moving average

In statistics, a moving average is a type of finite impulse response filter used to analyse a set of data points by creating a series of averages of different subsets of the full data set. Given a series of numbers and a fixed subset size, the first element of the moving average is obtained by taking the average of the initial fixed subset of the number series. Then the subset is modified by shifting forward, that is, excluding the first number of the series and including the next number following the original subset in the series. This creates a new subset of numbers, which is averaged. This process is repeated over the entire data series. The plot line connecting all the (fixed) averages is the moving average. A moving average may also use unequal weights for each datum value in the subset to emphasise particular values in the subset. A moving average is commonly used with time series data to smooth out short-term fluctuations and highlight longer-term trends or cycles. The threshold between short-term and long-term depends on the application, and the parameters of the moving average will be set accordingly. Mathematically, a moving average is a type of convolution and so it can be viewed as an example of a low-pass filter used in signal processing.

11.1.1 Simple moving average

In financial applications a SMA is the unweighted mean of the previous n datum points. As an example of a simple equally weighted running mean for a n -day sample of closing price is the mean of the previous n days' closing prices. Given the latest day $t = j\delta$, if these prices are $P_j, P_{j-1}, \dots, P_{j-(n-1)}$ then the formula for the moving average at index j and order n is

$$SMA(j, n) = \frac{1}{n} \sum_{i=0}^{n-1} P_{j-i}$$

where P_j is the price of the latest day.

Remark 11.1 Note, it correspond to the Savitzky-Golay low-pass smoothing filter²⁰ with $n_R = 0$ and $c_n = \frac{1}{n_L+1}$ (or $m = 0$).

When calculating successive values, a new value comes into the sum and an old value drops out, meaning a full summation each time is unnecessary for this simple case

$$SMA(j, n) = SMA(j-1, n) - \frac{P_{j-n}}{n} + \frac{P_j}{n}$$

The period selected depends on the type of movement of interest, such as short, intermediate, or long term. In financial terms MA levels can be interpreted as support in a rising market, or resistance in a falling market. If data used is not centred around the mean, a SMA lags behind the latest datum by half the sample width. Further, an SMA can be disproportionately influenced by old datum points dropping out or new data coming in. In order to choose the order n we need to evaluate suitable measures of forecast performance.

²⁰Assume we want to smooth a series of equally spaced data points $f_i = f(t_i)$ where $t_i = t_0 + i\Delta$ for some constant Δ and $i = \dots -2, -1, 0, 1, 2, \dots$. This filter replaces each data value f_i by a linear combination g_i of itself and some number of nearby neighbours

$$g_i = \sum_{n=-n_L}^{n_R} c_n f_{i+n} \quad (11.54)$$

where n_L is the number of points used to the left of a data point i , and n_R is the number of points used to the right.

11.1.2 Weighted moving average

A weighted average is any average that has multiplying factors to give different weights to data at different positions in the sample window. The MA is the convolution of the datum points with a fixed weighting function. In finance, a WMA has the specific meaning of weights that decrease in arithmetical progression. In an n -day WMA the latest day $t = j\delta$ has weight n , the second latest $(n - 1)$ etc. down to one

$$WMA(j) = \frac{nP_j + (n-1)P_{j-1} + \dots + 2P_{j-n+2} + P_{j-n+1}}{n + (n-1) + \dots + 2 + 1}$$

The denominator is a triangle number equal to $\frac{n(n+1)}{2}$. In the more general case the denominator will always be the sum of the individual weights

$$WMA(j) = \frac{1}{norm} \sum_{i=0}^{n-1} \omega_i P_{j-i}$$

with $\omega_i = n - i$ and $norm = \sum_{i=0}^{n-1} \omega_i$. Hence, in the SMA the weights are $\omega_i = 1$ for $i = 0, \dots, (n - 1)$. The difference of the numerator at time $(t + 1)$ and t is

$$Num(j+1) - Num(j) = \sum_{i=0}^{n-1} \omega_i P_{(j+1)-i} - \sum_{i=0}^{n-1} \omega_i P_{j-i} = nP_{j+1} - P_j - \dots - P_{j-(n-1)}$$

Let's call $T(j) = P_j + \dots + P_{j-(n-1)}$ then

$$T(j+1) = T(j) + P_{j+1} - P_{j-(n-1)}$$

As a result

$$Num(j+1) = Num(j) + nP_{j+1} - T_j$$

so that the weighted average at time $t + 1$ becomes

$$WMA(j+1) = \frac{1}{norm} Num(j+1)$$

11.1.3 Exponential smoothing

Brown [1959] introduced the exponential smoothing as a forecasting method adjusting more smoothly over time than moving averages. We let $\bar{Y}(N)$ be the arithmetic mean of the past N observations where the updated mean $\bar{Y}(N+1)$ for the past $(N+1)$ observations is given by

$$\bar{Y}(N+1) = \frac{Y_1 + \dots + Y_{N+1}}{N+1} = \bar{Y}(N) + \frac{Y_{N+1} - \bar{Y}(N)}{N+1}$$

However, as the series length increases the forecasts generated by the latest mean, become increasingly unresponsive to fluctuations in recent values, since each observation has weight $\frac{1}{\text{sample size}}$. The update of the simple MA given in Appendix (11.1.1) avoided this problem and maintained a constant coefficient $\frac{1}{N}$ where N is the sample running period, but at the cost of dumping the oldest observation completely. To overcome this problem, we consider the updating relationship of the Exponential Smoothing (or exponentially weighted moving average EWMA) with basic equation being

$$\bar{Y}(N+1, \alpha) = \bar{Y}(N, \alpha) + \alpha[Y_{N+1} - \bar{Y}(N, \alpha)]$$

where $\alpha \in [0, 1]$ is the smoothing constant. The process involves comparing the latest observation with the previous weighted average and making a proportional adjustment, governed by the coefficient α , known as the smoothing constant. Note, the updates require only the latest observation and the previous mean. If we start with the expression for time $(N + 1)$ and substitute into it the comparable expression for time N , we obtain

$$\begin{aligned}\bar{Y}(N + 1, \alpha) &= (1 - \alpha)\bar{Y}(N, \alpha) + \alpha Y_{N+1} \\ &= (1 - \alpha)[(1 - \alpha)\bar{Y}(N - 1, \alpha) + \alpha Y_N] + \alpha Y_{N+1} \\ &= (1 - \alpha)^2 \bar{Y}(N - 1, \alpha) + \alpha[(1 - \alpha)Y_N + Y_{N+1}]\end{aligned}$$

Continuing to substitute the earlier means, we eventually arrive back at the start of the series with the expression

$$\bar{Y}(N + 1, \alpha) = (1 - \alpha)^{N+1} \bar{Y}(0, \alpha) + \alpha[Y_{N+1} + (1 - \alpha)Y_N + (1 - \alpha)^2 Y_{N-1} + \dots + (1 - \alpha)^N Y_1] \quad (11.55)$$

The right hand side contains a weighted average of the observations and the weights $\alpha, \alpha(1 - \alpha), \alpha(1 - \alpha)^2, \dots$ decay steadily over time in an exponential fashion. The decay is slower for small values of α . Note, the equation depends on a starting value $\bar{Y}(0, \alpha)$. Further, when N and α are both small, the weight attached to the starting value may be high.

We now need to convert these averages into forecasts where we assume that the average level for future observations is best estimated by our current weighted average. Thus, forecasts made at time t for all future time periods will be the same

$$F_{t+1|t} = F_{t+2|t} = \dots = F_{t+h|t} = \bar{Y}(t, \alpha)$$

so that as the lead time increases the forecasts will usually be progressively less accurate.

Remark 11.2 We note that forecasts based upon the mean or the simple moving average also have the same property of being equal for all future periods.

Of course, once the next observation is made, the common forecast value changes. The forecast equation for the EWMA is

$$F_{t+1|t} = F_{t|t-1} + \alpha[Y_t - F_{t|t-1}]$$

We define the observed error ϵ_t as the difference between the newly observed value of the series and its previous one-step-ahead forecast

$$\epsilon_t = Y_t - F_t$$

so that the forecast equation becomes

$$F_{t+1|t} = F_{t|t-1} + \alpha \epsilon_t \quad (11.56)$$

To set up these calculations, we need to specify the value for α and a starting value F_1 . Two principal options are commonly used to resolve the choice of starting values, either to use the first observation as F_1 , or to use an average of a number of observations (average of the first six observations). Earlier literature recommended a choice for α in the range $0.1 < \alpha < 0.3$ to allow the EWMA to change relatively slowly. However, we should not rely on an arbitrary pre-set smoothing parameter. One possibility is to minimise the mean squared error (MSE) for the one-step ahead forecasts. Given Equation (11.55) we can rewrite the forecast equation as

$$\hat{Y}_{T+\tau}(T) = (1 - \alpha)^T \hat{Y}_0 + \alpha[Y_T + (1 - \alpha)Y_{T-1} + (1 - \alpha)^2 Y_{T-2} + \dots + (1 - \alpha)^{N-1} Y_{T-(N-1)}]$$

where $\hat{Y}_0$ is an initial estimate of the smoothed series $\{\hat{Y}_t\}_{t \in \mathbb{Z}^+}$ which is simply the average of the first six observations or, if there are less than six observations, the average of all the observations. As a measure of error we can approximate the 95% prediction interval for $\hat{Y}_{T+\tau}(T)$ given by

$$\hat{Y}_{T+\tau}(T) \pm z_{0.25} 1.25 MAD(T)$$

where $z_{0.25} \approx 1.96$.

11.1.4 Exponential moving average revisited

An exponential moving average (EMA), also known as an exponentially weighted moving average (EWMA), is a type of infinite impulse response filter that applies weighting factors which decrease exponentially. The weighting for each older datum point decreases exponentially, never reaching zero. According to Hunter (1986), the EMA for a series P may be calculated recursively as

$$E_t = \alpha P_t + (1 - \alpha)E_{t-1} \text{ for } t > 1$$

with $E_1 = P_1$ and where P_t is the value at a time period t , and E_t is the value of the EMA at any time period t . The coefficient $\alpha \in [0, 1]$ represents the degree of weighting decrease. A higher α discounts older observations faster. Alternatively, α may be expressed in terms of N time periods, where $\alpha = \frac{2}{(N+1)}$. For example, if $N = 19$ it is equivalent to $\alpha = 0.1$, the half-life of the weights is approximately $\frac{N}{2.8854}$. Note, E_1 is undefined, and it may be initialised in a number of different ways, most commonly by setting E_1 to P_1 , though other techniques exist, such as setting E_1 to an average of the first 4 or 5 observations. The prominence of the E_1 initialisation's effect on the resultant moving average depends on α ; smaller α values make the choice of E_1 relatively more important than larger α values, since a higher α discounts older observations faster. By repeated application of this formula for different times, we can eventually write E_t as a weighted sum of the datum points P_t as:

$$E_t = \alpha(P_{t-1} + (1 - \alpha)P_{t-2} + (1 - \alpha)^2P_{t-3} + \dots + (1 - \alpha)^kP_{t-(k+1)}) + (1 - \alpha)^{k+1}E_{t-(k+1)}$$

for any suitable $k = 0, 1, 2, \dots$. The weight of the general datum point P_{t-i} is $\alpha(1 - \alpha)^{i-1}$. We can show how the EMA steps towards the latest datum point, but only by a proportion of the difference (each time)

$$E_t = E_{t-1} + \alpha(P_t - E_{t-1})$$

Expanding out E_{t-1} each time results in the following power series, showing how the weighting factor on each datum point $p_1, p_2, \dots$ decreases exponentially:

$$E_t = \alpha(p_1 + (1 - \alpha)p_2 + (1 - \alpha)^2p_3 + \dots)$$

where $p_1 = P_t, p_2 = P_{t-1}, \dots$. Note, the weights $\alpha(1 - \alpha)^t$ decrease geometrically, and their sum is unity. Using a property of geometric series, we get

$$\alpha \sum_{i=0}^{t-1} (1 - \alpha)^i = \alpha \sum_{i=1}^t (1 - \alpha)^{i-1} = \alpha \left[\frac{1 - (1 - \alpha)^t}{1 - (1 - \alpha)} \right] = 1 - (1 - \alpha)^t$$

and $\lim_{t \rightarrow \infty} (1 - \alpha)^t = 0$. As a result, we get

$$\alpha \sum_{i=1}^{\infty} (1 - \alpha)^{i-1} = 1$$

Focusing on the term α , we get

$$E_t = \frac{p_1 + (1 - \alpha)p_2 + (1 - \alpha)^2 p_3 + \dots}{norm} = \frac{1}{norm} \sum_{i=1}^{\infty} \omega_i p_i$$

where $\omega_i = (1 - \alpha)^{i-1}$ and

$$\frac{1}{\alpha} = norm = 1 + (1 - \alpha) + (1 - \alpha)^2 + \dots = \sum_{i=1}^{\infty} (1 - \alpha)^{i-1}$$

This is an infinite sum with decreasing terms. The N periods in an N-day EMA only specify the α factor. N is not a stopping point for the calculation in the way it is in an SMA or WMA. For sufficiently large N, the first N datum points in an EMA represent about 86% of the total weight in the calculation

$$\frac{\alpha(1 + (1 - \alpha) + (1 - \alpha)^2 + \dots + (1 - \alpha)^N)}{\alpha(1 + (1 - \alpha) + (1 - \alpha)^2 + \dots + (1 - \alpha)^{\infty})} = 1 - \left(1 - \frac{2}{N + 1}\right)^{N+1}$$

and

$$\lim_{N \rightarrow \infty} \left[1 - \left(1 - \frac{2}{N + 1}\right)^{N+1}\right] = 1 - e^{-2} \approx 0.8647$$

since $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$. This power formula can give a starting value for a particular day, after which the successive days formula above can be applied. The question of how far back to go for an initial value depends, in the worst case, on the data. Large price values in old data will affect on the total even if their weighting is very small. The weight omitted by stopping after k terms can be used in the fraction

$$\frac{\text{weight omitted by stopping after k terms}}{\text{total weight}} = (1 - \alpha)^k$$

For example, to have 99.9% of the weight, set the ratio equal to 0.1% and solve for k:

$$k = \frac{\log(0.001)}{\log(1 - \alpha)}$$

As the Taylor series of $\log(1 - \alpha) = -\alpha - \frac{1}{2}\alpha^2 - \dots$ tends to $-\alpha$ we get the limit $\lim_{N \rightarrow \infty} \log(1 - \alpha) = -\frac{2}{(N+1)}$, and the computation simplifies to

$$k = -\log(0.001) \frac{(N + 1)}{2}$$

for this example and $\frac{1}{2} \log(0.001) = -3.45$.

11.2 A few facts on distributions

A probability distribution is the mathematical function that gives the probabilities of occurrence of different possible outcomes for an experiment. We distinguish between discrete and absolutely continuous random variables. The probability density function (pdf) describes the infinitesimal probability of any given value, and the probability that the outcome lies in a given interval can be computed by integrating the pdf over that interval. The cumulative distribution function (cdf) describes the probability that the random variable is no larger than a given value. The probability function is defined as the element of a probability space $(\Omega, \mathcal{A}, P)$, where Ω is the set of possible outcomes, $\mathcal{A}$ is the set of all subsets $E \subset \Omega$ whose probability can be measured, and P is the probability function, or probability measure, that assigns a probability to each of these measurable subsets $E \in \Omega$. There is a large number of probability distributions grouped by the type of process that they are related to. For some distributions, the values cluster around a single point (univariate distributions), while others may cluster around multiple values. In the latter, they can be

modelled using a mixture distribution. Distribution fitting is the fitting of a probability distribution to a series of data concerning the repeated measurement of a variable phenomenon. It is used to predict the probability or to forecast the frequency of occurrence of the magnitude of the phenomenon in a certain interval. The distribution giving a close fit is supposed to lead to good predictions. Hence, we need to select a distribution that suits well the data.

See books by Fisz [1963], Grimmer et al. [1992], Evans et al. [2010], among others.

11.2.1 Sampling random numbers from a distribution

A continuous random variable x is a random variable that can take on infinitely many values, that is $x \in \mathbb{R}$. A 1D probability distribution function (PDF), or probability density function, $f(x)$ describes the likelihood that the value of the continuous random variable will take on a given value. The PDF is normalised if $\int_{-\infty}^{\infty} f(x)dx = 1$. A cumulative distribution function (CDF), $F(x)$, is the likelihood that the value of the continuous random variable lies in the interval $(-\infty, x)$. The PDF and CDF are related by integration

$$F(x) = \int_{-\infty}^x f(s)ds$$

Hence, it is clear that if the probability distribution function is normalised, then $\lim_{x \rightarrow \infty} F(x) = 1$. As a result, $F(x) \in (0, 1)$.

Selecting a value at random from a uniform distribution is usually quite easy. For non-uniform distribution we must rely on Inverse Transform Sampling. Let U be a uniformly distributed random number between zero and one. Then to sample a random number with a (possibly nonuniform) probability distribution function $f(x)$ we do the following:

1. Normalise the function $f(x)$ if it isn't already normalised.
2. Integrate the normalised PDF $f(x)$ to compute the CDF, $F(x)$.
3. Invert the function $F(x)$. The resulting function is the inverse cumulative distribution function, or quantile function, $F^{-1}(x)$. Since we have already normalised $f(x)$, we can call it the inverse normal cumulative distribution function, or simply the inverse normal CDF.
4. Substitute the value of the uniformly distributed random number U into the inverse normal CDF. As a result, if $U \in (0, 1)$, then $F^{-1}(U)$ is a random number x with a probability distribution function $f(x)$.

11.2.2 The NCDF

We let $N(x)$, with support $x \in \mathbb{R}$, be the standard normal cumulative distribution function (CDF) given by

$$N(x) = P(X \leq x) = \int_{-\infty}^x n(s)ds$$

where $N'(x) = n(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$ is the probability density function (pdf). The pdf has its pick $\frac{1}{\sqrt{2\pi}}$ at $x = 0$ and inflection points at $x = \pm 1$. The limits of the CDF are

$$\lim_{x \rightarrow -\infty} N(x) = 0, \lim_{x \rightarrow 0} N(x) = \frac{1}{2}, \lim_{x \rightarrow \infty} N(x) = 1$$

The CDF of the standard normal distribution can be expanded by integration by parts into a series:

$$N(x) = \frac{1}{2} + n(x)\left[x + \frac{x^3}{3} + \frac{x^5}{3 \cdot 5} + \cdots + \frac{x^{2n+1}}{(2n+1)!!}\right]$$

where !! denotes the double factorial.

We let $f(x) = N(x)$ which we approximate with a Taylor series around x_0 , getting

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{1}{2}(x - x_0)^2 f''(x_0) + \cdots + \frac{1}{n!}(x - x_0)^n f^n(x_0) + R_n$$

where R_n is the remainder term. Truncating the series to the second order, we get

$$N(x) = N(x_0) + (x - x_0)n(x_0) - \frac{1}{2}(x - x_0)^2 x_0 n(x_0)$$

where $N'(x) = n(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$ and $N''(x) = -xn(x)$. Hence, for $x_0 = 0$, we have $n(0) = \frac{1}{\sqrt{2\pi}}$ and the expansion of the CDF simplifies to

$$N(x) \approx \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \sum_{k=0}^n \frac{(-1)^k x^{2k+1}}{2^k k! (2k+1)}$$

In its simplest form, we get

$$N(x) \approx \frac{1}{2} + \frac{1}{\sqrt{2\pi}}x$$

We can express the integral $\int_a^b xn(x)dx = E[xI_{a < x < b}]$ in terms of $n(x)$ and $N(x)$. The density of the standard normal distribution being

$$n(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$$

then, $\frac{dn(x)}{dx} = -xn(x)$, and we can re-express the integral as

$$\int_a^b xn(x)dx = \int_a^b -\frac{dn(x)}{dx}dx = n(a) - n(b)$$

11.2.3 Some properties

11.2.3.1 Standardisation Every normal distribution is a version of the standard normal distribution, whose domain has been stretched by a factor σ (the standard deviation) and then translated by μ (the mean value):

$$f(x | \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sigma}n\left(\frac{x-\mu}{\sigma}\right)$$

The probability density of the standard Gaussian distribution is usually denoted ϕ or φ , and the normal distribution is often referred to as $N(\mu, \sigma^2)$.

If Z is a standard normal deviate, then

$$X = \mu + \sigma Z$$

has a normal distribution with expected value μ and standard deviation σ . That is, the standard normal distribution Z can be scaled/stretched by a factor of σ and shifted by μ to yield a different normal distribution, called X . Conversely, if X is a normal deviate with parameters μ and σ^2 , then this X distribution can be re-scaled and shifted via the formula $Z = \frac{(X-\mu)}{\sigma}$ to convert it to the standard normal distribution. This variate is also called the standardised form of X .

We can write the NCDF as

$$N(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right)\right] \text{ where } \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-s^2} ds$$

For a generic normal distribution, we get the CDF

$$F(x; \mu, \sigma^2) = \int_{-\infty}^x \frac{1}{\sigma} n\left(\frac{s - \mu}{\sigma}\right) ds = N\left(\frac{x - \mu}{\sigma}\right)$$

We can easily prove it by setting $z = \frac{s - \mu}{\sigma}$ so that $ds = \sigma dz$ and

$$\int_{-\infty}^x \frac{1}{\sigma} n\left(\frac{s - \mu}{\sigma}\right) ds = \int_{-\infty}^x \frac{1}{\sigma} n(z) \sigma dz = N\left(\frac{x - \mu}{\sigma}\right)$$

We display in Figure (69) a normal probability distribution centered on mean $\mu = 0$.

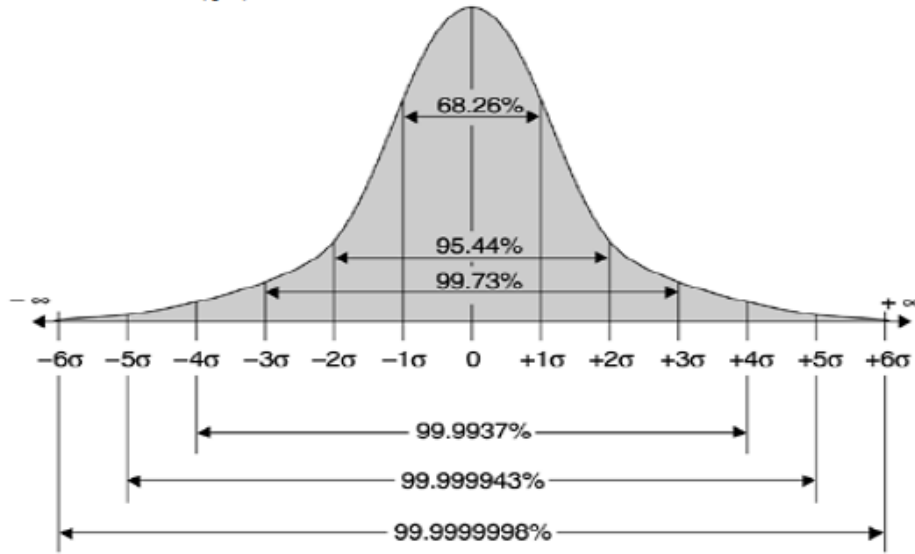


Figure 69: Normal distribution curve.

About 68% of values drawn from a normal distribution are within one standard deviation σ away from the mean. About 95% of the values lie within two standard deviations. More precisely, the probability that a normal deviate lies in the range $[\mu - k \cdot \sigma, \mu + k \cdot \sigma]$, for some constant k , is given by

$$p = F(\mu + k\sigma) - F(\mu - k\sigma) = N(k) - N(-k) = \text{erf}\left(\frac{k}{\sqrt{2}}\right)$$

We display in Table (2) the range of a normal deviate for $k = 1, 2, 3$.

k	p	1-p
1	0.682	0.317
2	0.954	0.045
3	0.997	0.002

Table 2: Range of a normal deviate

For large k , we can use the approximation $1 - p \approx \frac{e^{-\frac{k^2}{2}}}{k\sqrt{\frac{\pi}{2}}}$ (probability of being outside the above range).

11.2.3.2 The quantiles Quantiles are cut points dividing the range of a probability distribution into continuous intervals with equal probabilities. In discrete distributions, q-quantiles are values that partition a finite set of values into q subsets of (nearly) equal sizes. In continuous distributions, they provide a way of generalising rank statistics to continuous variables. This is illustrated in Figure (70). The quantile function of a distribution is the inverse of the cumulative distribution function. When the CDF of a random variable is known, the q-quantiles are the application of the quantile function to the values $\{\frac{1}{q}, \frac{2}{q}, \dots, \frac{(q-1)}{q}\}$.

The quantile function of the standard normal distribution is called the probit function, and can be expressed in terms of the inverse error function:

$$N^{-1}(p) = \sqrt{2} \operatorname{erf}^{-1}(2p - 1), p \in (0, 1)$$

For a normal random variable with mean μ and variance σ^2 , the quantile function is

$$F^{-1}(p) = \mu + \sigma N^{-1}(p) = \mu + \sigma \sqrt{2} \operatorname{erf}^{-1}(2p - 1), p \in (0, 1)$$

where F^{-1} is the inverse CDF. The quantile $N^{-1}(p)$ of the standard normal distribution is commonly denoted as z_p . A normal random variable X will exceed $\mu + \sigma z_p$ with probability $1 - p$ and will lie outside the interval $\mu \pm \sigma z_p$ with probability $2(1 - p)$.

For a population, of discrete values or for a continuous population density, the k-th q-quantile is the data value where the CDF crosses $\frac{k}{q}$. That is, x is a k-th q-quantile for a variable X if

$$P(X < x) \leq \frac{k}{q} \text{ and } P(X \leq x) \geq \frac{k}{q}$$

For a finite population of N equally probable values indexed $1, \dots, N$, from lowest to highest, the k-th q-quantile of this population can also be computed via the value of $I_p = \frac{Nk}{q}$. For any population probability distribution on finitely many values, and generally for any probability distribution with a mean μ and variance σ^2 , it is the case that

$$\mu - \sigma \cdot \alpha \leq Q(p) \leq \mu + \sigma \cdot \alpha, \alpha = \sqrt{\frac{1-p}{p}}$$

where $Q(p)$ is the value of the p-quantile for $0 < p < 1$. We can use this formula to get the lower bound

$$\mu + k \cdot \sigma \geq Q\left(\frac{k^2}{1+k^2}\right), k \geq 0$$

Hence, the value that is $k = 1$ standard deviation above the mean is always greater than or equal to the median $Q(p = \frac{1}{2})$. Conversely, when $k \leq 0$, we get the upper bound

$$\mu + k \cdot \sigma \leq Q\left(\frac{k^2}{1+k^2}\right), k \leq 0$$

If a distribution is symmetric, then the median is the mean (when it exists), otherwise the median differs from the mean (see Figure (71)). Hence, quantiles are useful measures because they are less susceptible than means to long-tailed distributions and outliers.

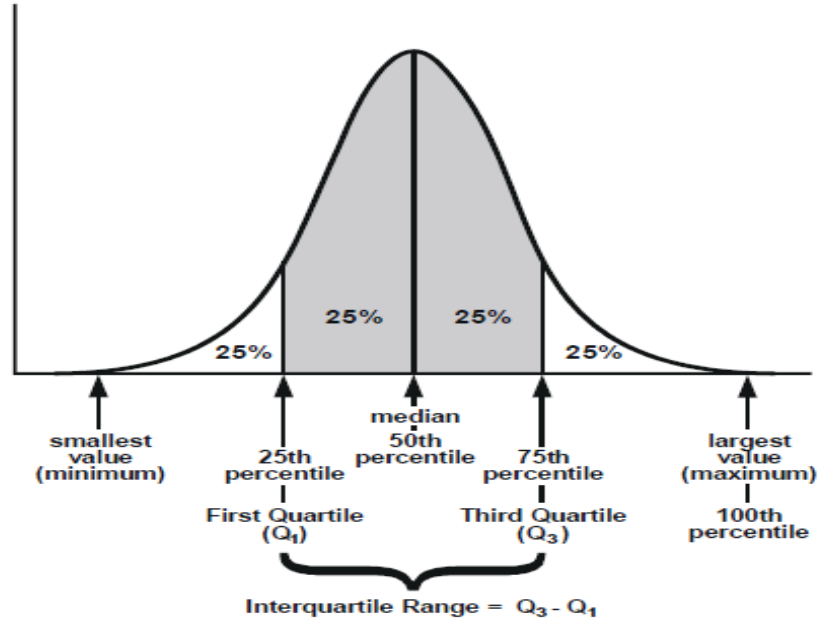


Figure 70: Distribution quantile.

Modern statistical packages rely on a number of techniques to estimate the quantiles for populations based on a finite sample of size N . For instance, Hyndman et al. [1996] compiled a taxonomy of nine algorithms. Computing approximate quantiles from data arriving from a stream can be done efficiently by using compressed data structures. For instance, t-Digests (see Dunning et al. [2019]) and KLL (see Karnin et al. [2016]) read a stream of values in a continuous fashion and can, at any time, be queried about the approximate value of a specified quantile. Both methods belong to the family of data sketches that are subsets of Streaming Algorithms, and can therefore be combined. Computing the sketch for a very large vector of values can be split into trivially parallel processes where sketches are computed for partitions of the vector in parallel and merged later.

11.2.4 Multivariate normal distribution

11.2.4.1 The first two moments Recall, the magnitude is the length of the vector which does not have a direction (positive or negative). Hence, magnitude cannot be negative. The dot product of two vectors $A \cdot B = ab \cos(C)$, where a, b are the magnitudes of A, B , and C is the angle between A and B , can be any real value, including negative and zero.

Let A and B be matrices, then $(A + B)^\top = A^\top + B^\top$, $(AB)^\top = B^\top A^\top$, $(A^\top)^{-1} = (A^{-1})^\top$.

Definition 11.1 A symmetric $n \times n$ matrix A is said to be positive definite (PD) if for all $x \in \mathbb{R}^n$,

$$x^\top A x > 0 \text{ if } x \neq 0$$

If $x^\top A x \geq 0$, then A is said to be positive semidefinite (PSD).

Theorem 11.1 Let A be a positive definite matrix. Then there exists a unique matrix $A^{\frac{1}{2}}$ such that $A^{\frac{1}{2}} A^{\frac{1}{2}} = A$.

The multivariate normal distribution of a k -dimensional random vector $X = (X_1, \dots, X_k)^\top$, where X_i are random variables, is $X \sim N(\mu, \Sigma)$ with k -dimensional mean vector

$$\mu = E[X] = (\mu_1, \dots, \mu_k)^\top, \mu_i = E[X_i]$$

and $k \times k$ covariance matrix

$$Cov(X) = Cov(X, X) = E[(X - \mu)(X - \mu)^\top] = E[XX^\top] - E[X](E[X])^\top$$

with elements

$$\Sigma_{ij} = E[(X_i - \mu_i)(X_j - \mu_j)] = Cov(X_i, X_j), 1 \leq i \leq k \text{ and } 1 \leq j \leq k$$

We let K_{XX} be the covariance matrix. It is sometimes called the variance of the random vector X . Since

$$\Sigma_{ii} = E[(X_i - \mu_i)(X_i - \mu_i)] = Cov(X_i, X_i) = Var(X_i)$$

is the variance of X_i (positive), K_{XX} is also called the variance-covariance matrix. The covariance ranges from negative values to positive values. $R_{XX} = E[XX^\top]$ is the autocorrelation matrix and $Cov^{-1}(X)$, if it exists, is the inverse covariance matrix (or precision matrix). Then, for $X = (X_1, \dots, X_n)^\top$ with $\mu_X = E[X] \in \mathbb{R}^n$ and $K_{XX} = Cov(X) \in \mathbb{R}^{n \times n}$, we get

- $K_{XX} = E[XX^\top] - \mu_X \mu_X^\top$
- K_{XX} is positive-semidefinite
- K_{XX} is symmetric

Similar to correlation, the autocorrelation can be either positive or negative. It refers to the degree of correlation between the values of the same variables across different observations in the data and can range from -1 (perfectly negative autocorrelation) to 1 (perfectly positive autocorrelation). A covariance matrix with all non-zero elements tells us that all the individual random variables are interrelated.

Let A denote a matrix of constants and X be a random vector with mean μ and variance-covariance Σ . Then,

$$E[A^\top X] = A^\top \mu, Cov(A^\top X) = A^\top \Sigma A, E[X^\top A X] = \mu^\top A \mu + tr(A \Sigma)$$

where $tr(A) = \sum_i A_{ii}$. The inverse of the covariance matrix is called the precision matrix, denoted by $Q = \Sigma^{-1}$.

Given two random vectors $X \in \mathbb{R}^n$ and $Y \in \mathbb{R}^m$, the joint mean $\mu \in \mathbb{R}^{n+m}$ and joint covariance matrix $\Sigma \in \mathbb{R}^{(n+m) \times (n+m)}$ can be written in block form

$$\mu = \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix} \text{ and } \Sigma = \begin{bmatrix} K_{XX} & K_{XY} \\ K_{YX} & K_{YY} \end{bmatrix}$$

where $K_{XY} = K_{YX}^\top = Cov(X, Y) \in \mathbb{R}^{n \times m}$, and $K_{XX} \in \mathbb{R}^{n \times n}$, $K_{YY} \in \mathbb{R}^{m \times m}$ are the variance matrices of the marginal distributions for X and Y , respectively. If X, Y are jointly normally distributed, then $X, Y \sim N(\mu, \Sigma)$. The conditional distribution for Y given X is given by $Y|X \sim N(\mu_{Y|X}, K_{Y|X})$ defined by the conditional mean $\mu_{Y|X} = \mu_Y + K_{YX} K_{XX}^{-1}(X - \mu_X)$ and conditional variance $K_{Y|X} = K_{YY} - K_{YX} K_{XX}^{-1} K_{XY}$. The matrix $K_{YX} K_{XX}^{-1}$ is called the matrix of regression coefficients.

More generally, we let $X = [X_1, \dots, X_m]^\top \in \mathbb{R}^{mn}$ be a random vector, where $X_i \in \mathbb{R}^n$ is a random vector with mean $\mu_i = (\mu_{i1}, \dots, \mu_{in})^\top \in \mathbb{R}^n$ and variance-covariance $\Sigma_i \in \mathbb{R}^{n \times n}$. The covariance of the random array X must account for the covariance between each pair its elements. Assume that all the elements of X are mutually independent, then $\Sigma_i = \sigma_i^2 I_n$ where $I_n \in \mathbb{R}^{n \times n}$ is the identity matrix. The mean $\mu_X = [\mu_1, \dots, \mu_m]^\top \in \mathbb{R}^{mn}$ and covariance $\Sigma_X \in \mathbb{R}^{mn \times mn}$ in block form are

$$\mu_X = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \dots \\ \mu_m \end{bmatrix} \text{ and } \Sigma_X = \begin{bmatrix} \Sigma_1 & \mathcal{O} & \dots & \mathcal{O} \\ \mathcal{O} & \Sigma_2 & \dots & \mathcal{O} \\ \dots & \dots & \dots & \dots \\ \mathcal{O} & \mathcal{O} & \dots & \Sigma_m \end{bmatrix}$$

where $\mathcal{O} \in \mathbb{R}^{n \times n}$ is a zero matrix.

Now, we let $X = [X_1, \dots, X_m]^\top \in \mathbb{R}^{m \times n}$ be a random array, where $X_i \in \mathbb{R}^{1 \times n}$ is a random vector with mean $\mu_i = (\mu_{i1}, \dots, \mu_{in}) \in \mathbb{R}^{1 \times n}$ and variance-covariance $\Sigma_i = E[X_i X_i^\top] \in \mathbb{R}$. The covariance of the random array X must account for the covariance between each pair its elements. Assume that all the elements of X are mutually independent, then $\Sigma_X = \sigma_i^2 I_m$, where $I_m \in \mathbb{R}^{m \times m}$ is the identity matrix. The mean $\mu_X = [\mu_1, \dots, \mu_m]^\top \in \mathbb{R}^{m \times n}$ and covariance $\Sigma_X \in \mathbb{R}^{m \times m}$ in block form are

$$\mu_X = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \dots \\ \mu_m \end{bmatrix} \text{ and } \Sigma_X = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \sigma_m^2 \end{bmatrix}$$

11.2.4.2 The Gaussian variables The vector $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ is called a standard normal random vector if all of its components X_i are independent and each is a zero-mean unit-variance normally distributed random variable, that is, $X_i \sim N(0, 1)$. Then, X is called a normal random vector if there exists a random d -vector Z which is a standard normal random vector, a n -vector μ , and a $n \times d$ matrix A such that

$$X = AZ + \mu$$

where the covariance matrix is $\Sigma = K_{XX} = AA^\top \in \mathbb{R}^{n \times n}$. Then, in matrix form we have

$$\begin{bmatrix} X_1 \\ X_2 \\ \dots \\ X_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1d} \\ a_{21} & a_{22} & \dots & a_{2d} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nd} \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \\ \dots \\ Z_d \end{bmatrix} + \begin{bmatrix} \mu_1 \\ \mu_2 \\ \dots \\ \mu_n \end{bmatrix}$$

The matrix $X = [X_1, \dots, X_m]^\top \in \mathbb{R}^{m \times n}$ is called a standard normal random matrix if all of its components $X_i \in \mathbb{R}^{1 \times n}$ are independent and each is a standard normal random vector. Then, X is called a normal random matrix if there exists a random $p \times n$ -matrix Z , with element $1 \times n$ -vector Z_i , which is a standard normal random matrix, a $m \times n$ -matrix μ , and a $m \times p$ matrix A , with element $1 \times p$ vector A_i , such that

$$X = AZ + \mu$$

where the covariance matrix is $\Sigma = AA^\top \in \mathbb{R}^{m \times m}$. Then, in block form we have

$$\begin{bmatrix} X_1 \\ X_2 \\ \dots \\ X_m \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ \dots \\ A_m \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \\ \dots \\ Z_p \end{bmatrix} + \begin{bmatrix} \mu_1 \\ \mu_2 \\ \dots \\ \mu_m \end{bmatrix}$$

11.2.4.3 The distribution The multivariate normal distribution is said to be non-degenerate when the symmetric covariance matrix Σ is positive definite (PD). In this case the distribution has density

$$f_X(x_1, \dots, x_k) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)}$$

where x is a real k -dimensional column vector and $|\Sigma| = \det \Sigma$ is the determinant of Σ also known as the generalized variance. The quantity $\sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}$ is known as the Mahalanobis distance, which represents the distance of the test point x from the mean μ . When $k = 1$, the Mahalanobis distance reduces to the absolute value of the standard score.

The notion of cumulative distribution function (cdf) in dimension 1 can be extended in two ways to the multidimensional case, based on rectangular and ellipsoidal regions: (1) define the cdf $F(x)$ of a random vector X as the probability that all components of X are less than or equal to the corresponding values in the vector x . That is

$$F(x) = P(X \leq x), X \sim N(\mu, \Sigma)$$

Though there is no closed form for $F(x)$, there are a number of algorithms that estimate it numerically. (2) define the cdf $F(r)$ as the probability that a sample lies inside the ellipsoid determined by its Mahalanobis distance r from the Gaussian, a direct generalisation of the standard deviation. In order to compute the values of this function, closed analytic formulae exist.

Consider a random vector X which follows a multivariate normal with zero means and Identity Covariance, $X \sim N_n(0, I)$. We can say that the individual variables $X_1, \dots, X_n$ are pair-wisely independent. Note that pair-wise independence does not generally imply mutual independence, as the latter is a stronger condition. In that case, the density is

$$p(X) = (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2} X^\top X}$$

If the variables are uncorrelated then the variance-covariance matrix will be a diagonal matrix with variances of the individual variables appearing on the main diagonal of the matrix and zeros everywhere else.

Theorem 11.2 Suppose $X \sim N_n(\mu, \Sigma)$. If $X = [X_1 X_2]$ and the corresponding off-diagonal of Σ_{12} is zero, then X_1 and X_2 are independent.

In particular, if Σ is a diagonal matrix, then $X_1, \dots, X_n$ are mutually independent. Suppose Σ is a diagonal matrix whose diagonal entries $\sigma_1^2, \dots, \sigma_n^2$ positive. Let $x = (x_1, \dots, x_n)^\top$. Then

$$e^{-\frac{1}{2} x^\top \Sigma^{-1} x} = e^{-\frac{1}{2} \sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}} = \prod_{i=1}^n e^{-\frac{1}{2} \frac{x_i^2}{\sigma_i^2}} = \prod_{i=1}^n \phi(x_i; 0, \sigma_i^2)$$

More generally, we get

$$f_X(x_1, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sigma_i \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x_i - \mu_i}{\sigma_i} \right)^2} = \prod_{i=1}^n \phi(x_i; \mu_i, \sigma_i^2)$$

which is the product of univariate normal distributions.

Theorem 11.3 Let b be a $k \times 1$ vector of constants, B a $k \times d$ matrix of constants, and $X \sim N_d(\mu, \Sigma)$. Then,

$$b + BX \sim N_k(B\mu + b, B\Sigma B^\top)$$

In the case where Σ is non-singular and positive definite, we get

Corollary 1 Let $X \sim N_d(\mu, \Sigma)$, Then,

$$\Sigma^{-\frac{1}{2}}(X - \mu) \sim N_d(0, I)$$

where $\Sigma^{-\frac{1}{2}}$ is the square-root of Σ^{-1} .

11.3 A few facts about moments

See books by Kendall et al. [1977], Vapnik [1998], and Kenney [2013], among others.

11.3.1 Defining the moments

We let X be an independent and identically distributed random variables. A central moment is a moment of the probability distribution of the random variable X around its mean. That is, the n th central moment of a real-valued random variable X is

$$\mu_n = E[(X - E[X])^n], n > 1 \quad (11.57)$$

where $\tilde{\mu}_n = E[X^n]$ is the n th non-central moment (or raw moment). We define the cumulant of the random variable X as

$$\kappa_n = \frac{d^n}{dz^n} \ln E[e^{zX}] \Big|_{z=0}, n = 1, 2, \dots \quad (11.58)$$

A fundamental task in many statistical analyses is to characterise the location and variability of a data set. The expectation is $\mu_1 = \mu_X = E[X]$, and we call the second central moment μ_2 , the variance, denoted σ_X^2 . The variance of a random variable X is

$$\sigma_X^2 = Var(X) = E[X^2] - (E[X])^2 = \mu_2 = \kappa_2$$

A further characterisation of the data includes skewness and kurtosis.

11.3.1.1 The skewness The skewness of a random variable X is the third standardised moment given by

$$\begin{aligned} S(X) &= E\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^3\right] = \frac{\mu_3}{\sigma_3} = \frac{\kappa_3}{\kappa_2^{3/2}} \\ &= \frac{E[X^3] - 3\mu_X(E[X^2] - \mu_X E[X]) - \mu_X^3}{\sigma_X^3} \\ &= \frac{E[X^3] - 3\mu_X\sigma_X^2 - \mu_X^3}{\sigma_X^3} \end{aligned}$$

where μ_X is the mean and σ_X is the standard deviation of X , μ_3 is the third central moment, and κ_k is the k th cumulant. Here, $E[X^3]$ is the third non-central moment. It is sometimes referred to as Pearson's moment coefficient of skewness, or simply the moment coefficient of skewness. Skewness is a measure of the lack of symmetry. The skewness for a normal distribution is zero, and any symmetric data should have a skewness near zero. Negative values for the skewness indicate data that are skewed left and positive values for the skewness indicate data that are skewed right. Note, Skewness can be infinite.

11.3.1.2 The kurtosis The kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails, or outliers. Data sets with low kurtosis tend to have light tails, or lack of outliers. The histogram is an effective graphical technique for showing both the skewness and kurtosis of data set. The kurtosis is the fourth standardised moment, defined as

$$K(X) = E\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^4\right] = \frac{\mu_4}{\sigma^4}$$

where μ_4 is the fourth central moment and σ is the standard deviation. For any outlier x_p , the value $(x_p - \mu)$ tends to be large. Here, the outliers contribution to the sum is raised to the power of four. As a result, outliers significantly impact the value of the kurtosis. The kurtosis is bounded below by the squared skewness plus one:

$$K(X) \geq \left(\frac{\mu_3}{\sigma_3}\right)^2 + 1$$

11.3.1.3 The general standardised summary statistics A standardised moment of a probability distribution is a moment (often a higher degree central moment) that is normalised, typically by a power of the standard deviation, rendering the moment scale invariant. The shape of different probability distributions can be compared using standardised moments. In the general case, we get the n th standardised moment as follows:

$$\gamma_n(X) = E\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^n\right] = \frac{\mu_n}{\sigma^n}, n \geq 3 \quad (11.59)$$

which is the ratio of the n th moment about the mean, μ_n , to the n th power of the standard deviation

$$\sigma^n = \mu_2^{\frac{n}{2}} = \left(\sqrt{E[(X - \mu_X)^2]}\right)^n = (\sigma^2)^{n/2}$$

Note, $\gamma_1(X) = 0$ and $\gamma_2(X) = 1$. As a result, we can compute the n th central moment as

$$\mu_n(X) = \gamma_n(X) \sigma_n(X), n \geq 3 \quad (11.60)$$

where $\sigma_n(X) = (\sigma_X^2)^{n/2}$, and $\mu_1 = \mu_X$ and $\mu_2 = \sigma_X^2$. We see that the third standardised moment is a measure of skewness, and the fourth standardised moment refers to the kurtosis.

11.3.2 Defining the cumulants

The cumulants κ_n of a probability distribution are a set of quantities that provide an alternative to the moments of the distribution. Any two probability distributions whose moments are identical will have identical cumulants as well, and vice versa. The cumulants of a random variable X are defined by using the cumulant-generating function $K(t)$ which is the natural logarithm of the moment-generating function:

$$K(t) = \log E[e^{tX}]$$

If it exists, the cumulant generating function $K(t)$ is infinitely differentiable and convex, and passes through the origin. The cumulants κ_n are obtained from a power series expansion of the cumulant generating function:

$$K(t) = \sum_{n=1}^{\infty} \kappa_n \frac{t^n}{n!} = \kappa_1 \frac{t}{1!} + \kappa_2 \frac{t^2}{2!} + \dots = \mu t + \sigma^2 \frac{t^2}{2} + \dots$$

which is a Maclaurin series. The n th cumulant can be obtained by differentiating the above expansion n times and evaluating the result at zero

$$\kappa_n = K^{(n)}(0)$$

where $K^{(n)}$ is the n th derivative of K . The n th cumulant $\kappa_n(X)$ a random variable X exhibits the following properties:

- The cumulant is translation-invariant: If $n > 1$ and c is constant, then $\kappa_n(X + c) = \kappa_n(X)$. When $n = 1$, we get $\kappa_1(X + c) = \kappa_1(X) + c$.
- The n th cumulant is homogeneous of degree n : If c is constant, then $\kappa_n(cX) = c^n \kappa_n(X)$.

- The cumulant is cumulative (hence the name): If the random variables $X_1, \dots, X_m$ are independent then

$$\kappa_n(X_1 + \dots + X_m) = \kappa_n(X_1) + \dots + \kappa_n(X_m)$$

It follows directly from the cumulant-generating function:

$$\begin{aligned} K_{X_1 + \dots + X_m}(t) &= \log E[e^{t(X_1 + \dots + X_m)}] \\ &= \log E[e^{t(X_1)}] + \dots + \log E[e^{t(X_m)}] \\ &= K_{X_1}(t) + \dots + K_{X_m}(t) \end{aligned}$$

The cumulants of degrees 2 and 3 are central moments. All of the higher cumulants are polynomial functions of the central moments with integer coefficients:

$$\begin{aligned} \kappa_1(X) &= E[X] \\ \kappa_2(X) &= \text{var}(X) = E[(X - E[X])^2] \\ \kappa_3(X) &= E[(X - E[X])^3] \\ \kappa_4(X) &= E[(X - E[X])^4] - 3(\text{var}(X))^2 \\ \kappa_5(X) &= E[(X - E[X])^5] - 10E[(X - E[X])^3]\text{var}(X) \end{aligned}$$

Note, the moment generating function is given by:

$$M(t) = 1 + \sum_{n=1}^{\infty} \frac{\tilde{\mu}_n}{t^n n!} = \sum_{n=1}^{\infty} \frac{\kappa_n}{t^n n!} = e^{K(t)}$$

Thus, the cumulant generating function is the logarithm of the moment generating function

$$K(t) = \log M(t)$$

The moments can be recovered in terms of cumulants by evaluating the n th derivative of $e^{K(t)}$ at $t = 0$

$$\tilde{\mu}_n = M^{(n)}(0) = \frac{d^n}{dt^n} e^{K(t)} \Big|_{t=0}$$

and the cumulants can be recovered in terms of moments by evaluating the n th derivative of $\log M(t)$ at $t = 0$

$$\kappa_n = K^{(n)}(0) = \frac{d^n}{dt^n} \log M(t) \Big|_{t=0}$$

The explicit expression for the n -th moment in terms of the first n cumulants, and vice versa, can be obtained by using Faa di Bruno's formula for higher derivatives of composite functions. In general, we have

$$\begin{aligned} \tilde{\mu}_n &= \sum_{k=1}^n B_{n,k}(\kappa_1, \dots, \kappa_{n-k+1}) \\ \kappa_n &= \sum_{k=1}^n (-1)^{k-1} (k-1)! B_{n,k}(\tilde{\mu}_1, \dots, \tilde{\mu}_{n-k+1}) \end{aligned}$$

where $B_{n,k}$ are incomplete (or partial) Bell polynomials. In combinatorics, the n th Bell number is the number of partitions of a set of size n . All of the cumulants of the sequence of Bell numbers are equal to 1. The Bell numbers

are the moments of the Poisson distribution with expected value 1.

Denoting the mean μ , the central moment generating function is given by

$$C(t) = E[e^{t(X-\mu)}] = e^{\mu t} M(t) = e^{K(t)-\mu t}$$

The n -th central moment is obtained in terms of cumulants as

$$\mu_n = C^{(n)}(0) = \frac{d^n}{dt^n} e^{K(t)-\mu t} \Big|_{t=0} = \sum_{k=1}^n B_{n,k}(0, \kappa_2, \dots, \kappa_{n-k+1})$$

and for $n > 1$, the n th cumulant in terms of the central moments is

$$\kappa_n = K^{(n)}(0) = \frac{d^n}{dt^n} (\log C(t) + \mu t) \Big|_{t=0} = \sum_{k=1}^n (-1)^{k-1} (k-1)! B_{n,k}(0, \mu_2, \dots, \mu_{n-k+1})$$

The cumulants can be related to the moments by differentiating the relationship $\log M(t) = K(t)$ with respect to t , giving $M'(t) = K'(t)M(t)$, which conveniently contains no exponentials or logarithms. Equating the coefficient of t^{n-1} on the left and right sides and using $\tilde{\mu}_0 = 1$, for $n \geq 1$, we get

$$\tilde{\mu}_n = \sum_{k=1}^{n-1} \binom{n-1}{k-1} \kappa_k \tilde{\mu}_{n-k} + \kappa_n$$

which is Equation (11.63) below. As a result, we can compute κ_n or $\tilde{\mu}_n$ from the other using knowledge of the lower-order cumulants and moments. The corresponding formulas for the central moments μ_n , for $n \geq 2$, are formed from these formulas by setting $\tilde{\mu}_1 = \kappa_1 = 0$ and replacing each $\tilde{\mu}_n$ with μ_n .

11.3.3 Computing the moments recursively

Using the central moments, Meucci [2010c] derived the noncentral moments $\tilde{\mu}_1, \dots, \tilde{\mu}_n$ for X . Given $\tilde{\mu}_1 = \mu_1$, we solve the recursive equation

$$\tilde{\mu}_n(X) = (-1)^{n+1} \mu_1^n(X) + \sum_{k=1}^{n-1} \binom{n}{k} (-1)^{n-k+1} \mu_1^{n-k}(X) \tilde{\mu}_k(X) + \mu_n(X), \quad n > 1 \quad (11.61)$$

where the binomial coefficient $\binom{n}{k}$ is the combination $C_k^n = \frac{n!}{(n-k)!k!}$.

Knowing the non-central moments, the author also derived the cumulants as follows: from the Taylor series we get the approximations

- $E[e^{zX}] \approx E[1 + zX] = 1 + z\tilde{\mu}_1$ for any small z .
- $\ln(1+x) \approx x$ for any small x .

Given $\kappa_1 = \tilde{\mu}_1$, we get the recursive equation

$$\kappa_n(X) = \tilde{\mu}_n(X) - \sum_{k=1}^{n-1} \binom{n-1}{k-1} \kappa_k(X) \tilde{\mu}_{n-k}(X), \quad n > 1 \quad (11.62)$$

Alternatively, we can reverse the process and compute all the moments starting from the cumulants. First, we compute the non-central moments $\tilde{\mu}_1(X), \dots, \tilde{\mu}_n(X)$ with the following recursive equation

$$\tilde{\mu}_n(X) = \kappa_n(X) + \sum_{k=1}^{n-1} \binom{n-1}{k-1} \kappa_k(X) \tilde{\mu}_{n-k}(X) \quad (11.63)$$

Further, using the noncentral moments, we can derive the central moments $\mu_1, \dots, \mu_n$ for X . We solve the recursive equation

$$\mu_n(X) = (-1)^n \mu_1^n(X) + \sum_{k=1}^{n-1} \binom{n}{k} (-1)^{n-k} \tilde{\mu}_k(X) \mu_1^{n-k}(X) + \tilde{\mu}_n(X) \quad (11.64)$$

11.4 The sample moments

We assume that the population is of size N and that associated with each member of the population is a numerical value of interest denoted by $x_1, x_2, \dots, x_N$. We take a sample with replacement (SWR) of n values $X_1, \dots, X_n$ from the population, where $n < N$ and such that X_i is a random variable. That is, X_i is the value of the i th member of the sample, and x_i is that of the i th member of the population.

While volatility is a parameter measuring the risk associated with the returns of the underlying price, local volatility is a parameter measuring the risk associated with the instantaneous variation of the underlying price. It can be deterministic or stochastic. On the other hand, historical volatility is computed by using historical data on observed values of the underlying price (opening, closing, highest value, lowest value etc). In general, one uses standard estimators of variance per unit of time of the logarithm of the underlying price, which is assumed to follow a non-centred Brownian motion. As discussed in Appendix (11.3), the projection (or annualisation) of statistics must be a sum to use the properties of the cumulants. In the case of log-returns, the continuously compounded multiperiod return is simply the sum of continuously compounded one-period returns (see Equation (12.79)).

We are now going to present conventional measures for location and dispersion as well as skewness and kurtosis. However, these measures are extremely sensitive to single outliers or small groups of outliers (see Kim et al. [2003]). While robust measures of location and dispersion are well known in the literature, it is more difficult to get robust measures of skewness and kurtosis. We also present some of these measures.

11.4.1 Summarising the data

Following Brockwell et al. [1991], we write the real-valued series of observations as $\dots, Y_{-2}, Y_{-1}, Y_0, Y_1, Y_2, \dots$ a doubly infinite sequence of real-valued random variables indexed by $\mathbb{Z}$. Given a set of n values $Y_1, Y_2, \dots, Y_n$, we place these values in ascending order written as $Y_{(1)} \leq Y_{(2)} \leq \dots \leq Y_{(n)}$. The median is the middle observation. When n is odd it can be written $n = 2m + 1$ and the median is $Y_{(m+1)}$, and when n is even we get $n = 2m$ and the median is $\frac{1}{2}(Y_{(m)} + Y_{(m+1)})$. It is possible to have two very different datasets with the same means and medians. For that reason, measures of the middle are useful but limited. Another important attribute of a dataset is its dispersion or variability about its middle. The most useful measures of dispersion (or variability) are the range, the percentiles, the mean absolute deviation, and the standard deviation. The **range** denotes the difference between the largest and smallest values in the sample

$$Range = Y_{(n)} - Y_{(1)}$$

Therefore, the more spread out the data values are, the larger the range will be. However, if a few observations are relatively far from the middle but the rest are relatively close to the middle, the range can give a distorted measure of dispersion. **Percentiles** are positional measures for a dataset that enable one to determine the relative standing of a single measurement within the dataset. In particular, the p th percentile is defined to be a number such that $p\%$ of the observations are less than or equal to that number and $(100 - p)\%$ are greater than that number. So, for example, an observation that is at the 75th percentile is less than only 25% of the data. In practice, we often can not satisfy the definition exactly. However, the steps outlined below at least satisfies the spirit of the definition:

1. Order the data values from smallest to largest, including ties.

2. Determine the position

$$k.ddd = 1 + \frac{p(n-1)}{100}$$

3. The p th percentile is located between the k th and the $(k+1)$ th ordered value. Use the fractional part of the position, $.ddd$ as an interpolation factor between these values. If $k = 0$, then take the smallest observation as the percentile and if $k = n$, then take the largest observation as the percentile.

The 50th percentile is the median and partitions the data into a lower half (below median) and upper half (above median). The 25th, 50th, 75th percentiles are referred to as quartiles. They partition the data into 4 groups with 25% of the values below the 25th percentile (lower quartile), 25% between the lower quartile and the median, 25th between the median and the 75th percentile (upper quartile), and 25% above the upper quartile. The difference between the upper and lower quartiles is referred to as the inter-quartile range. This is the range of the middle 50% of the data. Given $d_i = Y_i - \bar{Y}$ where $\bar{Y}$ is the arithmetic mean, the Mean Absolute Deviation (MAD) is the average of the deviations about the mean, ignoring the sign

$$MAD = \frac{1}{n} \sum_i |d_i|$$

The sample variance is an average of the squared deviations about the mean

$$S^2 = \frac{1}{n-1} \sum_i d_i^2$$

The population variance is given by

$$\sigma_p^2 = \frac{1}{n} \sum_i (Y_i - \mu_p)^2$$

where μ_p is the population mean. Note that the unit of measure for the variance is the square of the unit of measure for the data. For that reason (and others), the square root of the variance, called the standard deviation, is more commonly used as a measure of dispersion. Note that datasets in which the values tend to be far away from the middle have a large variance (and hence large standard deviation), and datasets in which the values cluster closely around the middle have small variance. Unfortunately, it is also the case that a dataset with one value very far from the middle and the rest very close to the middle also will have a large variance. Comparing the variance with the MAD, S gives greater weight to the more extreme observations by squaring them and it may be shown that $S > MAD$ whenever MAD is greater than zero. A rough relationship between the two is

$$S = 1.25 \cdot MAD$$

Chebychev's inequality (or Markov's inequality) guarantees that, for a wide class of probability distributions, no more than $\frac{1}{k^2}$ of the distribution's values can be more than k standard deviations away from the mean. Equivalently, at least $(1 - \frac{1}{k^2})$ of the distribution's values are within k standard deviations of the mean. That is, the standard deviation of a dataset can be interpreted by Chebychev's Theorem (see Tchebichef [1867]):

Theorem 11.4 *Chebychev's Theorem*

For any $k > 1$, the proportion of observations within the interval $\mu_p \pm k\sigma_p$ is at least $(1 - \frac{1}{k^2})$.

The inequality has great utility because it can be applied to any probability distribution in which the mean and variance are defined. Hence, knowing just the mean and standard deviation of a dataset allows us to obtain a rough picture of the distribution of the data values. Note that the smaller the standard deviation, the smaller is the interval that is guaranteed to contain at least 75% of the observations. Conversely, the larger the standard deviation, the more likely it is that an observation will not be close to the mean. Note that Chebychev's Theorem applies to all data and therefore

must be conservative. In many situations the actual percentages contained within these intervals are much higher than the minimums specified by this theorem. If the shape of the data histogram is known, then better results can be given. In particular, if it is known that the data histogram is approximately bell-shaped (like the Gaussian distribution in Figure (69)), then we can say

- $\mu_p \pm \sigma_p$ contains approximately 68%,
- $\mu_p \pm 2\sigma_p$ contains approximately 95%,
- $\mu_p \pm 3\sigma_p$ contains essentially all

of the data values. This set of results is called the empirical rule. Several extensions of the Chebyshev's inequality have been developed, among which is the asymmetric two-sided version given by

$$P(k_1 < Y < k_2) \geq \frac{4((\mu_p - k_1)(k_2 - \mu_p) - \sigma_p^2)}{(k_2 - k_1)^2}$$

11.4.2 The population mean and volatility

11.4.2.1 Description We let X_t be a Gaussian random variable with drift μ and volatility σ . We observe values uniformly allocated in time with time difference δ . That variable seen at time $t = j\delta$ for the time $t + 1 = (j + 1)\delta$ is given by

$$X(j, j + 1) = \mu\delta - \sigma(W_{(j+1)\delta} - W_{j\delta})$$

Given N period of time, and $X_{j\delta} = X(j, j + 1)$, the first two sample moments are

$$\hat{\mu}_N = \frac{1}{N} \sum_{j=0}^{N-1} X_{j\delta} \text{ and } \hat{\sigma}_N^2 = \frac{1}{N} \sum_{j=0}^{N-1} (X_{j\delta} - \hat{\mu}_N)^2 \quad (11.65)$$

Since the observations are selected randomly, both $\hat{\mu}_N$ and $\hat{\sigma}_N^2$ are random variables. Their expected values can be evaluated by averaging over the ensemble of all possible samples of size N from the population. We get

$$E[\hat{\sigma}_N^2] = \frac{n-1}{n} \sigma^2$$

with population variance $\sigma^2 = E[X_j^2] - \mu^2$ where $\mu^2 = E[X_i X_j] = E[X_i]E[X_j]$ due to independence of X_i and X_j . Note, $\hat{\sigma}_N^2$ gives an estimate of the population variance that is biased by a factor of $\frac{n-1}{n}$. Hence, it is a biased sample variance. Correcting for this bias yields the unbiased sample variance

$$S^2 = \frac{n-1}{n} \hat{\sigma}_N^2$$

These two estimators are simply referred to as the sample variance. Since the sample variance is a function of random variables, it is itself a random variable. Hence, we must study its distribution. Assuming independent observations from a normal distribution, given the population variance σ^2 , the unbiased sample variance S^2 follows a scaled chi-squared distribution

$$(n-1) \frac{S^2}{\sigma^2} \sim \chi_{n-1}^2$$

As a direct consequence, it follows that

$$E[S^2] = E\left[\frac{\hat{\sigma}^2}{n-1} \chi_{n-1}^2\right] = \hat{\sigma}^2 \text{ and } Var(S^2) = \frac{\hat{\sigma}^4}{(n-1)^2} Var(\chi_{n-1}^2) = \frac{2\hat{\sigma}^4}{n-1}$$

For non-normal samples, the variance of the sample variance depends on the kurtosis. If the observations are independent and identically distributed, but not necessarily normally distributed, then

$$E[S^2] = \hat{\sigma}^2 \text{ and } Var(S^2) = \frac{\hat{\sigma}^4}{n} \left(K - 1 + \frac{2}{n-1} \right) = \frac{1}{n} \left(\mu_4 - \frac{n-3}{n-1} \hat{\sigma}^4 \right)$$

where K is the kurtosis of the distribution (see Appendix (11.4.3.2)) and μ_4 is the fourth central moment. Note that the variance of the estimator tends asymptotically to zero (when n goes to infinity).

In order to use statistical techniques on market data, one must make sure that the data is stationary, and test the assumption of log-normality on the observed values of the underlying price. Note, the underlying prices rarely satisfy the Black-Scholes assumption. However, we are not trying to compute option prices in the risk-neutral measure, but we are estimating the moments of the stock returns under the historical measure.

11.4.2.2 Implementation Given the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, we compute the sample mean and sample variance recursively as follows: Starting from the identities

$$(n+1)\bar{X}_{n+1} = n\bar{X}_n + X_{n+1}$$

and

$$(n+1)(\sigma_{n+1}^2 + (\bar{X}_{n+1})^2) = n(\sigma_n^2 + (\bar{X}_n)^2) + X_{n+1}^2$$

we get the recursive sample mean

$$\bar{X}_{n+1} = \bar{X}_n + \frac{X_{n+1} - \bar{X}_n}{n+1} \quad (11.66)$$

and the recursive sample variance

$$\sigma_{n+1}^2 = \sigma_n^2 + (\bar{X}_n)^2 - (\bar{X}_{n+1})^2 + \frac{X_{n+1}^2 - \sigma_n^2 - (\bar{X}_n)^2}{n+1} \quad (11.67)$$

such that σ_{n+1}^2 only depends on σ_n^2 , $\bar{X}_n$, $\bar{X}_{n+1}$ and X_{n+1} . If we normalise the data with a constant K such that the i th value is $\frac{X_i}{K}$, we let $\tilde{X} = \frac{1}{n} \sum_{i=1}^n \frac{X_i}{K}$ be the sample mean and $\tilde{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{X_i}{K} - \tilde{X} \right)^2$ be the sample variance. Then, the recursive sample mean becomes

$$\tilde{X}_{n+1} = \tilde{X}_n + \frac{1}{n+1} \left(\frac{X_{n+1}}{K} - \tilde{X}_n \right) \quad (11.68)$$

and the recursive sample variance becomes

$$\tilde{\sigma}_{n+1}^2 = \tilde{\sigma}_n^2 + (\tilde{X}_n)^2 - (\tilde{X}_{n+1})^2 + \frac{1}{n+1} \left(\frac{X_{n+1}^2}{K^2} - \tilde{\sigma}_n^2 - (\tilde{X}_n)^2 \right) \quad (11.69)$$

11.4.3 The population skewness and kurtosis

For $t = j\delta$, we let $X_t = \ln \frac{S_{(j+1)\delta}}{S_{j\delta}}$, $t = 1, \dots, N$, be the log-returns. We assume that the X_t are independent and identically distributed with a cumulative distribution function F .

11.4.3.1 The population skewness As defined in Appendix (11.3), skewness is a statistical measure of the asymmetry of the probability distribution of a random variable (in this case the return of a stock) around its mean. It can be positive, zero, negative, or undefined. In the case of negative skew, the left tail is longer and the mass of the distribution is concentrated on the right of the figure. Conversely, in the case of positive skew, the right tail is longer and the mass of the distribution is concentrated on the left of the figure. This is illustrated in Figure (71). A distribution with negative skew can have its mean greater than or less than the median, and likewise for positive skew. Since a normal distribution is symmetrical, then the mean is equal to the median, and it exhibits exactly zero skewness. Hence, skewness indicates the direction and relative magnitude of a distribution's deviation from the normal distribution. That is, whether deviations from the mean are going to be positive or negative. Moreover, the more asymmetric the distribution, thus unlike a normal distribution, the larger the figure gets in absolute terms. With pronounced skewness, standard statistical inference procedures such as a confidence interval for a mean will be incorrect, in the sense that the true coverage level will differ from the nominal (such as 95%) level. In addition, they will also result in unequal error probabilities on each side.

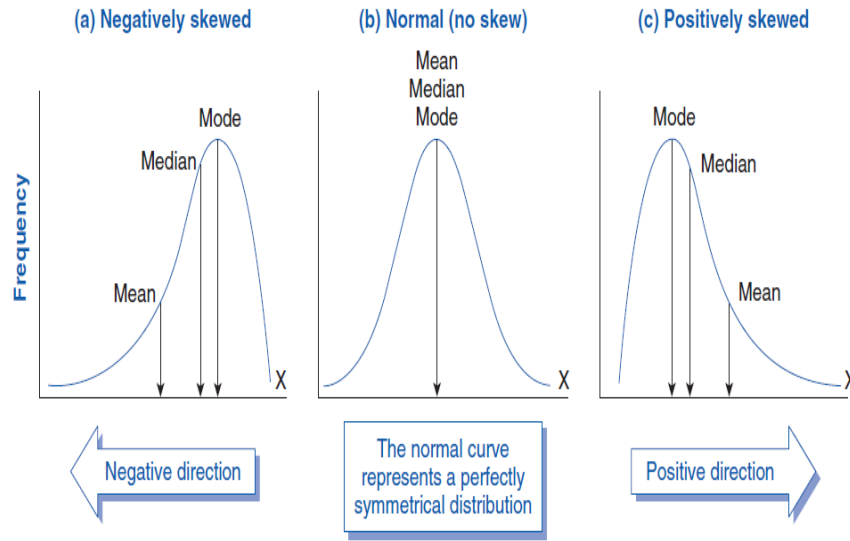


Figure 71: Distribution skew.

Given N period of time, the sample skewness on log-returns is

$$\begin{aligned}
 g_1 &= \hat{S}_N = \frac{m_3}{m_2^{3/2}} \\
 &= \frac{1}{N\hat{\sigma}_N^3} \sum_{j=0}^{N-1} \left(\ln \frac{S_{(j+1)\delta}}{S_{j\delta}} - \hat{\mu}_N \right)^3
 \end{aligned}$$

where $\hat{\sigma}_N$ is the sample standard deviation, m_2 is the (biased) sample second central moment, and m_3 is the sample third central moment. This is the Fisher-Pearson coefficient of skewness.

Remark 11.3 In the case of non-symmetric distribution, g_1 is not a good measure when N is small. it does move towards the limiting value as $N \rightarrow \infty$.

The adjusted Fisher-Pearson coefficient of skewness is given by

$$G_1 = \frac{\kappa_3}{\kappa_2^{3/2}} = \bar{S}_N = \frac{\sqrt{N(N-1)}}{N-2} \hat{S}_N$$

where κ_3 is the unique symmetric unbiased estimator of the third cumulant and κ_2 is the symmetric unbiased estimator of the second cumulant (i.e. the sample variance). The adjustment approaches 1 as N gets large. For reference, the adjustment factor is 1.49 for $N = 5$, 1.19 for $N = 10$, 1.08 for $N = 20$, 1.05 for $N = 30$, and 1.02 for $N = 100$. Under the assumption that the underlying random variable X is normally distributed, it can be shown that all these ratios (g_1 and G_1) are unbiased and consistent estimators of the population skewness. That is, their distributions converge to a normal distribution with mean 0 and variance 6. The variance of the sample skewness is thus approximately $\frac{6}{N}$ for sufficiently large samples. When we just have one sample drawn from a population we need a test statistic to measure how many standard errors separate the sample skewness from zero. We divide the sample skewness G_1 by the standard error of skewness (SES) (see Cramer [1997]), getting

$$Z_{g_1} = \frac{G_1}{SES} \text{ where } SES = \sqrt{\frac{6N(N-1)}{(N-2)(N+1)(N+3)}} \text{ or } SES = \sqrt{\frac{6}{N}}$$

This is a two-tailed test of skewness different from 0 at roughly the 0.05 significance level. It tells us whether the whole population is probably skewed, but not by how much. The bigger the number, the higher the probability. The critical value of Z_{g_1} is approximately 2. That is,

$$\begin{cases} \text{if } Z_{g_1} < -2 \text{ the population is very likely skewed negatively} \\ \text{if } Z_{g_1} > 2 \text{ the population is very likely skewed positively} \\ \text{for any values in } [-2, 2] \text{ we can't reach any conclusion about the skewness of the population} \end{cases}$$

We can get a 95% confidence interval of population skewness

$$G_1 \pm 2 \times SES$$

It should be noted that there are alternative definitions of skewness in the literature. For example, the Galton skewness, also known as Bowley's skewness (see Bowley [1920]), is defined as

$$S_G = \frac{Q_1 + Q_3 - 2Q_2}{Q_3 - Q_1}$$

where $Q_1 = F^{-1}(0.25)$ is the lower quartile, $Q_3 = F^{-1}(0.75)$ is the upper quartile, and $Q_2 = F^{-1}(0.5)$ is the median (see Appendix (11.2.3)). The denominator $Q_3 - Q_1$ re-scales the coefficient so that the maximum value for S_G is 1, representing extreme right skewness, and the minimum value is -1 , representing extreme left skewness.

Hinkley [1975] generalised the Bowley coefficient of skewness as follows:

$$S_H(\alpha) = \frac{F^{-1}(\alpha) + F^{-1}(1 - \alpha) - 2Q_2}{F^{-1}(1 - \alpha) - F^{-1}(\alpha)}$$

for $\alpha \in [0, \frac{1}{2}]$. We recover the Bowley coefficient of skewness for $\alpha = 0.25$. Since it is not not easy to estimate α , Groeneveld et al. [1984] proposed to integrate out α as follows:

$$\begin{aligned} S_{GM} &= \frac{\int_0^{0.5} (F^{-1}(\alpha) + F^{-1}(1 - \alpha) - 2Q_2) d\alpha}{\int_0^{0.5} (F^{-1}(1 - \alpha) - F^{-1}(\alpha)) d\alpha} \\ &= \frac{\mu_N - Q_2}{E[X_t - Q_2]} \end{aligned}$$

which is bounded by -1 and 1 . Note, the denominator $E[X_t - Q_2]$ is similar to a dispersion measure. The Pearson coefficient of skewness (mode skewness) (see Kendall et al. [1977]) is obtained by replacing the denominator with the standard deviation as follows:

$$S_{KS} = \frac{\mu_N - Q_2}{\sigma_N}$$

The Pearson 2 skewness coefficient (median skewness) is defined as

$$S_{\kappa_2} = 3(\mu_N - \tilde{X}_N)$$

where $\tilde{X}_N$ is the sample median.

11.4.3.2 The population kurtosis The kurtosis has been associated to measures of both the peakedness of the distribution and the heaviness of its tail. However, it isn't a measure of the peakedness of a probability distribution of random variables, but it is related to the tails of the distribution (see Balanda et al. [1988]). The only data values that contribute to kurtosis in any meaningful way are those outside the region of the peak: the outliers. Higher kurtosis means more of the variance is due to infrequent extreme deviations (fat tails) (outliers), lower kurtosis implies a variance composed of frequent modestly-sized deviations (thin tails) (see Westfall [2014]). Normally distributed asset returns exhibit a kurtosis of 3. To measure the excess kurtosis with regard to a normal distribution, a value of 3 is hence subtracted from the kurtosis value. Distributions with negative excess kurtosis is called platykurtic, having thinner tails producing fewer and/or less extreme outliers than the normal distribution. A distribution with positive excess kurtosis is called leptokurtic: it has tails that asymptotically approaches zero more slowly than a Gaussian (fatter tails), producing more outliers than the normal distribution. This is illustrated in Figure (72).

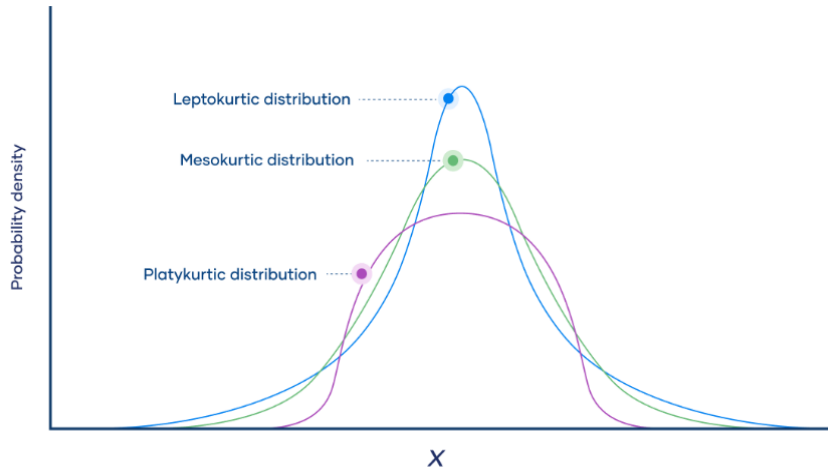


Figure 72: Distribution kurtosis.

Given a random variable X with mean μ and standard deviation σ , Moors set the variable $Z = \frac{X - \mu}{\sigma}$ and defined kurtosis as $\kappa = E[Z^4] = \frac{\mu_4}{\sigma^4} = \text{var}(Z^2) + 1$. Hence, it is a measure of the dispersion of Z^2 around its expectation. It can be seen as a measure of the dispersion of Z around ± 1 , attaining its minimal value in a symmetric two-point distribution. When expressed in terms of X , it is a measure of the dispersion of X around the two values $\mu \pm \sigma$. High value of the kurtosis occurs for two reasons:

1. The probability mass is concentrated around the mean and the data-generating process produces occasional values far from the mean.

2. The probability mass is concentrated in the tails of the distribution.

We can calculate the sample kurtosis of a single asset class with

$$\hat{K}_N = \frac{1}{N\hat{\sigma}_N^4} \sum_{j=0}^{N-1} (X_{j\delta} - \hat{\mu}_N)^4$$

where $\hat{\sigma}_N$ is the sample standard deviation computed using N in the denominator rather than $N - 1$. Under normality assumption, skew and kurtosis are distributed asymptotically as normal with mean equal to zero and variance being $\frac{6}{N}$ and $\frac{24}{N}$, respectively (see Snedecor et al. [1980]).

Remark 11.4 *In the case of non-symmetric distribution, $\hat{K}_N$ is not a good measure when N is small. it does move towards the limiting value as $N \rightarrow \infty$.*

Since the kurtosis for a standard normal distribution is 3, some sources use the following definition of kurtosis (often referred to as 'excess kurtosis'):

$$g_2 = \hat{K}_{3,N} = \hat{K}_N - 3 = \frac{m_4}{m_2^2} - 3$$

where m_4 is the fourth sample moment about the mean, m_2 is the second sample moment about the mean (that is, the sample variance). This definition is used so that the standard normal distribution has a kurtosis of zero. In addition, with the second definition positive kurtosis indicates a 'heavy-tailed' distribution and negative kurtosis indicates a 'light tailed' distribution. An upper bound for the sample kurtosis of n real numbers is

$$g_2 \leq \frac{1}{2} \frac{(n-3)}{(n-2)} g_1^2 + \frac{n}{2} - 3$$

where $g_1 = \frac{m_3}{m_2^{3/2}}$ is the corresponding sample skewness. The sample kurtosis is a useful measure of whether there is a problem with outliers in a data set. Larger kurtosis indicates a more serious outlier problem.

Given a sub-set of samples from a population, the sample excess kurtosis g_2 above is a biased estimator of the population excess kurtosis. An alternative estimator of the population excess kurtosis, which is unbiased in random samples of a normal distribution, is the adjusted Fisher-Pearson standardised moment coefficient. It is defined as follows:

$$\begin{aligned} G_2 &= \frac{\kappa_4}{\kappa_2^2} \\ &= \frac{(n-1)}{(n-2)(n-3)} ((n+1)g_2 + 6) \end{aligned}$$

where κ_4 is the unique symmetric unbiased estimator of the fourth cumulant, κ_2 is the unbiased estimate of the second cumulant (identical to the unbiased estimate of the sample variance), m_4 is the fourth sample moment about the mean, m_2 is the second sample moment about the mean.

When we just have one sample drawn from a population we need a test statistic to measure how many standard errors separate the sample excess kurtosis from zero. We divide the sample excess kurtosis G_2 by the standard error of kurtosis (SEK) (see Cramer [1997]), getting

$$Z_{g_2} = \frac{G_2}{SEK} \text{ where } SEK = 2 \times SES \times \sqrt{\frac{n^2 - 1}{(n-3)(n+5)}} \text{ or } SEK = \sqrt{\frac{24}{n}}$$

This is a two-tailed test of excess kurtosis different from 0 at roughly the 0.05 significance level. It tells us whether the whole population probably exhibits kurtosis, but not by how much. The bigger the number, the higher the probability. The critical value of Z_{g_2} is approximately 2. That is,

$$\begin{cases} \text{if } Z_{g_2} < -2 \text{ the population has very likely negative excess kurtosis} \\ \text{if } Z_{g_2} > 2 \text{ the population has very likely positive excess kurtosis} \\ \text{for any values in } [-2, 2] \text{ we can't reach any conclusion about the kurtosis of the population} \end{cases}$$

We can get a 95% confidence interval of population kurtosis

$$G_2 \pm 2 \times SEK$$

Crow et al. [1967] used quantiles to define the kurtosis as

$$K_{CS} = \frac{F^{-1}(\alpha) + F^{-1}(1 - \alpha)}{F^{-1}(1 - \beta) - F^{-1}(\beta)}, \alpha, \beta \in (0, 1)$$

Centring the kurtosis on the standard normal distribution, they chose $\alpha = 0.025$ and $\beta = 0.25$, obtaining $F^{-1}(0.975) = -F^{-1}(0.025) = 1.96$ and $F^{-1}(0.75) = -F^{-1}(0.25) = -0.68$, and $K_{CS} = 2.91$. Hence, the centred Crow et al. coefficient is given by

$$K_{CS} = \frac{F^{-1}(0.975) + F^{-1}(0.025)}{F^{-1}(0.75) - F^{-1}(0.25)} - 2.91$$

In order to detect heavy-tailed distributions, Hogg [1972] [1974] proposed the following measure of kurtosis:

$$K_H = \frac{U_\alpha - L_\alpha}{U_\beta - L_\beta}$$

where

- $U_\alpha = \frac{1}{\alpha} \int_{1-\alpha}^1 F^{-1}(x) dx$ is the average of the upper α quantiles
- $L_\alpha = \frac{1}{\alpha} \int_0^\alpha F^{-1}(x) dx$ is the average of the lower α quantiles

for $\alpha \in (0, 1)$. The best results were obtained with $\alpha = 0.05$ and $\beta = 0.5$. In that setting, to recover the standard normal distribution, we set $U_{0.05} = -L_{0.05} = 2.06$ and $U_{0.5} = -L_{0.5} = 0.8$, getting $K_H = 2.59$. Hence, the centred Hogg coefficient is given by

$$K_H = \frac{U_{0.05} - L_{0.05}}{U_{0.5} - L_{0.5}} - 2.59$$

Moors [1988] showed that the sample kurtosis $\hat{K}_N$ could be interpreted as a measure of the dispersion of a distribution around the two values $\mu_N \pm \sigma_N$. Thus, it can be large when the probability mass is concentrated either near the mean or in the tails of the distributions. He proposed the following robust alternative

$$K_M = \frac{(E_7 - E_5) + (E_3 - E_1)}{E_6 - E_2}$$

where E_i is the i th octile, that is, $E_i = F^{-1}(\frac{i}{8})$, for $i = 1, \dots, 7$. Here, $(E_7 - E_5)$ and $(E_3 - E_1)$ are large (small) if relatively little (much) probability mass is concentrated in the neighbourhood of E_6 and E_2 , corresponding to large (small) dispersion around $\mu_N \pm \sigma_N$. The denominator is a scaling factor which ensures that the statistic is invariant under linear transformation. Centring the Moors kurtosis on the standard normal distribution, we set $E_1 = -E_7 = -1.15$, $E_2 = -E_6 = -0.68$, $E_3 = -E_5 = -0.32$, and $E_4 = 0$ to get $K_M = 1.23$. Hence, the centred Hogg coefficient is given by

$$K_M = \frac{(E_7 - E_5) + (E_3 - E_1)}{E_6 - E_2} - 1.23$$

11.4.4 The impacts of skewness and kurtosis

While the histogram gives us a general idea of the shape of the distribution, skewness and kurtosis give a more precise evaluation. Note, the mean and standard deviation have the same units as the original data (the variance has the square of those units). However, both the skewness and the kurtosis have no units. They are pure numbers, like a z-score.

In order to study the impacts of skewness and kurtosis on distributions, we consider the four graphs in Figure (73) each showing two probability distribution functions. The blue curve represents a Normal distribution and the orange curve could be another one.

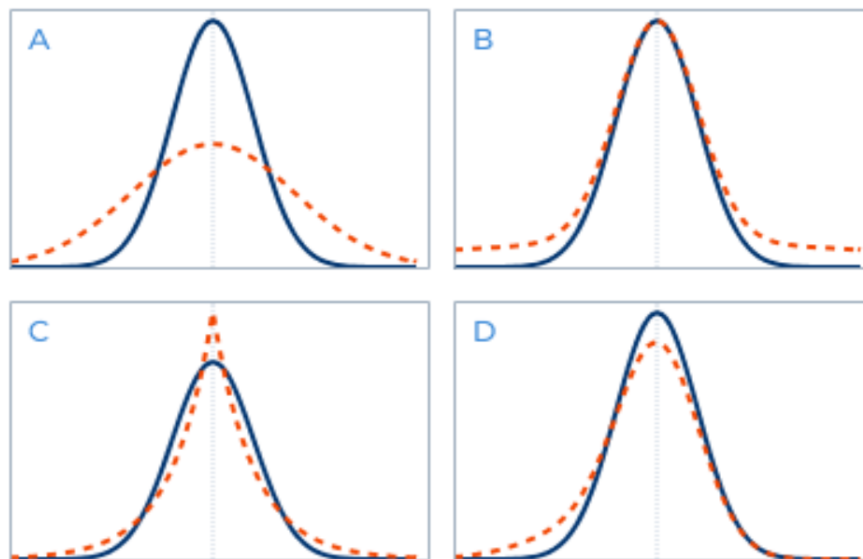


Figure 73: Impacts of skewness and kurtosis.

The orange curve in Graph A just shows a Normal distribution with a larger standard deviation. In graph D, the orange curve shows a negatively skewed distribution with same volatility as the Normal one. While Graph B shows obvious fat tails, statistically it does not make any sense. Since the probability of a large move is higher, the probability of a different move should be lower for the CDF to integrate to one. This is not the case. Graph C represents a distribution with exactly the same standard deviation as the blue curve, but with a larger kurtosis. Note, both distributions have the same mean and volatility. However, the higher kurtosis leads to a higher than normal probability of both a large move and small move, in combination with a lower than normal probability of a medium-sized move. That is, the intermediate values have become less likely and the central and extreme values have become more likely. The kurtosis increases while the standard deviation stays the same, because more of the variation is due to extreme values.

Focussing on the skew, we display in Figure (74) two distributions having the same mean and standard deviation, $\mu = 0.69$ and $\sigma = 0.168$, respectively, but different skewness. The first one is moderately skewed left with $S = -0.537$ and the other one is moderately skewed right with $S = 0.537$.

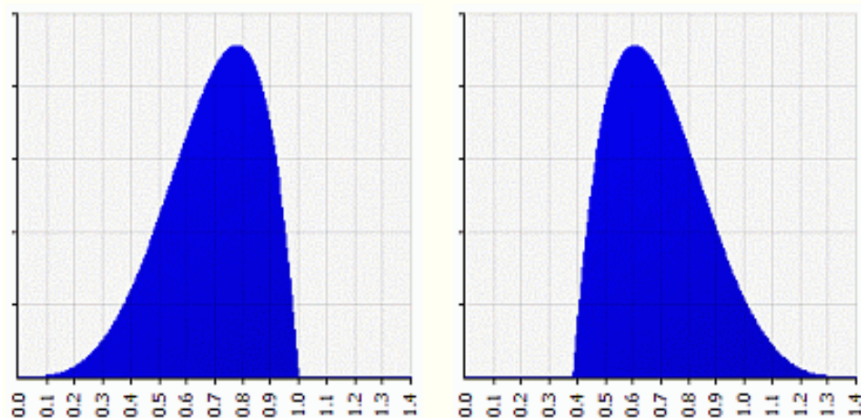


Figure 74: Two skewed distributions with $S = -0.537$ and $S = 0.537$.

Focussing on the kurtosis, we display in Figure (75) three distributions having the same mean and standard deviation, $\mu = 0$ and $\sigma = 1$, respectively, but different kurtosis. That is, $K = 1.8$ (excess kurtosis -1.2), $K = 3$ (excess kurtosis 0), and $K = 4.2$ (excess kurtosis 1.2). Moving from the illustrated uniform distribution to a normal distribution, we observe that the shoulders have transferred some of their mass to the center and the tails of the distribution. The smallest possible kurtosis is 1 (excess kurtosis -2), and the largest is ∞ .

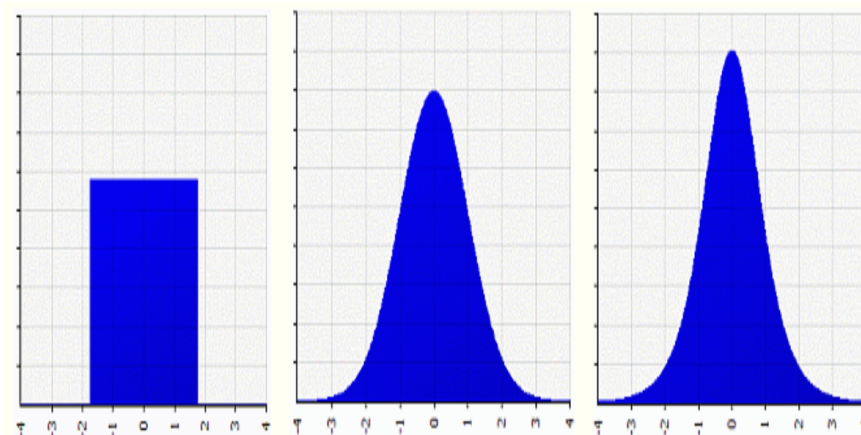


Figure 75: Three kurtosis distributions with $K = 1.8$, $K = 0$, and $K = 4.2$.

As another example, we combine a normal distribution, skewed one, and one with kurtosis in a single graph displayed in Figure (76). The mean and standard deviation are denoted μ and σ , respectively. *A* represents a normal distribution with corresponding mean (solid vertical line) and standard deviation (horizontal arrowed lines). *B* illustrates a positively skewed distribution and corresponding shift in mean (dotted vertical line), and *C* illustrates a peaked distribution with a corresponding positive kurtosis. Note that distributions *A* and *C* have the same mean.

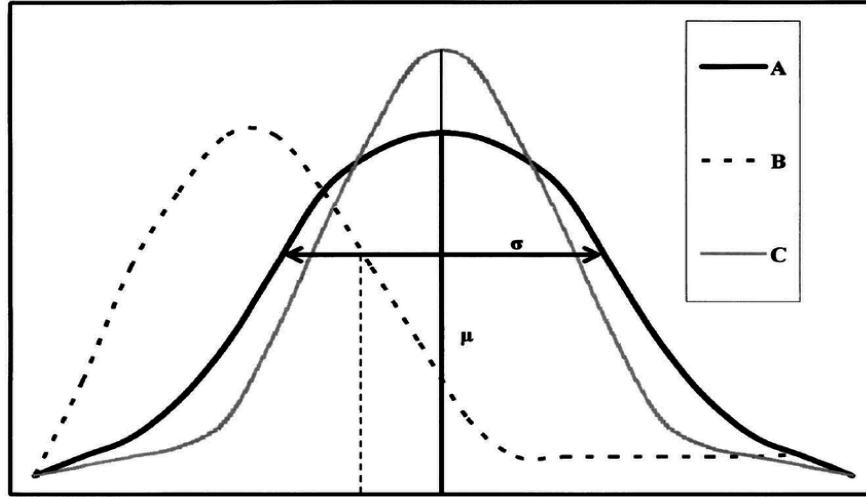


Figure 76: Impacts of skewness and kurtosis.

These examples show that risk and standard deviation are not the same thing. We need to account for higher skew and higher kurtosis. For instance, in Graph D, there is the exchange of standard deviation for kurtosis. The risk stays the same but we get lower standard deviation and higher kurtosis. As a result, stock markets may show low volatility but exhibit high kurtosis with fat tails and high peakedness, hence high risk. Trend following strategies usually struggle to generate positive returns in high peaks (sideway markets). Nonetheless, in periods of fat tails, these strategies are able to profit from the positive autocorrelation.

11.5 Annualisation of the moments

11.5.1 The first two moments

The sample estimates of the first two moments presented in Appendix (11.4) are often based on monthly, weekly or daily data, but all quantities are usually quoted in annualised terms. As discussed in Appendix (11.5.3), the annualisation problem can be expressed as computing the statistics of the sum Y of n i.i.d. random variables, where n is the horizon period. Thus, the annualisation is often performed on the sample estimates under the assumption that the random variables (returns) are i.i.d. We let the annualised volatility σ be the standard deviation of the instrument's yearly logarithmic returns. The generalised volatility σ_T , for time horizon T in years, is expressed as $\sigma_T = \sigma\sqrt{T}$ (see Meucci [2010b]). Therefore, if the daily logarithmic returns of a stock have a standard deviation of σ_{SD} and the time period of returns is P (or Δt), then the annualised volatility is

$$\sigma = \frac{\sigma_{SD}}{\sqrt{P}}$$

A common assumption for daily returns is that $P = 1/252$ (there are 252 trading days in any given year). Then, if $\sigma_{SD} = 0.01$, the annualised volatility is

$$\sigma = \frac{0.01}{\sqrt{\frac{1}{252}}} = 0.01\sqrt{252}$$

More generally, we have

$$Y \times \frac{F}{N}$$

where Y is the sum of all values referenced, F is the base rate of return (time period frequency) with 12 monthly, 252 daily, 52 weekly, 4 quarterly, and N is the total number of periods. For example, setting $P = \frac{1}{F}$, the annualised mean and variance becomes

$$\begin{aligned}\mu &= \frac{\mu_{SD}}{P} = \mu_{SD} 252 \\ \sigma^2 &= \frac{\sigma_{SD}^2}{P} = \sigma_{SD}^2 252\end{aligned}$$

This formula to convert returns or volatility measures from one time period to another assume a particular underlying model or process. These formulas are accurate extrapolations of a random walk, or Wiener process, whose steps have finite variance. However, if portfolio returns are autocorrelated, the standard deviation does not obey the square-root-of-time rule. Again, F is the time period frequency (number of returns per year), then the annualised mean return is still F times the mean return, but the standard deviation of returns should be calculated using the scaling factor

$$\sqrt{F + 2\frac{Q}{(1-Q)^2}[(F-1)(1-Q) - Q(1-Q^{F-1})]} \quad (11.70)$$

where Q is the first order autocorrelation of the returns (see Alexander [2008]). If the autocorrelation of the returns is positive, then the scaling factor is greater than the square root of F . More generally, for natural stochastic processes, the precise relationship between volatility measures for different time periods is more complicated. Some researchers use the Levy stability exponent α (linked to the Hurst exponent) to extrapolate natural processes

$$\sigma_T = T^{\frac{1}{\alpha}} \sigma$$

If $\alpha = 2$ we get the Wiener process scaling relation, but some people believe $\alpha < 2$ for financial activities such as stocks, indexes and so on. Mandelbrot [1967] followed a Levy alpha-stable distribution with $\alpha = 1.7$. Given our previous example with $P = 1/252$ we get

$$\sigma = 0.01(252)^{\frac{1}{\alpha}}$$

We let $\alpha_W = 2$ for the Wiener process and $\alpha_M = 1.7$ for the Levy alpha-stable distribution. Since $\frac{1}{\alpha_M} > \frac{1}{\alpha_W}$ we get $\hat{\alpha}_M = \frac{1}{\alpha_M} = \hat{\alpha}_W + \xi$ we get $(252)^{\hat{\alpha}_M} = (252)^{\hat{\alpha}_W} (252)^\xi$ so that

$$\sigma_M = 0.01(252)^{\hat{\alpha}_M} (252)^\xi$$

with $\xi = 0.09$ and $(252)^\xi = 1.64$. Hence, we get the Mandelbrot annualised volatility

$$\sigma_M = \sigma_{SD} \sqrt{252} (1 + \epsilon_S) = \sigma_W (1 + \epsilon_S)$$

where $\epsilon_S \geq 0$ is the adjusted volatility.

11.5.2 How the moments scale

A basic property of cumulants (see Appendix (11.3.2)) is that the cumulants of all orders are additive under convolution. So, assume $X_1, X_2, \dots, X_n$ are i.i.d. random variables and consider their sum

$$Y_n = \sum_{i=1}^n X_i$$

All the cumulants of Y_n scale linearly with n , that is,

$$\kappa_k(Y_n) = n \kappa_k(Y_1), \quad k = 1, 2, \dots$$

We now normalise the sum, getting

$$Z_n = \frac{Y_n}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$$

Another basic property of cumulants is that the k th cumulant is homogeneous of order k as to scale. Using both properties together we have

$$\kappa_k(Z_n) = \left(\frac{1}{\sqrt{n}}\right)^k \kappa_k(Y_n) = \left(\frac{1}{\sqrt{n}}\right)^k n \kappa_k(Y_1) = \frac{\kappa_k(Z_1)}{n^{(k-2)/2}}$$

One can show that the statistics scale as follows:

- variance: $\kappa_2(Z_n) = \kappa_2(Z_1) \propto 1$
- skewness: $\frac{\kappa_3(Z_n)}{\kappa_2^{3/2}(Z_n)} = \frac{\frac{1}{\sqrt{n}} \kappa_3(Z_1)}{\kappa_2^{3/2}(Z_1)} \propto \frac{1}{\sqrt{n}}$
- excess-kurtosis: $\frac{\kappa_4(Z_n)}{\kappa_2^2(Z_n)} = \frac{\frac{1}{n} \kappa_4(Z_1)}{\kappa_2^2(Z_1)} \propto \frac{1}{n}$

The one period Cornish-Fisher VaR is given by

$$\mu - \sigma z^*$$

where

$$z^* = z + \frac{(z^2 - 1)S}{6} + \frac{(z^3 - 3z)K}{24} - \frac{(2z^3 - 5z)S^2}{36}$$

and z is the desired standard normal quantile, S is the skewness, and K is the excess-kurtosis.

11.5.3 General projection

We assume that the population is of size n_p , taking values $X = \{X_1, X_2, \dots, X_{n_p}\}$, and that X is independent and identically distributed. Since X is invariant, the distribution of X_t does not depend on t . We consider the sum

$$Y = X_1 + X_2 + \dots + X_{n_p}$$

Given the distribution of X , we want to compute the statistics of Y . If the population size n_p corresponds to a horizon of one year, this is the annualisation problem in finance.

If we let X be a normal invariant $X \sim N(\mu_X, \sigma_X^2)$, then the sum of independent normal variables is normal and we get $Y \sim N(\mu_Y, \sigma_Y^2)$ where

$$\mu_Y = n_p \mu_X \text{ and } \sigma_Y^2 = n_p \sigma_X^2$$

Since the skewness, kurtosis, and all the other standardised summary statistics don't depend the expectation and the standard deviation, they are independent of the projection horizon (see Meucci [2010a]). We get

$$\gamma_n(Y) = \begin{cases} 0 & \text{if } n \geq 3 \text{ is odd} \\ 1 \times 3 \times \dots \times (n-1) & \text{if } n \geq 3 \text{ is even} \end{cases}$$

Thus, the skewness $\gamma_3(Y)$ is 0 and the kurtosis $K = \gamma_4(Y)$ is 3. In the case where X is invariant with a fully general distribution with finite variance, from the central limit theorem, when the projection horizon n_p is large, then the summary statistics of Y must approach the normal statistics. In the case of non-normal variables and short to medium horizon n_p , the central limit approximation no-longer applies. Meucci [2010c] proposed an algorithm to handle this case. Given the invariant X , the algorithm is as follows:

1. Compute the first n statistics $\mu_X, \sigma_X, S(X), K(X), \gamma_5, \dots, \gamma_n$.
2. Using the above statistics, we compute the central moments of X , $\mu_1, \dots, \mu_n$ from Equation (11.60).
3. Using the above central moments we compute the noncentral moments $\tilde{\mu}_1, \dots, \tilde{\mu}_n$ from Equation (11.61).
4. Given the above non-central moments, we compute the cumulants for X , $\kappa_1(X), \dots, \kappa_n(X)$, from Equation (11.62).
5. We now reverse the process. Given the above cumulants for X , we compute the cumulants $\kappa_1(Y), \dots, \kappa_n(Y)$ for the projection Y . For an independent variables $X_1, \dots, X_\tau$, where τ is an arbitrary horizon, we have

$$E[e^{z(X_1 + \dots + X_\tau)}] = E[e^{zX_1}] \dots E[e^{zX_\tau}]$$

Replacing in the definition of the cumulants, we get

$$\kappa_{X_1 + \dots + X_\tau} = \kappa_{X_1} + \dots + \kappa_{X_\tau}$$

Since X_t is invariant, all the X_t are identically distributed and the projected cumulants become (see Section (11.5.2))

$$\kappa_n(Y) = \tau \kappa_n(X)$$

6. From the cumulants $\kappa_1(Y), \dots, \kappa_n(Y)$, we compute the non-central moments $\tilde{\mu}_1(Y), \dots, \tilde{\mu}_n(Y)$ using the recursive relation in Equation (11.63) applied to Y .
7. From the above non-central moments, we compute the central moments $\mu_1(Y), \dots, \mu_n(Y)$ using the recursive relation in Equation (11.64) applied to Y .
8. From the above central moments, $\mu_1(Y) = \mu_Y, \mu_2(Y) = \sigma_Y^2$ and applying Equation (11.59) for $n \geq 3$, we compute the standardised summary statistics for Y to the arbitrary horizon τ :

$$\mu_Y, \sigma_Y, S(Y), K(Y), \gamma_5(Y), \dots, \gamma_n(Y)$$

This algorithm preserves the linear and square-root projection rules for the first two moments. Further, when X is a normal invariant, it recover the normal statistics. More generally, it allows to compute the statistics for arbitrary distributions of X . Note, when X is not an invariant we must account for the autocorrelations in the projection. When the standard deviation, the skewness and the other statistics are not defined, we must find alternative ways to summarise the features of X .

11.6 Generating data using families of distributions

Non-Gaussian distributions being defined by all moments, by focusing only the first four moments, we can only approximate them. Hence there are numerous ways of combining the first four moments to approximate non-Gaussian distributions. There are several families of distributions that can account for a variety of skewness and kurtosis levels (see Jones [2015]). These include:

- Skew-Symmetric distributions: It is the construction of a continuous probability distribution obtained by applying a certain form of perturbation to a symmetric density function (see Wang et al. [2004]).
- Two piece distributions: It is a collection of unimodal (single-peaked) parametric distributions that can account for a variety of tail weights and asymmetry obtained by joining two half-normal densities with different scale parameters. It contains members with three or four parameters that control the tails and asymmetry level of the probability density function.

- Sinh-arcsinh distributions
- Alpha-stable family: Use the Stable (Levy) distribution with its four parameters (mean μ , s.d. c , skewness β , and kurtosis α). The stable distribution can represent normal, skew-normal, logistic, Rayleigh, Cauchy etc.
- Many other families of distributions obtained by transforming the pdf/cdf or using parametric changes of variables. It includes the Pearson and Johnson systems of distributions.
- the Fleishman function

11.6.1 The Pearson and Johnson systems of distributions

Pearson [1895] devised a system, or family, of distributions that includes a unique distribution corresponding to every valid combination of mean, standard deviation, skewness, and kurtosis. If we compute sample values for each of these moments from data, it is easy to find the distribution in the Pearson system that matches these four moments and to generate a random sample. Note, there is no need to have a priori or a posteriori knowledge of the data-producing process. Each system is a flexible parametric family of distributions that includes a wide range of distribution shapes. The Pearson system embeds seven basic types of distribution together in a single parametric framework. It includes common distributions such as the normal and t distributions, simple transformations of standard distributions such as a shifted and scaled beta distribution and the inverse gamma distribution, and one distribution, the Type IV, that is not a simple transformation of any standard distribution. The Pearson types are characterized by two quantities, commonly referred to as β_1 and β_2 . The first one is the skewness $\beta_1 = y_1$ where y_1 is the skewness (third standardized moment). The second one is the traditional kurtosis (fourth standardised moment) $\beta_2 = y_2 + 3$. A Pearson density p is defined to be any valid solution to the differential equation

$$\frac{p'(x)}{p(x)} + \frac{a + (x - \mu)}{b_0 + b_1(x - \mu) + b_2(x - \mu)^2} = 0$$

with $\psi = 10\beta_2 - 12\beta_1 - 18$ and

$$b_0 = \frac{4\beta_2 - 3\beta_1}{\psi} \mu_2, a = b_1 = \sqrt{\mu_2} \sqrt{\beta_1} \frac{\beta_2 + 3}{\psi}, b_2 = \frac{2\beta_2 - 3\beta_1 - 6}{\psi}$$

Its solution is given by

$$p(x) \propto e^{-\int \frac{x+a}{f(x)} dx} \text{ where } f(x) = b_2x^2 + b_1x + b_0$$

Pearson distinguished two main cases determined by the sign of the discriminant (and hence the number of real roots) of the quadratic function $f(x)$.

Johnson devised a different system of distributions that also includes a unique distribution for every valid combination of mean, standard deviation, skewness, and kurtosis. However, since it is more natural to describe distributions in the Johnson system using quantiles, working with this system is different than working with the Pearson system. The Johnson system is based on three possible transformations of a normal random variable, plus the identity transformation. The three nontrivial cases are known as SL, SU, and SB, corresponding to exponential, logistic, and hyperbolic sine transformations. All three can be written as

$$X = \gamma + \delta \cdot \Gamma\left(\frac{Z - \xi}{\lambda}\right)$$

where Z is a standard normal random variable, Γ is the transformation, and γ , δ , ξ , and λ are scale and location parameters. The fourth case, SN, is the identity transformation.

11.6.2 Two piece distributions

11.6.2.1 Description Fechner [1897] proposed to construct a pdf by joining two half-normal densities with different scale parameters. Let f be a unimodal symmetric (about 0) pdf from the location-scale family, possibly including a shape parameter δ . Then, the pdf of a member of the univariate two-piece family of distributions (see John [1982]) is:

$$s(X; \mu, \sigma_1, \sigma_2, \delta) = \begin{cases} w \cdot f(z_1; \delta) & \text{if } X < \mu \\ w \cdot f(z_2; \delta) & \text{if } X \geq \mu \end{cases}$$

where $w = \frac{2}{\sigma_{1,2}}$, $\sigma_{1,2} = \sigma_1 + \sigma_2$, and $z_i = \frac{X - \mu}{\sigma_i}$ for $i = 1, 2$, where μ is the mode. The parameters of s have a clear interpretation:

- μ is the (unique) mode of s
- σ_1 controls the scale (spread) of s on the left-hand side of μ , while σ_2 controls the scale on the right-hand side of μ .
- The interpretation of δ is the same as that in f .

The mass allocated to both sides of the mode is only a function of the ratio of σ_1 and σ_2 . In particular, $P(x < \mu) = \frac{\sigma_1}{\sigma_{1,2}}$ for any choice of f . Some examples of pdf f are:

- If f is the normal pdf, then s is the two-piece normal distribution proposed by Fechner.
- If f is the Laplace distribution, then s is the asymmetric (or two-piece) Laplace distribution.
- If f is the Student-t distribution with $\delta > 0$ degrees of freedom, then we obtain the two-piece t-distribution.

Other interpretable parameterisations of these distributions exist (see Rubio et al. [2015]). In the special case where the random variable X has a two-piece normal distribution (split normal) (see Wallis [2013]), its pdf is

$$s(X; \mu, \sigma_1, \sigma_2, \delta) = \begin{cases} w \cdot \tilde{N}(\mu, \sigma_1) & \text{if } X < \mu \\ w \cdot \tilde{N}(\mu, \sigma_2) & \text{if } X \geq \mu \end{cases} \quad (11.71)$$

with modified density function $\tilde{N}(\mu, \sigma_i) = n(z_i) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z_i^2}$ for $i = 1, 2$. At the mode μ , we get the common value $s(\mu; \mu, \sigma_1, \sigma_2, \delta) = \frac{1}{\sqrt{2\pi}} w = \sqrt{\frac{2}{\pi}} \sigma_{1,2}^{-1}$. That is, the density results from joining at the mode the corresponding halves of two normal distributions with the same mode but different standard deviations. Recall, the density function is $N(\mu, \sigma) = \frac{1}{\sigma} n(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$. Hence, we can rewrite the pdf as

$$s(X; \mu, \sigma_1, \sigma_2, \delta) = \begin{cases} w_1 \cdot N(\mu, \sigma_1) & \text{if } X < \mu \\ w_2 \cdot N(\mu, \sigma_2) & \text{if } X \geq \mu \end{cases}$$

where $w_i = \sigma_i \cdot w$ for $i = 1, 2$. Now, the scaling factor applied to the left half of $N(\mu, \sigma_1)$ is $\sigma_1 \cdot w$, while that applied to the right half of $N(\mu, \sigma_2)$ is $\sigma_2 \cdot w$. As a result, the probability mass under the left or right piece is $\frac{1}{2}\sigma_1 \cdot w$ or $\frac{1}{2}\sigma_2 \cdot w$, respectively. When $\sigma_1 < \sigma_2$ the two piece normal distribution is positively skewed, while when $\sigma_1 > \sigma_2$ it is negatively skewed. The skewness becomes extreme as $\sigma_i \rightarrow 0$, $i = 1, 2$, and the distribution collapses to the half normal distribution. On the other hand, the skewness is reduced as $\sigma_1 \rightarrow \sigma_2$, reaching zero when $\sigma_1 = \sigma_2$ (the normal distribution).

11.6.2.2 Properties The mean and variance of the distribution are

$$E^{TPD}[X] = \mu + \sqrt{\frac{2}{\pi}}(\sigma_2 - \sigma_1) \text{ and } Var^{TPD}(X) = (1 - \frac{2}{\pi})(\sigma_2 - \sigma_1)^2 + \sigma_1\sigma_2 \quad (11.72)$$

The variance can be rewritten as

$$Var^{TPD}(X) = b\sigma_1^2 \text{ where } b = (1 - \frac{2}{\pi})(\frac{\sigma_2}{\sigma_1} - 1)^2 + \frac{\sigma_2}{\sigma_1}$$

Note, skewness is interpreted either in terms of the ratio of the areas under the two pieces of the pdf, that is, $\frac{\sigma_1}{\sigma_2}$, or in terms of a monotone transformation such as $\frac{(\sigma_2 - \sigma_1)}{\sigma_{1,2}}$. The skewness is given by

$$m_3 = \sqrt{\frac{2}{\pi}}(\sigma_2 - \sigma_1) \left[\left(\frac{4}{\pi} - 1 \right) (\sigma_2 - \sigma_1)^2 + \sigma_1\sigma_2 \right]$$

where m_3 is the third central moment. We divide it by $(Var^{TPD})^{\frac{3}{2}}$ to get the skewness. The moment can be rewritten as

$$m_3 = \sigma_1^3 \sqrt{\frac{2}{\pi}} \left(\frac{\sigma_2}{\sigma_1} - 1 \right) \left[\left(\frac{4}{\pi} - 1 \right) \left(\frac{\sigma_2}{\sigma_1} - 1 \right)^2 + \frac{\sigma_2}{\sigma_1} \right]$$

The sign of its third central moment is determined by the difference $(\sigma_2 - \sigma_1)$. If it is positive, the distribution is skewed to the right and if it is negative, then it is skewed to the left.

We let $F(x)$ be the cdf and $\Phi(x)$ be the scaled cdf. Then, $x_p = F^{-1}(p)$ and $z_p = \Phi^{-1}(p)$ are their respective quantiles. Then, in the left piece of the distribution we have $x_\alpha = \sigma_1 z_\beta$ where $\beta = \frac{\alpha\sigma_{1,2}}{2\sigma_1}$. In the right piece of the distribution, defining quantiles with reference to their upper tail probabilities, we have $x_{1-\alpha} = \sigma_2 z_{1-\delta}$ where $\delta = \frac{\alpha\sigma_{1,2}}{2\sigma_2}$.

11.6.2.3 Parameters estimation John [1982] proposed to estimate the parameters by using maximum likelihood method (MLE). The likelihood function can be expressed in an intensive form, in which the scale parameters σ_1 and σ_2 are a function of the location parameter μ . It is given by

$$L(\mu) = -g(\mu) = -s_1^{\frac{1}{3}} - s_2^{\frac{1}{3}} \text{ where } s_1 = \sum_{x_i: x_i < \mu} (x_i - \mu)^2 \text{ and } s_2 = \sum_{x_i: x_i > \mu} (x_i - \mu)^2 \quad (11.73)$$

It has only to be maximised numerically with respect to the single parameter μ . Given the maximum likelihood estimator $\hat{\mu}$, the estimated scale parameters are given by

$$\hat{\sigma}_1^2 = \frac{g(\mu)}{n} s_1^{\frac{2}{3}} \text{ and } \hat{\sigma}_2^2 = \frac{g(\mu)}{n} s_2^{\frac{2}{3}} \quad (11.74)$$

Villani [2006] derived properties of the multivariate split normal distribution. They also considered the MLE method and the Bayesian estimation to provide some analytical results for the univariate and multivariate case, respectively. They used the following reformulation: for $x \sim SN(\mu, \lambda^2, \tau^2)$, the density is

$$f(x) = \begin{cases} w \cdot \tilde{N}(\mu, \lambda) & \text{if } x < \mu \\ w \cdot \tilde{N}(\mu, \tau\lambda) & \text{if } x \geq \mu \end{cases}$$

where $\sigma_1 = \lambda$ and $\sigma_2 = \tau\lambda$ so that $\tau = \frac{\sigma_2}{\sigma_1}$. Hence, $\sigma_2 - \sigma_1 = \lambda(\tau - 1)$ and $\sigma_1 + \sigma_2 = \lambda(1 + \tau)$. In this formulation, the likelihood function is given by

$$\hat{L}(\mu) = \left(\frac{2n}{\pi e} \right)^{\frac{n}{2}} g(\mu)^{-\frac{3n}{2}}$$

and the scale parameters are as above.

11.7 Optimal stopping problem

We follow the description given by Oksendal [1998].

11.7.1 Formulation of the problem

Let $U \subset \mathbb{R}^n$ be open. Then the first exit time $\tau_U = \inf\{t > 0; X_t \notin U\}$ is a stopping time (it generalises to any Borel set). Let X_t be an Ito diffusion on $\mathbb{R}^n$ and let the reward function $g \in \mathbb{R}^n$ satisfies

1. $g(\xi) \geq 0$ for all $\xi \in \mathbb{R}^n$
2. g is continuous

Find an optimal stopping time $\tau^* = \tau^*(x, \omega)$ for $\{X_t\}$ such that

$$E_x[g(X_{\tau^*})] = \sup_{\tau \in \mathcal{T}} E_x[g(X_\tau)], \forall x \in \mathbb{R}^n \quad (11.75)$$

and the corresponding optimal expected reward $g^*(x) = E_x[g(X_{\tau^*})]$. Here, $\mathcal{T}$ is the set of all stopping times for $\{X_t\}$ and τ^* gives the biggest expected reward. Further, the process X_t starts at $X_0 = x \in \mathbb{R}^n$. This problem also covers the more general time-inhomogeneous problem

$$g^*(s, x) = \sup_{\tau \in \mathcal{T}} E_{(s,x)}[g(\tau, X_\tau)] = E_{(s,x)}[g(\tau^*, X_{\tau^*})]$$

and problem involving an integral

$$\begin{aligned} G^*(s, x) &= \sup_{\tau \in \mathcal{T}} E_{(s,x)} \left[\int_0^\tau f(t, X_t) dt + g(\tau, X_\tau) \right] \\ &= E_{(s,x)} \left[\int_0^{\tau^*} f(t, X_t) dt + g(\tau^*, X_{\tau^*}) \right] \end{aligned}$$

where f is a reward function satisfying certain conditions.

11.7.2 The time-inhomogeneous problem

We want to find $g_0(x)$ and τ^* such that $g_0(x) = \sup_{\tau} E_x[g(\tau, X_\tau)] = E_x[g(\tau^*, X_{\tau^*})]$. Consider the process $X_t = X_t^x$ with dynamics

$$dX_t = b(X_t)dt + \sigma(X_t)dB(t), t \geq 0, X_0 = x$$

where $b : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are functions satisfying certain conditions. We define the Ito diffusion $Y_t = Y_t^{(s,x)}$ in $\mathbb{R}^{n+1}$ where $Y_t = [s + t, X_t^x]^\top$ with dynamics

$$dY_t = \begin{cases} dt \\ b(X_t)dt + \sigma(X_t)dB(t) \end{cases} = \hat{b}(Y_t)dt + \hat{\sigma}(Y_t)dB(t)$$

so that Y_t is an Ito diffusion starting at $y = (s, x)$. Let $\mathbb{R}^y = \mathbb{R}^{(s,x)}$ denotes the probability law of $\{Y_t\}$ and $E_y = E_{(s,x)}$ denotes the expectation with respect to $\mathbb{R}^y$. Thus, we can rewrite the problem as

$$g_0(x) = g^*(0, x) = \sup_{\tau} E_{(0,x)}[g(Y_\tau)] = E_{(0,x)}[g(Y_{\tau^*})]$$

which corresponds to the time-homogeneous case. The characteristic operator $\mathcal{A}_Y$ of Y_t is

$$\mathcal{A}_Y \phi(s, x) = \partial_s \phi(s, x) + \mathcal{A} \phi(s, x), \phi \in C^2(\mathbb{R} \times \mathbb{R}^n)$$

where $\mathcal{A}$ is the characteristic operator of X_t .

11.7.2.1 Buying and selling stocks We consider the problem of finding the right time to sell stocks. It is assumed that the investor own the asset independently from an existing trade with an entry level. Let the stock price X_t seen at time t have the dynamics

$$\frac{dX_t}{X_t} = rdt + \sigma dB(t), X_0 = x > 0$$

where $B(t)$ is a one-dimensional Brownian motion, and r and σ are known constants. Let's assume a fixed fee (or transaction cost), $a > 0$, when selling the asset, so that the discounted net of the sale ²¹ at time t is

$$e^{-\rho t}(X_t - a)$$

where $\rho > 0$ (in general ρ is the sum of interest rate, inflation and tax). Note, if we want to guarantee a positive payoff when selling the asset, we can impose

$$e^{-\rho t}(X_t - a)^+$$

in which case we get an American call option (we can sell the asset at any time). In the case where we want to buy an asset, the discounted net of the purchase at time t is

$$e^{-\rho t}(-X_t - a)$$

Back to the original problem, when selling the asset, we want to find a stopping time τ maximising the discounted net of the sale (it should be the discounted return)

$$E_{(s,x)}[e^{-\rho\tau}(X_\tau - a)] = E_{(s,x)}[g(\tau, X_\tau)]$$

where $g(t, x) = e^{-\rho t}(x - a)$. The characteristic operator $\mathcal{A}_Y$ of the process $Y_t = [s + t, X_t]^\top$ is given by

$$\mathcal{A}_Y f(s, x) = Lf(s, x) = \partial_s f + rx\partial_x f + \frac{1}{2}\sigma^2 x^2 \partial_{xx}^2 f, f \in C^2(\mathbb{R}^2)$$

Differentiating, we get $\partial_s g = -\rho e^{-\rho s}(x - a)$, $\partial_x g = e^{-\rho s}$, and $\partial_{xx}^2 g = 0$, and the PDE becomes

$$\mathcal{A}_Y g(s, x) = -\rho e^{-\rho s}(x - a) + rx e^{-\rho s} = e^{-\rho s}((r - \rho)x + \rho a)$$

So,

$$U := \{(s, x); \mathcal{A}_Y g(s, x) > 0\} = \begin{cases} \mathbb{R} \times \mathbb{R}_+ & \text{if } r \geq \rho \\ \{(s, x); x < \frac{\rho a}{\rho - r}\} & \text{if } r < \rho \end{cases}$$

So, if $r \geq \rho$ we have $U = D = \mathbb{R} \times \mathbb{R}_+$ so that τ^* does not exists. If $r > \rho$ then $g^* = \infty$, while if $r = \rho$ then

$$g^*(s, x) = x e^{-\rho s}$$

We are left with considering the case $r < \rho$. In that setting, the connected component of D that contains U must have the form

$$D(x_0) = \{(t, x); 0 < x < x_0\}$$

for some $x_0 \geq \frac{\rho a}{\rho - r}$ (because of the definition of U if $r < \rho$). Put $\tau(x_0) = \tau_{D(x_0)}$ (first exit time of the domain $D(x_0)$) and compute

$$\tilde{g}(s, x) = \tilde{g}_{x_0}(s, x) = E_{(s,x)}[g(Y_{\tau(x_0)})]$$

²¹Gross sales measures the total sales of a company, unadjusted for the costs related to generating those sales. Net sales are calculated by deducting sales allowances, sales discounts, and sales returns from gross sales.

Then, $f = \tilde{g}$ is the bounded solution of the boundary value problem (BVP)

$$\begin{aligned} Lf &= 0 \text{ for } 0 < x < x_0 \\ f(s, x_0) &= e^{-\rho s}(x_0 - a) \end{aligned}$$

If we try the solution

$$f(s, x) = e^{-\rho s} \phi(x)$$

we get a 1-dimensional problem

$$\begin{aligned} L\phi - \rho\phi &= 0 \text{ for } 0 < x < x_0 \\ \phi(x_0) &= x_0 - a \end{aligned}$$

with general solution

$$\phi(x) = C_1 x^{\gamma_1} + C_2 x^{\gamma_2}$$

where $C_i, i = 1, 2$, are arbitrary constants and

$$\gamma_i = \sigma^{-2} \left[\frac{1}{2} \sigma^2 - r \pm \sqrt{\left(r - \frac{1}{2} \sigma^2\right)^2 + 2\rho\sigma^2} \right], i = 1, 2, \gamma_2 < 0 < \gamma_1$$

Since $\phi(x)$ is bounded as $x \rightarrow 0$, we must have $C_2 = 0$, and the boundary requirement $\phi(x_0) = x_0 - a$ gives $C_1 = x_0^{-\gamma_1}(x_0 - a)$. Thus, the bounded solution f in the above PDE is

$$\tilde{g}_{x_0}(s, x) = f(s, x) = e^{-\rho s}(x_0 - a) \left(\frac{x}{x_0}\right)^{\gamma_1}$$

If we fix (s, x) , the value of x_0 maximising $\tilde{g}_{x_0}(s, x)$ is given by

$$x_0 = x_{max} = \frac{a\gamma_1}{\gamma_1 - 1}$$

Thus, we have a candidate $\tilde{g}_{x_{max}}(s, x)$ for $g^*(s, x)$. In conclusion, one should sell the assets the first time their price reach the value x_{max} given above. The expected discounted profit for this strategy is

$$g^*(s, x) = \tilde{g}_{x_{max}}(s, x) = e^{-\rho s} \left(\frac{\gamma_1 - 1}{a}\right)^{\gamma_1 - 1} \left(\frac{x}{\gamma_1}\right)^{\gamma_1}$$

12 Risk return analysis

12.1 Defining financial returns

We consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where $\mathcal{F}_t$ is a right continuous filtration including all $\mathbb{P}$ negligible sets in $\mathcal{F}$. For simplicity, we let the market be complete and assume that there exists an equivalent martingale measure $\mathbb{Q}$ as defined in a mixed diffusion model by Bellamy et al. [2000]. In the presence of continuous dividend yield, that unique probability measure equivalent to $\mathbb{P}$ is such that the discounted stock price plus the cumulated dividends are martingale when the riskless asset is the numeraire. In a general setting, we let the underlying process $(S_t)_{t \geq 0}$ be a one-dimensional Ito process valued in the open subset D .

12.1.1 Asset returns

Return series are easier to handle than price series due to their more attractive statistical properties and to the fact that they represent a complete and scale-free summary of the investment opportunity (see Campbell et al. [1997]). Expected returns need to be viewed over some time horizon, in some base currency, and using one of many possible averaging and compounded methods. Holding the asset for one period from date t to date $(t + 1)$ would result in a simple gross return

$$1 + R_{t,t+1} = \frac{S_{t+1}}{S_t} \quad (12.76)$$

where the corresponding one-period simple net return, or simple return, $R_{t,t+1}$ is given by

$$R_{t,t+1} = \frac{S_{t+1}}{S_t} - 1 = \frac{S_{t+1} - S_t}{S_t}$$

More generally, we let

$$R_{t-d,t} = \frac{\nabla_d S_t + D_{t-d,t}}{S_{t-d}}$$

be the discrete return of the underlying process where $\nabla_d S_t = S_t - S_{t-d}$ with period d and where $D_{t-d,t}$ is the dividend over the period $[t-d, t]$. For simplicity we will only consider dividend-adjusted prices with discrete dividend-adjusted returns $R_{t-d,t} = \frac{\nabla_d S_t}{S_{t-d}}$. Hence, holding the asset for d periods between dates $t-d$ and t gives a d -period simple gross return

$$\begin{aligned} 1 + R_{t-d,t} &= \frac{S_t}{S_{t-d}} = \frac{S_t}{S_{t-1}} \times \frac{S_{t-1}}{S_{t-2}} \times \dots \times \frac{S_{t-d+1}}{S_{t-d}} \\ &= (1 + R_{t-1,t})(1 + R_{t-2,t-1}) \dots (1 + R_{t-d,t-d+1}) = \prod_{j=1}^d (1 + R_{t-j,t-j+1}) \end{aligned} \quad (12.77)$$

so that the d -period simple gross return is just the product of the d one-period simple gross returns, which is called a compound return. Holding the asset for d years, then the annualised (average) return is defined as

$$R_{t-d,t}^A = \left(\prod_{j=1}^d (1 + R_{t-j,t-j+1}) \right)^{\frac{1}{d}} - 1$$

which is the geometric mean of the d one-period simple gross returns involved and can be computed by

$$R_{t-d,t}^A = e^{\frac{1}{d} \sum_{j=1}^d \ln(1 + R_{t-j,t-j+1})} - 1$$

It is simply the arithmetic mean of the logarithm returns $(1 + R_{t-j,t-j+1})$ for $j = 1, \dots, d$ which is then exponentiated to return the computation to the original scale. As it is easier to compute arithmetic average than geometric mean, and since the one-period returns tend to be small, one can use a first-order Taylor expansion²² to approximate the annualised (average) return

$$R_{t-d,t}^A \approx \frac{1}{d} \sum_{j=1}^d R_{t-j,t-j+1}$$

Note, the arithmetic mean of two successive returns of +50% and -50% is 0%, but the geometric mean is -13% since $[(1 + 0.5)(1 - 0.5)]^{\frac{1}{2}} = 0.87$ with $d = 2$ periods. While some financial theory requires arithmetic mean as

²²since $\log(1 + x) \approx x$ for $|x| \leq 1$

inputs (single-period Markowitz or mean-variance optimisation, single-period CAPM), most investors are interested in wealth compounding which is better captured by geometric means.

In general, the net asset value A of continuous compounding is

$$A = Ce^{r \times n}$$

where r is the interest rate per annum, C is the initial capital, and n is the number of years. Similarly,

$$C = Ae^{-r \times n}$$

is referred to as the present value of an asset that is worth A dollars n years from now, assuming that the continuously compounded interest rate is r per annum. If the gross return on a security is just $1 + R_{t-d,t}$, then the continuously compounded return or logarithmic return is

$$r_L(t-d, t) = \ln(1 + R_{t-d,t}) = L_t - L_{t-d} \quad (12.78)$$

where $L_t = \ln S_t$. Note, on a daily basis we get $R_t = R_{t-1,t}$ and $r_L(t) = \ln(1 + R_t)$. The change in log price is the yield or return, with continuous compounding, from holding the security from trading day $t-1$ to trading day t . As a result, the price becomes

$$S_t = S_{t-1}e^{r_L(t)}$$

Further, the return $r_L(t)$ has the property that the log return between the price at time t_1 and at time t_n is given by the sum of the $r_L(t)$ between t_1 and t_n

$$\log \frac{S_n}{S_1} = \sum_{i=1}^n r_L(t_i)$$

which implies that

$$S_n = S_1 e^{\sum_{i=1}^n r_L(t_i)}$$

so that if the $r_L(t)$ are independent random variables with finite mean and variance, the central limit theorem implies that for very large n , the summand in the above equation is normally distributed. Hence, we would get a log-normal distribution for S_n given S_1 . In addition, the variability of simple price changes for a given security is an increasing function of the price level of the security, whereas this is not necessarily the case with the change in log price. Given Equation (12.77), then Equation (12.78) becomes

$$r_L(t-d, t) = r_L(t) + r_L(t-1) + r_L(t-2) + \dots + r_L(t-d+1) \quad (12.79)$$

so that the continuously compounded multiperiod return is simply the sum of continuously compounded one-period returns. Note, statistical properties of log returns are more tractable. Moreover, in the cross section approach aggregation is done across the individual returns.

Remark 12.1 That is, simple returns $R_{t-d,t}$ are additive across assets but not over time (see Equation (12.77)), whereas continuously compounded returns $r_L(t-d, t)$ are additive over time but not across assets (see Equation (12.79)).

12.2 Performance measures

When considering the performance evaluation of mutual funds, one need to assess whether these funds are earning higher returns than the benchmark returns (portfolio or index returns) in terms of risk. Three measures developed in the framework of the Capital Asset Pricing Model (CAPM) proposed by Treynor [1965], Sharpe [1964] and Lintner [1965] directly relate to the beta of the portfolio through the security market line (SML). Jensen's [1968] alpha is defined as the portfolio excess return earned in addition to the required average return, while the Treynor ratio and the Information ratio are defined as the alpha divided by the portfolio beta and by the standard deviation of the portfolio residual returns. More recent performance measures developed along hedge funds, such as the Sortino ratio, the M2 and the Omega, focus on a measure of total risk, in the continuation of the Sharpe ratio applied to the capital market line (CML). In the context of the extension of the CAPM to linear multi-factor asset pricing models, the development of measures has not been so prolific (see Hubner [2007]).

The Sharpe ratio or Reward to Variability and Sterling ratio have been widely used to measure commodity trading advisor (CTA) performance. One can group investment statistics as Sharpe type combining risk and return in a ratio, or descriptive statistics (neither good nor bad) providing information about the pattern of returns. Examples of the latter are regression statistics (systematic risk), covariance and R^2 . Additional risk measures exist to accommodate the risk concerns of different types of investors. Some of these measures have been categorised in Table (3).

Table 3: List of measure

Type	Combined Return and Risk Ratio
Normal	Sharpe, Information, Modified Information
Regression	Apraisal, Treynor
Partial Moments	Sortino, Omega, Upside Potential, Omega-Sharpe, Prospect
Drawdown	Calmar, Sterling, Burke, Sterling-Calmar, Pain, Martin
Value at Risk	Reward to VaR, Conditional Sharpe, Modified Sharpe

12.2.1 The Sharpe ratio

The Sharpe ratio measures the excess return per unit of deviation in an investment asset or a trading strategy defined as

$$M_{SR} = \frac{E[R_a - R_b]}{\sigma} \quad (12.80)$$

where R_a is the asset return and R_b is the return of a benchmark asset such as the risk free rate or an index. Hence, $E[R_a - R_b]$ is the expected value of the excess of the asset return over the benchmark return, and σ is the standard deviation of this expected excess return. More formally, following Appendix (12.1), we let $R_t = R_{t-1,t}$ denote the one-period simple return between dates $t - 1$ and t , and μ and σ^2 its mean and variance

$$\mu \equiv E[R_t] \text{ and } \sigma^2 \equiv Var(R_t)$$

The Sharpe ratio becomes

$$M_{SR} \equiv \frac{\mu - R_b}{\sigma} \quad (12.81)$$

It characterise how well the return of an asset compensates the investor for the risk taken. If we graph the risk measure with a the measure of return in the vertical axis and the measure of risk in the horizontal axis, then the Sharpe ratio simply measures the gradient of the line from the risk-free rate to the combined return and risk of each asset (or portfolio). Thus, the steeper the gradient, the higher the Sharpe ratio, and the better the combined performance of risk and return.

Remark 12.2 The ex-post Sharpe ratio uses the above equation with the realised returns of the asset and benchmark rather than expected returns.

$$M_{SR} = \frac{r_P - r_F}{\sigma_P}$$

where r_P is the asset/portfolio return (annualised), r_F is the annualised risk-free rate, and σ_P is the portfolio risk or standard deviation of return.

This measure can be compared with the Information ratio in finance defined in general as mean over standard deviation of a series of measurements. The Sharp ratio is directly computable from any observed series of returns without the need for additional information surrounding the source of profitability. While the Treynor ratio only works with systemic risk of a portfolio, the Sharp ratio observes both systemic and idiosyncratic risks. The SR has some shortcomings because all volatility is not equal, and the volatility taken in the measure ignores the distinction between systematic and diversifiable risks. Further, volatility does not distinguish between losses occurring in good or bad time or even between upside and downside surprises.

Remark 12.3 The returns measured can be any frequency (daily, weekly, monthly or annually) as long as they are normally distributed, as the returns can always be annualised. However, not all asset returns are normally distributed.

The SR assumes that assets are normally distributed or equivalently that the investors' preferences can be represented by the quadratic (exponential) utility function. That is, the portfolio is completely characterised by its mean and volatility. As soon as the portfolio is invested in technology stocks, distressed companies, hedge funds or high yield bonds, this ratio is no-longer valid. In that case, the risk comes not only from volatility but also from higher moments like skewness and kurtosis. Abnormalities like kurtosis, fatter tails and higher peaks or skewness on the distribution can be problematic for the computation of the ratio as standard deviation does not have the same effectiveness when these problems exist. As a result, we can get very misleading measure of risk-return. In addition, the Sharp ratio being a dimensionless ratio it may be difficult to interpret the measure of different investments. This weakness was well addressed by the development of the Modigliani risk-adjusted performance measure, which is in units of percent returns. One need to consider a proper risk-adjusted return measure to get a better feel of risk-adjusted out-performance such as M^2 defined as

$$M^2 = (r_P - r_F) \frac{\sigma_M}{\sigma_P} + r_F$$

where σ_M is the market risk or standard deviation of a benchmark return (see Modigliani et al. [1997]). It can also be rewritten as

$$M^2 = r_P + M_{SR}(\sigma_M - \sigma_P)$$

where the variability can be replaced by any measure of risk and M^2 can be calculated for different types of risk measures. This statistic introduces a return penalty for asset or portfolio risk greater than benchmark risk and a reward if it is lower.

12.2.1.1 The annualised Sharpe ratio For Sharpe on intraday strategies, you need to take your results on a daily basis. That is, just take your daily P&L and compute the Sharpe from that. For example, if you have prices for each stock, with a 10 second resolution, roll them into daily returns and then compute the Sharpe on daily returns.

Alternatively, instead of using the conventional Sharpe calculation we can calculate the trade level Sharpe to get a better view of the strategy's performance. In this case, the risk-free rate can be considered to be 0 since we don't roll over positions, there is no interest charge. Assuming N number of trades in a day, Sharpe ratio can be calculated by the following steps:

- The P&L for each trade

- The mean of P&Ls for all trades
- The standard deviation for P&Ls for all trades

The numerator of the ratio, which is the mean of PnL, gets multiplied by N when you consider all trades. In the denominator, the standard deviation gets multiplied by the square root of N . Thus, we get $\frac{N}{\sqrt{N}} = \sqrt{N}$. So effectively, the annualised Sharpe ratio is $M_{ASR} = \sqrt{N}M_{SR}$, or

$$M_{ASR} = \sqrt{N} \frac{E[R_a - R_b]}{\sigma}$$

Note that the Sharpe ratio itself must be calculated based on the Sharpe of that particular time period type. For a strategy based on trading period of days $N = 252$ (as there are 252 trading days in a year), and R_a and R_b must be the daily returns. Similarly for hours $N = 6.5 \times 252 = 1638$ since there are only 6.5 hours in a trading day. In the case of minute trading, we do $60 \times$ trading hours in a day of trading $\times$ number of trading days in a year, that is, $N = 60 \times 6.5 \times 252 = 98280$. For high-frequency strategies, a large number of small successful trades for specific amounts smoothen the PnL curve and the standard deviation approaches to zero which significantly spikes the Sharpe ratio, such that it might range in double digits.

In general, intraday/market-making strategies don't have a reasonable return metric. The highest risks with HF are usually operational, not financial. For this reason you can't characterise them with the Sharpe Ratio, which depends on a capital basis and how that basis is leveraged. P&L is more related to strategy capacity than it is to any capital invested metric. In order to characterise the performance of a daily stream of dollar income that doesn't have a capital basis, we could start with mean, standard deviation, and skewness. Further, we can ask for % Winning days and AvgWin/AvgLoss, or Profit Factor.

12.2.2 More measures of risk

Treynor proposed a risk adjusted performance measure associated with abnormal returns in the CAPM. The Treynor ratio or reward to volatility is a Sharpe type ratio where the numerator (or vertical axis graphically speaking) is identical but the denominator (horizontal axis) replace total risk with systematic risk as calculated by beta

$$M_{TR} = \frac{r_P - r_F}{\beta_P}$$

where β_P is the market beta (see Treynor [1965]). Although well known, the Treynor ratio is less useful precisely because it ignores specific risk. It will converge to the Sharpe ratio in a fully diversified portfolio with no specific risk. The Appraisal ratio suggested by Treynor & Black [1973] is a Sharpe ratio type with excess return adjusted for systematic risk in the numerator, and specific risk and not total risk in the denominator

$$M_{AR} = \frac{\alpha}{\sigma_\epsilon}$$

where α is the Jensen's alpha. It measures the systematic risk adjusted reward for each unit of specific risk taken. While the Sharpe ratio compares absolute return and absolute risk, the Information ratio compares the excess return and tracking error (the standard deviation of excess return). That is, the Information ratio is a Sharpe ratio type with excess return on the vertical axis and tracking error or relative risk on the horizontal axis. As we are not using the risk-free rate, the information ratio lines radiate from the origin and can be negative indicating underperformance.

12.2.3 Incorporating tail risk

While the problem of fat tails is everywhere in financial risk analysis, there is no solution, and one should only consider partial solutions. Hence, one practical strategy for dealing with the messy but essential issues related to measuring and managing risk starts with the iron rule of never relying on one risk metric. Even though all of the standard metrics

have well-known flaws, that does not make them worthless, but it is a reminder that we must understand where any one risk measure stumbles, where it can provide insight, and what are the possible fixes, if any.

The main flaw of the Sharpe ratio (SR) is that it uses standard deviation as a proxy for risk. This is a problem because standard deviation, which is the second moment of the distribution, works best with normal distributions. Therefore, investors must have a minimal type of risk aversion to variance alone, as if their utility function was exponential. Even though normality has some validity over long periods of time, in the short run it is very unlikely (see short maturity smile on options). Note, when moving away from the normality assumption for the stock returns, only the denominator in the Sharpe ratio is modified. Hence, all the partial solutions are attempts at expressing one way or another the proper noise of the stock returns. While extensions of the SR to normality assumption have been successful, extension to different types of utility function have been more problematic.

The problem is that extreme losses occur more frequently than one would expect when assuming that price changes are always and forever random (normally distributed). Put another way, statistics calculated using normal assumption might underestimate risk. One must therefore account for higher moments to get a better understanding of the shape of the distribution of returns in view of assessing the relative qualities of portfolios. Hodges [1997] introduced an extension of the SR accounting for non-normality and incorporating utility function (see Appendix (12.3.2)). Assuming that investors are able to find the expected maximum utility $E[u^*]$ associated with any portfolio, the generalised Sharpe ratio (GSR) of the portfolio is

$$M_{GSR} = (-2 \ln(-E[u^*]))^{\frac{1}{2}}$$

One can avoid the difficulty of computing the maximum expected utility by assuming the investor has an exponential utility function. Using the fourth order Taylor approximation of the certain equivalent return (CER), and approximating the multiplicative factor Pezier [2004] and Pezier et al. [2006] obtained the maximum expected utility function in that setting and showed that the GSR simplifies to the ASR (see Appendix (12.3)). Thus, when the utility function is exponential and the returns are normally distributed, the GSR is identical to the SR. Otherwise a negative skewness and high positive kurtosis will reduce the GSR relative to the SR. Investors should prefer high average returns, lower variance or standard deviation, positive skewness, and lower kurtosis. The adjusted Sharpe ratio explicitly rewards positive skewness and low kurtosis (below 3, the kurtosis of a normal distribution) in its calculation

$$M_{ASR} = M_{SR} \left[1 + \frac{S}{6} M_{SR} - \frac{K-3}{24} M_{SR}^2 \right] \quad (12.82)$$

where S is the skew and K is the kurtosis. This adjustment will tend to lower the SR if there is negative skewness and positive excess kurtosis in the returns. Hence, it potentially removes one of the possible criticisms of the Sharpe ratio.

12.3 Mean-variance-skewness-kurtosis approach

We are going to give some hints at the way the Adjusted Sharpe Ratio in Equation (12.82) was derived. First, we review the Taylor series expansion of expected utility function. Then, we discuss the Generalised Sharpe Ratio introduced by Hodges [1998]. Finally, we follow Maillard [2018] in his derivation of the formula.

12.3.1 The Taylor expansion

As the infinite Taylor expansion is not suitable for numerical implementation, an approximation of the expected utility function by truncating the Taylor expansion at a given value, k , proves an adequate solution. A truncated expansion to k provides an exact solution when the utility value is defined by a polynomial function of order k . Jurczenko et al. [2003] applied this approach to the fourth order of utility functions. In that setting, the exact Taylor series expansion of expected utility can be approximated as follows:

$$E[u(R)] = u(E(R)) + \frac{1}{2}u^{(2)}(E[R])\sigma^2(R) + \frac{1}{6}u^{(3)}(E[R])S(R) + \frac{1}{24}u^{(4)}(E[R])K(R)$$

where $\sigma^2(x) = E[(x - E[x])^2]$, $S(x) = E[(x - E[x])^3]$, $K(x) = E[(x - E[x])^4]$ are the variance, the skew, and the kurtosis, respectively, of probability distribution of x . The differential of expected utility over $E[x]$, $\sigma^2(x)$, $S(x)$, and $K(x)$ is

$$\begin{aligned}\frac{\partial E[u(x)]}{\partial E[x]} &= E[u^{(1)}(x)] > 0, \quad \frac{\partial E[u(x)]}{\partial \sigma^2(x)} = \frac{1}{2} E[u^{(2)}(x)] < 0 \\ \frac{\partial E[u(x)]}{\partial S(x)} &= \frac{1}{6} E[u^{(3)}(x)] > 0, \quad \frac{\partial E[u(x)]}{\partial K(x)} = \frac{1}{24} E[u^{(4)}(x)] < 0\end{aligned}$$

This relationship illustrates investor's preferences to mean and asymmetry (positive) and aversion to variance and kurtosis of probability distribution. It is necessarily to impose additional assumptions on the nature of trader's preferences, or on the characteristics of risky asset distributions. Asset returns distributions must belong to four parameter probability distributions which the first four moments exist. Further, all investors should have a polynomial Von Neumann-Morgenstern four order utility function. Under these assumptions, we can introduce centered moments of order three and four on uncertain decision criterion and justify it theoretically and empirically.

12.3.2 The Generalised Sharpe Ratio

12.3.2.1 Maximising utility The main idea is to relate the performance measure to the utility function. Following Hodges [1998], consider an investor maximising $E[u(w)]$ with the utility $u(w) = -e^{-\lambda w}$ for the wealth w . For simplicity, assume $w_0 = 0$ and $\lambda > 0$ is the risk aversion. Assume the investor faces a single opportunity to maximise his utility function. Assume forward investment opportunities are normally distributed after time T , that is, the future outcomes are distributed as $N(\mu T, \sigma^2 T)$. Since the utility function is an exponential function and the returns are normal, we can maximise $E[u(w)]$ by solving

$$\max_x E[u] = \max_x -e^{-\lambda E[xr] + \frac{1}{2} \text{Var}(\lambda xr)} = e^{-\lambda(\mu x T - \frac{1}{2} \lambda \sigma^2 T x^2)}$$

where r is the return. Hence, the first order condition is

$$\mu T - \lambda \sigma^2 T x = 0$$

so that $x^* = \frac{\mu}{\lambda \sigma^2}$. Substituting into the expression for $E[u]$, we get

$$u^* = \max_x E[u] = -e^{-\frac{1}{2} \frac{\mu^2}{\sigma^2} T}$$

Taking the logarithm, we get

$$\log(-u^*) = -\frac{1}{2} \frac{\mu^2}{\sigma^2} T$$

and the Sharpe ratio becomes

$$\frac{\mu}{\sigma} = \sqrt{\frac{-2}{T} \log(-u^*)}$$

The Generalised Sharpe Ratio (GSR) is defined as

$$GSR = \sqrt{\frac{-2}{T} \log(-u^*)}$$

where u^* is the optimal expected utility obtainable with the above exponential function. Since u^* is independent from the risk aversion, we can set $\lambda = 1$. We can also ignore the time dimension and set $T = 1$.

12.3.2.2 Maximising CER The Certainty Equivalent Return (CER) satisfies $u(c) = E[u(r)]$, where $r = \mu + \sigma y$ with $y \sim N(0, 1)$. Note, if a share x of wealth was invested in the risky asset and the rest in the risk-free asset, the portfolio's CER would write:

$$c_x = \mu x T - \frac{1}{2} \lambda \sigma^2 T x^2$$

This function is first increasing and then decreasing. Hence, it holds a maximum. There is an optimal investment share in the risky asset obtained by equating to zero the first derivative (with respect to x) of the above function. We get

$$\mu T - \lambda \sigma^2 T x = 0$$

so that $x^* = \frac{\mu}{\lambda \sigma^2}$. Substituting into the expression for c_x , we get

$$c_x^* = \frac{\mu^2}{\lambda \sigma^2} T - \frac{1}{2} T \frac{\mu^2}{\lambda \sigma^2} = \frac{1}{2\lambda} \frac{\mu^2}{\sigma^2} T$$

Thus, we deduce that

$$M_{SR} = \frac{\mu}{\sigma} = \sqrt{\frac{2\lambda}{T} c_x^*}$$

Again, we can ignore the time dimension by setting $T = 1$. Then, we get a relationship between the SR and CER given by $\frac{\mu}{\sigma} = \sqrt{2\lambda c_x^*}$. We can compute a CER for non Gaussian returns, getting the Generalised Sharpe Ratio (GSR)

$$GSR = \sqrt{2\lambda c_x^*} \quad (12.83)$$

In that setting, it is assumed that the optimal portfolio made of the risky asset and the risk-free asset used to compute the CER (for non Gaussian returns) will have the same weights as the optimal portfolio in the case of Gaussian returns. That is, it has the same mean and volatility given by

$$\mu^* = x^* \mu = \frac{M_{SR}^2}{\lambda} \text{ and } \sigma^* = x^* \sigma = \frac{M_{SR}}{\lambda}$$

The optimal CER will verify (for $x_0 = 1$)

$$u(c^*) = E[u(r)] = -e^{-\lambda \mu^*} E[e^{-\lambda \sigma^* y}] \quad (12.84)$$

Since there is no closed-form solution to this expectation, we need to replace the utility function by a fourth-order Taylor expansion around the mean return of the optimal portfolio μ^* .

12.3.3 Derivation of the formula

An insurer having initial wealth W and utility function u is said to have exponential utility if the function u can be expressed as follows

$$u(x) = -ce^{-cx}$$

Differentiating with respect to x , we get

$$u'(x) = c^2 e^{-cx} = -cu(x), u''(x) = c^2 u(x), u'''(x) = -c^3 u(x), u''''(x) = c^4 u(x)$$

Note, if the utility function is $u(x) = -e^{-cx}$, we get the same derivative terms. Do a Taylor expansion around a and get

$$u(x) = u(a) + u'(a)(x-a) + \frac{1}{2} u''(a)(x-a)^2 + \frac{1}{6} u'''(a)(x-a)^3 + \frac{1}{24} u''''(a)(x-a)^4 + R$$

where R is the residual term. Replace with the derivatives, we get

$$u(x) = u(a) - cu(a)(x - a) + \frac{1}{2}c^2u(a)(x - a)^2 - \frac{1}{6}c^3u(a)(x - a)^3 + \frac{1}{24}c^4u(a)(x - a)^4 + R$$

Take the expectation, we get

$$E[u(x)] = u(a) - cu(a)E[(x - a)] + \frac{1}{2}c^2u(a)E[(x - a)^2] - \frac{1}{6}c^3u(a)E[(x - a)^3] + \frac{1}{24}c^4u(a)E[(x - a)^4] + R$$

Let $a = E[x] = \mu$, then the expected utility simplifies to

$$E[u(x)] = u(a) + \frac{1}{2}c^2u(a)\sigma^2(x) - \frac{1}{6}c^3u(a)m_3(x) + \frac{1}{24}c^4u(a)m_4(x) + R$$

where $\sigma^2(x) = E[(x - a)^2]$, $m_3(x) = E[(x - a)^3]$, $m_4(x) = E[(x - a)^4]$ are the variance, the third central moment, and the fourth central moment, respectively. Let $S(x) = \frac{m_3}{\sigma^3}$ be the Skew and $K(x) = \frac{m_4}{\sigma^4}$ be the Kurtosis. Then, we can rewrite the above equation as

$$E[u(x)] = u(a) + \frac{1}{2}c^2u(a)\sigma^2(x) - \frac{1}{6}c^3u(a)\sigma^3S(x) + \frac{1}{24}c^4u(a)\sigma^4K(x) + R$$

Factorising, we get

$$E[u(x)] = u(a)\left(1 + \frac{1}{2}c^2\sigma^2(x) - \frac{1}{6}c^3\sigma^3S(x) + \frac{1}{24}c^4\sigma^4K(x)\right) + R$$

Set $c = \frac{\mu}{\sigma^2}$, we get

$$E[u(x)] = u(a)\left(1 + \frac{1}{2}\frac{\mu^2}{\sigma^2} - \frac{1}{6}\frac{\mu^3}{\sigma^3}S(x) + \frac{1}{24}\frac{\mu^4}{\sigma^4}K(x)\right) + R$$

We now perform a Taylor series expansion of $u(c^*)$ around $a = \mu$ to the first order, getting

$$u(c^*) = u(a) + u'(a)(c^* - a) + R$$

Replace u' with its derivative, we get

$$\begin{aligned} u(c^*) &= u(a) - cu(a)(c^* - a) + R \\ &= u(a)(1 - c(c^* - a)) + R \\ &= u(a)\left(1 + \frac{\mu^2}{\sigma^2} - cc^*\right) + R \end{aligned}$$

since $\mu = \frac{M_{SR}^2}{c}$. From Equation (12.84), we can equate the two expressions above, getting

$$1 + \frac{\mu^2}{\sigma^2} - cc^* = 1 + \frac{1}{2}\frac{\mu^2}{\sigma^2} - \frac{1}{6}\frac{\mu^3}{\sigma^3}S(x) + \frac{1}{24}\frac{\mu^4}{\sigma^4}K(x)$$

The optimum CER is

$$cc^* = \frac{1}{2}\frac{\mu^2}{\sigma^2} + \frac{1}{6}\frac{\mu^3}{\sigma^3}S(x) - \frac{1}{24}\frac{\mu^4}{\sigma^4}K(x)$$

Here, the CER is increased by a third-order term and decreased by a fourth-order term. From Equation (12.83), we get

$$GSR = \frac{\mu}{\sigma} \sqrt{\left[1 + \frac{1}{3}\frac{\mu}{\sigma}S(x) - \frac{1}{12}\frac{\mu^2}{\sigma^2}K(x)\right]}$$

Using the first order Taylor series on the square root²³, it simplifies to

²³ $\sqrt{1+x} \approx 1 + \frac{x}{2}$

$$GSR \approx M_{SR} \left[1 + \frac{1}{6} M_{SR} S(x) - \frac{1}{24} M_{SR}^2 K(x) \right]$$

Instead, to justify the modified kurtosis term, we rewrite the expected utility as

$$E[u(x)] = u(a) + \frac{1}{2} c^2 u(a) \sigma^2(x) - \frac{1}{6} c^3 u(a) \sigma^3 S(x) + \frac{1}{24} c^4 u(a) \sigma^4 (K(x) - 3) + \tilde{R}$$

where $\tilde{R} = R + \frac{3}{24} c^4 u(a) \sigma^4$. In that setting, the GSR becomes

$$GSR \approx M_{SR} \left[1 + \frac{1}{6} M_{SR} S(x) - \frac{1}{24} M_{SR}^2 (K(x) - 3) \right]$$

and we recover the Adjusted Sharpe Ratio given in Equation (12.82).

13 Technical indicators

13.1 Presenting a few trading indicators

Any mechanical trading system (MTS) must have some consistent method or trigger for entering and exiting the market based on some type of indicator or mathematical statistics with price forecasting capability. Anything that indicates what the future may hold is an indicator. In general, time series analysis is used to examine data for repeating patterns by collecting data on a system over time in order to analyse a system or to predict future trends. For instance, researchers consider serial correlation and run-tests to test if there are significant price patterns (see Damodaran [2003]).

Most system traders and developers spend 90% of their time developing entry and exit technique, and the rest of their time is dedicated to the decision process determining profitability. Some of the most popular indicators include moving average (MA), weighted moving average (WMA), rate of change, momentum, stochastic (Lane 1950), Relative Strength Index (RSI) (see Wilder [1978]), moving average convergence divergence (MACD) (see Appel [1999]), Donchian breakout, Bollinger bands (BB) (see Bollinger [1992]), Keltner bands (see Keltner [1960]), Commodity Channel Index (CCI) (see Lambert [1980]), Chaikin accumulation/distribution (ChaikinAD), triple smoothed exponential oscillator (TRIX). However, indicators can not stand alone, and should be used in concert with other ideas and logic. There is a large list of indicators with price forecasting capability and we are now going to describe a few of them. For more details we refer the readers to Murphy [1999], Hill et al. [2000], Brabazon [2000], among other.

13.1.1 Stable levels

Technical Analysis tries to derive profitable buy and sell signals by isolating upward and downward price trends from oscillations around a stable level (see Schulmeister [2005]). We let C_t be the closing price at time t , O_t is the opening price, L_t is the low price, H_t the high price, and V_t is the volume of the trade at time t . LL_n is the lowest low and HH_n is the highest high in the look back period with n discrete events. Some of these levels are as follows:

- Range: $R_t = H_t - L_t$. Large or increasing ranges suggest traders are prepared to continue to bid up or sell down a stock through the course of the day.

- The Typical Price

$$M_t = \frac{H_t + L_t + C_t}{3} \tag{13.85}$$

- Momentum: $C_t - C_{t-n}$, where n is the number of discrete events in the look back period.

- Support and resistance: the former is a price level that a share has reached but not fallen below, the latter is a price level that the market has reached but has not risen above. A breach of either of these levels is sometimes considered to indicate a significant change in a share's price, with the level breached then forming a new support or resistance level respectively.
- Retracing: it assumes that prices trends will eventually tend to reverse to visit a support or resistance level. Thus, attempting to predict a future price.
- UpDown Length: It is the number of consecutive days that a security price has either closed up (higher than previous day) or closed down (lower than previous days). Closing up values represented in positive numbers and closing down is represented with negative numbers. If a security closes at the same price on back to back days, the UpDown Length is 0.
- Accumulation / Distribution (A/D) oscillator: $\frac{(H_t - O_t) + (C_t - L_t)}{2(H_t - L_t)} \times 100$
- Moving average indicator: the simple moving average (SMA), or just MA, is

$$MA_t(n) = \frac{1}{n} \sum_{i=0}^{n-1} C_{t-i} \quad (13.86)$$

A buy signal is generated when a share's price exceeds the moving average and a sell signal when the moving average exceeds the share price. A variation of this indicator is the moving average convergence divergence (MACD) oscillator calculated by taking the difference of a short run and a long run moving averages of differing length. If the result is positive, this is taken as a signal that the market is trending upward.

- Disparity in n days: $\frac{C_t}{MA(n)} \times 100$
- Bias in n days: $BIAS(n) = \frac{C_t - MA(n)}{MA(n)} \times 100$
- OSCP price oscillator: $\frac{MA(short) - MA(long)}{MA(short)}$
- Average return in the last n days: $ASY(n) = \frac{1}{n} \sum_{i=1}^n SY_{t-i+1}$, where $SY_t = \ln \frac{C_t}{C_{t-1}} \times 100$
- Ratio of the number of rising periods over the n day period: $PSY(n) = \frac{1}{n} A \times 100$, where A is the number of rising days in the last n days.
- Exponential moving average (EMA) indicator (also known as an exponentially weighted moving average (EWMA)): It gives more weight to the most recent periods, making it more reliable than the SMA and a better representation of the recent performance of the security. Hence, it can be used to create a better moving average strategy. At time t , given a window of k periods, it is given by the formula

$$EMA(k)_t = EMA(k)_{t-1} + \alpha \times (C_t - EMA(k)_{t-1}) \quad (13.87)$$

where $EMA(k)_{t-k+1} = C_{t-k+1}$, $\alpha = \frac{2}{k+1}$ is a smoothing factor, and k is the time period of k -day EMA. Note, it can be rewritten as

$$EMA(k)_t = \alpha \times C_t + (1 - \alpha)EMA(k)_{t-1} \quad (13.88)$$

The weightage to the most recent data is greater for a shorter period EMA than for a longer period. Each term in the moving average period has an exponentially greater weightage than its preceding term, making the EMA faster to react than the SMA.

- Variable Moving Average (VMA) indicator: In 1991 Tushar Chande suggested that the performance of an EMA could be improved by using a Volatility Index (VI) to adjust the smoothing period when market conditions change. Volatility is the measure of how quickly or slowly prices change over time. The volatility index shows the market's volatility predictions for the next 30 days. The main idea is to slow down the average when prices are in the consolidation period to avoid unavailing trading signals and to speed up the average when the market is trending so as to make the most out of the trending prices. Its formula is obtained by modifying the smoothing factor α in the EMA in Equation (13.88), getting

$$VMA(k)_t = \hat{\alpha} \times C_t + (1 - \hat{\alpha})VMA(k)_{t-1}$$

where $\hat{\alpha} = VI * \alpha$ and VI is a measure of volatility or trend strength.

- On Balance Volume (OBV) indicator: it is based on a cumulative total volume and is intended to relate price and volume in the stock market (see Granville [1976]). The idea is that volume is higher on days where the price move is in the dominant direction, for example in a strong uptrend there is more volume on up days than down days. The formula is as follows:

$$OBV(k)_t = OBV(k)_{t-1} + \theta \times V_t \quad (13.89)$$

where

$$\theta = \begin{cases} +1 & \text{if } C_t \geq C_{t-1} \\ -1 & \text{if } C_t < C_{t-1} \end{cases}$$

- The Volume-Weighted Average Price (VWAP): Based both on volume and price, it represents the average price adjusted for volume at which a security has been trading throughout the day. The main idea being to give more weights to prices exhibiting high volume. These are special prices which should be overweighted. It is used on intraday charts and resets at the start of every new trading session. Given an intraday time frame (for instance bars of 5 minutes), let $\mathcal{T} = \{t_1, \dots, t_n\}$ be the time set with $t_i = i \times dt$, and $\mathcal{B} = \{B_1, \dots, B_n\}$ be the set of trading bars for that day. Given the time index k , the VWAP for that bar is given by

$$V_{wap}^{(k)} = M_k \times V_k$$

where M_k is the Typical Price (see Equation (13.85)) and V_k is the volume for that bar. Then, at any discrete time $t_i \in \mathcal{T}$, the VWAP is

$$V_{wap}(t_i) = \frac{\sum_{k=1}^i V_{wap}^{(k)}}{\sum_{k=1}^i V_k}, i \in \mathcal{I}$$

We can rewrite it as a weighted average by setting the weight as $w_k = \frac{V_k}{\sum_{k=1}^i V_k}$. That is,

$$V_{wap}(t_i) = \sum_{k=1}^i w_k M_k, i \in \mathcal{I}$$

It looks similar to a moving average line but smoother. However, since the cumulated volume is reset each day, we usually observe a jump in the indicator at the start of the day. Many VWAP indicators also come with an upper and lower trend line that is similar to a Bollinger Band. Note, the VWAP can be anchored at important points such as reversal area.

- Stochastic oscillators: In the 50s Lane introduced an oscillating type of indicator called Stochastics that compares the current market close to the range of prices over a specific time period, indicating when a market is overbought or oversold (see Lane [1984]). The stochastic oscillator is a momentum indicator that uses support and resistance levels. It is based on the assumption that when an up-trend/down-trend approaches a turning

point, the closing prices start moving away from the high/low price of a specific range. The number generated by this indicator is a percentage in the range $[0, 100]$ such that for a reading of 70 or more it indicates that the close is near the high of the range. A reading of 30 or below indicates the close is near the low of the range. Note, in general the values of the indicator are smoothed with a moving average (MA).

1. Stochastic $K\%$ oscillators: $\frac{C_t - LL_n}{HH_n - LL_n} \times 100$ where C_t is the current price, LL_n is the lowest price in the last n days and HH_n is the highest price in the last n days. A value nearer 0 is considered to indicate a market which is oversold (which will tend to rise) and a value near 100 indicates a market which is overbought.
 2. Stochastic $D\%$ oscillators: It is the moving average of the stochastic $K\%$ oscillators. The formula is $\frac{1}{n} \sum_{i=0}^{n-1} K_{t-i}\%$, where $K_t\%$ is the above oscillator.
 3. Stochastic slow $D\%$ oscillators: $\frac{1}{n} \sum_{i=0}^{n-1} D_{t-i}\%$ where $D_t\%$ is the above oscillator.
- Larry William $R\%$ (LWR): $\frac{HH_n - C_t}{HH_n - LL_n} \times (-100)$
 - The average true range (ATR): It measures market volatility by decomposing the entire range of an asset price for that period (see Welles Wilder [1978]). The true range indicator is taken as the greatest of the following: current high less the current low; the absolute value of the current high less the previous close; and the absolute value of the current low less the previous close. The ATR is then a moving average, generally using 14 days, of the true ranges. The true range is given by

$$TR_t = \max\left((H_t - L_t), |H_t - C_{t-1}|, |L_t - C_{t-1}|\right)$$

Note, the range $H_t - L_t$ measures the range/size of one candlestick. The larger the candle, the higher the ATR is, while the smaller the candle, the lower the ATR is. Hence, large or increasing ranges suggest traders are prepared to continue to bid up or sell down a stock through the course of the day. Then, we compute the arithmetic mean of the TR over n periods (excluding TR_t), getting $\overline{TR_t}(n) = \frac{1}{n} \sum_{i=1}^n TR_{t-i}$. Finally the ATR is given by

$$ATR_t(n) = \frac{1}{n} \sum_{i=0}^{n-1} TR_{t-i} = \frac{\overline{TR_t}(n-1) \times (n-1) + TR_t}{n} = \frac{ATR_{t-1}(n-1) \times (n-1) + TR_t}{n} \quad (13.90)$$

Thus, the ATR is roughly an average measure of the range/size of the past n candlesticks. A stock experiencing a high level of volatility has a higher ATR, and a lower ATR indicates lower volatility for the period evaluated.

13.1.2 Technical indicators

We now describe more complex technical indicators:

- The Chande momentum oscillator (CMO) is a technical momentum indicator (see Chande et al. [1994]) that identifies relative strength or weakness in a market. It measures momentum on both up and down days. It is created by calculating the difference between the sum of all recent gains and the sum of all recent losses and then dividing the result by the sum of all price movement over the period.
We consider $U_t(n) = \sum_{i=0}^{n_u-1} UP_{t-i}$ (UP_t is upward price changes) where n_u is the number of up periods characterised by the close being higher than the previous close, that is, $UP_t = C_t - C_{t-1} > 0$.
We also consider $D_t(n) = \sum_{i=0}^{n_d-1} DW_{t-i}$ (DW_t is downward price changes) where n_d is the number of down periods characterised by the close being lower than the previous close, that is, $DW_t = C_{t-1} - C_t > 0$.
The formula is given by

$$CMO = \frac{U_t(n) - D_t(n)}{U_t(n) + D_t(n)} \times 100 \quad (13.91)$$

Note, $U_t(n) + D_t(n)$ is the sum of all price movement over a given time period. The oscillator is in the range $[-100, +100]$.

- **Commodity channel index (CCI):** It is an oscillator originally introduced by Donald Lambert [1980] used for identifying cyclical trends. CCI measures a security's variation from the statistical mean, fitting into the momentum category of oscillators. It is calculated as the difference between the typical price of the underlying and its simple moving average, divided by the mean absolute deviation of the typical price. It is an unbounded indicator so that the index is usually scaled by an inverse factor of 0.015 to ensure that approximately 70 to 80 percent of CCI values would fall between -100 and $+100$. The four steps to compute the formula are as follow:

1. Compute the Typical Price M_t in Equation (13.85).
2. Compute the simple moving average (SMA) price of the last n Typical prices

$$SM_t(n) = \frac{1}{n} \sum_{i=1}^n M_{t-i+1}$$

3. Compute the mean absolute deviation of the last n Typical prices

$$D_t(n) = \frac{1}{n} \sum_{i=1}^n |M_{t-i+1} - SM_t(n)|$$

4. Compute the CCI

$$CCI = \frac{M_t - SM_t(n)}{0.015 \times D_t(n)}$$

The percentage of CCI values falling in the range $[-100, +100]$ will depend on the number of periods used. A shorter CCI will be more volatile with a smaller percentage of values in that range. Conversely, the more periods used to calculate the CCI, the higher the percentage of values in that range.

- Wilder [1978] developed an indicator that measures the velocity of directional movement in prices, making it a momentum²⁴ oscillator. It provides a relative evaluation of the strength of a security's recent price performance. As such, it can be used to evaluate overvalued or undervalued conditions in the price of that security. The Relative Strength Index (RSI) is a ratio of the upward price movement to the total price movement over a given period of time (days). The average gain (AU) and average loss (AD) are calculated using an n -period smoothed or modified moving average (SMMA or MMA,) which is an exponentially smoothed Moving Average with $\alpha = \frac{1}{\text{period}}$. Note, one can also use a standard exponential moving average (EMA) as the average instead of the SMMA. Default smoothing period is 14 periods (or days), $\alpha = \frac{1}{14}$. It can be lowered to increase sensitivity or raised to decrease sensitivity. Wilder originally formulated the calculation of the moving average as:

$$\text{newval} = \frac{1}{\text{period}} (\text{prevval} \times (\text{period} - 1) + \text{newdata})$$

A variation called Cutler's RSI (RSI) is based on a simple moving average (SMA) of U and D , instead of the exponential average above. Cutler found that since Wilder used a smoothed moving average to calculate RSI, the value of the RSI depended upon where in the data file his calculations started. Cutler termed this Data Length

²⁴Momentum is the rate of the rise or fall in price.

Dependency. His RSI is not data length dependent, and returns consistent results regardless of the length of, or the starting point within a data file. Losses are expressed as positive values, not negative values. That is, periods with price losses are counted as zero in the calculations of average gain, and periods with price increases are counted as zero in the calculations of average loss. The formula is given by

$$RSI = \frac{AU(n)}{AU(n) + AD(n)} \times 100 \quad (13.92)$$

where

- $AU(n) = \frac{1}{n_u} \sum_{i=0}^{n_u-1} UP_{t-i} = SMA(U, n)$ (UP_t is upward price changes) is the average of n_u days up closes. Up periods are characterised by the close being higher than the previous close. That is, $UP_t = C_t - C_{t-1} > 0$.
- $AD(n) = \frac{1}{n_d} \sum_{i=0}^{n_d-1} DW_{t-i} = SMA(D, n)$ (DW_t is downward price changes) is the average of n_d days down closes. Down periods are characterised by the close being lower than the previous close. That is, $DW_t = C_{t-1} - C_t > 0$.

The ratio takes values in the range $[0, 100]$. When $AD = 0$ the RSI is 100 (only going up) and when $AU = 0$ the RSI is 0. Traditionally, an RSI reading of 70 or above indicates an overbought situation, and a reading of 30 or below indicates an oversold condition.

The formula is sometime written as

$$RSI = 100 - \frac{100}{(1 + RS(n))} \quad (13.93)$$

where

$$RS(n) = \frac{AU(n)}{AD(n)} = \frac{SMA(U, n)}{SMA(D, n)}$$

is the average gain over the average loss. If we choose the EMA as the average, we get

$$RS(n) = \frac{EMA(U, n)}{EMA(D, n)} \quad (13.94)$$

In general the formula will need at least 250 data points. A plot of RS looks exactly the same as a plot of RSI. The normalisation step makes it easier to identify extremes because RSI is range bound.

- The Donchian channel (or breakout) is an envelope indicator involving two lines that are plotted above and below the market. The top line represents the highest high of n days back (the last n days) (or weeks), and conversely the bottom line represents the lowest low of n days back (the last n days). This centreline is simply the mean of the upper and lower values. While it is an indicator of volatility of a market price, it is used for providing signals for long and short positions. If a security trades above its highest n periods high, then a long is established, otherwise if it trades below the lowest n periods, a short is established. Alternatively, when the day's high penetrates the highest high of four weeks back and selling when the day's low penetrates the lowest low of four weeks back. The channels are wider when there are heavy price fluctuations and narrow when prices are relatively flat.
- The Moving Average Crossover involves two or more moving averages usually consisting of a longer-term and shorter-term average. When the short-term MA crosses from below the long-term MA, it usually indicates a buying opportunity, and selling opportunities occur when the shorter-term MA crosses from above the longer-term MA. Moving averages can be calculated as simple, exponential, and weighted average. Exponential and weighted MAs tend to skew the moving averages toward the most recent prices, increasing volatility.

- In the 70s, Appel [1974] developed another price oscillator called the moving average convergence divergence (MACD). The **MACD line** is the difference between a fast (short term) EMA and a slow (long term) EMA of the closing price of a particular security. The formula is given by

$$DIFF_t = EMA(n_S)_t - EMA(n_L)_t \quad (13.95)$$

where n_S is the number of observations in the short period, n_L is the number of observations in the long period, and $EMA(k)_t$ is given in Equation (13.87). Practically, it is calculated by subtracting the value of a 26-period exponential moving average ($EMA(26)_t$) from a 12-period exponential MA $EMA(12)_t$. It corresponds to a Double Moving Average Crossover (DMAC) on EMA, where $EMA(12)_t$ is the fast average and $EMA(26)_t$ is the slow average.

In general, it is compared with its 9-day exponential moving average, which is called **the signal line**

$$SL_t = EMA_{DIFF_t}(9) \quad (13.96)$$

In that case, it corresponds to a Triple Moving Average Crossover (DMAC) on EMA, where $EMA(9)_t$ is the fast average, $EMA(12)_t$ is the medium average, and $EMA(26)_t$ is the slow average. The intersection of these two signals indicates a buying signal if the MACD line comes from the bottom (below), and a selling signal if this line comes from the top (above). The formula is given by

$$MACD_t = DIFF_t - SL_t$$

We buy if $MACD_t > 0$ and sell if $MACD_t < 0$.

Variants exist. For instance, it can be derived from three different exponentially smoothed moving averages (see Appel et al. [2008]). It is plotted as two different lines, the first line (**the MACD line**) being the difference between the two MAs (long-term and short-term MAs), and the second line (**the signal or trigger line**) being an exponentially smoothed MA of the MACD line. The formula is given by

$$MACD(n; n_S, n_L)_t = MACD(n; n_S, n_L)_{t-1} + \alpha(DIFF_t - MACD(n; n_S, n_L)_{t-1}) \quad (13.97)$$

with smoothing factor $\alpha = \frac{2}{n+1}$.

Remark 13.1 By analogy with the EMA formula in Equation (13.87), the MACD line $DIFF_t$ plays the role of the closing price C_t and the signal $MACD(n)_{t-1}$ plays the role of the EMA $EMA(n)_{t-1}$. That is, the signal line is the EMA of the MACD line. Hence, the MACD is an EMA applied to the MACD line instead of the price.

The above formula can be rewritten as

$$MACD(n; n_S, n_L)_t = \alpha DIFF_t + (1 - \alpha) MACD(n; n_S, n_L)_{t-1}$$

Note, the difference (or divergence) at time t between the MACD line and the signal line, given by

$$Hist_t = (DIFF_t - MACD(n; n_S, n_L)_t) \quad (13.98)$$

is shown as a bar graph. It is called the MACD histogram. The purpose being to try to eliminate the lag associated with MA type systems. This is done by anticipating the MA crossover and taking action before the actual crossover. The system buys when the MACD line crosses the signal line from below and sells when the MACD line crosses the signal line from above.

- The rate of change (ROC) momentum involves the analysis of the rate of price change rather than the price level. It is a momentum-based technical indicator that measures the percentage change in price between the current price and the price a certain number of periods ago. It measures the strength of price momentum by the rate of change. The speed of price movement and the rate at which prices are moving up or down provide clues to the amount of strength the bulls or bears have at a given point in time. It is calculated by dividing the change in the days' closing price C_t , over n periods, by the closing price n number of days (or weeks) ago C_{t-n} , and then multiplying the ratio by 100. The formula is

$$\begin{aligned} ROC &= \left(\frac{\text{Today's close} - \text{close } n \text{ periods ago}}{\text{close } n \text{ periods ago}} \right) \times 100 \\ &= \frac{C_t - C_{t-n}}{C_{t-n}} \times 100 \end{aligned}$$

It is sometimes given by the difference in prices

$$ROC = \text{Today's close} - \text{close } n \text{ periods ago}$$

Generally, the ROC is calculated based on 14-periods for input n , but can be modified to any trader preferred period. The ROC indicator is plotted against zero: when prices are rising or advancing, ROC values remain above the Zero Line (positive) and when they are falling or declining, they remain below the Zero Line (negative). Note, it is not bounded to a set range since there is no limit to how far a security can advance in price even if there is a limit to how far it can decline.

- Connors RSI (CRSI): The indicator is a composite of three separate components where Wilder's RSI is used in two of the indicator's three components. The RSI, UpDown Length, and Rate-of-Change (ROC), combine to form a momentum oscillator. CRSI outputs a value in the range $[0, 100]$, which is then used to identify short-term overbought and oversold conditions. A value over 90 should be considered overbought and a value under 10 should be considered oversold. Note, have a tendency to produce false signals.
- As an example of channel trading, Keltner [1960] proposed a system called the 10-day moving average rule using a constant width channel to time buy-sell signals with the following rules

1. Compute the daily average price (or typical price): $M_t = \frac{H_t + L_t + C_t}{3}$.
2. Compute a 10-day average of the daily average price: $SM_t(n) = \frac{1}{n} \sum_{i=1}^n M_{t-i+1}$.
3. Compute a 10-day average of the daily range: given the range $R_t = H_t - L_t$, we get $AR_t(n) = \frac{1}{n} \sum_{i=1}^n R_{t-i+1}$.
4. Add and subtract the daily average range from the 10-day moving average to form a band or channel:

$$BotBand_t = SM_t(n) - AR_t(n) \text{ and } TopBand_t = SM_t(n) + AR_t(n)$$

5. Buy when the market penetrates the upper band and sell when it breaks the lower band.

While this system is buying on the strength and selling on weakness, some practitioners have modified the rules as follows:

1. instead of buying at the upper band, you sell and vice versa.
2. the number of days are changed to a three-day average with bands around that average.

A variation of the Keltner channel (in the 80s by Linda Raschke) consists in using the exponential moving average (EMA) with bands generated by the average-true range (ATR). Signals are produced when the price action breaks above or below the channel bands. The formula is

$$\begin{aligned} MidBand_t &= EMA(m)_t \\ BotBand_t &= EMA(m)_t - \alpha \times ATR_t(n), TotBand_t = EMA(m)_t + \alpha \times ATR_t(n) \end{aligned} \quad (13.99)$$

where $\alpha = 2$. In general, $m = 10$ days and $n = 20$ days. The Keltner Channels use ATR to calculate the upper and lower bands, while the Bollinger Bands use standard deviation.

- The Bollinger bands (BB), also called alpha-beta bands, focuses on price and volatility. Assuming prices are Gaussian $N(\mu, \sigma)$, it is a measure of deviation from the mean. Given $P(t)$ the price seen at time t and the maturity $T = t + \Delta t$, we get

$$P(T) = P(t) + \mu\Delta t \pm \alpha \times \sigma\sqrt{\Delta t}$$

where α is the number of standard deviation. It gives the dispersion for the price distribution. Note, in financial models, prices are log-normally distributed and returns are normally distributed.

It usually uses 20 or more days in its calculations and does not oscillate around a fixed point (see Bollinger [1992]). It consist of three lines (bands), where the middle line is a simple moving average (SMA) and the outside lines are plus or minus two (that number can vary) standard deviations above and below the SMA. Note, the first two steps are the same as in the CCI. The formula is as follows:

1. Compute the Typical Price

$$M_t = \frac{H_t + L_t + C_t}{3}$$

2. Compute the simple moving average (SMA) price of the last n Typical prices

$$SM_t(n) = \frac{1}{n} \sum_{i=1}^n M_{t-i+1}$$

3. Compute the (daily) standard deviation²⁵ of the last n Typical prices

$$SD_t(n) = \sqrt{\frac{1}{n} \sum_{i=1}^n (M_{t-i+1} - SM_t(n))^2}$$

4. Calculate the upper band

$$TopBand = SM_t(n) + \alpha \times SD_t(n) \quad (13.100)$$

where $\alpha = 2$ (two standard deviations).

5. Calculate the middle band

$$MidBand = SM_t(n) \quad (13.101)$$

6. Calculate the bottom band

$$BotBand = SM_t(n) - \alpha \times SD_t(n) \quad (13.102)$$

²⁵The standard deviation measures how spread out prices are from the average. It quantifies the volatility of the price series.

Note, one can use a 20-day MA with one and two standard deviations above and below the MA. In the Bollinger bands (BB), looking at the chart we can deduce trend, volatility, and overbought/oversold conditions. A market above one standard deviation is overbought, and it becomes extremely overbought above two standard deviation. That is, most underlyings will pullback to the average even in strongly trending markets. Further, with narrow bands we should buy volatility (calls and puts) and when the bands are widening we should sell volatility. Bollinger [1992] also described two other indicators, the $B\%$ (in relation to the stochastic oscillator $K\%$). The former tells where the prices are within the bands, and it can be above or below one and zero. The formula is

$$B\% = \frac{C_t - \text{BotBand}}{\text{TopBand} - \text{BotBand}}$$

Harrington [2005] normalised this equation to map the bands to $[0, 100]$ instead of $[0, 1]$

$$NB\% = 200 \times B\% - 100$$

In the $B\%$, the 100% line is the TopBand and the -100% line is the BotBand of the Bollinger bands. Recall, the bandwidth measures the width of the band as a percentage of the moving average

$$BW = \frac{\text{TopBand} - \text{BotBand}}{\text{MidBand}}$$

When the bands narrow drastically, a sharp expansion in volatility usually occurs in the very near future.

- Since equities alternate between periods of low volatility and high volatility, BBs can be used to identify volatility squeeze.

$$S_t(n) = \frac{\text{TopBand} - \text{BotBand}}{SC_t(n)} \quad (13.103)$$

where $SC_t(n)$ is the SMA of close prices (as opposed to typical prices). The squeeze measures the width of the band. When BBs are far apart, volatility is high, and when they are close together, it is low. A Squeeze is triggered when volatility reaches a six-month low and is identified when BBs reach a six-month minimum distance apart.

- The Bollinger oscillator (BOS) informs whether the market is overbought or oversold. Derived from the Bollinger band, the formula is

$$BOS_t = \frac{M_t - SM_t(n)}{SD_t(n)} \quad (13.104)$$

where M_t is the Typical price. Note, the numerator $M_t - SM_t(n)$ is a Moving Average Crossover between the Typical price and its Moving Average, indicating a positive or negative trend. This quantity is normalised by the standard deviation of the Typical price, which is a measure of dispersion (risk). That is, the trend is risk-adjusted. The absolute value of the BOS is high when the standard deviation is low.

- The Bollinger double bands (BDB): It generates two sets of Bollinger bands, one set using the parameter of one standard deviation and the other using the typical setting of two standard deviations. Given the SMA $SM_t(n)$, the one-SD band is

$$\text{TopBand1} = SM_t(n) + \alpha_1 \times SD_t(n) \text{ and } \text{BotBand1} = SM_t(n) - \alpha_1 \times SD_t(n) \quad (13.105)$$

where α_1 is one-SD, and the two-SD band is

$$\text{TopBand2} = SM_t(n) + \alpha_2 \times SD_t(n) \text{ and } \text{BotBand2} = SM_t(n) - \alpha_2 \times SD_t(n) \quad (13.106)$$

where α_2 is two-SD. While these formulas use a simple moving average (SMA) for the middle band, we can also use other types of moving averages, such as exponential moving averages (EMA) or weighted moving averages (WMA). These alternative moving averages give more weight to recent price data, making them more responsive to changes in price action.

13.2 Trading with technical indicator

13.2.1 Trading rules

A trader must decide on a specific entry technique, stop point, and take profit. For instance, whenever a buy signal is triggered, the investor takes a long position in the respective share until a sell signal is generated, at which point the trader unwind his position (sell). He could have simultaneously unwind his position and go short. This position is held until the next buy signal. These are trading strategies discussed in the following section. For now we focus on the entry point and stopping point. When trading on a daily basis at market close, it is assumed that one can estimate the asset price that would trigger the buy and sell signal just before the day's close and initiate a conditional limit order at the close of the market to follow various trading rules. In the case of high frequency trading, buying and selling occur until the end of the trading day, since no positions are held overnight.

Given the technical indicators defined in Appendix (13.1), their associated trading rules are as follows:

- In the case of a simple moving average (SMA) and exponential moving average (EMA), a buy signal (+1) is emitted when the asset price moves from below to above the SMA/EMA considered, and a sell signal (−1) is emitted when the asset price penetrates the SMA/EMA from above to below.
- In the case of stochastic oscillators, when the indicator is increasing, the asset prices are likely to go up and vice versa. In the case of an up-trend, prices tend to make higher highs, and the settlement price usually tends to be in the upper end of that time period's trading range. When the momentum starts to slow, the settlement prices will start to retreat from the upper boundaries of the range, causing the stochastic indicator to turn down at or before the final price high. A value nearer 0 is considered to indicate a market which is oversold (which will tend to rise) and a value near 100 indicates a market which is overbought. An alert or set-up is present when the $D\%$ line is in an extreme area and diverging from the price action. The actual signal takes place when the faster $K\%$ line crosses the $D\%$ line.
- In the case of two moving averages, say a 10 days and 20 days MA, when the 10D-MA moves up through the 20D-MA a buy sign is generated and when the 10D-MA moves down through the 20D-MA a sell sign is generated. The logic being that the price will continue to move in that direction so that the value of shares continues to increase or decrease. See details in Section (3).
- In the case of a moving average convergence divergence (MACD), then $MACD > 0$ represents the rule such that if MACD is greater than zero we choose to be in the market, otherwise we stay out of the market. See details in Section (3).
- For variants of MACD, we have $MACD > \text{signal line}$ is the trading rule such that if MACD is above the signal line we choose to be in the market, otherwise we stay out of the market.
- The Chande momentum oscillator (CMO) is in the range $[-100, +100]$. A security is overbought when the CMO is above +50 and oversold when it is below −50. We can add a 10-period MA to this oscillator to act as a signal line. The oscillator generates a bullish signal when it crosses above the MA and a bearish signal when it drops below the MA.

- Traders and investors use the commodity channel index (CCI) to help identify price reversals, price extremes and trend strength. It is often used for detecting divergences from price trends as an overbought/oversold indicator, and to draw patterns on it and trade according to those patterns. Readings above +100 imply an overbought condition (showing strength), while readings below -100 imply an oversold condition (showing weakness). It means that there is a large probability that the price will correct to more representative levels. Thus, assuming momentum, when movements are above +100 we buy (+1) and when they are below -100 we sell (-1). It becomes a coincident indicator. The position should be closed when the CCI moves back below +100, in the former. The position should be closed when the CCI moves back above -100, in the latter. Since about 70 to 80 percent of the CCI values are between +100 and -100, a buy or sell signal will be in force only 20 to 30 percent of the time. Alternatively, traders have also found the CCI valuable for identifying reversals and divergences. Assuming a mean-reverting market, when price movements are above +100 we sell (-1) and when they are below -100 we buy (+1). It becomes a leading indicator, where the trader look for overbought or oversold conditions.
- Traditional interpretation and usage of the RSI is that RSI values of 70 or above indicate that a security is becoming overbought or overvalued, and therefore, may be primed for a trend reversal or corrective pull-back in price (-1). An RSI reading of 30 or below is commonly interpreted as indicating an oversold or undervalued condition that may signal a trend change or corrective price reversal to the upside (+1). Note, the RSI indicator can stay in the overbought region for extended periods while the stock is in an uptrend. Conversely, it may also remain in oversold territory for a long time when the stock is in a downtrend. Sudden large price movements can create false buy or sell signals in the RSI. It is, therefore, best used with refinements to its application or in conjunction with other, confirming technical indicators.

In addition to Wilder's original theories of RSI interpretation, Cardwell and Brown developed several new interpretations of RSI to help determine and confirm trend. Brown [1999] suggested that oscillators do not travel between 0 and 100 and identified a bull market range and a bear market for RSI. RSI tends to fluctuate between 40 and 90 in a bull market (uptrend) with the 40 – 50 zones acting as support. These ranges may vary depending on RSI parameters, strength of trend and volatility of the underlying security. RSI tends to fluctuate between 10 and 60 in a bear market (downtrend) with the 50 – 60 zone acting as resistance. Andrew Cardwell developed positive and negative reversals for RSI, which are the opposite of bearish and bullish divergences. A positive reversal forms when RSI forges a lower low and the security forms a higher low. A negative reversal is the opposite of a positive reversal. RSI forms a higher high, but the security forms a lower high.
- The Rate of Change indicator (ROC) might be used to confirm price moves or detect divergences and might be used as a guide for determining overbought and oversold conditions. If it is positive we choose to be in the market (buy signal +1) and, if it is negative, then it is a sell signal (-1). It applies to both ROC formula. For instance we can consider $ROC(30)$, the 30-day rate of change of the asset price.
- Donchian channels (DC) are mainly used to identify the breakout of a stock or any traded entity enabling traders to take either long or short positions. Traders can take a long position (+1), if the stock is trading higher than the Donchian channels n period and book their profits/short the stock (-1) if it is trading below the DC channels n period.

The middle band is the average of the upper and lower bands. The middle band in Donchian channels could also be used as a breakout indicator. If the stock rises above the middle band of the DCs, then you can open a long position (+1). On the contrary, if the stock is trading below the middle band of the DCs, then a trader can open a short position (-1).
- A security is deemed to be overbought when the Chande momentum oscillator (CMO) is above +50 and over-sold when it is below -50. Trend strength can also be measured using the Chande momentum oscillator. The higher the oscillator's value, the stronger the trend, the lower the value, the weaker the trend.

Traders can use the Chande momentum oscillator to spot positive and negative price divergence between the indicator and the underlying security. A negative divergence occurs if the underlying security is trending upward

and the Chande momentum oscillator is moving downwards. A positive divergence occurs if price is declining, but the oscillator is rising.

Many technical traders add a 10-period moving average to this oscillator to act as a signal line. The oscillator generates a bullish signal (+1) when it crosses above the moving average, and a bearish signal (−1) when it moves below the moving average.

- The Bollinger bands (BB) can be used for determining overbought and oversold levels, as a trend following tool, and for monitoring for breakouts. When the price of an asset is trading near the upper BB (TopBand), it is considered overbought and may indicate that the price is likely to fall. Conversely, when the price is trading near the lower BB (BotBand), it is considered oversold and may indicate that the price is likely to rise. Hence, a typical system buys (+1) when price reaches the bottom (BotBand) and liquidates (−1) as the price moves up past the MA. The sell side is simply the opposite. BB identifies asset prices that have deviated from the mean. It is assumed that when a price goes beyond two standard deviations it should revert to the MA.
- In the Bollinger double bands (BDB), whenever price holds between the upper TopBand1 and TopBand2 away from mean, the trend is up. We call it the buy zone. Conversely, whenever price holds between the lower BotBand1 and BotBand2 away from mean, the trend is down. We call it the sell zone. If price meanders between the upper TopBand1 and the lower BotBand1, we are in a neutral state (or zone). In general, we take a long position once the price enters the buy zone from below and unwind the position when it leaves that zone from above.

13.2.2 Trading strategies

Accounting for the predictive power of trading rules, Metghalchi et al. [2015] proposed some trading strategies to beat the profitability of the buy and hold (B&H) strategy. Contrary to the B&H strategy, which is only long, the trader can still invest in cash or short the market. The authors detail some actions to be taken when the trader is out of the market. The strategies are as follows:

- 1= the trader will be in the market when a trading rule emits buy signals and be in the money market (cash) when a trading rule emits sell signals (long/money).
- 2= the trader will be in the market when a trading rule emits buy signals and would short the market when the rule emits sell signals (long/short).
- 3= the trader will borrow at the money market rate (cash rate) and double stock investment when a trading rule emits buy signals and be in the money market when the rule emits sell signals (leverage/ money).
- 4= the trader will borrow at the money market rate and double stock investment when a trading rule emits buy signals and would short the market when the rule emits sell signals (leverage/short).

In case of leverage, the total return on buy days is estimated as twice the market return minus the daily money market return.

13.2.3 Discretising technical indicators

Note, one can directly use continuous technical indicators (actual time series) or one can choose to represent them as trend deterministic data (discrete in nature). In the former, for comparison purposes, the values of all technical indicators are normalised in the range $[-1, +1]$ (see Kara et al. [2011]). Since each technical indicator has its own inherent property used by traders to predict up movement or down movement, a mapping function can be used to convert the continuous values to discrete values representing the trend (see Patel et al. [2015]). That is, +1 indicates an up movement and −1 shows a down movement. To do so, we use the trading rules in Appendix (13.2.1). For example, when SMA/EMA at time t is greater than at time $t - 1$, it suggests an up movement (+1) and vice versa for a down movement (−1). In the case of stochastic oscillators, they are clear trend indicators for any asset. MACD, RSI, CCI and A/D oscillators also follow the asset trend.

13.2.4 Trading methodology

Since pattern is defined as a predictable route or movement, all trading systems are pattern recognition systems. For instance, a long-term moving average cross over system uses pattern recognition, the crossover, in its decision to buy or sell. Similarly, an open range breakout is pattern recognition given by the movement from the open to the breakout point. All systems look for some type of reoccurring event and try to capitalise on it. Hill demonstrated the success of pattern recognition when used as a filter. The system was developed around the idea of a pattern consisting of the last four days' closing prices. A buy or sell signal is not generated unless the range of the past four days' closes is less than the 30-day average true range, indicating that the market has reached a state of rest and any movement, up or down, from this state will most likely result in a significant move.

Eventually, we want to develop an approach into a comprehensive, effective, trading methodology that combines analytical sophistication with tradable rules and principles. One way forward is to consider a multiple time frame approach. A time frame is any regular sampling of prices in a time series, from the smallest such as one minute up to the longest capped out at ten year. The multiple time frame approach has proven to be a fundamental advance in the field of TA allowing for significant improvement in trading results. For instance, if market analysis is coordinated to show the interaction of these time frames, then the trader can monitor what happens when the support and resistance lines of the different time frames coincide. Assuming we are interested in analysing the potential of the integration of time frames, then we need to look at both a higher time frame and a lower time frame.

13.3 Analysing the trading rules

Sweeney [1986] applied filter rules on ten currencies, and Lukac et al. [1988] applied moving average and other technical rules to twelve US futures markets. They both showed that trading rules have predictive power. Brock et al. [1992] analysed the Dow Jones industrial index for a period of 89 years and reported consistent and positive results about the forecasting power of technical trading rules and in particular the Moving Average (MA) rules (see Section (3)).

Since these seminal studies, numerous studies have examined MA rules in various world markets and generally found a good measure of support for their effectiveness (see Chen et al. [2009], Metghalchi et al. [2012]). However, it was observed that the rules tend to vary in their effectiveness over time (see Shynkevich et al. [2012]). Further, adding trading costs and other market frictions several articles found that these adjustments eliminate risk-adjusted excess profits (see Bessembinder et al. [1998], Day et al. [2002]). Some new insights were added to the literature that do not necessarily depend on the ability to make direct trading profits. For instance, Han et al. [2013] showed that applying a moving average strategy to portfolios sorted by volatility substantially outperforms a buy-and-hold strategy.

It is not clear as to why MA rules can predict future price moves (see Section (3)). Some explanations considered short term market behaviour while other focused on the long term behaviour. It can be summarised as follows:

1. short term trends occur because of
 - (a) rational factors: the existence of short term market frictions
 - (b) behavioural factors: behavioural biases
2. the existence of long term effects tend to depend on behavioural biases such as under-reaction to news over a substantial period of time or the existence of extended periods where a particular type of market sentiment dominates.

Since the moving average rules draw on relatively long term data to make predictions, it is clear whether short term or long term models are more appropriate. In any case, the poor performance of conventional econometric models (random walk, $AR(1)$, GARCH-M and EGARCH) (see Appendix (14)) are not supportive of short term rational factors. Note that an MA rule can be explained by an almost infinite variety of underlying processes.

Hudson et al. [2017] proposed to model MA rule simple trend following rules such as a basic Markov chain. They showed that the ability of the rules to predict market direction is largely based on the existence of very short term trends in the data in which positive/negative daily returns are more likely to be followed by positive/negative returns. They concluded that short term market frictions were the main explanation for the success of the MA rules.

Metghalchi et al. [2018] tested the profitability of four trading rules (MA, RSI, MACD, ROC momentum) (see Appendix (13.2.1)) in the case of the OMX Iceland All-Share Index (market close from 1999 till 2016) and showed that they had predictive power. Then, they designed four trading strategies (see Appendix (13.2.2)) for each trading rule (a total of ten: 5 MA, 1 RSI, 2 MACD, 2 ROC). They obtained very good results for strategy (1) where the average of all 10 trading rules have a mean daily return of 0.0067 and annual return of 16.03%, compared with 0.00001 and 0.37% of the Buy and Hold (B&H) strategy. Strategy (2) has higher daily and annual returns than Strategy (1), but it also has higher risk. For Strategy (3) the rules have higher daily and annual returns than Strategy (1), implying that leveraging on buy days improves returns, but at higher risk (almost the double of Strategy (1)). However, these trading rules imply many times in and out of the market resulting in transaction costs and reduced net returns. In the case of Strategies (2) and (4) they estimated the one-way break-even transaction costs (BEC). The best rules are those with high BEC and low risk. Considering risk and transaction costs, they showed that three trading rules, namely MA200, MA150, and MACD, could beat the B&H strategy for the entire period.

14 Introducing the volatility process

14.1 An overview of risk and volatility

14.1.1 The need to forecast volatility

When visualising financial time series, one can observe heteroskedasticity²⁶, with periods of high volatility and periods of low volatility, corresponding to periods of high and low risks, respectively. We also observe returns having very high absolute value compared with their mean, suggesting fat tail distribution for returns, with large events having a larger probability to appear when compared to returns drawn from a Gaussian distribution. Hence, besides the return series introduced in Appendix (12.1), we must also consider the volatility process and the behaviour of extreme returns of an asset (the large positive or negative returns). The negative extreme returns are important in risk management, whereas positive extreme returns are critical to holding a short position. Volatility is important in risk management as it provides a simple approach to calculating the value at risk (VaR) of a financial position. Further, modelling the volatility of a time series can improve the efficiency in parameter estimation and the accuracy in interval forecast. As returns may vary substantially over time and appear in clusters, the volatility process is concerned with the evolution of conditional variance of the return over time. When using risk management models and measures of preference, users must make sure that volatilities and correlations are predictable and that their forecasts incorporate the most useful information available. As the forecasts are based on historical data, the estimators must be flexible enough to account for changing market conditions. One simple approach is to assume that returns are governed by the random walk model, and that the sample standard deviation $\hat{\sigma}_N$ or the sample variance $\hat{\sigma}_N^2$ of returns for N periods of time can be used as a simple forecast of volatility of returns, r_t , over the future period $[t + 1, t + h]$ for some positive integer h . However, volatility has some specific characteristics such as

- volatility clusters: volatility may be high for certain time periods and low for other periods.
- continuity: volatility jumps are rare.
- mean-reversion: volatility does not diverge to infinity, it varies within some fixed range so that it is often stationary.
- volatility reacts differently to a big price increase or a big price drop.

²⁶Heteroskedastic means that a time series has a non-constant variance through time.

These properties play an important role in the development of volatility models. As a result, there is a large literature on econometric models available for modelling the volatility of an asset return, called the conditional heteroscedastic (CH) models. Some univariate volatility models include the autoregressive conditional heteroscedastic (ARCH) model of Engle [1982], the generalised ARCH (GARCH) model of Bollerslev [1986], the exponential GARCH (EGARCH) of Nelson [1991], the stochastic volatility (SV) models and many more. Tsay [2002] discussed the advantages and weaknesses of each volatility model and showed some applications of the models. Unfortunately, stock volatility is not directly observable from returns as in the case of daily volatility where there is only one observation in a trading day. Even though one can use intraday data to estimate daily volatility, accuracy is difficult to obtain. The unobservability of volatility makes it difficult to evaluate the forecasting performance of CH models and heuristics must be developed to estimate volatility on small samples.

14.1.2 A first decomposition

As risk is mainly given by the probability of large negative returns in the forthcoming period, risk evaluation is closely related to time series forecasts. The desired quantity is a forecast for the probability distribution (pdf) $\tilde{p}(r)$ of the possible returns r over the risk horizon ΔT . This problem is generally decomposed into forecasts for the mean and variance of the return probability distribution

$$r_{\Delta T} = \mu_{\Delta T} + a_{\Delta T}$$

with

$$a_{\Delta T} = \sigma_{\Delta T} \epsilon$$

where the return $r_{\Delta T}$ over the period ΔT is a random variable, $\mu_{\Delta T}$ is the forecast for the mean return, and $\sigma_{\Delta T}$ is the volatility forecast. The term $a_{\Delta T} = r_{\Delta T} - \mu_{\Delta T}$ is the mean-corrected asset return. The residual ϵ , which corresponds to the unpredictable part, is a random variable distributed according to a pdf $p_{\Delta T}(\epsilon)$. The standard assumption is to let $\epsilon(t)$ be an independent and identically distributed (iid) random variable. In general, a risk methodology will set the mean μ to zero and concentrate on σ and $p(\epsilon)$. To validate the methodology, we set

$$\epsilon = \frac{r - \mu}{\sigma}$$

compute the right hand side on historical data, and obtain a time series for the residual. We can then check that ϵ is independent and distributed according to $p(\epsilon)$. For instance, we can test that ϵ is uncorrelated, and that given a risk threshold α (say, 95%), the number of exceedance behaves as expected. However, when the horizon period ΔT increases, it becomes very difficult to perform back testing due to the lack of data. Alternatively, we can consider a process to model the returns with a time increment δt of one day, computing the forecasts using conditional averages. We can then relate daily data with forecasts at any time horizon, and the forecasts depend only on the process parameters, which are independent of ΔT and are consistent across risk horizon. The quality of the volatility forecasts is the major determinant factor for a risk methodology. The residuals can then be computed and their properties studied.

14.2 The structure of volatility models

The above heuristics being poor estimates of the future volatility, one must rely on proper volatility models such as the conditional heteroscedastic (CH) models. Since the early 80s, volatility clustering spawned a large literature on a new class of stochastic processes capturing the dependency of second moments in a phenomenological way. As the lognormal assumption is not consistent with all the properties of historical stock returns, Engle [1982] first introduced the autoregressive conditional heteroscedasticity model (ARCH) which has been generalised to GARCH by Bollerslev [1986]. We let r_t be the log return of an asset at time t , and assume that $\{r_t\}$ is either serially uncorrelated or with minor lower order serial correlations, but it is dependent. Volatility models attempt at capturing such dependence in the return series. We consider the conditional mean and conditional variance of r_t given the filtration $\mathcal{F}_{t-1}$ defined by

$$\mu_t = E[r_t | \mathcal{F}_{t-1}], \sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1}) = E[(r_t - \mu_t)^2 | \mathcal{F}_{t-1}]$$

Since we assumed that the serial dependence of a stock return series was weak, if it exists at all, μ_t should be simple and we can assume that r_t follows a simple time series model such as a stationary $ARMA(p, q)$ model. That is

$$r_t = \mu_t + a_t, \mu_t = \phi_0 + \sum_{i=1}^p \phi_i r_{t-i} - \sum_{i=1}^q \theta_i a_{t-i} \quad (14.107)$$

where a_t is the shock or mean-corrected return of an asset return²⁷. The model for μ_t is the mean equation for r_t , and the model for σ_t^2 is the volatility equation for r_t .

Remark 14.1 Some authors use h_t to denote the conditional variance of r_t , in which case the shock becomes $a_t = \sqrt{h_t} \epsilon_t$.

The parameters p and q are non-negative integers, and the order (p, q) of the ARMA model may depend on the frequency of the return series. The excess kurtosis values, measuring deviation from the normality of the returns, are indicative of the long-tailed nature of the process. Hence, one can then compute and plot the autocorrelation functions for the returns process r_t as well as the autocorrelation functions for the squared returns r_t^2 .

1. If the securities exhibit a significant positive autocorrelation at lag one and higher lags as well, then large (small) returns tend to be followed by large (small) returns of the same sign. That is, there are trends in the return series. This is evidence against the weakly efficient market hypothesis which asserts that all historical information is fully reflected in prices, implying that historical prices contain no information that could be used to earn a trading profit above that which could be attained with a naive buy-and-hold strategy which implies further that returns should be uncorrelated. In this case, the autocorrelation function would suggest that an autoregressive model should capture much of the behaviour of the returns.
2. The autocorrelation in the squared returns process would suggest that large (small) absolute returns tend to follow each other. That is, large (small) returns are followed by large (small) returns of unpredictable sign. It implies that the returns series exhibits volatility clustering where large (small) returns form clusters throughout the series. As a result, the variance of a return conditioned on past returns is a monotonic function of the past returns, and hence the conditional variance is heteroskedastic and should be properly modelled.

The conditional heteroscedastic (CH) models are capable of dealing with this conditional heteroskedasticity. The variance in the model described in Equation (14.107) becomes

$$\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1}) = \text{Var}(a_t | \mathcal{F}_{t-1})$$

Since the way in which σ_t^2 evolves over time differentiate one volatility model from another, the CH models are concerned with the evolution of the volatility. Hence, modelling conditional heteroscedasticity (CH) amounts to augmenting a dynamic equation to a time series model to govern the time evolution of the conditional variance of the shock. We distinguish two types or groups of CH models, the first one using an exact function to govern the evolution of σ_t^2 , and the second one using a stochastic equation to describe σ_t^2 . For instance, the (G)ARCH model belongs to the former, and the stochastic volatility (SV) model belongs to the latter. In general, we estimate the conditional mean and variance equations jointly in empirical studies.

²⁷since $a_t = r_t - \mu_t$

14.2.1 Benchmark volatility models

The ARCH model, which is the first model providing a systematic framework for volatility modelling, states that

1. the mean-corrected asset return a_t is serially uncorrelated, but dependent
2. the dependence of a_t can be described by a simple quadratic function of its lagged values.

Specifically, setting $\mu_t = 0$ for simplicity, an $ARCH(p)$ model assumes that

$$r_t = h_t^{\frac{1}{2}} \epsilon_t, h_t = \alpha_0 + \alpha_1 r_{t-1}^2 + \dots + \alpha_p r_{t-p}^2$$

where $\{\epsilon_t\}$ is a sequence of i.i.d. random variables with mean zero and variance 1, $\alpha_0 > 0$, and $\alpha_i \geq 0$ for $i > 0$. In practice, ϵ_t follows the standard normal or a standardised Student-t distribution. Generalising the ARCH model, the main idea behind (G)ARCH models is to consider asset returns as a mixture of normal distributions with the current variance being driven by a deterministic difference equation

$$r_t = h_t^{\frac{1}{2}} \epsilon_t \text{ with } \epsilon_t \sim N(0, 1) \quad (14.108)$$

and

$$h_t = \alpha_0 + \sum_{i=1}^p \alpha_i r_{t-i}^2 + \sum_{j=1}^q \beta_j h_{t-j}, \alpha_0 > 0, \alpha_i, \beta_j > 0 \quad (14.109)$$

where $\alpha_0 > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and $\sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) < 1$. The latter constraint on $\alpha_i + \beta_i$ implies that the unconditional variance of r_t is finite, whereas its conditional variance h_t evolves over time. In general, empirical applications find the $GARCH(1,1)$ model with $p = q = 1$ to be sufficient to model financial time series

$$h_t = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 h_{t-1}, \alpha_0 > 0, \alpha_1, \beta_1 > 0 \quad (14.110)$$

When estimated, the sum of the parameters $\alpha_1 + \beta_1$ turns out to be close to the non-stationary case, that is, mostly only a constraint on the parameters prevents them from exceeding 1 in their sum, which would lead to non-stationary behaviour. Different extensions of GARCH were developed in the literature with the objective of better capturing the financial stylised facts. Among them are the Exponential GARCH (EGARCH) model proposed by Nelson [1991] accounting for asymmetric behaviour of returns, the Threshold GARCH (TGARCH) model of Rabemananjara et al. [1993] taking into account the leverage effects, the regime switching GARCH (RS-GARCH) developed by Cai [1994], and the Integrated GARCH (IGARCH) introduced by Engle et al allowing for capturing high persistence observed in returns time series. Nelson [1990] showed that Ito diffusion or jump-diffusion processes could be obtained as a continuous time limit of discrete GARCH sequences. In order to capture stochastic shocks to the variance process, Taylor [1986] introduced the class of stochastic volatility (SV) models whose instantaneous variance is driven by

$$\ln(h_t) = k + \phi \ln(h_{t-1}) + \tau \xi_t \text{ with } \xi_t \sim N(0, 1) \quad (14.111)$$

This approach has been refined and extended in many ways. The SV process is more flexible than the GARCH model, providing more mixing due to the co-existence of shocks to volatility and return innovations. However, one drawback of the GARCH models and extension to Equation (14.111) is their implied exponential decay of the autocorrelations of measures of volatility which is in contrast to the very long autocorrelation. Both the GARCH and the baseline SV model are only characterised by short-term rather than long-term dependence. In order to capture long memory effects, the GARCH and SV models were expanded by allowing for an infinite number of lagged volatility terms instead of the limited number of lags present in Equations (14.109) and (14.111). To obtain a compact characterisation of the long memory feature, a fractional differencing operator was used in both extensions, leading to the fractionally integrated GARCH (FIGARCH) model introduced by Baillie et al. [1996], and the long-memory stochastic volatility model of Breidt et al. [1998]. As an intermediate approach, Dacorogna et al. [1998] proposed the heterogeneous

ARCH (HARCH) model, considering returns at different time aggregation levels as determinants of the dynamic law governing current volatility. In this model, we need to replace Equations (14.109) with

$$h_t = c_0 + \sum_{j=1}^n c_j r_{t,t-\Delta t_j}^2$$

where $r_{t,t-\Delta t_j} = \ln(p_t) - \ln(p_{t-\Delta t_j})$ are returns computed over different frequencies. This model was motivated by the finding that volatility on fine time scales can be explained to a larger extent by coarse-grained volatility than vice versa (see Muller et al. [1997]). Thus, the right hand side of the above equation covers local volatility at various lower frequencies than the time step of the underlying data ($\Delta t_j = 2, 3, \dots$). Note, multifractal models have a closely related structure but model the hierarchy of volatility components in a multiplicative rather than additive format.

14.2.2 Some practical considerations

As an example, we are briefly discussing the (G)ARCH models, which are assumed to be serially uncorrelated. However, before these models are used one must remove the autocorrelation present in the returns process. One can remove the autocorrelation structure from the returns process by fitting Autoregressive models. For instance, we can consider the $AR(p)$ model

$$r_t - \mu = \phi_1(r_{t-1} - \mu) + \phi_2(r_{t-2} - \mu) + \dots + \phi_p(r_{t-p} - \mu) + u_t$$

where μ is the sample mean of the series $\{r_t\}$ and the error $\{u_t\}$ are assumed to be from an i.i.d. process. One can use the Yule-Walker method to estimate the model parameters. To discern between models we use the AIC criterion which while rewarding a model fitting well (large maximum likelihood) also penalises for the inclusion of too many parameter values, that is, overfitting. We then obtain models of order p with associated standard errors of the parameter values. Again one has to study the normality of the errors by computing kurtosis statistics as well as the autocorrelation function. One expects the residuals to be uncorrelated not contradicting the Gaussian white noise hypothesis. However, it may happen that the squares of the residuals are autocorrelated. That is, the $AR(p)$ model did not account for the volatility clustering present in the returns process. Even if the residuals from returns are uncorrelated, but still show some evidence of volatility clustering, we can attempt to model the residuals with (G)ARCH processes. Hence, we can remove the autocorrelation by modelling the first order moment with $AR(p)$ models, and we can model the second order moments using (G)ARCH models. (G)ARCH models can explain the features of small autocorrelations, positive and statistically significant autocorrelations in the squares and excess kurtosis present in the residuals of the AR model. For an adequate fit, the residuals from the (G)ARCH models should have white noise properties. Hence, we fit (G)ARCH models to the residuals of the AR models $\{u_t\}$ getting

$$\mu_t - \nu = h_t^{\frac{1}{2}} \epsilon_t$$

where $\{\epsilon_t\}$ is Gaussian with zero mean and constant variance. Letting ν be the mean, we get

$$h_t = \alpha_0 + \sum_{i=1}^p \alpha_i (u_{t-i} - \nu)^2 + \sum_{j=1}^q \beta_j h_{t-j}$$

For $q = 0$ the process reduces to the $ARCH(p)$ process and for $p = q = 0$ the process is a Gaussian white noise. In the $ARCH(p)$ process the conditional variance is specified as a linear function of past sample variances only, whereas the $GARCH(p, q)$ process allows lagged conditional variances to enter as well. Once the (G)ARCH models have been fitted, we can discern between (nested) competing models using the Likelihood Ratio Test. To discern between two competing models where neither is nested in the other, we resort to examining the residuals of the models. For an adequate model, the residuals should resemble a white noise process being uncorrelated with constant variance. In addition, the autocorrelation of the squared residuals should also be zero. We should favour simpler models when two models appear adequate from an examination of the residuals.

14.3 Forecasting volatility with RiskMetrics methodology

14.3.1 The exponential weighted moving average

As volatility of financial markets changes over time, we saw that forecasting volatility could be of practical importance when computing the conditional variance of the log return of the underlying asset. The historical sample variance computed in Appendix (11.4) assigns weights of zero to squared deviations prior to a chosen cut-off date and an equal weight to observations after the cut-off date. In order to benefit from closing auction prices, one should consider close-to close volatility, with a large number of samples to get a good estimate of historical volatility. Letting D be the number of past trading days used to estimate the volatility, the daily log-return at time t is $r(t) = r_{t-1,t} = C(t) - C(t-1)$ where $C(t)$ is the closing log-price at the end of day t , and the annualised D-day variance of return is given by

$$\sigma_{STDEV}^2(t; D) = C_{sf} \frac{1}{D} \sum_{i=0}^{D-1} (r(t-i) - \bar{r}(t))^2$$

where the scalar C_{sf} scales the variance to be annual, and where $\bar{r}(t) = \frac{1}{D} \sum_{i=0}^{D-1} r(t-i)$ is the sample mean. Some authors set $C_{sf} = 252$ while others set it to $C_{sf} = 261$.

While for large sample sizes the standard close to close estimator is best, it can obscure short-term changes in volatility. That is, volatility reacts faster to shocks in the market as recent data carry more weight than data in the distant past. Further, following a shock (a large return), the volatility declines exponentially as the weight of the shock observation falls. As a result, the equally weighted moving average leads to relatively abrupt changes in the standard deviation once the shock falls out of the measurement sample, which can be several months after it occurs. On the other hand, the weighting scheme for GARCH(1,1) in Equation (14.110) and the Exponential Weighted Moving Average (EWMA) (see details in Appendix (11.1)) are such that weights decline exponentially in both models. Therefore, RiskMetrics (see Longerstaey et al. [1996]) let the ex-ante volatility σ_t be estimated with the exponentially weighted lagged squared daily returns (similar to a simple univariate GARCH model). It assigns the highest weight to the latest observations and the least to the oldest observations in the volatility estimate. The assignment of these weights enables volatility to react to large returns (jumps) in the market, so that following a jump the volatility declines exponentially as the weight of the jump falls. Specifically, the ex-ante annualised variance σ_t^2 is calculated as follows:

$$\sigma_t^2(t; D) = C_{sf} \sum_{i=0}^{\infty} \omega_i (r_{t-1-i} - \bar{r}_t)^2$$

where the weights $\omega_i = (1 - \delta)\delta^i$ add up to one, and where

$$\bar{r}_t = \sum_{i=0}^{\infty} \omega_i r(t-i)$$

is the exponentially weighted average return computed similarly. The parameter δ with $0 < \delta < 1$ is called the decay factor and determines the relative weights that are assigned to returns (observations) and the effective amount of data used in estimating volatility. It is chosen so that the center of mass of the weights $\sum_{i=0}^{\infty} (1 - \delta)\delta^i i = \frac{\delta}{(1 - \delta)}$ is equal to 30 or 60 days (since $\sum_{i=0}^{\infty} (1 - \delta)\delta^i = 1$). The volatility model is the same for all assets at all times. To ensure no look-ahead bias contaminates our results, we use the volatility estimates at time $t - 1$ applied to time t returns throughout the analysis, that is, $\sigma_t = \sigma_{t-1|t}$.

The formula above being an infinite sum, in practice we estimate the volatility σ with the Exponential Weighted Moving Average model (EWMA) for a given sequence of k returns as

$$\sigma_{EWMA}^2(t; D) = C_{sf} \sum_{i=0}^{k-1} \omega_i (r_{t-i} - \bar{r})^2$$

The latest return has weight $(1 - \delta)$ and the second latest $(1 - \delta)\delta$ and so on. The oldest return appears with weight $(1 - \delta)\delta^{k-1}$. The decay factor δ is chosen to minimise the error between observed volatility and its forecast over some sample period.

14.3.2 Forecasting volatility

One advantage of the exponentially weighted estimator is that it can be written in recursive form, allowing for volatility forecasts. If we assume the sample mean $\bar{r}$ is zero and that infinite amounts of data are available, then by using the recursive feature of the exponential weighted moving average (EWMA) estimator, the one-day variance forecast satisfies

$$\sigma_{1,t+1|t}^2 = \delta\sigma_{1,t|t-1}^2 + (1 - \delta)r_{1,t}^2$$

where $\sigma_{1,t+1|t}^2$ denotes 1-day time $t + 1$ forecast given information up to time t . It is derived as follow

$$\begin{aligned}\sigma_{1,t+1|t}^2 &= (1 - \delta) \sum_{i=0}^{\infty} \delta^i r_{1,t-i}^2 \\ &= (1 - \delta)(r_{1,t}^2 + \delta r_{1,t-1}^2 + \delta^2 r_{1,t-2}^2 + \dots) \\ &= (1 - \delta)r_{1,t}^2 + \delta(1 - \delta)(r_{1,t-1}^2 + \delta r_{1,t-2}^2 + \delta^2 r_{1,t-3}^2 + \dots) \\ &= \delta\sigma_{1,t|t-1}^2 + (1 - \delta)r_{1,t}^2\end{aligned}$$

Taking the square root of both sides of the equation we get the one day volatility forecast $\sigma_{1,t+1|t}$. For two return series, the EWMA estimate of covariance for a given sequence of k returns is given by

$$\sigma_{1,2}^2(t; D) = C_{sf} \sum_{i=0}^{k-1} \omega_i (r_{1,t-i} - \bar{r}_1)(r_{2,t-i} - \bar{r}_2)$$

Similarly to the variance forecast, the covariance forecast can also be written in recursive form. The 1-day covariance forecast between two return series $r_{1,t}$ and $r_{2,t}$ is

$$\sigma_{12,t+1|t}^2 = \delta\sigma_{12,t|t-1}^2 + (1 - \delta)r_{1,t}r_{2,t}$$

In order to get the correlation forecast, we apply the corresponding covariance and volatility forecast. The 1-day correlation forecast is given by

$$\rho_{12,t+1|t} = \frac{\sigma_{12,t+1|t}^2}{\sigma_{1,t+1|t}\sigma_{2,t+1|t}}$$

Using the EWMA model, we can also construct variance and covariance forecast over longer time horizons. The T -period forecasts of the variance and covariance are, respectively,

$$\sigma_{1,t+T|t}^2 = T\sigma_{1,t+1|t}^2$$

and

$$\sigma_{12,t+T|t}^2 = T\sigma_{12,t+1|t}^2$$

implying that the correlation forecasts remain unchanged irrespective of the forecast horizon, that is, $\rho_{t+T|t} = \rho_{t+1|t}$. We observe that in the EWMA model, multiple day forecasts are simple multiples of one-day forecasts. Note, the square root of time rule results from the assumption that variances are constant. However, the above derivation of volatilities and covariances vary with time. In fact, the EWMA model implicitly assume that the variance process is

non-stationary. In the literature, this model is a special case the integrated GARCH model (IGARCH). In practice, scaling up volatility estimates prove problematic when

- rates/prices are mean-reverting.
- boundaries limit the potential movements in rates and prices.
- estimates of volatilities optimised to forecast changes over a particular horizon are used for another horizon.

The cluster properties are measured by the lagged correlation of volatility, and the decay of that correlation quantifies the memory shape and magnitude. Based on statistical analysis, Zumbach [2006a] [2006b] found the lagged correlation of volatility to decay logarithmically as $1 - \frac{\log \Delta T}{\log \Delta T_0}$ in the range from 1 day to 1 year for all assets. That is, the memory of the volatility decays very slowly, so that a volatility model must capture its long memory. He considered a multi-scales long memory extension of IGARCH called the Long-Memory ARCH (LMARCH) process. The main idea being to measure the historical volatilities with a set of exponential moving averages (EMA) on a set of time horizons chosen according to a geometric series. The feed-back loop of the historical returns on the next random return is similar to a $GARCH(1, 1)$ process, but the volatilities are measured at multiple time scales. Computing analytically the conditional expectations related to the volatility forecasts, we get

$$\sigma_{\Delta T|t}^2 = \frac{\Delta T}{\delta t} \sum_i \lambda\left(\frac{\Delta T}{\delta t}, i\right) r_{t-i\delta t}^2$$

with weights $\lambda(\frac{\Delta T}{\delta t}, i)$ derived from the process equations and satisfying $\sum_i \lambda(\frac{\Delta T}{\delta t}, i) = 1$. We see that the leading term of the forecast is given by $\sigma_{\Delta T|t} \approx \sqrt{\frac{\Delta T}{\delta t}}$ which is the Normal square root scaling, and $\sum_i \lambda(i) r^2(i)$ is a measure of the past volatility constructed as a weighted sum of the past return square. We saw in Appendix (14.2) that more complicated nonlinear modelling of volatility exists such as GARCH, stochastic volatility, applications of chaotic dynamics etc.

14.3.3 Assuming zero-drift in volatility calculation

Given the logarithmic return in Equation (12.78) and the k -days period, the sample mean return $\bar{r} = \frac{1}{k} \sum_{i=0}^{k-1} r_{t-i}$ is the estimate of the mean μ . An important issue arising in the estimation of the historical variance is the noisy estimate of the mean return. This is due to the fact that the mean logarithmic return depends on the range (length) of the return series in the sense that

$$\bar{r} = \frac{1}{k} \sum_{i=0}^{k-1} r_{t-i} = \frac{1}{k} \sum_{i=0}^{k-1} (L_t - L_{t-i})$$

Thus, the mean return does not take into account the price movements or the number of prices within the period. Therefore, while a standard deviation measures the dispersion of the observations around its mean, in practice, it may be difficult to obtain a good estimate of the mean. As a result, some authors suggested to measure volatility around zero rather than the sample mean. Assuming a EWMA model to estimate volatility, Longerstaey et al. [1995a] proposed to study the difference between results given by the sample mean and zero-mean centred estimators. They considered the one-day volatility forecast referred to as the estimated mean estimator, and given by

$$\hat{\sigma}_t^2 = \delta \hat{\sigma}_{t-1}^2 + \delta(1 - \delta)(R_t - \bar{R}_{t-1})^2$$

where R_t is the percentage change return and $\bar{R}_{t-1}$ is an exponentially weighted mean. The zero-men estimator derived in Appendix (14.3.2) is given by

$$\tilde{\sigma}_t^2 = (1 - \delta)R_t^2 + \delta \tilde{\sigma}_{t-1}^2$$

Setting up a Monte Carlo experiment, they studied the forecast difference of the two models $\hat{\sigma}_t^2$ and $\tilde{\sigma}_t^2$ at any time t . Defining the arithmetic difference Δ_t as

$$\Delta_t = \tilde{\sigma}_t^2 - \hat{\sigma}_t^2 \big|_{\bar{R}_i=0} \text{ for } i = t, t-1, \dots$$

we get

$$\Delta_t = R_t^2(1 - \delta)^2$$

so that the one-day volatility forecast for $\delta = 0.94$ becomes $\delta_t = 0.0036R_t^2$. Assuming zero sample mean, for sufficiently small percentage return R_t , the difference Δ_t is negligible, and one should not expect significant differences between the two models. Considering a database consisting of eleven time series, Longerstaey et al. [1995a] concluded that the relative differences between the two models are quite small. Further, investigating the differences of the one-day forecasted correlation between 1990 and 1995, they found very small deviations. They extended the analysis of the difference between the estimated mean and zero-mean estimators to one month horizons and obtained relatively small differences between the two estimates. As a result, the zero-mean estimator is a viable alternative to the estimated mean estimator, which is simpler to compute and not sensitive to short-term trends.

Note, assuming a conditional zero mean of returns is consistent with the financial theory of the efficient market hypothesis (EMH). However, financial markets are multifractal and that conditional mean of returns experience long term trends. In fact, Zumbach [2006a] [2006b] showed that neglecting the mean return forecast was not correct, particularly for interest rates and stock indexes. For the former, the yields can follow a downward or upward trend for very long periods, of the order of a year or more, and the latter can follow an overall upward trend related to interest rates. These long trends introduce correlations, equivalent to some predictability in the rates themselves. Even though these effects are quantitatively small, they introduce clear deviations from the random walk with ARCH effect on the volatility.

14.3.4 Estimating the decay factor

We now need to estimate the sample mean and the exponential decay factor δ . The largest sample size available should be used to reduce the standard error. Choosing a suitable decay factor is a necessity in forecasting volatility and correlations. One important issue in this estimation is the determination of an effective number of days (k) used in forecasting. It is postulated in RiskMetrics that the volatility model should be determined by using the metric

$$\Omega_k = (1 - \delta) \sum_{t=k}^{\infty} \delta^t$$

where Ω_k is set to the tolerance level α . We can now solve for k . Expanding the summation we get

$$\alpha = \delta^k(1 - \delta)[1 + \delta + \delta^2 + \dots]$$

and taking the natural logarithms on both sides we get

$$\ln \alpha = k \ln \delta + \ln(1 - \delta) + \ln[1 + \delta + \delta^2 + \dots]$$

Since $\log(1 \pm x) \approx \pm x - \frac{1}{2}x^2$ for $|x| < 1$ we get

$$k \approx \frac{\ln \alpha}{\ln \delta}$$

In principle, we can find a set of optimal decay factors, one for each covariance can be determined such that the estimated covariance matrix is symmetric and positive definite. RiskMetrics presented a method for choosing one optimal decay factor to be used in estimation of the entire covariance matrix. They found $\delta = 0.94$ to be the optimal decay factor for one-day forecast and $\delta = 0.97$ to be optimal for one month (25 trading days) forecast.

14.4 Computing historical volatility

Computing historical volatility is not an easy task as it depends on two parameters, the length of time and the frequency of measurement. As volatility mean revert over a period of months, it is difficult to define the best period of time to obtain a fair value of realised volatility. Further, in presence of rare events, the best estimate of future volatility is not necessary the current historical volatility. While historical volatility can be measured monthly, quarterly or yearly, it is usually measured daily or weekly. In presence of independent stock price returns, then daily and weekly historical volatility should on average be the same. However, when stock price returns are not independent there is a difference due to autocorrelation. If we assume that daily volatility should be preferable to weekly volatility because there are five times as many data points available then intraday volatility should always be preferred. However, intraday volatility is not constant as it is usually greatest just after the market open and just before the market close and falling in the middle of the day, leading to noisy time series. Hence, traders taking into account intraday prices should depend on advanced volatility measures. One approach is to use exponential weighted moving average (EWMA) model described in Appendix (14.3) which avoids volatility collapse of historic volatility. It has the advantage over standard historical volatility (STDEV) to gradually reduce the effect of a spike on volatility. However, EWMA models are rarely used, partly due to the fact that they do not properly handle regular volatility driving events such as earnings. That is, previous earnings jumps will have least weight just before an earnings date (when future volatility is most likely to be high), and most weight just after earnings (when future volatility is most likely to be low).

The availability of intraday data allows one to consider volatility estimators making use of intraday information for more efficient volatility estimates. Using the theory of quadratic variation, Andersen et al. [1998] and Barndorff-Nielsen et al. [2002] introduced the concept of integrated variance and showed that the sum of squared high-frequency intraday log-returns is an efficient estimator of daily variance in the absence of price jumps and serial correlation in the return series. However, market effects such as lack of continuous trading, bid/ask spread, price discretisation, swamp the estimation procedure, and in the limit, microstructure noise dominates the result. The research on high-frequency volatility estimation and the effects of microstructure noise being extremely active, we will just say that intervals between 5 to 30 minutes tend to give satisfactory volatility estimates. We let $N_{day}(t)$ denote the number of active price quotes during the trading day t , so that $S_1(t), \dots, S_{N_{day}(t)}(t)$ denotes the intraday quotes. Assuming 30 minutes quotes, the realised variance (RV) model is

$$\hat{\sigma}_{RV}^2(t) = \sum_{j=2}^{N_{day}} (\log S_j(t) - \log S_j(t-1))^2$$

One simple heuristic to define advanced volatility measures making use of intraday information is to consider range estimators that use some or all of the open (O), high (H), low (L) and close (C). In that setting we define

- opening price: $O(t) = \log S_1(t)$
- closing price: $C(t) = \log S_{N_{day}}(t)$
- high price: $H(t) = \log (\max_{j=1, \dots, N_{day}} S_j(t))$
- low price: $L(t) = \log (\min_{j=1, \dots, N_{day}} S_j(t))$
- normalised opening price: $o(t) = O(t) - C(t-1) = \log \frac{S_1(t)}{S_{N_{day}}(t-1)}$
- normalised closing price: $c(t) = C(t) - O(t) = \log \frac{S_{N_{day}}(t)}{S_1(t)}$
- normalised high price: $h(t) = H(t) - O(t) = \log (\max_{j=1, \dots, N_{day}} \frac{S_j(t)}{S_1(t)})$
- normalised low price: $l(t) = L(t) - O(t) = \log (\min_{j=1, \dots, N_{day}} \frac{S_j(t)}{S_1(t)})$

We are going to discuss a few of them below, introduced by Parkinson [1980], Garman et al. [1980], Rogers et al. [1994], Yang et al. [2000]. We refer the readers to Baltas et al. [2012b] and Bennett [2012] for a more detailed list with formulas.

- close to close (C): the most common type of calculation benefiting only from reliable prices at closing auctions.
- Parkinson (HL): it is the first to propose the use of intraday high and low prices to estimate daily volatility

$$\hat{\sigma}_{PK}^2(t) = \frac{1}{4 \log 2} (h(t) - l(t))^2$$

The model assumes that the asset price follows a driftless diffusion process and it is about 5 times more efficient than STDEV.

- Garman-Klass (OHLC): it is an extension of the PK model including opening and closing prices. It is the most powerful estimate for stocks with Brownian motion, zero drift and no opening jumps. The GK estimator is given by

$$\hat{\sigma}_{GK}^2(t) = \frac{1}{2} (h(t) - l(t))^2 - (2 \log 2 - 1) c^2(t)$$

and it is about 7.4 times more efficient than STDEV.

- Rogers-Satchell (OHLC): being similar to the GK estimate, it benefits from handling non-zero drift in the price process. However, opening jumps are not well handled. The estimator is given by

$$\hat{\sigma}_{RS}^2(t) = h(t)(h(t) - c(t)) + l(t)(l(t) - c(t))$$

Rogers-Satchell showed that GK is just 1.2 times more efficient than RS.

- Garman-Klass Yang-Zhang extension (OHLC): Yang-Zhang extended the GK method by incorporating the difference between the current opening log-price and the previous day's closing log-price. The estimator becomes robust to the opening jumps, but still assumes zero drift. The estimator is given by

$$\hat{\sigma}_{GKYZ}^2 = \sigma_{GK}^2 + (O(t) - C(t-1))^2$$

- Yang-Zhang (OHLC): Having a multi-period specification, it is an unbiased volatility estimator that is independent of both the opening jump and the drift of the price process. It is the most powerful volatility estimator with minimum estimation error. It is a linear combination of the RS estimator, the standard deviation of past daily log-returns (STDEV), and a similar estimator using the normalised opening prices instead of the close-to-close log-returns.

Since these estimators provide daily estimates of variance/volatility, an annualised D-day estimator is therefore given by the average estimate over the past D days

$$\sigma_l^2(t; D) = \frac{C_{sf}}{D} \sum_{i=0}^{D-1} \hat{\sigma}_l^2(t-i)$$

for $l = \{RV, PK, GK, GKYZ, RS\}$. The Yang-Zhang estimator is given by

$$\sigma_{YZ}^2(t; D) = \sigma_{open}^2(t; D) + k \sigma_{STDEV}^2(t; D) + (1-k) \sigma_{RS}^2(t; D)$$

where $\sigma_{open}^2(t; D) = \frac{C_{sf}}{D} \sum_{i=0}^{D-1} (o(t-i) - \bar{o}(t))^2$. The parameter k is chosen so that the variance of the estimator is minimised. YZ showed that for the value

$$k = \frac{0.34}{1.34 + \frac{D+1}{D-1}}$$

their estimator is $1 + \frac{1}{k}$ times more efficient than the ordinary STDEV estimator. Brandt et al. [2005] showed that the range-based volatility estimates are approximately Gaussian, whereas return-based volatility estimates are far from Gaussian, which is an advantage when calibrating stochastic volatility models with likelihood procedure. Bennett [2012] defined two measures to determine the quality of a volatility measure, namely, the efficiency and the bias. The former is defined by

$$(\sigma_e^2) = \frac{\sigma_{STDEV}^2}{\sigma_l^2}$$

where σ_l is the volatility of the estimate and σ_{STDEV} is the volatility of the standard close to close estimate. It describes the volatility of the estimate, and decreases as the number of samples increases. The latter is the difference between the estimated variance and the average volatility. It depends on the sample size and the type of distribution of the underlying. Generally, for small sample sizes the Yang-Zhang estimator is best overall, and for large sample sizes the standard close to close estimator is best. Setting aside the realised variance, the Yang-Zhang estimator is the most efficient estimator, it exhibits the smallest bias when compared to the realised variance, and it generates the lowest turnover. While the optimal choice for volatility estimation is the realised variance (RV) estimator, the Yang-Zhang estimator constitute an optimal tradeoff between efficiency, turnover, and the necessity of high frequency data, as it only requires daily information on opening, closing, high and low prices.

15 A few facts about option trading

15.1 Option pricing

15.1.1 The assumption of lognormality

Assuming that the returns between two periods are measured by the difference between the logarithms of the asset prices, and that returns are modelled with a Brownian motion with volatility σ and a drift $\mu - \frac{1}{2}\sigma^2$, then for $\{S_t; t \in [0, T]\}$ the returns $\log S_t - \log S_s$ are normally distributed with mean $(\mu - \frac{1}{2}\sigma^2)(t-s)$ and variance $\sigma^2(t-s)$. Further, given $0 < t_1 < \dots < t_n$, then the relative prices $\{\frac{S_{t_{i+1}}}{S_{t_i}}; 0 \leq i \leq n-1\}$ are independent and follow the same law. That is, there exists a Brownian motion $\widehat{W}$ such that

$$S_t = f(t, \widehat{W}_t) = xe^{\mu t + \sigma \widehat{W}_t - \frac{1}{2}\sigma^2 t}, S_0 = x$$

Applying Ito's lemma to the function $f(t, z) = xe^{\mu t + \sigma z - \frac{1}{2}\sigma^2 t}$, we compute the time and space derivatives

$$f'_t(t, z) = f(t, z)(\mu - \frac{1}{2}\sigma^2), f'_z(t, z) = f(t, z)\sigma, f''_{zz}(t, z) = f(t, z)\sigma^2$$

and we get the dynamics of the stock price as

$$\frac{dS_t}{S_t} = \mu dt + \sigma d\widehat{W}_t$$

Given the initial condition $S_0 = x$, the first two moments of the stock price are given by

$$\begin{aligned} E[S_t] &= xe^{\mu t} \text{ and } E[S_t^2] = x^2 e^{(2\mu + \sigma^2)t} \\ \text{Var}(S_t) &= x^2 e^{2\mu t} (e^{\sigma^2 t} - 1) \end{aligned}$$

and the Sharpe ratio (see Sharpe [1970]) becomes

$$R_S = \frac{E[S_t] - x}{\sqrt{\text{Var}(S_t)}} = \frac{e^{(2\mu + \sigma^2)t}}{\sqrt{e^{2\mu t}(e^{\sigma^2 t} - 1)}}$$

which is independent from the initial value x . Given a bounded positive function h , the option price is

$$E[h(S_t^x)] = \int h(y)\phi(t, x, y)dy = \int h(xe^{(\mu - \frac{1}{2}\sigma^2)t + \sigma\sqrt{t}u})\psi(u)du$$

where $\psi(z) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}$ is the Gaussian density. The market compares the rate of return per unit of time μ to the return a risk-free asset, so that the market reference is $\mu - r$. In that setting, the Sharpe ratio is assumed to be the market price of risk (MPR), denoted λ , that the market assign to the noise $\widehat{W}$ as

$$R_S = \lambda = \frac{\frac{1}{dt}E[\frac{dS_t}{S_t}] - r}{\sqrt{\frac{1}{dt}\text{Var}(\frac{dS_t}{S_t})}} = \frac{\mu - r}{\sigma} \quad (15.112)$$

15.1.2 The Black-Scholes formula

In the special case where we assume that the stock prices $(S_t)_{t \geq 0}$ are lognormally distributed and pay a continuous dividend, we can derive most of the contract market values in closed-form solution. For example, Black and Scholes [1973] derived the price of a call option seen at time t with strike K and maturity T as

$$C_{BS}(t, x, K, T) = xe^{-q(T-t)}N(d_1(T-t, F(t, T), K)) - Ke^{-r(T-t)}N(d_2(T-t, F(t, T), K)) \quad (15.113)$$

where $F(t, T) = xe^{(r-q)(T-t)}$ is the forward price and $N(\cdot)$ is the standard normal cumulative distribution function (CDF). Further, we have

$$d_2(t, x, y) = \frac{1}{\sigma\sqrt{t}} \log \frac{x}{y} - \frac{1}{2}\sigma\sqrt{t} \text{ and } d_1(t, x, y) = d_2(t, x, y) + \sigma\sqrt{t}$$

15.1.3 The put-call parity

We now consider the put-call parity where we are long a European call option and short a European put option. In the presence of repo rates and discrete dividends, we can always buy one stock and borrow at time t the quantity $\sum_{i=1}^N d_i \frac{P(t, t_i)}{Re(t, t_i)} + KP(t, T)$ to recover the flux $(S_T - K)$ at maturity T . Further, the put-call parity must satisfies

$$C(t, S_t, K, T) - P(t, S_t, K, T) = P(t, T)E_t[S_T - K]$$

and using the definition of S_T in our model, it simplifies to

$$\begin{aligned} C(t, S_t, K, T) - P(t, S_t, K, T) &= P(t, T)(S_t C(t, T) - D(t, T)) - KP(t, T) \\ &= S_t Re(t, T) - P(t, T)D(t, T) - KP(t, T) \end{aligned} \quad (15.114)$$

Dividing Equation (15.114) by the zero-coupon bond $P(t, T)$, we get the put-call parity expressed as

$$\overline{C}(t, S_t, K, T) - \overline{P}(t, S_t, K, T) = F(t, T) - K \quad (15.115)$$

where $F(t, T) = S_t \frac{Re(t, T)}{P(t, T)} - D(t, T)$. Further, $\overline{C}(t, S_t, K, T) = \frac{C(t, S_t, K, T)}{P(t, T)}$ and $\overline{P}(t, S_t, K, T) = \frac{P(t, S_t, K, T)}{P(t, T)}$ are the discounted call price and discounted put price, respectively.

15.1.4 The digital prices

We can calculate the prices of some special options via the derivation of call price $C(t, S_t, T, K)$

$$\begin{aligned}
 \text{Digital} &= -\partial_K C(t, S_t, K, T) = 1 - \partial_K P(t, S_t, K, T) = P(t, T) E_t[I_{S_T > K}] \\
 &= P(t, T) P(S_T > K) = P(t, T) \int_K^\infty \phi(T, S) dS \\
 \text{Complement Digital} &= 1 + \partial_K C(t, S_t, K, T) = \partial_K P(t, S_t, K, T) = P(t, T) E_t[I_{S_T \leq K}] \\
 &= P(t, T) P(S_T \leq K) = P(t, T) \int_0^K \phi(T, S) dS \\
 \text{Density} &= \partial_{KK}^2 C(t, S_t, T, K) = \partial_{KK}^2 P(t, S_t, T, K) = P(t, T) E_t[\delta(S_T - K)] = P(t, T) \phi(T, K)
 \end{aligned} \tag{15.116}$$

Note, these equations are also true for the put price $P(t, S_t, T, K)$. Now, the price of a call option can be expressed in terms of those quantities as

$$\begin{aligned}
 C(t, S_t, K, T) &= P(t, T) E_t[(S_T - K)^+] = P(t, T) E_t[(S_T - K) I_{S_T > K}] \\
 &= P(t, T) (E_t[S_T I_{S_T > K}] - K E_t[I_{S_T > K}])
 \end{aligned}$$

which gives

$$\frac{C(t, S_t, T, K)}{P(t, T)} = E_t[S_T I_{S_T > K}] - K E_t[I_{S_T > K}] = S_t \Delta_S + K \Delta_K$$

so that the Digital Bond is

$$\Delta_K = P(S_T > K) = N[d_2(T - t, \frac{S_t}{P(t, T)}, K)]$$

and the Digital Stock is

$$S_t \Delta_S = E_t[S_T I_{S_T > K}] = \frac{C(t, S_t, T, K)}{P(t, T)} + K E_t[I_{S_T > K}]$$

Now, from the digital option in Equation (15.116), we have $E_t[I_{S_T > K}] = -\frac{1}{P(t, T)} \frac{\partial C(t, S_t, \dots)}{\partial K}$, so that the Digital Stock becomes

$$E_t[S_T I_{S_T > K}] = \frac{C(t, S_t, K, T)}{P(t, T)} - K \frac{1}{P(t, T)} \frac{\partial C(t, S_t, K, T)}{\partial K} \tag{15.117}$$

Note,

$$\Delta_S = \frac{1}{S_t} E_t[S_T I_{S_T > K}]$$

is the delta hedge of the call option.

More generally, following Ingersoll [1998] [2000], we call $D_B(S, t, T; \xi)$ the Digital Bond and $D_S(S, t, T; \xi)$ the Digital Stock for the event ξ .

- Digital Bond: We let $D(S, t, T; A)$ be the value at time t of receiving \$1 at the maturity T if and only if the event A occurs. Such a contract is called a binary option or a cash-or-nothing option.

- Digital Stock: we let $S(S, t, T; A)$ be the value at time t of receiving one share of the stock at the maturity T (excluding any intervening dividends) if and only if the event A occurs. Such a contract is called a all-or-nothing share.

In the special case where the rate, repo and volatility are constants, Cox et al. [1976] showed that the Digital Bond and Digital Share with event $\xi = \{S_T > K\}$, where K is the strike price, could be valued under the risk-neutral measure $\mathbb{Q}$ with the Black-Scholes formula (see Equation (15.113)). That is,

$$\begin{aligned} D(S, t, T; \xi) &= E^Q[e^{-\int_t^T r_s ds} I_{S_T > K} | \mathcal{F}_t] = e^{-r(T-t)} N(d_2(T-t, S_t e^{(r-q)(T-t)}, K)) \\ S(S, t, T; \xi) &= E^Q[e^{-\int_t^T r_s ds} S_T I_{S_T > K} | \mathcal{F}_t] = S_t e^{-q(T-t)} N(d_1(T-t, S_t e^{(r-q)(T-t)}, K)) \end{aligned} \quad (15.118)$$

European contracts are piecewise linear in the spot price on the maturity date. The value of any of these contracts can be represented

$$\sum_i a_i D(S, t, T; \xi_i) + \sum_j b_j S(S, t, T; \xi_j)$$

for some weights a_i and b_j . For example, if we let the events $\xi_C = \{S_T > K\}$ and $\xi_P = \{K > S_T\}$ correspond to the call and the put option, respectively, the call option is recovered by taking a Digital Stock and subtracting the strike multiplied by the Digital Bond. That is,

$$C(S, t; K, T) = S(S, t, T; \xi_C) - K D(S, t, T; \xi_C) \quad (15.119)$$

and the put option price is

$$P(S, t; K, T) = K D(S, t, T; \xi_P) - S(S, t, T; \xi_P) \quad (15.120)$$

15.2 Forward starting option

15.2.1 The framework

In a general setting, we let the underlying process $(S_t^x)_{t \geq 0}$ be a one-dimensional Ito process valued in the open subset D . Given the absence of arbitrage opportunity (AAO), the price of a European call option $C(t, x)$ on $[0, T] \times [0, +\infty[$ under the risk-neutral measure $\mathbb{Q}$ is

$$\begin{aligned} C(t, S_t) &= E^Q[e^{-\int_t^T r_s ds} h(S_T) | \mathcal{F}_t] \\ &= P(t, T) \int_{-\infty}^{\infty} h(x) \phi_S(t, x) dx \end{aligned} \quad (15.121)$$

where h is a sufficiently smooth payoff function and $\phi_S(t, \cdot)$ is a continuous density function. Furthermore, we assume that these price functions are of class $\mathcal{C}^2$, such that for $T > t$ the law of S_T knowing $S_t = x$ has a density $\phi(T, K)$ given by

$$\phi(T, K) = E_t^Q[\delta(S_T - K)] = \frac{1}{E_t^Q[e^{-\int_t^T r_s ds} | S_T = K]} \partial_{KK} C(t, S_t; K, T) \quad (15.122)$$

where $\delta(\cdot)$ is the Dirac function.

15.2.2 The pricing

Let's denote $C^F(t_0, t_1, k, T)$ the value at time t_0 of forward start call option starting at time t_1 with risky strike kS_{t_1} and maturity T . The No-Arbitrage conditions lead to the pricing formula

$$C^F(t_0, t_1, k, T) = e^{-r(T-t_0)} \mathbb{E}((S_T - kS_{t_1})^+ | \mathcal{F}_{t_0})$$

if we look at forward start call option with *absolute* pay-off²⁸ $(S_T - kS_{t_1})^+$. By conditioning the forward start option with information available at time t_1 , i.e. by the filtration $\mathcal{F}_{t_1}$, we get

$$\begin{aligned} C^F(t_0, t_1, k, T) &= e^{-r(T-t_0)} \mathbb{E}((S_T - kS_{t_1})^+ | \mathcal{F}_{t_0}) \\ &= e^{-r(T-t_0)} \mathbb{E}(\mathbb{E}((S_T - kS_{t_1})^+ | \mathcal{F}_{t_1}) | \mathcal{F}_{t_0}) \end{aligned}$$

The problem with pricing such derivatives is actually that today's market prices do not provide us with the future anticipations of stock process evolution. This is because the implied volatility surface is not stationary and not Markovian. It means that if we denote by t_0 today's date and t_1 and t_2 two future dates with $t_2 > t_1 > t_0$, market prices do not directly give us the probability density function (*pdf*)

$$\phi(S_{t_2} \in dx | \mathcal{F}_{t_1})$$

where $\mathcal{F}_{t_1}$ can be considered all available information at time t_1 . Yet, the knowledge of this function is of fundamental importance if we want to price forward start call options.

15.2.2.1 Point on Kolmogorov-Compatibility Dealing with implied volatility surface (IVS) evolution always rises the question of whether the smile surface allows for arbitrage opportunities. It is natural to impose that future smile surface are compatible with today's prices of calls and puts. The problem is that defining a smile surface $\sigma(t, T, K, S_t)$ means defining a future density ϕ for stock process. For instance, assuming a deterministic smile surface, a natural condition would be to impose that the future pdf is actually a conditional density. This is formalised as the Kolmogorov-Compatibility.

First of all, let's remind the relation between prices and pdf. Without doing any assumptions apart from that of no arbitrage opportunity we have

$$\begin{aligned} C(t, K, t_1 - t_0) &= e^{-r(t_1-t_0)} \mathbb{E}((S_{t_1} - K)^+ | \mathcal{F}_{t_1}) = e^{-r(t_1-t_0)} \int_K^{+\infty} (x_1 - K) \phi(x_1, t_1, S_{t_0}, t_0) d\lambda(x_1) \\ \frac{\partial C}{\partial K}(t, K, t_1 - t_0) &= -e^{-r(t_1-t_0)} \int_K^{+\infty} \phi(x_1, t_1, S_{t_0}, t_0) d\lambda(x_1) \\ \frac{\partial^2 C}{\partial K^2}(t, K, t_1 - t_0) &= e^{-r(t_1-t_0)} \phi(K, t_1, S_{t_0}, t_0) \end{aligned}$$

By definition $C(t, K, t_1 - t_0) = C_{BS}(S_{t_0}; K, t_1 - t_0; r, \Sigma(S_{t_0}; K, t_1 - t_0))$ we have

$$\frac{\partial^2 C}{\partial K^2} = \frac{\partial^2 C_{BS}}{\partial K^2} + \frac{\partial^2 C_{BS}}{\partial K \partial \Sigma} \frac{\partial \Sigma}{\partial K} + \left[\frac{\partial^2 C_{BS}}{\partial K \partial \Sigma} + \frac{\partial^2 C_{BS}}{\partial K^2} \frac{\partial \Sigma}{\partial K} \right] \frac{\partial \Sigma}{\partial K} + \frac{\partial C_{BS}}{\partial \Sigma} \frac{\partial^2 \Sigma}{\partial K^2}$$

For simplicity, we denote by Θ the right-hand-side operator.

²⁸There exists forward start call options which pay their owner the difference, if positive, between the performance and the relative strike k i.e. $(\frac{S_T}{S_{t_1}} - k)^+$

Definition 15.1 (Kolmogorov-Compatibility) Any future deterministic conditional density or smile surface such that

$$\Theta(S_0; K, T - t_0; r, \sigma(t_0, T, K, S_{t_0})) = \int \Theta(S_0; K, t_1 - t_0; r, \sigma(t_0, t_1, K, S_{t_0})) \Theta(x_1; K, T - t_1; r, \sigma(t_1, T, K, x_1)) d\lambda(x_1)$$

is satisfied defines a Kolmogorov-compatible density.

Property 15.1 Given a current admissible²⁹ smile surface, if the future smile surface is Kolmogorov-compatible no model-independent strategy can generate arbitrage profits.

In general, there is an infinity of solutions for

$$\Theta(x_1; K, T - t_1; r, \sigma(t_1, T, K, x_1))$$

such that Definition (15.1) is satisfied. Therefore, even if we require the smile surface to be deterministic, there still exists an infinity of future smile surfaces compatible with today's prices of calls and puts.

15.2.2.2 The pricing formula We assume that the IVS is only driven by the stock process S and time to maturity $T - t$.

$$(\mathcal{H}) \quad \forall T \quad \forall t < T, \quad \forall K \quad \Sigma_t(K, T) = \Sigma(S_t; K, T - t) \quad (15.123)$$

Therefore, we can rewrite our assumption in terms of the BS-formula as

Assumption 15.1

$$(\mathcal{H}) \quad \forall T \quad \forall t < T, \quad \forall K \quad C(t, S_t, K, T - t) = C_{BS}(S_t, K, T - t, \Sigma(S_t, K, T - t)) \quad (15.124)$$

Note that this hypothesis implies that European call price are not time-dependent *i.e.* do not depend on current time but depend on time to maturity $(T - t)$ and the stock price. We can notice that the assumption (15.124) implies that all European calls can be priced at any time. In deed for any t_0, t_1 and T such that $t_0 < t_1 < T$ we have

$$\begin{aligned} C(t_0, K, T - t_0) &= e^{-r(T-t_0)} \mathbb{E}((S_T - K)^+ | S_{t_0}) \\ &= e^{-r(t_1-t_0)} \mathbb{E}(e^{-r(T-t_1)} \mathbb{E}((S_T - K)^+ | S_{t_1}) | S_{t_0}) \\ &= e^{-r(t_1-t_0)} \mathbb{E}(C(t_1, K, T - t_1) | S_{t_0}) \\ &= e^{-r(t_1-t_0)} \mathbb{E}(C_{BS}(S_{t_1}; K, T - t_1; r, \Sigma(S_{t_1}; K, T - t_1)) | S_{t_0}) \\ &= e^{-r(T-t_0)} \mathbb{E} \left(\int (S_T - K)^+ \phi(S_T, T, S_{t_1}, t_1) d\lambda(S_T) | S_{t_0} \right) \\ &= e^{-r(T-t_0)} \iint (S_T - K)^+ \phi(S_T, T, S_{t_1}, t_1) \phi(S_{t_1}, t_1, S_{t_0}, t_0) d\lambda(S_T) d\lambda(S_{t_1}) \end{aligned}$$

As we have $C(t_0, K, T - t_0) = e^{-r(T-t_0)} \int (S_T - K)^+ \phi(S_T, T, S_{t_0}, t_0) d\lambda(S_T)$ then

$$\int (S_T - K)^+ \phi(S_T, T, S_{t_0}, t_0) d\lambda(S_T) = \iint (S_T - K)^+ \phi(S_T, T, S_{t_1}, t_1) \phi(S_{t_1}, t_1, S_{t_0}, t_0) d\lambda(S_T) d\lambda(S_{t_1})$$

²⁹*i.e.* such that the associated call prices satisfy $\frac{\partial C_{Call}(t, K, T-t)}{\partial K} < 0$, $\frac{\partial^2 C_{Call}(t, K, T-t)}{\partial K^2} > 0$, $\frac{\partial C_{Call}(t, K, T-t)}{\partial T} > 0$, $\frac{\partial Put(t, K, T-t)}{\partial K} > 0$, $C_{Call}(t, K, T-t)|_{K=0} = S_t$ and $\lim_{K \rightarrow \infty} C_{Call}(t, K, T-t) = 0$

Differentiating twice with respect to K leads to

$$\phi(S_T, T, S_{t_0}, t_0) = \int (S_T - K)^+ \phi(S_T, T, x_1, t_1) \phi(x_1, t_1, S_{t_0}, t_0) d\lambda(x_1)$$

which naturally ensures that the deterministic volatility shape we take allows no arbitrage opportunities.

15.3 Some traded contracts

15.3.1 The digital option

Given the stock price $(S_t)_{t \geq 0}$, a Digital option $D(K, T) = P(t, T) \Delta_K$ for strike K and maturity T pays \$1 when the stock price S_T is greater than the strike K , and zero otherwise (see Appendix (15.1.4)). From a static replication argument, the price of the Digital option is given by

$$D(K, T) = \lim_{\Delta K \rightarrow 0} \frac{C(K, T) - C(K + \Delta K, T)}{\Delta K} = -\frac{\partial}{\partial K} C(K, T)$$

where $C(K, T)$ is a market call option. Hence, the digital options are uniquely priced from the volatility surface. The payoff is displayed in Figure (77).



Figure 77: Digital call option on strike $K = 100$.

Given $C(K, T) = C_{BS}(K, T; \Sigma_{BS}(K, T))$ where $\Sigma_{BS}(K, T)$ is the BS implied volatility for strike K and maturity T , and using the chain rule, the Digital option becomes

$$D(K, T) \equiv -\frac{\partial}{\partial K} C(K, T) = -\frac{\partial}{\partial K} C_{BS}(K, T; \Sigma_{BS}) - \frac{\partial}{\partial \Sigma(K, T)} C_{BS}(K, T; \Sigma(K, T)) \frac{\partial}{\partial K} \Sigma(K, T)$$

We can express the Digital option in terms of the Vega and the Skew terms as

$$D(K, T) = -\frac{\partial}{\partial K} C_{BS}(K, T; \Sigma_{BS}) - Vega(K, T) Skew(K, T) \quad (15.125)$$

where $Vega(K, T; \Sigma_{BS}(K, T))$ is the Black-Scholes vega for the strike K and maturity T , and $\partial_K C_{BS}(K, T; \Sigma_{BS})$ is the BS digital price for the volatility $\Sigma_{BS}(K, T)$.

In the special case where $r = q = 0$, $T = 1$ and for $S_0 = 100$ and $K = 100$, then $\eta = 1$ and we get $d_2 = -\frac{1}{2}\Sigma_{BS}\sqrt{T}$. Given a skew of 2.5% per 10% change in the strike and an ATM volatility $\Sigma_{ATM} = 25\%$, the Digital option simplifies to

$$\begin{aligned} D(100, 1) &= N\left[-\frac{\Sigma_{ATM}}{2}\right] - S_0 n\left[\frac{\Sigma_{ATM}}{2}\right] \frac{-0.025}{0.1S_0} \\ &\approx 0.45 + 0.25 \times 0.4 = 0.55 \end{aligned}$$

In this example, the skew has an impact of 10% of notional on the price. Ignoring the skew, the price is 45% of notional which is significantly lower than 55% of notional when the skew is included.

15.3.2 The butterfly option

Assuming that the volatility surface has been constructed from European option prices, we consider a butterfly strategy centered at K where we are long a call option with strike $K - \Delta K$, long a call option with strike $K + \Delta K$, and short two call options with strike K . The value of the butterfly for strike K and maturity T is

$$B(t_0, K, T) = C(K - \Delta K, T) - 2C(K, T) + C(K + \Delta K, T) \approx P(t_0, T)\phi(t_0; K, T)(\Delta K)^2$$

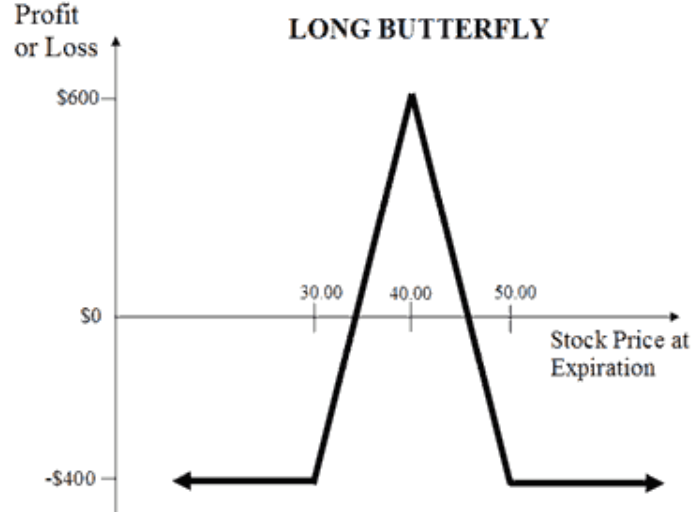
where $\phi(t_0; K, T)$ is the probability density function (PDF) of S_T evaluated at strike K . As a result, we have

$$\phi(t_0; K, T) \approx \frac{1}{P(t_0, T)} \frac{C(K - \Delta K, T) - 2C(K, T) + C(K + \Delta K, T)}{(\Delta K)^2}$$

where

$$A_{t_0}(K, T) = P(t_0, T)\phi(t_0; K, T)$$

can be seen as the price at time t_0 of a security paying \$1 at state K and future time T , called the Arrow-Debreu security. The payoff is displayed in Figure (78).

Figure 78: Long Butterfly option on strike $K = 40$.

Letting $\Delta K \rightarrow 0$ to be in continuous time, the density becomes

$$\phi(t_0; T, K) = \frac{1}{P(t_0, T)} \frac{\partial^2}{\partial K^2} C(K, T) \quad (15.126)$$

Hence, for any time T , one can recover the marginal risk-neutral distribution of the stock price from the volatility surface. However, it tells us nothing about the joint distribution of the stock price at multiple times $T_1, \dots, T_n$. This is because the volatility surface is constructed from European options prices which only depend on the marginal distribution of S_T .

15.4 Volatility estimation

For simplicity, we are going to consider a simple diffusion model for the dynamics of the stock price in Equation (6.26) with a one-factor volatility σ_t . We can approximate the local volatility σ_t with the mean volatility over $\tau = T - t$, that is, $\frac{1}{T-t} \int_t^T \sigma_s ds$ on historical data. However,

- The variance of the estimator decrease with τ
- the bias of of the estimator increase with τ

15.4.1 The realised variance

We let the dynamics of the log-price $L_t = \log S_t$ be given by

$$dL_t = \left(\mu_t - \frac{1}{2} \sigma_t^2 \right) dt + \sigma_t dW_S(t)$$

with quadratic variation

$$\langle L \rangle_t = \int_0^t \sigma_s^2 ds$$

We discretise the time interval $[0, t]$ into N uniform time steps $\delta = \frac{t}{N}$. We let $r_L(t, \delta) = \log \frac{S_{t+\delta}}{S_t}$ be the log-return at time t for the frequency δ . Given the quadratic variation of a semimartingale, we get

$$\langle L \rangle_t = \lim_{N \rightarrow \infty} \sum_{i=1}^N r_L^2((i-1)\delta, \delta)$$

Thus, the estimator for the realised variance is

$$\bar{V}(t, N) \equiv \sum_{i=1}^N r_L^2((i-1)\delta, \delta)$$

For simplicity, we assume:

Assumption 3 • The volatility σ_t is independent from W_S .

• The drift μ_t is deterministic.

We denote E^σ the conditional expectation with respect to the trajectory of σ . Then,

$$R_i \equiv r_L(a_i, b_i), a_i = (i-1)\delta \text{ and } b_i = i\delta$$

is a Gaussian variable with mean and variance

$$m_i = \int_{a_i}^{b_i} (\mu_s - \frac{1}{2}\sigma_s^2) ds \text{ and } v_i = \int_{a_i}^{b_i} \sigma_s^2 ds$$

and the variable R_i and R_j are independent for all $i \neq j$.

We can compute the expected value of the realised variance as follows:

$$\begin{aligned} E^\sigma[V_R(t, N)] &= \sum_{i=1}^N E^\sigma[R_i^2] = \sum_{i=1}^N m_i^2 + v_i \\ &\approx \frac{t}{N} \int_0^t (\mu_s - \frac{1}{2}\sigma_s^2) ds + \int_0^t \sigma_s^2 ds \end{aligned}$$

Thus, the bias of the estimator is of the order $O(\frac{t}{N})$. We can compute the variance of the realised variance as follows:

$$\begin{aligned} Var^\sigma(V_R(t, N)) &= \sum_{i=1}^N E^\sigma[(R_i^2 - m_i^2 - v_i)^2] = \sum_{i=1}^N (4v_i m_i^2 + 2v_i^2) \\ &\approx \frac{4t^2}{N^2} \int_0^t \sigma_s^2 (\mu_s - \frac{1}{2}\sigma_s^2)^2 ds + \frac{2t}{N} \int_0^t \sigma_s^4 ds \end{aligned}$$

Thus, the dominating term in the error of the estimator, to the first order, is

$$\sqrt{\frac{2t}{N} \int_0^t \sigma_s^4 ds}$$

15.4.2 The ARCH model

The ARCH (autoregressive conditional heteroscedasticity) method smooth the data, reducing the variance of the estimator at the cost of increasing the bias. This is a discrete model where returns are given by

$$R_i = \mu_i + \sigma_i \epsilon_i$$

where ϵ_i is a series of random variables adapted to a discrete filtration $\{\mathcal{F}_t\}$, and such that, for all i , ϵ_{i+1} is independent from $\mathcal{F}_i$, with law $N(0, 1)$. The series $\{\mu_i\}$ and $\{\sigma_i\}$ are adapted to the filtration $\{\mathcal{F}_t\}$. Thus, μ_i and σ_i are $\mathcal{F}_{i-1}$ -measurable. In that model

$$\sigma_i^2 = E[(R_i^2 - \mu_i)^2 | \mathcal{F}_{i-1}]$$

Note, from Assumption (3), the dynamics of log-returns L_t can be expressed as an ARCH model by setting $\mathcal{F}_t = \sigma\{\epsilon_1, \dots, \epsilon_t\} \vee \sigma\{\sigma_s; s \leq (t+1)\delta\}$ and

$$\mu_i = \int_{a_i}^{b_i} (\mu_s - \frac{1}{2}\sigma_s^2) ds \text{ and } \sigma_i^2 = \int_{a_i}^{b_i} \sigma_s^2 ds \text{ with } a_i = (i-1)\delta \text{ and } b_i = i\delta$$

In the $ARCH(n)$ model, we can approximate σ_i^2 with a moving average of the square of the centred observed returns

$$\sigma_i^2 \approx \omega + \alpha_1(R_{i-1} - \mu_{i-1})^2 + \dots + \alpha_n(R_{i-n} - \mu_{i-n})^2$$

We can use the method of maximum likelihood to estimate the parameters $\omega, \alpha_1, \dots, \alpha_n$ and then filter the volatility with the above equation. For example, in a $ARCH(1)$ model, we get $\mu_i \equiv \mu$ and the density of $(R_1, \dots, R_N)$ satisfies

$$p(R_1, \dots, R_N | \theta) = p(R_N | R_{N-1}, \theta) \dots p(R_2 | R_1, \theta) p(R_1)$$

where $\theta = (\mu, \omega, \alpha_1)$ is a vector of model parameters, and

$$\begin{aligned} p(R_i | R_{i-1}, \theta) &= \frac{1}{\sigma_i \sqrt{2\pi}} e^{-\frac{(R_i - \mu)^2}{2\sigma_i^2}} \\ &= \frac{1}{\sqrt{2\pi(\omega + \alpha_1(R_{i-1} - \mu)^2)}} e^{-\frac{(R_i - \mu)^2}{2(\omega + \alpha_1(R_{i-1} - \mu)^2)}} \end{aligned}$$

We let $p(R_1)$ be Gaussian with volatility set a priori. The vector θ can be estimated as follows:

$$\hat{\theta} = \arg \max_{\theta} \log p(R_1, \dots, R_N | \theta) = \arg \max_{\theta} \left(\sum_{i=1}^N \log \sigma_i - \sum_{i=1}^N \frac{(R_i - \mu)^2}{2\sigma_i^2} \right)$$

Note, to obtain a robust $ARCH(n)$ estimator, we must choose n sufficiently large. As a generalisation, the GARCH (generalised ARCH) model allows for the volatility σ_i to depend on past returns as well as on past volatility itself. In the GARCH(m,n), we get

$$\sigma_i^2 \approx \omega + \sum_{i=1}^n \alpha_i (R_{t-i} - \mu_{t-i})^2 + \sum_{j=1}^m \beta_j \sigma_{t-j}^2$$

15.5 Some volatility products

15.5.1 The straddle

A straddle is an option spread where one buy (sell) both put and call options of the same strike price and maturity on the same underlying. A long straddle involves going long both a call option and a put option on the same stock with strike K , and satisfies

$$S_{BS}(t, x, K, T) = C_{BS}(t, x, K, T) + P_{BS}(t, x, K, T) \quad (15.127)$$

The payoff is displayed in Figure (79).

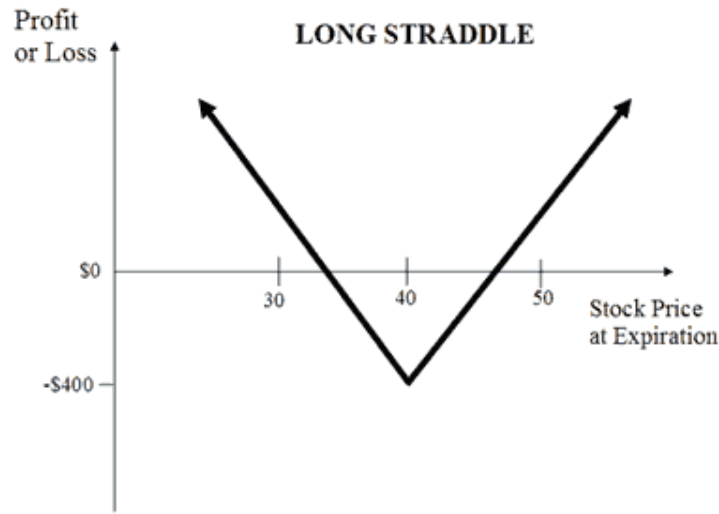


Figure 79: Long Straddle on strike $K_i = 40$.

Given the definition of the BS-formula in Appendix (15.1.2), we can write the price of a straddle as

$$S_{BS}(t, x, K, T) = xe^{-q(T-t)}(2N(d_1(T-t, F(t, T), K)) - 1) - Ke^{-r(T-t)}(2N(d_2(T-t, F(t, T), K)) - 1)$$

From the digital prices in Appendix (15.1.4), we can rewrite the straddle in terms of Digital Bond and Digital Stock as follows

$$\begin{aligned} S_{BS}(t, x, K, T) &= S(x, t, T; \xi_C) - KD(x, t, T; \xi_C) + KD(x, t, T; \xi_P) - S(x, t, T; \xi_P) \\ &= 2C_{BS}(t, x, K, T) - (F(t, T) - K) \end{aligned}$$

where $F(t, T) = \frac{x}{P(t, T)}$ is the forward price. Further, we can compute the limit case of a long straddle when the volatility tends to zero as

$$\lim_{\sigma \rightarrow 0} S_{BS}(t, x, K, T) = (xe^{-q(T-t)} - Ke^{-r(T-t)})^+ + (Ke^{-r(T-t)} - xe^{-q(T-t)})^+$$

which gives

$$\lim_{\sigma \rightarrow 0} S_{BS}(t, x, K, T) = \begin{cases} xe^{-q(T-t)} - Ke^{-r(T-t)} & \text{if } xe^{-q(T-t)} > Ke^{-r(T-t)} \\ Ke^{-r(T-t)} - xe^{-q(T-t)} & \text{if } xe^{-q(T-t)} < Ke^{-r(T-t)} \\ 0 & \text{if } xe^{-q(T-t)} = Ke^{-r(T-t)} \end{cases}$$

Hence, we can rewrite the limit of a long straddle when the volatility tends to zero as

$$\lim_{\sigma \rightarrow 0} S_{BS}(t, x, K, T) = |xe^{-q(T-t)} - Ke^{-r(T-t)}|$$

which is the absolute value of the payoff. When the volatility tends to infinity, we get

$$\lim_{\sigma \rightarrow \infty} S_{BS}(t, x, K, T) = xe^{-q(T-t)} + Ke^{-r(T-t)}$$

Assuming that the options in the straddle are ATM forward, we get the approximation

$$S_{BS}(t, x, K, T)|_{K=F(t,T)} \approx 0.8xe^{-q(T-t)}\sigma\sqrt{T-t}$$

From the above equation we can easily compute the delta of an ATM forward straddle option as

$$\Delta_S(t, x, K, T)|_{K=F(t,T)} \approx 0.8e^{-q(T-t)}\sigma\sqrt{T-t}$$

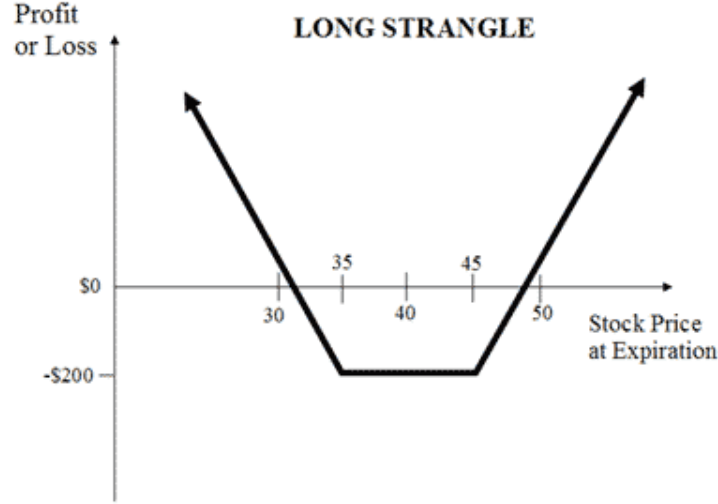
Straddles are a good strategy to pursue if an investor believes that a stock's price will move significantly, but is unsure as to which direction it will go. The stock price must move significantly if the investor is to make a profit. This position has a limited risk, since the purchaser of the strategy may lose at most the cost of both options, and there is unlimited profit potential. However, when shorting a straddle the profit is limited to the premium received from the sale of put and call. The risk is virtually unlimited as large moves of the underlying security's price either up or down will cause losses proportional to the magnitude of the price move. A maximum profit upon expiration is achieved if the underlying security trades exactly at the strike price of the straddle. In that case both puts and calls comprising the straddle expire worthless. Note, on stocks that are expected to jump, the market tends to price options at a higher premium, which ultimately reduces the expected payoff should the stock move significantly.

15.5.2 The strangle

A strangle option spread is similar to a straddle as one still buy (sell) both put and call options with the same maturity and on the same underlying, but different strikes are used. The investor use put options having a strike below the call options, and vice versa. Assuming a long strangle with strikes $K_1 < K_2$, we get

$$\hat{S}_{BS}(t, x, K_1, K_2, T) = P_{BS}(t, x, K_1, T) + C_{BS}(t, x, K_2, T) \quad (15.128)$$

The payoff is displayed in Figure (80).

Figure 80: Long Strangle on strike $K_i = 40$.

Given the definition of the BS-formula in Appendix (15.1.2), we can compute the limit case of a long strangle when the volatility tends to zero as

$$\lim_{\sigma \rightarrow 0} \hat{S}_{BS}(t, x, K_1, K_2, T) = (K_1 e^{-r(T-t)} - x e^{-q(T-t)})^+ + (x e^{-q(T-t)} - K_2 e^{-r(T-t)})^+$$

which gives

$$\lim_{\sigma \rightarrow 0} \hat{S}_{BS}(t, x, K_1, K_2, T) = \begin{cases} x e^{-q(T-t)} - K_2 e^{-r(T-t)} & \text{if } x e^{-q(T-t)} > K_2 e^{-r(T-t)} \\ K_1 e^{-r(T-t)} - x e^{-q(T-t)} & \text{if } x e^{-q(T-t)} < K_1 e^{-r(T-t)} \\ 0 & \text{if } K_1 e^{-r(T-t)} \leq x e^{-q(T-t)} \leq K_2 e^{-r(T-t)} \end{cases}$$

and when the volatility tends to infinity

$$\lim_{\sigma \rightarrow \infty} \hat{S}_{BS}(t, x, K_1, K_2, T) = K_1 e^{-r(T-t)} + x e^{-q(T-t)}$$

Assuming that the options in the strangle are close to ATM forward, with $K_1 \approx F(t, T)$ and $K_2 \approx F(t, T)$, we get the approximation

$$\hat{S}_{BS}(t, x, K_1, K_2, T) \approx K_1 e^{-r(T-t)} \left(\frac{1}{2} + \frac{1}{5} \sigma \sqrt{T-t} \right) + x e^{-q(T-t)} \frac{2}{5} \sigma \sqrt{T-t} - K_2 e^{-r(T-t)} \left(\frac{1}{2} - \frac{1}{5} \sigma \sqrt{T-t} \right)$$

Setting $K_1 = K_2 = F(t, T)$, we recover the formula, $S_{BS}(t, x, K, T)|_{K=F(t, T)}$, for the ATM forward straddle. In general, the strategy involves buying an out-of-the-money call and an out-of-the-money put option so that its intrinsic value is zero and stays there if the underlying asset hardly moves. A strangle is generally less expensive than a straddle as the contracts are purchased out of the money. The owner of a long strangle makes a profit if the underlying price moves far enough away from the current price, either above or below.

15.6 Strategies using options

15.6.1 Combining straddle and strangle

Given a stock price S_t , we let $C(T, K)$ and $P(T, K)$ be a call price and put price, respectively, with strike K and maturity T . We assume the existence of a strike chain $\mathcal{K} = \{K_1, \dots, K_n\}$ uniformly spaced, where the change in strike is denoted ΔK . Denote $\Phi(T)$ the expected stock move before time T .

15.6.1.1 The algorithm The strategy is as follows:

1. Choose an expiry date T in the future and pull up an options chain for that stock, $\{C(T, K_1), \dots, C(T, K_n)\}$.
2. Look at the option strike prices on the chain and identify the one that is closest to the current market value of our stock. Given the distances $d_j = |S_t - K_j|$, for all j , choose d^* such that the distance is minimum. Denote the resulting strike K_i . Thus, $K_i \approx S_t$.
3. Add the purchase price (premium) for both the call and put option of strike K_i , getting $\phi_1 = C_A(T, K_i) + P_A(T, K_i)$.
4. Then, identify both the first call option with strike higher than the strike K_i , denoted $C(T, K_{i+1})$, and the first put option with strike lower than K_i , denoted $P(T, K_{i-1})$. Add those two prices, getting $\phi_2 = C_A(T, K_{i+1}) + P_A(T, K_{i-1})$.
5. Compute the expected stock move before time T : calculate the sum $\phi_1 + \phi_2$ and divide by two, getting

$$\Phi(T) = \frac{C_A(T, K_i) + P_A(T, K_i) + C_A(T, K_{i+1}) + P_A(T, K_{i-1})}{2} \quad (15.129)$$

We get the dollar amount of the expected move of the stock over the time period $[t, T]$.

15.6.1.2 A numerical example For example, at time t , the close of business on April 11, 2018, Apple stock was trading at $S_{t,c} = 172.44$ dollars a share. Assume we want to know how much the stock could move between time t and the end of May. Pull up the options chain for T on May 25th 2018. The closest price to $S_{t,c}$ on the strike column is $K_i = 172.5$. Thus, we will use this row as our strike price for step 2. The ask call price is $C_A(T, K_i) = 6.60$ and the ask put price is $P_A(T, K_i) = 6.80$. The sum is

$$\phi_1 = C_A(T, K_i) + P_A(T, K_i) = 13.40$$

The asking price for the first call option higher than the strike $K_i = 172.5$ is $C_A(T, K_{i+1}) = 5.35$ where $K_{i+1} = 175$. The ask price for the first put option lower than the strike $K_i = 172.5$ is $P_A(T, K_{i-1}) = 5.70$ where $K_{i-1} = 170$. Summing up these prices, we get

$$\phi_2 = C_A(T, K_{i+1}) + P_A(T, K_{i-1}) = 11.05$$

Combining these prices, the total price in Equation (15.129) is $\Phi(T) = \frac{24.45}{2} = 12.23$ dollars. It means that the options market is pricing in a \$12.23 move in the stock over the next six weeks. Dividing \$12.23 by the closing price $S_{t,c} = 172.44$, we can see that the data calls for an estimated fluctuation of 7% away from the current price $S_{t,c}$ over the next 6 weeks.

This expected move is simply an average price of an at-the-money straddle and the first out-of-the-money strangle (see Appendix (15.5)). Note, this trading strategy and analysis tells us how much it is expected to move, but not where the stock will go (direction).

15.6.2 A generalisation

We rewrite ϕ_1 and ϕ_2 in terms of a straddle and strangle, respectively. We then put terms together to express the strategy in terms of straddle and strangle.

Given Equation (15.127), we get

$$\phi_1 = C_A(T, K_i) + P_A(T, K_i) = S_{BS}(t, x, K_i, T)$$

The Straddle assess if the stock price will move significantly, irrespective of its direction. The stock price must move significantly if the investor is to make a profit.

Given Equation (15.128), we get

$$\phi_2 = C_A(T, K_{i+1}) + P_A(T, K_{i-1}) = \hat{S}_{BS}(t, x, K_{i-1}, K_{i+1}, T)$$

The strategy involves buying an out-of-the-money call and an out-of-the-money put option so that its intrinsic value is zero and stays there if the underlying asset hardly moves. The owner of a long strangle makes a profit if the underlying price moves far enough away from the current price, either above or below. Putting terms together, we get

$$\Phi(T) = \frac{1}{2} (S_{BS}(t, x, K_i, T) + \hat{S}_{BS}(t, x, K_{i-1}, K_{i+1}, T))$$

Here, the price Φ is the average between a straddle with strike K_i and the nearest strangle around that strike. These options are mostly comprised of time value, and this value takes into account the expected move at the current volatility level.

This approach can be generalised as follows: Given $K_{i \pm L} \in \mathcal{K}$, define the strangles

$$\phi_i^{(l)}(T) = \hat{S}_{BS}(t, x, K_{i-l}, K_{i+l}, T) \text{ for } l = 0, \dots, L$$

with the special case $\phi_i^{(0)}(T) = \hat{S}_{BS}(t, x, K_i, K_i, T) = S_{BS}(t, x, K_i, T)$. We can express the expected stock move Φ as the weighted sum of strangle prices

$$\Phi(T) = \sum_{l=0}^L w_l \phi_i^{(l)} \quad (15.130)$$

We are left with estimating the weights $\{w_l; l = 0, \dots, L\}$. An example could be a decreasing function with respect to $l \times \Delta K$ such that the exponential function

$$w_l = e^{-\alpha \times l \times \Delta K}$$

Then, renormalise the weights so that they sum to one.

16 The linear equation

16.1 Solving linear equations

Following Karatzas and Shreve [1991], we consider the linear stochastic differential equation (SDE)

$$dX_t = (K_0(t) + K_1(t)X_t)dt + \Sigma_X(t)dW_t \text{ with } X_0 = \xi$$

where $X_t \in \mathbb{R}^n$, $K_0 \in \mathbb{R}^n$ and $K_1 \in \mathbb{R}^{n \times n}$ with the corresponding deterministic differential equation

$$\dot{Q}(t) = K_0(t) + K_1(t)Q(t) \text{ with } Q_0 = Q$$

where $Q_t \in \mathbb{R}^n$. It has the associated homogeneous equation

$$\dot{Q}(t) = K_1(t)Q(t)$$

where standard calculus guarantee a unique, absolutely continuous solution to this initial value problem. In the special case where the matrix $K_1(t)$ is constant, the solution to the linear system is

$$Q(t) = e^{K_1 t} Q(0)$$

Moler et al. [2003] showed that there are many different ways to compute the exponential of a matrix and tried to classify them according to some criterions such as accuracy and efficiency. We are now going to use Q_t as an integrating factor to solve for X_t . We first take its inverse $Q^{-1}(t)$ and differentiate it with respect to time t , getting

$$\frac{dQ^{-1}(t)}{dt} = -Q^{-1}(t) \frac{dQ(t)}{dt} Q^{-1}(t)$$

We then apply Ito's lemma to get the dynamic of the product of the multi-factor Ornstein-Uhlenbeck process with the matrix $Q^{-1}(t)$

$$dQ^{-1}(t)X_t = Q^{-1}(t)K_0(t)dt + Q^{-1}(t)\Sigma_X dW_X^T(t)$$

Integrating between $[t, T]$, we get the solution

$$X_T = Q^{-1}(t)X_t + \int_t^T Q^{-1}(s)K_0(s)ds + \int_t^T Q^{-1}(s)\Sigma_X(s)dW_X^T(s) \quad (16.131)$$

where $\int_t^T Q^{-1}(s)\Sigma_X(s)dW_X(s)$ is a $(n, 1)$ stochastic integral. In that case, the conditional mean vector $M(t, T)$ and variance matrix $V(t, T)$ of the process X_t are

$$\begin{aligned} M(t, T) &= Q^{-1}(t)X_t - \int_t^T Q^{-1}(s)K_0(s)ds \\ V(t, T) &= \int_t^T Q^{-1}(s)\Sigma_X\Sigma_X^T(s)(Q^{-1}(s))^T ds \end{aligned}$$

16.2 A simple example

We consider a multivariate Ornstein-Uhlenbeck process $X = (X_t \in \mathbb{R}^2)_{t \in [0, T]}$ with initial value $X_0 \in \mathbb{R}^2$ defined by the SDE

$$dX_t = (K_0(t) - K_1 X_t)dt + \Sigma(t)dW_t$$

with $K_0(t) \in \mathbb{R}^{2 \times 2}$, $\Sigma(t) \in \mathbb{R}^{2 \times 2}$ and $W_t \in \mathbb{R}^2$. The solution given in Equation (16.131) can be rewritten as

$$X_t = e^{-(t-s)K_1} X_s + \int_s^t e^{-(t-u)K_1} K_0(u)du + \int_s^t e^{-(t-u)K_1} \Sigma(u)dW_u$$

16.2.1 Covariance matrix

We have

$$Var(X_t|X_s) = Var(e^{-(t-s)K_1} X_s + \int_s^t e^{-(t-u)K_1} K_0(u)du + \int_s^t e^{-(t-u)K_1} \Sigma(u)dW_u)$$

and then

$$Var(X_t|X_s) = Var(e^{-(t-s)K_1} X_s + \int_s^t e^{-(t-u)K_1} K_0(u) du) + Var(\int_s^t e^{-(t-u)K_1} \Sigma(u) dW_u)$$

which simplifies to

$$Var(X_t|X_s) = Var(\int_s^t e^{-(t-u)K_1} \Sigma(u) dW_u)$$

From the definition of the variance we get

$$Var(X_t|X_s) = E[(\int_s^t e^{-(t-u)K_1} \Sigma(u) dW_u)(\int_s^t e^{-(t-u)K_1} \Sigma(u) dW_u)^\top]$$

From Itô's isometry we get

$$Var(X_t|X_s) = E[\int_s^t (e^{-(t-u)K_1} \Sigma(u))(e^{-(t-u)K_1} \Sigma(u))^\top du]$$

which becomes

$$Var(X_t|X_s) = \int_s^t (e^{-(t-u)K_1} \Sigma(u))(e^{-(t-u)K_1} \Sigma(u))^\top du$$

16.2.2 Expectation

From the Equation (16.131) we have the conditional expectation

$$E[X_t|X_s] = E[e^{-(t-s)K_1} X_s + \int_s^t e^{-(t-u)K_1} K_0(u) du + \int_s^t e^{-(t-u)K_1} \Sigma(u) dW_u]$$

which we split as

$$E[X_t|X_s] = E[e^{-(t-s)K_1} X_s + \int_s^t e^{-(t-u)K_1} K_0(u) du] + E[\int_s^t e^{-(t-u)K_1} \Sigma(u) dW_u]$$

The conditional expectation becomes

$$E[X_t|X_s] = e^{-(t-s)K_1} X_s + \int_s^t e^{-(t-u)K_1} K_0(u) du$$

16.2.3 Distribution and probability

We have proven that X_t is a normal multivariate random variable with mean $M_t = E[X_t|X_0] \in \mathbb{R}^2$ and variance $V_t = Var(X_t|X_0) \in \mathbb{R}^2$ (We can write $X_t \sim \mathcal{N}(M_t, V_t)$). In order to compute the probability of the first element, we assume that V_t is diagonalisable. We can then decompose it as $V_t = U\Lambda U^{-1}$, with $U \in \mathbb{R}^{2 \times 2}$ being the unit eigenvectors and $\Lambda \in \mathbb{R}^{2 \times 2}$ the diagonal matrix of the eigenvalues. Since $X_t \sim \mathcal{N}(M_t, V_t)$, we have:

$$X_t = M_t + U\Lambda^{\frac{1}{2}}Y$$

with $Y \sim \mathcal{N}(0, I)$. Let's have the following matrix representation:

$$X_t = \begin{pmatrix} X \\ \bar{X} \end{pmatrix}, A = U\Lambda^{\frac{1}{2}} = \{A_{ij}\}_{i=1,2,j=1,2}, M_t = \{M_i\}_{i=1,2} \text{ and } Y = \{y_i\}_{i=1,2}, \text{ then:}$$

$$\begin{pmatrix} X \\ \bar{X} \end{pmatrix} = \begin{pmatrix} M_1 \\ M_2 \end{pmatrix} + \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$$

Then we have the first element of our process corresponding to:

$$X = M_1 + A_{11}Y_1 + A_{12}Y_2$$

Let us have $\bar{Y} \sim \mathcal{N}(0, 1)$, then from the independence of Y_1 and Y_2 , we get

$$X = M_1 + \sqrt{A_{11}^2 + A_{12}^2} \bar{Y}$$

Let us define $m_{X_t} = M_1$ and $\sigma_X = \sqrt{A_{11}^2 + A_{12}^2}$, then we have $X_t \sim \mathcal{N}(m_{X_t}, \sigma_X^2)$ and we can then compute the probability:

$$\begin{aligned} P(X_t < a) &= P(m_{X_t} + \sigma_X \bar{Y} < a) = P\left(\bar{Y} < \frac{a - m_{X_t}}{\sigma_X}\right) \\ P(X_t < a) &= \Phi\left(\frac{a - m_{X_t}}{\sigma_X}\right) \end{aligned}$$

where Φ being the cumulative distribution function of a standard normal random variable. As a result, the probability becomes

$$P(X_t \geq a) = 1 - P(X_t < a) = 1 - \Phi\left(\frac{a - m_{X_t}}{\sigma_X}\right) = \Phi\left(\frac{m_{X_t} - a}{\sigma_X}\right)$$

16.3 From OU to AR(1) process

ARMA models are mathematical models of the persistence or autocorrelation in a time series. One subset of ARMA models are the autoregressive (AR) models and express a time series as a linear function of its past values. The order of the AR model tells how many lagged past values are included. Discretising an Ornstein-Uhlenbeck process, we recover the $AR(1)$ model.

16.3.1 The Ornstein-Uhlenbeck process

Assuming $(X_t)_{t \geq 0}$ is an Ornstein-Uhlenbeck process with positive constant parameters ρ and σ , the dynamics of the model are

$$dX_t = \rho(\bar{X} - X_t)dt + \sigma dW_t \quad (16.132)$$

where $\bar{X} = \frac{b}{\rho}$ is the long-term mean. Note, X mean-reverts to $\bar{X}$ at infinity.

In that model, the distribution of future values depends on the current value. The solution of the SDE is

$$\begin{aligned} X_t &= X_0 e^{-\rho t} + \rho \int_0^t \bar{X} e^{-\rho(t-s)} ds + Z_X(0, t) \\ &= X_0 e^{-\rho t} + \bar{X}(1 - e^{-\rho t}) + Z_X(0, t) \end{aligned}$$

where $Z_X(0, t) = \int_0^t \sigma e^{-\rho(t-s)} dW(s)$ is normally distributed with mean equal to zero and variance $V_{Z_X}(0, t) = \int_0^t \sigma^2 (e^{-\rho(t-s)})^2 ds = \frac{\sigma^2}{2\rho} (1 - e^{-2\rho t})$. The mean and variance are

$$\begin{aligned} E[X_t] &= X_0 e^{-\rho t} + \bar{X}(1 - e^{-\rho t}) \\ Var(X_t) &= \frac{\sigma^2}{2\rho} (1 - e^{-2\rho t}) \end{aligned}$$

The stationary (or unconditional) mean and variance are computed in the limit by letting the time to infinity

$$\begin{aligned} E[X_t] &= \bar{X} \\ \text{Var}(X_t) &= \frac{\sigma^2}{2\rho} \end{aligned} \tag{16.133}$$

Without loss of generality, we set $\bar{X} = 0$ since for any $\alpha \in \mathbb{R}$ the process $(X_t - \alpha)$ is also an OU process. Hence, if $X_0 = x$ with probability 1, then X_t has the distribution

$$(X_t | X_0 = x) \sim N\left(xe^{-\rho t}, \frac{\sigma^2}{2\rho}(1 - e^{-2\rho t})\right)$$

and we recover our Gaussian distribution with $\alpha(0, t) = xe^{-\rho t}$ and variance $V_{Z_X}(0, t)$. The covariance functions are given by

$$\text{Cov}(X_t, X_s | X_0 = x) = \frac{\sigma^2}{2\rho}(e^{-\rho|t-s|} - e^{-\rho(t+s)})$$

16.3.2 Deriving the discrete model

Given the dynamics of the OU process in Equation (16.132), we let $T = N\Delta t$ and assume that historical data satisfies the discretised dynamics

$$X_n = \rho\bar{X}\Delta t + (1 - \rho\Delta t)X_{n-1} + \sigma\sqrt{\Delta t}y, \quad n = 1, \dots, N$$

where $y \sim N(0, 1)$, that is, the distribution of y is time independent. Given the initial condition X_0 , the parameters ρ and σ are unknown and must be estimated. We consider a parametric model where the conditional mean and variance of X_n given $X_{n-1} = x_{n-1}$ belong to the families

$$\begin{aligned} &\{\rho\bar{X}\Delta t + (1 - \rho\Delta t)x_{n-1}, \rho \in \mathbb{R}\} \\ &\{\sigma^2\Delta t, \sigma^2 \in \mathbb{R}^+\} \end{aligned}$$

Setting $K_1 = e^{-\rho\Delta t} \approx (1 - \rho\Delta t)$ and $\bar{\sigma} = \sigma\sqrt{\Delta t}$, we rewrite the process as

$$X_n = C + K_1X_{n-1} + \bar{\sigma}y_n \tag{16.134}$$

for $C = (1 - K_1)\bar{X}$. The sequence $X_0, \dots, X_N$ is a first order autoregressive sequence with lag-one correlation coefficient K_1 . Positive autocorrelation might be considered a specific form of persistence, a tendency for a system to remain in the same state from one observation to the next. Interpolating linearly the values $X_n = X(n\Delta t)$ for $n \in [1, N]$ we recover the desired path. The equation for X_n is the recursive representation of the $AR(1)$ with conditional mean and variance

$$\begin{aligned} E[X_n | X_{n-1}] &= C + K_1X_{n-1} \\ \text{Var}(X_n | X_{n-1}) &= \bar{\sigma}^2 \end{aligned}$$

If we lag that equation by p periods where $p = 1$ is the original equation and substitute each time the result back in the first equation, we get

$$X_n = C \sum_{j=0}^{p-1} K_1^j + K_1^p X_{n-p} + \bar{\sigma} \sum_{j=0}^{p-1} K_1^j y_{n-j} \tag{16.135}$$

where

$$C \sum_{j=0}^{p-1} K_1^j = (1 - K_1) \bar{X} \sum_{j=0}^{p-1} K_1^j = \bar{X} \sum_{j=0}^{p-1} K_1^j - K_1 \bar{X} \sum_{j=0}^{p-1} K_1^j = \bar{X} \sum_{j=0}^{p-1} K_1^j - \bar{X} \sum_{j=0}^{p-1} K_1^{j+1}$$

Assuming stationarity, that is $|K_1| < 1$, and taking the limit $p \rightarrow \infty$ then K_1^p will approach zero and from the infinite geometric series³⁰, we get

$$X_n = \bar{X} + \bar{\sigma} \sum_{j=0}^{\infty} K_1^j y_{n-j}$$

since $\frac{C}{1-K_1} = \bar{X}$, which is an infinite order moving average. Therefore, the variable X_n can be written as an infinite sum of past shocks where most distant shocks get smaller and smaller weights. In fact, the coefficients are geometrically declining since $|K_1| < 1$. Hence, when $K_1 = 1$ we get the unit root case and X_n has infinite memory, that is past shocks never dies out. As a result, the distant past matters more the closer K_1 is from unity. Since each y_{n-j} is an i.i.d standard normal and since $|K_1| < 1$ we compute the unconditional mean and variance as

$$\begin{aligned} E[X_n] &= \bar{X} \\ Var(X_n) &= \bar{\sigma}^2 \sum_{j=0}^{\infty} K_1^{2j} Var(y_{n-j}) = \frac{\bar{\sigma}^2}{1 - K_1^2} \end{aligned} \quad (16.136)$$

where the unconditional variance is larger than the conditional one when $K_1 \neq 0$. The further we look into the future the larger the number of shocks which come into play so that the set of possibilities or variance grows as a function of K_1 .

16.4 Some facts about AR series

16.4.1 Persistence

Autocorrelation refers to the correlation of a time series with its own past and future values. Positively autocorrelated series are referred to as persistent implying a positive dependence on the past and show up in a time series plot as unusually long runs or stretches of several consecutive observations above or below the mean. In the $AR(1)$ model, the higher K_1 , the more persistence and the longer the series stay above or below the mean. When $K_1 = 1$ we get a random walk and the speed of mean reversion ρ in the continuous counterpart is zero. But as K_1 decreases to zero, the time series mean revert faster and ρ increases. Because the departures for computing autocorrelation are computed relative to the mean, a horizontal line plotted at the sample mean is useful in evaluating autocorrelation with the time series plot. Hence, the equations on the process X_t are somewhat simpler if the time series is first reduced to zero-mean by subtracting the sample mean, that is $V_n = X_n - \bar{X}$ for $n = 1, \dots, N$ where X_n is the original time series, $\bar{X}$ is its sample mean and V_n is the mean-adjusted series. In that case, the $AR(1)$ model for the mean-adjusted series in year n becomes

$$V_n = K_1 V_{n-1} + \bar{\sigma} y_n \quad (16.137)$$

We can deduce the e-folding time constant τ of the $AR(1)$ model at lag Δt from the relation

$$K_1 = e^{-\rho \Delta t} = e^{-\frac{\Delta t}{\tau}}$$

such that $\rho = \frac{1}{\tau}$.

³⁰ $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$

16.4.2 Prewhitening and detrending

Prewhitening refers to the removal of autocorrelation from a time series prior to using the time series in some application. For a series with positive autocorrelation, prewhitening acts to damp those time series features that are characteristic of persistence. In general, a trend is a long term change in the mean but can also refer to change in other statistical properties. Detrending is the statistical operation of removing trend from the series. It can be applied to remove a feature thought to distort the relationships of interest or can be used as a preprocessing step to prepare time series for analysis by methods that assume stationarity. The variance at low frequencies is diminished relative to the variance of high frequencies resulting in lower spectrum (at lowest frequencies) of the detrended series compared with the spectrum of the original data. Simple linear trend in mean can be removed by subtracting a least-square fit straight line.

16.4.3 Simulation and prediction

Simulation is the generation of synthetic time series with the same persistence properties as the observed series. In our example, it effectively mimics the low-frequency behaviour of the observed series. That is, given Equation (16.137) we estimate the autoregressive parameter by modelling the time series as an $AR(1)$ process. Then we generate a time series of random noise ω by sampling from an appropriate distribution. Assuming some starting values for V_{n-1} , we recursively generate a time series of V_n . Usually, we assume that the noise is normally distributed with mean zero and variance equal to the variance of the residuals from fitting the $AR(1)$ model to the data. We can use simulation to generate empirical confidence intervals of the relationship between the observed time series and the climate variable.

Prediction is the extension of the observed series into the future based on past and present values. It differs from simulation in that the objective of prediction is to estimate the future value of the time series as accurately as possible from the current and past values. A prediction form of the $AR(1)$ model in Equation (16.137) is

$$\hat{V}_n = \hat{K}_1 V_{n-1}$$

where the hat indicates an estimate. The equation can be applied one step ahead to get estimate $\hat{V}_n$ from the observed V_{n-1} while k -step ahead prediction can be made by applying recursively the above equation. Note, because the modelling of V_n assume a departure from the mean, the convergence in terms of the original time series is a convergence toward the mean. Therefore, in the limit for large enough k , the predictions will eventually converge to zero. Besides the randomness of the residuals, we are concerned with the statistical significance of the model coefficients. Significance of the $AR(p)$ coefficients can be evaluated by comparing the estimated parameters with the standard deviation. In the $AR(1)$ model, the estimated first order autoregressive coefficient $\hat{K}_1$, is normally distributed with variance

$$Var(\hat{K}_1) = \frac{(1 - \hat{K}_1^2)}{N} \quad (16.138)$$

Therefore, the 95% confidence interval for $\hat{K}_1$ is two standard deviations around $\hat{K}_1$, that is

$$95\% \text{ CI} = \hat{K}_1 \pm 2\sqrt{Var(\hat{K}_1)}$$

17 Introduction to information theory

Shannon, for the first time, introduced the qualitative and quantitative model of communication as a statistical process underlying information theory. That is, information theory is based on probability theory and statistics. It concerns itself with measures of information of the distributions associated with random variables. Important quantities of information are entropy, a measure of information in a single random variable, and mutual information, a measure of information in common between two random variables. Given a discrete random variable X , with x a particular

outcome occurring with probability (mass function) $p(x)$. Then, we assign to that event, x , the information that it conveys via the uncertainty measure

$$\text{uncertainty} = -\log p(x) \quad (17.139)$$

where the base of the logarithms determines the units of information. The units of $\log_2$ are in bits, that of $\log_e$ are in nats.

17.1 Some facts on entropy in information theory

The entropy functional, defined on a Markov diffusion process, plays an important roll in the theory of information and statistical physics, informational macrodynamics and control systems. In information theory (IT), entropy is the average amount of information contained in each message received. That is, entropy is a measure of unpredictability of information content (see Equation (17.139)). Named after Boltzmann's H-theorem, Shannon [1948] defined the entropy H of a discrete random variable X with possible values $\{x_1, \dots, x_n\}$ and probability mass function $P(X)$ as

$$H(X) = E[I(X)] = E[-\log P(X)] \quad (17.140)$$

where $I(X)$ is the information content of X , which is itself a random variable. When taken from a finite sample, the entropy can explicitly be written as

Definition 17.1 The entropy $H(X)$ of a discrete random variable X taking values $\{x_1, \dots, x_n\}$ is

$$H(X) = \sum_{i=1}^n p(x_i) I(x_i) = - \sum_{i=1}^n p(x_i) \log_b p(x_i)$$

where b is the base of the logarithm used.

The unit of entropy is bit for $b = 2$, nat for $b = e$ (where e is Euler's number), and dit (or digit) for $b = 10$. We are now interested in the entropy of pairs of random variables (X, Y) .

Definition 17.2 If X and Y are jointly distributed according to $p(X, Y)$, then the joint entropy $H(X, Y)$ is

$$H(X, Y) = -E[\log p(X, Y)] = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y)$$

Definition 17.3 If $(X, Y) \sim p(x, y)$, the conditional entropy of two events X and Y is given by

$$H(Y|X) = -E_{p(x,y)}[\log p(Y|X)] = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x)$$

This quantity should be understood as the amount of randomness in the random variable Y given that you know the value of X . The conditional entropy can also be written as

$$\begin{aligned} E[Y|X] &= - \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log p(y|x) \\ &= - \sum_{x \in \mathcal{X}} p(x) H(Y|X = x) \end{aligned}$$

Theorem 17.1 chain rule

$$H(X, Y) = H(X) + H(Y|X)$$

We can also have a joint entropy with a conditioning on it.

Corollary 2

$$H(X, Y|Z) = H(X|Z) + H(Y|X, Z)$$

The inspiration for adopting the word entropy in information theory came from the close resemblance between Shannon's formula and very similar known formulae from statistical mechanics. In statistical thermodynamics the most general formula for the thermodynamic entropy S of a thermodynamic system is the Gibbs entropy

$$S = -k_B \sum_i p_i \ln p_i$$

where k_B is the Boltzmann constant, and p_i is the probability of a microstate. The Gibbs entropy translates over almost unchanged into the world of quantum physics to give the von Neumann entropy, introduced by John von Neumann in 1927

$$S = -k_B \text{Tr}(\rho \ln \rho)$$

where ρ is the density matrix of the quantum mechanical system and $\text{Tr}()$ is the trace. Note, at a multidisciplinary level, connections can be made between thermodynamic and informational entropy. In the view of Jaynes (1957), thermodynamic entropy, as explained by statistical mechanics, should be seen as an application of Shannon's information theory: the thermodynamic entropy is interpreted as being proportional to the amount of further Shannon information needed to define the detailed microscopic state of the system, that remains uncommunicated by a description solely in terms of the macroscopic variables of classical thermodynamics, with the constant of proportionality being just the Boltzmann constant.

17.2 Entropy measures and mutual information

17.2.1 Relative entropy

Another useful measure of entropy that works equally well in the discrete and the continuous case is the relative entropy of a distribution. Given a random variable with true distribution p , which we do not know due to incomplete information, instead we assume the (reference) distribution q . Then, the code will need more bits to represent the random variable, and the difference in bits is denoted by $D(p||q)$. The relative entropy is defined as the Kullback-Leibler divergence from the distribution to the reference measure q as follows.

Definition 17.4 *The relative entropy or Kullback-Leibler distance between two probability mass functions $p(x)$ and $q(x)$ is defined as*

$$D_{KL}(p||q) = E_p[\log \frac{p(x)}{q(x)}] = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)}$$

Relative entropy is always non-negative, that is, $D_{KL}(p||q) \geq 0$. This form is not symmetric, and q appears only in the denominator. Alternatively, assume that a probability distribution (density) p is absolutely continuous with respect to a measure q , that is, it is of the form $p(dx) = f(x)q(dx)$ for some non-negative q -integrable function f with q -integral 1, then the relative entropy can be defined as

$$D_{KL}(p||q) = \int \log f(x) p(dx) = \int f(x) \log f(x) q(dx)$$

In this form the relative entropy generalises (up to change in sign) both the discrete entropy, where the measure q is the counting measure, and the differential entropy, where the measure q is the Lebesgue measure. If the measure q is itself a probability distribution, the relative entropy is non-negative, and zero if $p = q$ as measures. It is defined for

any measure space, hence coordinate independent and invariant under co-ordinate reparametrisations if one properly takes into account the transformation of the measure q . The relative entropy, and implicitly entropy and differential entropy, do depend on the reference measure q .

17.2.2 Symmetrised divergence

Kullback-Leibler also considered the symmetrised function

$$D_{KL}(p||q) + D_{KL}(q||p)$$

which is called the KL divergence. It is symmetric and nonnegative, and was defined and used by Harold Jeffreys in 1948. Hence, it is also called the Jeffreys divergence. It has been used for feature selection in classification problems, where p and q are the conditional pdfs of a feature under two different classes.

An alternative formula is given via the λ divergence

$$D_\lambda(p||q) = \lambda D_{KL}(p||\lambda p + (1-\lambda)q) + (1-\lambda) D_{KL}(q||\lambda p + (1-\lambda)q)$$

which can be interpreted as the expected information gain about X from discovering which probability distribution X is drawn from, P or Q , if they currently have probabilities λ and $1-\lambda$, respectively. As an example, for $\lambda = \frac{1}{2}$, we get the Jensen-Shannon divergence, defined by

$$D_{JS} = \frac{1}{2} D_{KL}(p||M) + \frac{1}{2} D_{KL}(q||M)$$

where M is the average of the two distributions, $M = \frac{1}{2}(p+q)$. D_{JS} can also be interpreted as the capacity of a noisy information channel with two inputs giving the output distributions p and q . It is locally proportional to the Fisher information metric. It is also similar to the Hellinger metric (in the sense that it induces the same affine connection on a statistical manifold).

17.2.3 Mutual information

Another important concept, called mutual information, describes the amount of information a random variable tells about another one. That is, observing the output of a channel, we want to know what information was sent. The channel coding theorem is a statement about mutual information.

Definition 17.5 Let X and Y be random variables with joint distribution $p(X, Y)$ and marginal distributions $p(x)$ and $p(y)$. The mutual information $I(X; Y)$ is the relative entropy between the joint distribution and the product distribution

$$\begin{aligned} I(X; Y) &= D(p(x, y)||p(x)p(y)) \\ &= \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \end{aligned}$$

Note, when X and Y are independent, $p(x, y) = p(x)p(y)$, and we get $I(X; Y) = 0$. That is, in case of independence, Y , can not tell us anything about X . An important interpretation of mutual information comes from the following theorem.

Theorem 17.2

$$I(X; Y) = H(X) - H(X|Y)$$

which states that the information that Y tells us about X is the reduction in uncertainty about X due to the knowledge of Y . By symmetry we get

$$I(X; Y) = H(Y) - H(Y|X) = I(Y; X)$$

Using $H(X, Y) = H(X) + H(Y|X)$, we get

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

The information that X tells about Y is the uncertainty in X plus the uncertainty about Y minus the uncertainty in both X and Y . We can summarise statements about entropy as follows

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) \\ I(X; Y) &= H(Y) - H(Y|X) \\ I(X; Y) &= H(X) + H(Y) - H(X, Y) \\ I(X; Y) &= I(Y; X) \\ I(X; X) &= H(X) \end{aligned}$$

When dealing with the sequence of random variables $X_1, \dots, X_n$ drawn from the joint distribution $p(x_1, \dots, x_n)$, a variety of chain rules have been developed.

Theorem 17.3 *The joint entropy of $X_1, \dots, X_n$ is*

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1)$$

The chain rule for entropy leads us to a chain rule for mutual information.

Theorem 17.4

$$I(X_1, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y | X_{i-1}, \dots, X_1)$$

18 Integral probability metrics

We review a few integral probability metrics (see Muller [1997], Dudley [2002]).

Let x and y be random variables defined on a topological space $\mathcal{X}$ with respective Borel probability measures p and q . Given observations $X = \{x_1, \dots, x_m\}$ and $Y = \{y_1, \dots, y_n\}$ independent and identically distributed (i.i.d.) from p and q , respectively, we want to measure the disparity between p and q . That is, we want to decide whether the null hypothesis $p = q$ or not. Denote $E_x[f(x)] = E_{x \sim p}[f(x)]$ and $E_y[f(y)] = E_{y \sim q}[f(y)]$ the expectations with respect to p and q , respectively.

18.1 The maximum mean discrepancy

We briefly present the maximum mean discrepancy (MMD) proposed Gretton et al. [2012].

Lemma 18.1 *Let $(\mathcal{X}, d)$ be a metric space, and let p, q be two Borel probability measures defined on $\mathcal{X}$. Then, $p = q$ if and only if $E_x[f(x)] = E_y[f(y)]$ for all $f \in C(\mathcal{X})$, where $C(\mathcal{X})$ is the space of bounded continuous functions on $\mathcal{X}$.*

The authors defined a more general class of statistic, for the unspecified function classes $\mathcal{F}$, to measure the disparity between p and q .

Definition 18.1 Let $\mathcal{F}$ be a class of functions $f : \mathcal{X} \rightarrow \mathbb{R}$. We define the maximum mean discrepancy (MMD) as

$$MMD[\mathcal{F}, p, q] = \sup_{f \in \mathcal{F}} (E_x[f(x)] - E_y[f(y)])$$

This is an integral probability metric. A biased empirical estimate of the MMD is obtained by replacing the population expectations with empirical expectations computed on the samples X and Y ,

$$MMD_b[\mathcal{F}, X, Y] = \sup_{f \in \mathcal{F}} \left(\frac{1}{m} \sum_{i=1}^m f(x_i) - \frac{1}{n} \sum_{i=1}^n f(y_i) \right)$$

We must therefore identify a function class that is rich enough to uniquely identify whether $p = q$, yet restrictive enough to provide useful finite sample estimates. The authors considered the unit ball in a reproducing kernel Hilbert space (RKHS), denoted $\mathcal{H}$. From the Moore-Aronszajn theorem, the function $f : \mathcal{X} \rightarrow \mathbb{R}$ equipped with the inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and norm $\| \cdot \|_{\mathcal{H}}$ satisfies the reproducing property

$$\forall f \in \mathcal{H}, \forall x \in \mathcal{X} \quad \langle f, k(x, \cdot) \rangle_{\mathcal{H}} = f(x)$$

where $k(x, \cdot)$ is a kernel function. Since $\mathcal{H}$ is an RKHS, the operator of evaluation δ_x mapping $f \in \mathcal{H}$ to $f(x) \in \mathbb{R}$ is continuous. Thus, by the Riesz representation theorem, there is a feature mapping $\phi(x)$ from $\mathcal{X}$ to $\mathbb{R}$ such that $f(x) = \langle f, \phi(x) \rangle_{\mathcal{H}}$. It takes the canonical form $\phi(x) = k(x, \cdot)$, where the kernel $k(x_1, x_2) : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is positive definite. In particular $\langle \phi(x), \phi(y) \rangle_{\mathcal{H}} = k(x, y)$. We define an element $\mu_p \in \mathcal{H}$ such that $E_x[f] = \langle f, \mu_p \rangle_{\mathcal{H}}$, for all $f \in \mathcal{H}$. It is called the mean embedding of p .

Lemma 18.2 Assume the existence of μ_p and μ_q . Then,

$$MMD^2[\mathcal{F}, p, q] = \|\mu_p - \mu_q\|_{\mathcal{H}}^2$$

Theorem 18.1 Let $\mathcal{F}$ be a unit ball in a universal RKHS $\mathcal{H}$, defined on the compact metric space $\mathcal{X}$, with associated continuous kernel $k(\cdot, \cdot)$. Then, $MMD[\mathcal{F}, p, q] = 0$ if and only if $p = q$.

Thus, the MMD refers to the fact that computing the MMD is equivalent to finding the RKHS function that maximises the difference in expectations between the two probability distributions

$$MMD[\mathcal{F}, p, q] = \sup_{\|f\|_{\mathcal{H}} \leq 1} (E_x[f(x)] - E_y[f(y)])$$

Note, we can obtain MMD^2 in terms of kernel functions, and a corresponding unbiased finite sample estimate.

Lemma 18.3 Given x and x' independent random variables with distribution p , and y and y' independent random variables with distribution q , the squared population MMD is

$$MMD^2[\mathcal{F}, p, q] = E_{x, x'}[k(x, x')] - 2E_{x, y}[k(x, y)] + E_{y, y'}[k(y, y')]$$

where x' is an independent copy of x with the same distribution, and y' is an independent copy of y .

An unbiased empirical estimate is a sum of two U-statistics and a sample average:

$$MMD^2[\mathcal{F}, p, q] = \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{j \neq i}^m k(x_i, x_j) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(x_i, y_j) + \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n k(y_i, y_j)$$

18.2 Binary classification

The analysis of distributions is fundamental in machine learning (ML) and statistics, and many algorithms in these fields rely on information theoretic approaches such as entropy, mutual information, or Kullback-Leibler divergence (see Appendix (17)). However, to estimate these quantities, one must first either perform density estimation (either parametrically or nonparametrically), or employ sophisticated space-partitioning/bias-correction strategies which are typically infeasible for high-dimensional data. Methods for modelling complex distributions rely on parametric assumptions that may be unfounded or computationally challenging (such as Gaussian mixture models), while non-parametric methods like kernel density estimation or characteristic function representation (via the Fourier transform of the distribution) break down in high-dimensional settings. In ML, the kernel embedding of distributions is a class of nonparametric methods in which a probability distribution is represented as an element of a RKHS. They avoid the above problems. As an example, the MMD is easily estimated as an empirical mean which is concentrated around the true value of the MMD. For instance, Friedman [2003] showed that any classifier which maps a set of observations $\{z_i, l_i\}$ with $z_i \in \mathcal{X}$ on some domain $\mathcal{X}$ and labels $l_i \in \{\pm 1\}$, for which uniform convergence bounds exist on the convergence of the empirical loss to the expected loss, can be used to obtain a similarity measure on distributions. Just assign $l_i = 1$ if $z_i \in X$ and $l_i = -1$ for $z_i \in Y$ and find a classifier which is able to separate the two sets. Here, the maximisation of $E_x[f] - E_y[f]$ is obtained by ensuring that as many $z \sim p(z)$ as possible correspond to $f(z) = 1$, whereas for as many $z \sim q(z)$ as possible we have $f(z) = -1$. As a result, neural networks, decision trees, boosted classifiers and other objects for which uniform convergence bounds can be obtained can be used for the purpose of distribution comparison. Thus, metrics and divergences on distributions can also be defined explicitly starting from classifiers. For instance, Sriperumbudur et al. [2009] showed that the MMD minimises the expected risk of a classifier with linear loss on the samples X and Y .

18.3 Non-RKHS function classes

There are other function spaces $\mathcal{F}$ that can be used to define the MMD.

18.3.1 The Kolmogorov-Smirnov statistic

Denote $F_p(x)$ the cumulative distribution function of p and let $F_X(x)$ be its empirical counterpart

$$F_p(z) = P_{x \sim p}(x \leq z) \text{ and } F_X(z) = \frac{1}{|X|} \sum_{i=1}^m I_{\{z \leq x_i\}}$$

Clearly, F_p captures the properties of p . The Kolmogorov metric is the L_∞ distance $\|F_X - F_Y\|_\infty$ for two sets of observations X and Y . Smirnov (1939) showed that, for $p = q$, the limiting distribution of the empirical cumulative distribution functions satisfies

$$\lim_{m, n \rightarrow \infty} P_{X, Y} \left(\frac{mn}{m+n} \|F_X - F_Y\|_\infty > x \right) = 2 \sum_{j=1}^{\infty} (-1)^{j-1} e^{-2j^2 x^2}, \quad x \geq 0$$

which is distribution independent. It allows for an efficient characterisation of the distribution under the null hypothesis $\mathcal{H}_0$. Note, the distribution under the alternative $p \neq q$ is unknown. The Kolmogorov metric is a special instance of $MMD[\mathcal{F}, p, q]$ for a certain Banach space.

Proposition 3 *Let $\mathcal{F}$ be the class of functions $\mathcal{X} \rightarrow \mathbb{R}$ of bounded variation 1. Then,*

$$MMD[\mathcal{F}, p, q] = \|F_p - F_q\|_\infty$$

18.3.2 The Earth-Mover distances

We assume $(\mathcal{X}, \rho)$ is a separable metric space, and define $\mathcal{P}_1(\mathcal{X})$ to be the space of probability measures on $\mathcal{X}$ for which $\int \rho(x, z) dp(z) < \infty$ for all $p \in \mathcal{P}_1(\mathcal{X})$ and $x \in \mathcal{X}$. These are the probability measures for which $E_x[\|x\|] < \infty$ when $\mathcal{X} = \mathbb{R}$. We have the following definition:

Definition 18.2 *Monge-Wasserstein metric*

Let $p \in \mathcal{P}_1(\mathcal{X})$ and $q \in \mathcal{P}_1(\mathcal{X})$. The Monge-Wasserstein distance is defined as

$$W(p, q) = \inf_{\mu \in M(p, q)} \int \rho(x, y) d\mu(x, y)$$

where $M(p, q)$ is the set of joint distributions on $\mathcal{X} \times \mathcal{X}$ with marginals p and q .

This is the cost (represented by the metric $\rho(x, y)$) of transferring mass distributed according to p to a distribution in accordance with q , where μ is the movement schedule. Note, a large variety of costs of moving mass from x to y can be used. The following theorem provides the link with the MMD.

Theorem 18.2 *Kantorovich-Rubinstein*

Let $p \in \mathcal{P}_1(\mathcal{X})$ and $q \in \mathcal{P}_1(\mathcal{X})$, where $\mathcal{X}$ is separable. Then a metric on $\mathcal{P}_1(\mathcal{X})$ is defined as

$$W(p, q) = \|p - q\|_L^* = \sup_{\|f\|_L \leq 1} \left\| \int f d(p - q) \right\|$$

where

$$\|f\|_L = \sup_{x \neq y \in \mathcal{X}} \frac{|f(x) - f(y)|}{\rho(x, y)}$$

is the Lipschitz seminorm for real valued f on $\mathcal{X}$.

As an example, let $\mathcal{X} = \mathbb{R}$ with associated $\rho(x, y) = |x - y|$. Then, given f such that $\|f\|_L \leq 1$, we use integration by parts to obtain

$$\left\| \int f d(p - q) \right\| = \left\| \int (F_p - F_q)(x) f'(x) dx \right\| \leq \int |F_p - F_q|(x) dx$$

where the maximum is attained for the function g with derivative $g' = 2I_{\{F_p > F_q\}} - 1$ and for which $\|g\|_L = 1$. We recover the L_1 distance between distribution functions

$$W(p, q) = \int |F_p - F_q|(x) dx$$

One can further generalise the above Theorem to the set of all laws $\mathcal{P}(\mathcal{X})$ on arbitrary metric spaces $\mathcal{X}$.

Definition 18.3 *Bounded Lipschitz metric*

Let p and q be laws on a metric space $\mathcal{X}$. Then,

$$\beta(p, q) = \sup_{\|f\|_{BL} \leq 1} \left\| \int f d(p - q) \right\|$$

is a metric on $\mathcal{P}(\mathcal{X})$, where f belongs to the space of bounded Lipschitz functions with norm

$$\|f\|_{BL} = \|f\|_L + \|f\|_\infty$$

Sriperumbudur et al. [2010] provided empirical estimates of the Monge-Wasserstein and Bounded Lipschitz metrics on $\mathbb{R}^d$.

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